Topics on Electrical Engineering

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M.Sc Non-Linear Electronics

Lecture Book

2025.WS



2025.WS

Non-Linear Electronics

Lecture Book

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MCI

(2025, D. T. McGuines, Ph.D)

Current version is 2025.WS.

This document includes the contents of Non Linear Electronics, official name being *Nichtlineare Elektronik*, taught at MCI in the Mechatronik Smart Technologies. This document is the part of the module MECH-M-1-EDY-NLE-VO taught in the M.Sc degree.

All relevant code of the document is done using *SageMath* where stated and Python v3.13.7.

This document was compiled with LuaTEX v1.22.0, and all editing were done using GNU Emacs v30.1 using AUCTEX and org-mode package.

This document is based on the following books and resources shown in no particular order:

Linear And Nonlinear Circuits by L. Chua , Mc Graw Hill India *Introduction to Electric Circuits (9th Edition)* by J. A. Svoboda, R. C. Dorf , Wiley *Microelectronic Circuits (7th Edition)* by K. Smith , Oxford *Electrical and Electronic Technology (10th Edition)* by E. Hughes , Pearson *Electronic Amplifier Circuits: Theory and Design* by J. M. Petttit, et. al , McGraw-Hil

The document is designed with no intention of publication and has only been designed for education purposes.

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Part I

Circuit Fundamentals

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It has today occurred to me that an amplifier using semiconductors rather than vacuum is in principle possible.

(William Shockley, Laboratory notebook, 29 Dec 1939.)

Chapter 1

Signals and Amplifiers

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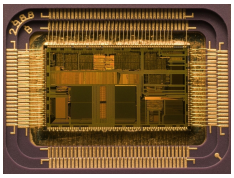
The main idea of this lecture is **modern electronics**, a field that has come to be known as microelectronics.[1] Microelectronics refers to the Integrated Circuit (IC) technology, which is capable of producing circuits containing billions of components in a small piece of silicon whose area is on the order of 100 mm^2. The microelectronic circuits we will learn in this lecture are used in almost every device we encounter in our daily lives:

> in appliances we use in our homes, in vehicles and transportation systems we use to travel, in cell phones we use to communicate, in medical equipment we need to care for our health, in the computers we use to do our work, and in audio and video systems, the radio and Television sets, and the multitude of other digital devices we use.

It is difficult to conceive of modern life without microelectronic circuits.

In this lecture we will study electronic devices which can be used individually[2] or as components of an IC chip. We shall study the design and analysis of interconnections of these devices, which form discrete and integrated circuits of varying complexity and perform a wide variety of functions. We shall also learn about available IC chips and their application in the design of electronic systems.

In this introductory chapter, we will learn about signals and about one of the most important signal-processing functions electronic circuits are designed to perform, namely, signal amplification. We shall then look at circuit representations or models for linear amplifiers. These models will be employed in later chapters in the design and analysis of actual amplifier circuits.



[1] **Microelectronics** is a sub-field of electronics, which relates to the study and manufacture of very small electronic designs and components. Usually, this means micrometre-scale or smaller and are typically made from semiconductor materials. Many components of a normal electronic design are available in a microelectronic equivalent. These include transistors, capacitors, inductors, resistors, diodes and insulators and conductors can all be found in microelectronic devices.

[2] in the design of discrete circuits



Figure 1.1
Two alternative representations of a signal source: (a) the Thévenin form; (b) the Norton form.

1.1 Fundamentals of Signals

1.1.1 Proper Definition

Signals contain information about a variety of things and activities in our physical world.

- Information about the weather is contained in signals representing the air temperature, pressure, wind speed, etc.
- The voice of a radio announcer reading the news into a microphone provides an acoustic signal that contains information about world affairs.
- To monitor the status of a nuclear reactor, instruments are used to measure a multitude of relevant parameters, each instrument producing a signal.

To extract required information from a set of signals, the observer[3] invariably needs to **process** these signals in some predetermined manner. This signal processing is usually most conveniently performed by *electronic systems*. For this to be possible, however, the signal must first be converted into an electrical signal, that is, either to **voltage** or a **current**.

[3] this can be a human or a machine.

> This process is accomplished by devices known as **transducers**.[4]

[4] This conversion process has been covered extensively in **B.Sc Drive Technology**.

A variety of transducers exist, each suitable for one of the various forms of physical signals.

> For instance, the sound waves generated by a human can be converted into electrical signals by using a microphone, which is in effect a pressure transducer.

However, it is **NOT** our purpose here to study transducers. For simplicity and keep things relevant to the lecture, we will assume the signals of interest already exist in the electrical domain and just represent them by one of the two (2) equivalent forms shown in **Fig.** 1.1.

In **Fig.** 1.1a the signal is represented by a voltage source $v_s(t)$ having a source resistance R_s. In the

alternate representation of **Fig.** 1.1b the signal is represented by a current source $i_s(t)$ having a source resistance R_s.

Although the two (**2**) representations are equivalent:

1. **Fig.** 1.1a, known as the Thévenin form, is preferred when R_s is low.
2. **Fig.** 1.1b, known as the Norton form, is preferred when R_s is high.

1.1 Information

Representation is Important

It is important to be familiar with Thévenin's and Norton's theorems and to note that for the two (**2**) representations in **Fig.** 1.1 to be equivalent, their parameters are related by the following relation:

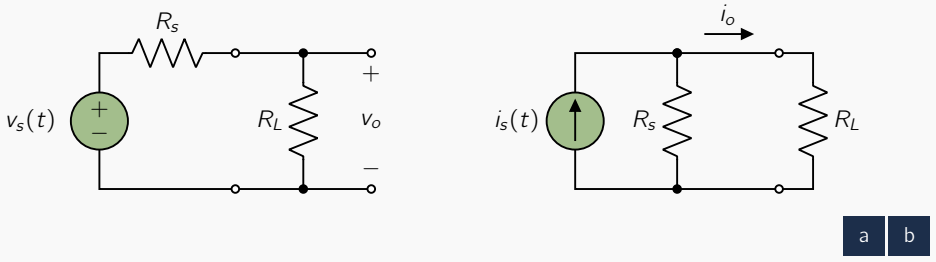
$$v_s(t) = R_s i_s(t)$$

1.1 Exercise

A Simple Thevenin Analysis

The output resistance of a signal source, although inevitable, is an imperfection that limits the ability of the source to deliver its full signal strength to a **load**. To see this point more clearly, consider the signal source when connected to a load resistance R_L as shown in **Fig.** 1.2.

Figure 1.2
Circuits for the example



For the case in which the source is represented by its Thévenin equivalent form, find the voltage v_o that appears across R_L, and therefore the condition that R_s must satisfy for v_o to be close to the value of v_s.

Repeat for the Norton-represented source. In this case finding the current i_o that flows through R_L and therefore the condition that R_s must satisfy for i_o to be close to the value of i_s.

For the Thévenin-represented signal source shown in **Fig.** 1.2a, the output voltage v_o that appears across the load resistance R_L can be found from the ratio of the voltage divider formed by R_s and R_L,

$$v_o = v_s \frac{R_L}{R_L + R_s}$$

From this equation we see that for 5

$$v_o \simeq v_s$$

the source resistance R_s must be much lower than the load resistance R_t,

$$R_s \ll R_L$$

Continues on next page →

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Therefore, for a source represented by its Thévenin equivalent, ideally $R_s = 0$, and as R_s is increased, relative to the load resistance R_L with which this source is intended to operate, the voltage v_o that appears across the load becomes smaller, not a desirable outcome.

Next, we consider the Norton-represented signal source in **Fig.** 1.2b. To obtain the current i_o which flows through the load resistance R_L, we utilize the ratio of the current divider formed by R_s and R_L,

$$i_o = i_s \frac{R_s}{R_s + R_t}$$

From this relationship we see that for

$$i_o \simeq i_s$$

the source resistance R_s must be much larger than R_L,

$$R_s \gg R_L$$

Thus for a signal source represented by its Norton equivalent, ideally $R_s = \infty$, and as R_s is reduced, relative to the load resistance R_L with which this source is intended to operate, the current i_o that flows through the load becomes smaller, not a desirable outcome.

Finally, we note that although circuit designers cannot usually do much about the value of R_s, they may have to devise a circuit solution that minimizes or eliminates the loss of signal strength that results when the source is connected to the load.

From the discussion above, it should be obvious to an engineer that a signal is a **time-varying quantity** which can be represented by a graph such as that shown in **Fig.** 1.3. In fact, the information content of the signal is represented by the **changes in its magnitude** as time progresses.

That is, the information is contained in the **wiggles** in the signal waveform.

In general, such waveforms are difficult to characterise mathematically. In other words, it is **NOT** easy to describe clearly an arbitrary-looking waveform such as that of **Fig.** 1.3. Of course, such a description is of great importance for the purpose of designing appropriate signal-processing circuits that perform desired functions on the given signal.[5]

[5] Of course, this limitation can be overcome with proper numerical approximations.

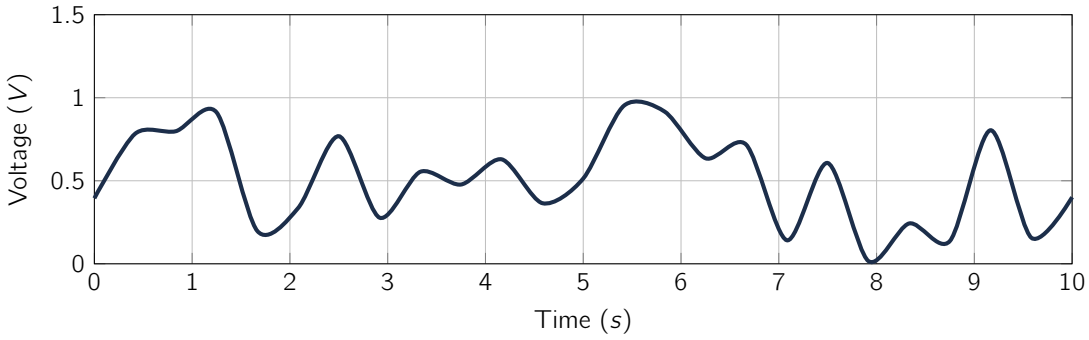


Figure 1.3
A random signal. This kind of signal could be generated from a wide variety of sources

1.1.2 Frequency Spectrum of Signals

An extremely useful characterisation of a signal, and for that matter of any arbitrary function of time, is in terms of its **frequency spectrum**. Such a description of signals is obtained through the mathematical tools of *Fourier series* and *Fourier transform*. We are not interested here in the details of these transformations.[6] To give a brief explanation, they provide the means for representing a voltage signal $v_s(t)$ or a current signal $i_s(t)$ as the sum of sine-wave signals of different frequencies and amplitudes.

[6] The mathematical intricacies and details are discussed in detail in **M.Sc Higher Mathematics II** which is also given by the author.

> This makes the **sine wave** a very important signal in the analysis, design, and testing of electronic circuits.

Therefore, we shall briefly review the properties of the sinusoid.

Figure 1.4 shows a sine-wave voltage signal $v_a(t)$,

$$v_a(t) = V_a \sin \omega t$$

where:

- V_a denotes the **peak value**[7] in volts,
- ω denotes the angular frequency in radians per second; where $\omega = 2\pi f$ rad$\cdot$s^{-1}, f is frequency in hertz, $f = 1/T$ Hz, and T is the period in seconds.

[7] or amplitude

The sine-wave signal, as we can observe, is characterised by its peak value V_a, its frequency ω, and its phase with respect to an arbitrary reference time. In the case depicted in Fig. 1.4, the time origin has been chosen so that the phase angle is 0.

It should be mentioned that it is common to express the amplitude of a sine-wave signal in terms of its root-mean-square (rms) value, which is equal to the peak value divided by $\sqrt{2}$.[8] Therefore the rms value of the sinusoid $v_a(t)$ of Fig. 1.4 is $V_a/\sqrt{2}$.

[8] A simple look at the signal would show that the average value of AC would give zero. Therefore we use RMS to compare both alternating and direct currents (or voltage). In other words (as an example), the RMS value of AC (current) is the direct current which when passed through a resistor for a given period of time would produce the same heat as that produced by alternating current when passed through the same resistor for the same time.

> For instance, when we speak of the wall power supply in our homes as being 120 V, we mean that it has a sine waveform of $120\sqrt{2}$ volts peak value.

Returning now to the representation of signals as the sum of sinusoids, we note that the Fourier series is utilized to accomplish this task for the special case of a signal that is a **periodic function** of time.[9] The Fourier series allows us to express a given periodic function of time as the sum of an infinite number of sinusoids whose frequencies are harmonically related.

[9] On the other hand, the Fourier transform is more general and can be used to obtain the frequency spectrum of a signal whose waveform is an arbitrary function of time.

For instance, the symmetrical square-wave signal can be expressed as:

$$\begin{aligned} v(t) &= \frac{4V}{\pi}\left(\sin \omega_0 t + \frac{1}{3}\sin 3\omega_0 t + \frac{1}{5}\sin 5\omega_0 t + \cdots\right) \\ &= \frac{4V}{\pi}\sum_{k=1}^{n}\frac{1}{2k-1}\sin\left((2k-1)\,\omega_0\right) \end{aligned} \tag{1.1}$$

where V is the amplitude of the square wave and $\omega_0 = 2\pi/T$ is[10] called the **fundamental frequency**. Note that because the amplitudes of the harmonics progressively decrease, the infinite series can be truncated, with the truncated series providing an approximation to the square waveform.

[10] T is the period of the square wave.

The sinusoidal components in the series of Eq. (**1.1**) constitute the frequency spectrum of the square-wave signal where the horizontal axis represents the angular frequency ω in radians per second.

The Fourier transform can be applied to a non-periodic function of time, such as that depicted in **Fig.** 1.3, and provides its frequency spectrum as a continuous function of frequency. Unlike the case of periodic signals, where the spectrum consists of discrete frequencies,[11] the spectrum of a nonperiodic signal contains in general all possible frequencies. Nevertheless, the essential parts of the spectra of practical signals are usually confined to relatively short segments of the frequency (ω) axis-an observation that is very useful in the processing of such signals.

[11] at ω_0 and its harmonics.

For instance, the spectrum of audible sounds such as speech and music extends from about 20 Hz to about 20 kHz, a frequency range known as the **audio band**.

It is worth noting that, although some musical tones have frequencies above 20 kHz, the human ear is **incapable** of hearing frequencies that are much above 20 kHz.

As another example, analog video signals have their spectra in the range of 0 MHz to 4,5 MHz.

We conclude this section by noting that a signal can be represented either by the manner in which its waveform varies with time, as for the voltage signal $v_a(t)$ shown in **Fig.** 1.3, or in terms of its **frequency spectrum**. The two alternative representations are known as the time-domain representation and the frequency-domain representation, respectively.

The frequency-domain representation of $v_a(t)$ will be denoted by the symbol $V_a(\omega)$.

1.1.3 Analog and Digital Signals

To clarify, the voltage signal depicted in **Fig.** 1.3 is called an **analog signal**. The name derives from the fact that such a signal is **analogous** to the physical signal that it represents.

The magnitude of an analog signal can take on **any value**. That is, the amplitude of an analogue signal exhibits a **continuous variation over its range of activity**.

The vast majority of signals in the world around us are analog. Electronic circuits that process such signals are known as **analog circuits**.

An alternative form of signal representation is that of a sequence of numbers, each number representing the signal magnitude at an instant of time. The resulting signal is called a **digital**

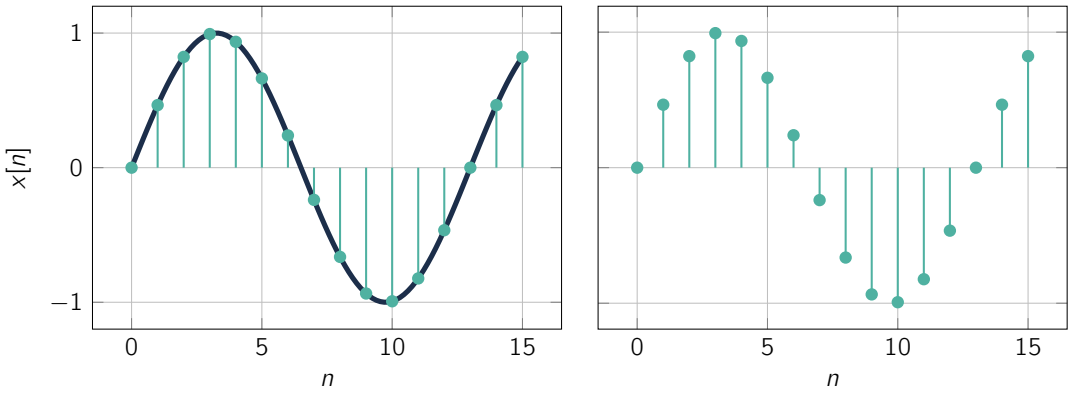


Figure 1.4
Sampling the continuous-time analog signal in (a) results in the discrete-time signal in (b).

signal. To see how a signal can be represented in this form-that is, how signals can be converted from analog to digital form-consider **Fig.** 1.4a. Here the curve represents a voltage signal. At equal intervals along the time axis, we have marked the time instants t_0, t_1, t_2, and so on. At each of these time instants, the magnitude of the signal is measured, a process known as **sampling**. **Fig.** 1.4b shows a representation of the signal of **Fig.** 1.4a in terms of its samples. The signal of **Fig.** 1.4b is defined only at the sampling instants; it no longer is a continuous function of time; rather, it is a **discrete-time** signal.

However, since the magnitude of each sample can take any value in a continuous range, the signal in **Fig.** 1.4b is still an analog signal.

Now if we represent the magnitude of each of the signal samples in **Fig.** 1.4b by a number having a finite number of digits, then the signal amplitude will no longer be continuous; rather, it is said to be **quantized**, **discretized**, or **digitized**. The resulting digital signal then is simply a sequence of numbers that represent the magnitudes of the successive signal samples.

The choice of number system to represent the signal samples affects the type of digital signal produced and has a profound effect on the complexity of the digital circuits required to process the signals. It turns out the **binary** number system results in the simplest possible digital signals and circuits.

In a binary system, each digit in the number takes on one of only two (**2**) possible values:

denoted 0 and 1.

Correspondingly, the digital signals in binary systems need have only two voltage levels, which can be labeled **low** and **high**.

As an example, in some of the digital circuits we will study, the levels are 0 V and +5 V. In this representation we can see that:

- the waveform is a pulse train with 0 V representing a 0 signal, or logic 0, and

- +5 V representing logic 1.

If we use N binary digits (bits) to represent each sample of the analog signal, then the digitised sample value can be expressed as:

$$D = b_0 2^0 + b_1 2^1 + b_2 2^2 + \cdots + b_{N-1} 2^{N-1} \tag{1.2}$$

where $b_0, b_1, \ldots, b_{N-1}$, denote the N bits and have values of 0 or 1. Here bit b_0 is the **least** significant bit (LSB), and bit b_{N-1} is the most significant bit (MSB). Conventionally, this binary number is written as b_{N-1}, b_{N-2}, ..., b_0. We observe that such a representation quantizes the analog sample into one of 2^N levels. Obviously the greater the number of bits,[12] the closer the digital word D approximates the magnitude of the analog sample. That is, increasing the number of bits reduces the **quantization error** and increases the resolution of the analog-to-digital conversion.

[12] i.e., the larger the N.

This improvement is, however, usually obtained at the expense of **more complex** and hence more **costly circuit implementations**.

It is **NOT** our purpose here to delve into this topic any deeper as we merely want to appreciate the nature of analog and digital signals.

Nevertheless, it is an excellent time to introduce a very important circuit building block of modern electronic systems which is the **analog-to-digital converter** (A/D or ADC) shown in block form in **Fig.** 1.5.

The ADC accepts at its input the samples of an analog signal and provides for each input sample the corresponding N-bit digital representation, according to Eq. (**1.2**), at its N output terminals. Therefore although the voltage at the input might be, say, 6,51 V, at each of the output terminals, i.e., at the ith terminal, the voltage will be either low (0 V) or high (5 V) if b_i is supposed to be 0 or 1, respectively.

The dual circuit of the ADC is the digital-to-analog converter (D/A or DAC). It converts an N-bit **digital input to an analog output voltage**.

Once the signal is in digital form, it can be processed using **digital circuits**. Of course, it goes without saying, digital circuits can deal also with signals which has no analogue origin, such as the signals that represent the various instructions of a digital computer.

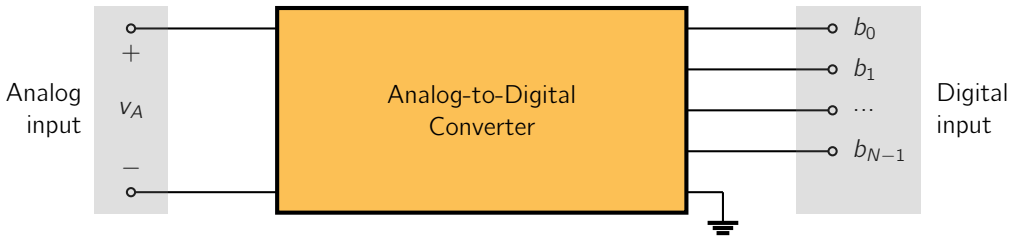


Figure 1.5 Block-diagram representation of the analog-to-digital converter (ADC).

Given digital circuits deal exclusively with binary signals, their design is simpler than that of analog circuits. In addition, digital systems can be designed using a relatively few different kinds of digital circuit blocks. However, a large number[13] of each of these blocks are usually needed. Therefore the design of digital circuits poses its own set of challenges to the designer but provides reliable and economic implementations of a great variety of signal-processing functions, many of which are not possible with analog circuits.

[13] e.g., hundreds of thousands or even millions.

At the present time, more and more of the signal-processing functions are being performed digitally. Examples are all around us:

> from the digital watch and the calculator to digital audio systems, digital cameras, and digital television.

Moreover, some long-standing analog systems such as the telephone communication system are now almost entirely digital. We should not forget the most important of all digital systems, the **digital computer**.

Although the digital processing of signals is at present all-pervasive, there remain many signal-processing functions that are **best performed by analog circuits**.

Many electronic systems include both **analog** and **digital** parts. It follows that a good electronics engineer must be proficient in the design of both analog and digital circuits, or mixed-signal or mixed-mode design as it is currently known.

1.2 Information

Analog v. Digital Circuit Engineers

As digital became the preferred implementation of more and more signal-processing functions, the need arose for greater numbers of digital circuit design engineers. Yet despite predictions made periodically that the demand for analog circuit design engineers would lessen, this has not been the case. Rather, the demand for analog engineers has, if anything, increased.

What is true, however, is that the skill level required of analog engineers has risen. Not only are they asked to design circuits of greater sophistication and tighter specifications, but they also have to do this using technologies that are optimized for digital (and not analog) circuits. This is dictated by economics, as digital usually constitutes the larger part of most systems.

1.2 Amplifiers as a Concept

In this section, we will look at the most fundamental signal-processing function, one that is implemented in some shape or form in almost every electronic system, namely, **signal amplification**. We will study the amplifier as a circuit building block. That is, we will consider its external characteristics for now and focus on operational amplifiers in Chapter 2.

1.2.1 Signal Amplification

From a conceptual point of view the simplest signal-processing task is **signal amplification**. The need for amplification is required as transducers provide signals that are said to be "weak", that is, in the microvolt (μV) or millivolt (mV) range and possessing minuscule energy. Such signals are too small for reliable processing, and processing is much easier if the signal magnitude is made larger. The functional block that accomplishes this task is the signal amplifier.

It is useful at this point to discuss the need for **linearity** in amplifiers. Special care needs to be taken in the amplification of a signal, so the information contained in the signal is **not changed and no new information is introduced**. Therefore when we feed the signal shown in **Fig.** 1.3 to an amplifier, we want to make sure the output signal of the amplifier to be an **exact replica of the input**, except of course for having larger magnitude.

> In other words, the "wiggles" in the output waveform needs to be identical to those in the input waveform. Any change in waveform is considered to be distortion and is obviously undesirable.

An amplifier which preserves the details of the signal waveform is characterised by the relationship:

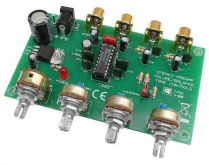
$$v_o(t) = Av_i(t) \tag{1.3}$$

where $v_i(t)$ and $v_o(t)$ are the input and output signals, respectively, and A is a constant representing the magnitude of amplification, known as **amplifier gain**.

As can be seen, Eq. (**1.3**) is of a **linear** relationship, therefore the amplifier it describes is a **linear amplifier**. It should be trivial to reason that if the relationship between v_a and v_i contains higher powers of v_i, then the waveform of v_a will no longer be identical to that of v_i.

> The amplifier is then said to exhibit **non-linear** distortion.

The amplifiers we have discussed up to now are primarily intended to operate on very small input signals. Their primary purpose is to make the signal magnitude larger, and therefore they are thought of as **voltage amplifiers**.[14]



[14] The **preamplifier** in the home stereo system is an example of a voltage amplifier.

At this time we wish to mention another type of amplifier, namely, the **power amplifier**. Such an amplifier may provide only a modest amount of voltage gain but substantial current gain. Therefore, while absorbing little power from the input signal source to which it is connected, often a pre-amplifier, it delivers large amounts of power to its load.

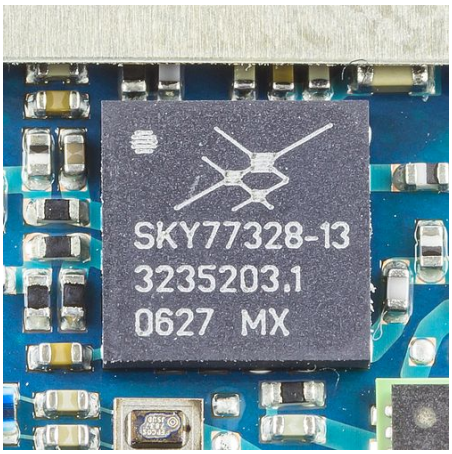


Figure 1.6 Power amplifier by Skyworks Solutions in a Smartphone.

> An example is found in the power amplifier of the home stereo system, whose purpose is to provide sufficient power to drive the loudspeaker, which is the amplifier load.

Here we should be observant that the loudspeaker is the output transducer of the stereo system; it converts the electric output signal of the system into an acoustic signal. A further appreciation of the need for linearity can be acquired by reflecting on the power amplifier. A linear power amplifier causes both soft and loud music passages to be reproduced without distortion.

1.2.2 Amplifier Circuit Symbol

The signal amplifier is obviously a two-port circuit. Its function is conveniently represented by the circuit symbol of **Fig.** 1.7a. This symbol clearly distinguishes the input and output ports and indicates the direction of signal flow.

Therefore, in subsequent diagrams it will **NOT** be necessary to label the two (2) ports as **input** and **output**. For generality we have shown the amplifier to have two input terminals that are distinct from the two output terminals. A more common situation is illustrated in **Fig.** 1.7b, where a common

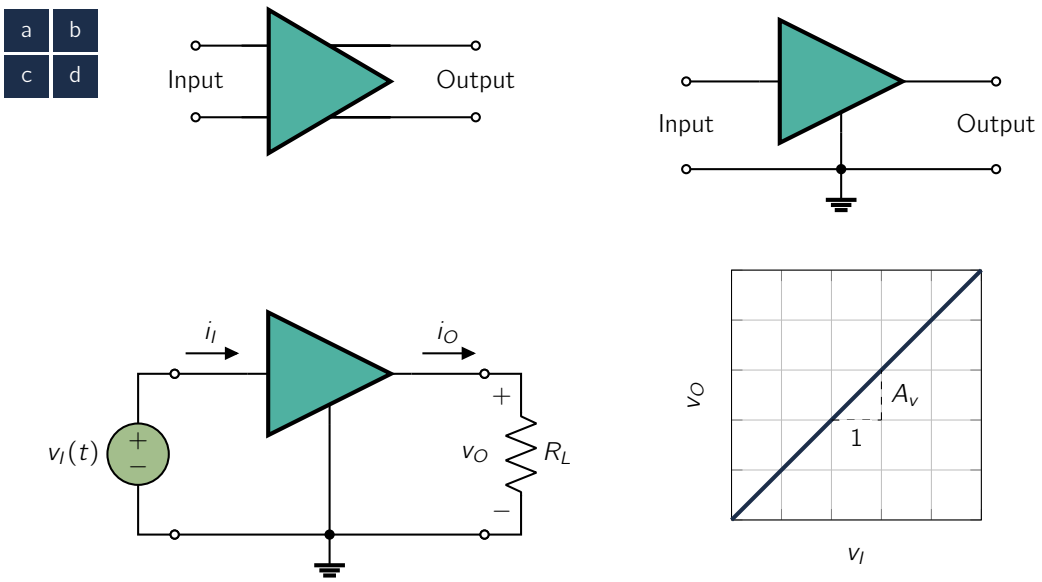


Figure 1.7 (a) Circuit symbol for amplifier (b) An amplifier with a common terminal (ground) between the input and output ports (c) A voltage amplifier fed with a signal $v_I(t)$ and connected to a load resistance R_L (d) Transfer characteristic of a linear voltage amplifier with voltage gain A_v.

terminal exists between the input and output ports of the amplifier. This common terminal is used as a reference point and is called the **circuit ground**.

1.2.3 Voltage Gain

A linear amplifier accepts an input signal $vi(t)$ and provides at the output, across a load resistance R_t as shown in **Fig.** 1.7c, an output signal $v_O(t)$ that is a **magnified replica** of $v_I(t)$. The voltage gain of the amplifier is therefore defined by:

$$\text{Voltage gain } (A_v) = \frac{v_O}{v_I} \tag{1.4}$$

Fig. 1.7d shows the **transfer characteristic** of a linear amplifier. If we apply to the input of this amplifier a sinusoidal voltage of amplitude $\hat{V}$, we obtain at the output a sinusoid of amplitude $A_v\hat{V}$.

1.2.4 Power Gain and Current Gain

> An amplifier **increases the signal power**. This is an important feature that distinguishes an amplifier from a transformer.

In the case of a transformer, while the voltage delivered to the load could be greater than the voltage feeding the input side, called the **primary**, the power delivered to the load[15] is less than or at most equal to the power supplied by the signal source.

[15] from the secondary side of the transformer.

On the other hand, an amplifier provides the load with power greater than that obtained from the signal source. That is, amplifiers have **power gain**.

The **power gain** of the amplifier in **Fig.** 1.7c is defined as:

$$\text{Power gain} = (A_p) \equiv \frac{\text{load power } (P_L)}{\text{input power}(P_I)} = \frac{v_O i_O}{v_I i_I}$$

where i_O is the current that the amplifier delivers to the load (R_L), $i_O = v_O/R_L$, and i_I is the current the amplifier draws from the signal source. The current gain of the amplifier is defined as

$$\text{Current gain}(A_i) \equiv \frac{i_O}{i_I} \tag{1.5}$$

From Eq. (**1.4**) to Eq. (**1.5**) we note that:

$$A_p = A_v A_i$$

1.2.5 Expressing Gain in Decibels

The amplifier gains defined previously are ratios of similarly dimensioned quantities. Therefore they will be expressed either as dimensionless numbers or, for emphasis,

- as V/V for the voltage gain,
- A/A for the current gain, and
- W/W for the power gain.

Alternatively, for a number of reasons, some of them historic, electronics engineers express amplifier gain with a **logarithmic** measure. Specifically the voltage gain A_v can be expressed as:

$$\text{Voltage gain in decibels} = 20\log|A_v|dB$$

and the current gain A_i can be expressed as:

$$\text{Current gain in decibels} = 20\log|A_i|dB$$

Since power is related to voltage (or current) squared, the power gain A_p can be expressed in decibels as:

$$\text{Power gain in decibels} = 10logA_pdB$$

The absolute values of the voltage and current gains are used because in some cases A_v or A_i will be a negative number.

> A negative gain A_v simply means that there is a 180° phase difference between input and output signals. It does **NOT** imply that the amplifier is **attenuating** the signal.

On the other hand, an amplifier whose voltage gain is, say, -20 dB is in fact attenuating the input signal by a factor of 10 (i.e., $A_v = 0.1$ V/V).

1.2.6 The Amplifier Power Supplies

Given the power delivered to the load is greater than the power drawn from the signal source, and important question to ask here is:

> where does the additional power come from?

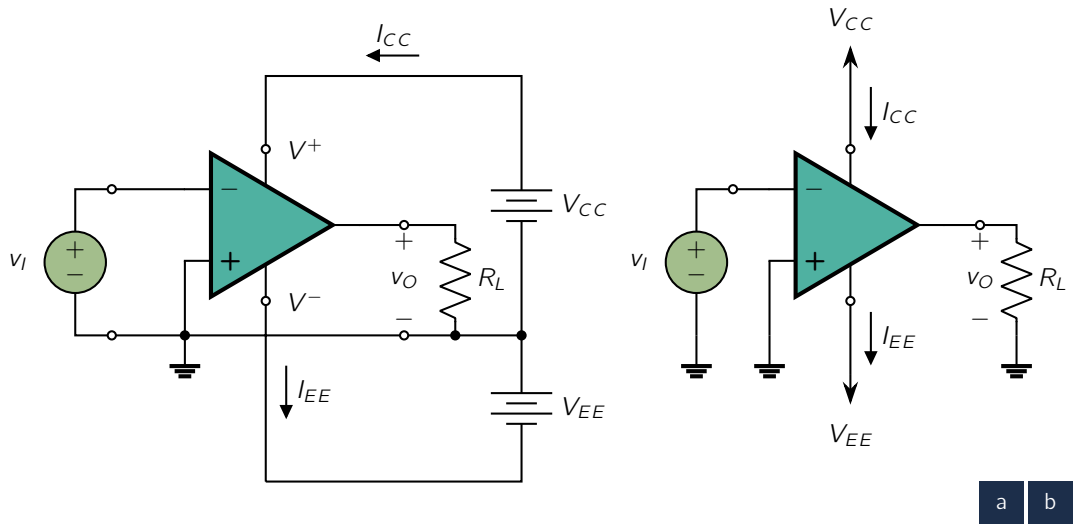
The answer is found by observing that amplifiers need Direct Current (DC) power supplies for their operation. These DC sources supply the extra power delivered to the load as well as any power that might be dissipated in the internal circuit of the amplifier.[16] In **Fig.** 1.7c we have not explicitly shown these DC sources.

[16] such power is converted to heat.

Fig. 1.8a shows an amplifier which requires two (2) DC sources:

1. one positive of value V_{CC}, and
2. one negative of value V_{EE}.

Figure 1.8
An amplifier that requires two DC supplies (shown as batteries) for operation.



The amplifier has two terminals, labeled V^+ and V^-, for connection to the DC supplies. For the amplifier to operate, the terminal labeled V^+ has to be connected to the **positive side** of a DC source whose voltage is V_{CC} and whose negative side is connected to the **circuit ground**. Also, the terminal labeled V^- has to be connected to the **negative side** of a DC source whose voltage is V_{EE} and whose positive side is connected to the circuit ground.

Now, if the current drawn from the positive supply is denoted I_{CC} and that from the negative supply is I_{EE}, shown in **Fig.** 1.8a, then the DC power delivered to the amplifier is

$$P_{dc} = V_{CC} I_{CC} + V_{EE} I_{EE}.$$

If the power dissipated in the amplifier circuit is denoted as $P_{dissipated}$, the power-balance equation for the amplifier can be written as

$$P_{dc} + P_I = P_L + P_{dissipated},$$

where P_I is the power drawn from the signal source and P_L is the power delivered to the load. Given the power drawn from the signal source is usually small, the amplifier power efficiency is defined as

$$\eta \equiv \frac{P_L}{P_{dc}} \times 100$$

The power efficiency is an important performance parameter for amplifiers that handle large amounts of power.

> Such amplifiers, called **power amplifiers**, are used, for example, as output amplifiers of stereo systems.

to simplify circuit diagrams, we will adopt the convention illustrated in **Fig.** 1.8b. Here the V^+ terminal is shown connected to an arrowhead pointing upward and the V^- terminal to an arrowhead pointing downward. The corresponding voltage is indicated next to each arrowhead.

In many cases we will **NOT** explicitly show the connections of the amplifier to the DC power sources and that some amplifiers **require only one power supply**.

A Simple Amplifier 1.2 Exercise

Consider an amplifier operating from $\pm 10\,\text{V}$ power supplies. It is fed with a sinusoidal voltage having 1 V peak and delivers a sinusoidal voltage output of 9 V peak to a $1\,\text{k}\Omega$ load. The amplifier draws a current of 9,5 mA from each of its two power supplies. The input current of the amplifier is found to be sinusoidal with 0,1 mA peak.

Find the voltage gain, the current gain, the power gain, the power drawn from the DC supplies, the power dissipated in the amplifier, and the amplifier efficiency.

$$A_v = \frac{9}{1} = 9$$

which can be written as:

$$A_v = 20\log 9 = 19{,}1\,\text{dB} \qquad \hat{I}_O = \frac{9\,\text{V}}{1\,\text{k}\Omega} = 9\,\text{mA} \qquad A_i = \frac{\hat{I}_O}{\hat{I}_i} = \frac{9}{0.1} = 90\,\text{A}\cdot\text{A}^{-1}$$

or

$$A_i = 20\log 90 = 39{,}1\,\text{dB} \qquad P_L = V_{o_{\text{rms}}} I_{o_{\text{rms}}} = \frac{9}{\sqrt{2}}\frac{9}{\sqrt{2}} = 40{,}5\,\text{mW} \qquad A_p = \frac{P_L}{P_L}$$

$$P_I = V_{i_{\text{rms}}} I_{i_{\text{rms}}} = \frac{1}{\sqrt{2}}\frac{0.1}{\sqrt{2}} = 0{,}05\,\text{mW} \qquad = \frac{40.5}{0.05} = 810\,\text{W}\cdot\text{W}^{-1}$$

and finally

$$A_n = 10\log 810 = 29{,}1\,\text{dB}$$

$$P_{\text{dc}} = 10\times 9.5 + 10\times 9.5 = 190\,\text{mW}$$

$$P_{\text{dissipated}} = P_{\text{dc}} + P_I - P_L = 190 + 0.05 - 40.5 = 149{,}6\,\text{mW}$$

$$\eta = \frac{P_L}{P_A}\times 100 = 21.3\% \quad \blacksquare$$

From the above example we observe that the amplifier converts some of the DC power it draws from the power supplies to signal power that it delivers to the load.

1.2.7 Amplifier Saturation

From a practical point-of-view, the amplifier transfer characteristic remains linear over **only a limited range of input and output voltages**. For an amplifier operated from two power supplies the output voltage cannot exceed a specified positive limit and cannot decrease below a specified negative limit.

The resulting transfer characteristic is shown in Fig. 1.14, with the positive and negative saturation levels denoted L_+ and L_-, respectively. Each of the two saturation levels is usually within a fraction of a volt of the voltage of the corresponding power supply.

Obviously, to avoid distorting the output signal waveform, the input signal swing must be kept within the linear range of operation,

$$\frac{L_-}{A_n} \leqslant v_I \leqslant \frac{L_+}{A_n}$$

In Fig. 1.14, which shows two input waveforms and the corresponding output waveforms, the peaks

of the larger waveform have been clipped off because of amplifier saturation.

1.2.8 Symbol Convention

At this point, it is important we set up some standard terminology we shall use throughout the lecture which a diagram is given in **Fig.** 1.9.

To begin, a current ($i_c(t)$) consists of a DC component I_c on which is superimposed a sinusoidal component (i.e., Alternating Current (AC)) $i_c(t)$ whose peak amplitude is I_c. Please observe that at a time t, the **total instantaneous** current $i_c(t)$ is the **sum** of the DC current I_c and the signal current $i_c(t)$:

$$i_c(t) = I_c + i_c(t),$$

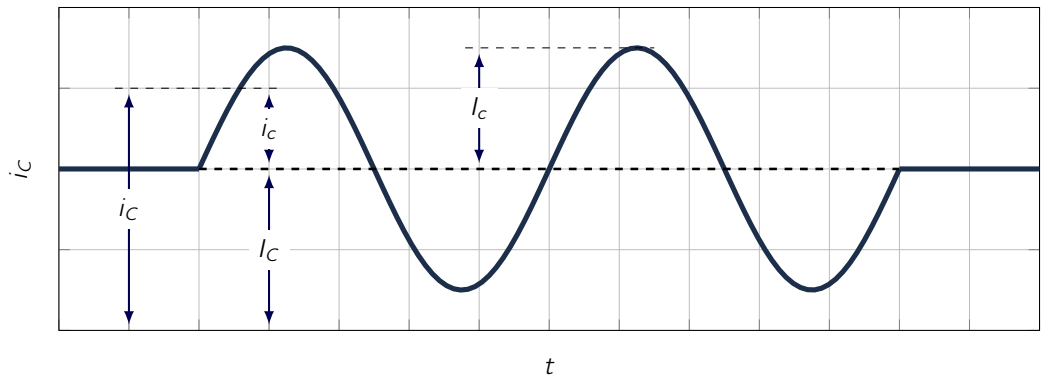
where the signal current is given by

$$i_s(t) = I_s \sin \omega t$$

Therefore, we state some conventions:

- Total instantaneous quantities are denoted by a lowercase symbol with uppercase subscript(s), for example, $i_c(t)$, $v_{DS}(t)$.
- Direct-current (dc) quantities are denoted by an uppercase symbol with uppercase subscript(s), for example, I_c, V_{DS}.
- Incremental signal quantities are denoted by a lowercase symbol with lowercase subscript(s), for example, $i_c(t)$, $v_{gs}(t)$.
- If the signal is a sine wave, then its amplitude is denoted by an uppercase symbol with lowercase subscript(s), for example, I_c, V_{gs}.
- DC power supplies are denoted by an uppercase letter with a double-letter uppercase subscript, for example, V_{CC}, V_{DD}. A similar notation is used for the DC current drawn from the power supply, for example, I_{cc}, I_{DD}.

Figure 1.9
Symbol conventions used.



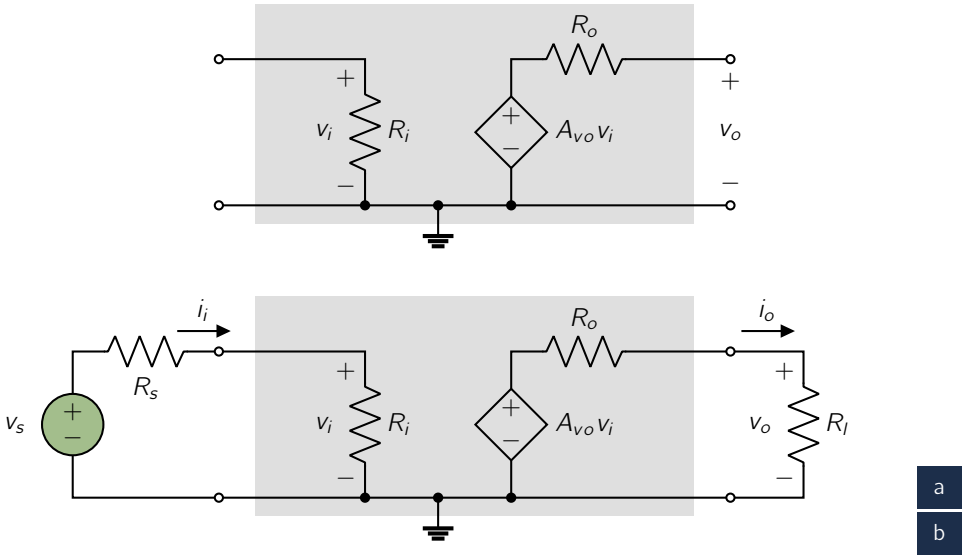


Figure 1.10
(a) Circuit model for the voltage amplifier. (b) The voltage amplifier with input signal source and load.

1.3 Circuit Models for Amplifiers

A substantial part of this lecture will focus on the design of amplifier circuits which use **transistors** of different types.[17] Such circuits will vary in complexity from those using a single transistor to those with 20 or more devices. To apply the resulting amplifier circuit as a building block in a system, we need to be able to characterise, or model., its terminal behavior.

[17] These transistors will be studied in sufficient detail in later lectures.

In this section, we will study simple but effective amplifier models. These models apply irrespective of the complexity of the internal circuit of the amplifier. The values of the model parameters can be found either by analyzing the amplifier circuit or by performing measurements at the amplifier terminals.

1.3.1 Voltage Amplifiers

Fig. 1.10a shows a circuit model for the voltage amplifier. The model consists of a voltage-controlled voltage source having a gain factor A_{vo}, an input resistance R_i that accounts for the fact that the amplifier draws an input current from the signal source, and an output resistance R_o, which accounts for the change in output voltage as the amplifier is called upon to supply output current to a load.

To be specific, we show in **Fig.** 1.10b the amplifier model fed with a signal voltage source v_s having a resistance R_s and connected at the output to a load resistance R_L.

The nonzero output resistance R_o causes only a fraction of $A_{vo}v_i$ to appear across the output.

Using the voltage-divider rule we obtain

$$v_o = A_{vo} v_i \frac{R_L}{R_L + R_o} \qquad \text{therefore Voltage Gain is} \qquad A_v \equiv \frac{v_o}{v_i} = A_{vo} \frac{R_L}{R_L + R_o} \tag{1.6}$$

We can see that to **NOT** lose gain in coupling the amplifier output to a load, the output resistance (R_0) should be **much smaller** than the load resistance (R_L).

For a given R_L one must design the amplifier so R_o is much smaller than R_L.

Applications with Changing Load 1.3 Information

Furthermore, there are applications in which R_L is known to vary over a certain range. To keep the output voltage v_a as constant as possible, the amplifier is designed with R_o much smaller than the lowest value of R_L. An ideal voltage amplifier is one with $R_o = 0$.

Eq. (**1.6**) indicates also that for $R_L = \infty$, $A_v = A_{vo}$. Therefore A_{vo} is the voltage gain of the **unloaded amplifier**, or the open-circuit voltage gain. It should also be clear in specifying the voltage gain of an amplifier, one must also specify the value of load resistance at which this gain is measured or calculated.

If a load resistance is not specified, it is normally assumed that the given voltage gain is the open-circuit gain A_{vo}.

The finite input resistance R_i introduces another voltage-divider action at the input, with the result that only a fraction of the source signal v_s actually reaches the input terminals of the amplifier:

$$v_i = v_s \frac{R_i}{R_i + R_s} \tag{1.7}$$

It follows that, to not lose a significant portion of the input signal in coupling the signal source to the amplifier input, the amplifier must be designed to have an input resistance R_i much greater than the resistance of the signal source, $R_i \gg R_s$.

Furthermore, there are applications in which the source resistance is known to vary over a certain range. To minimize the effect of this variation on the value of the signal that appears at the input of the amplifier, the design ensures that R_i is much greater than the largest value of R_s. An ideal voltage amplifier is one with $R_i = \infty$. In this ideal case both the current gain and power gain become infinite.

The overall voltage gain (v_o/v_s) can be found by combining Eq. (**1.6**) and Eq. (**1.7**):

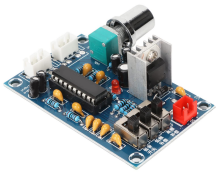
$$\frac{v_o}{v_s} = A_{vo} \frac{R_i}{R_i + R_s} \frac{R_L}{R_i + R_o}$$

There are situations in which one is interested not in voltage gain but only in a significant power gain.

For example, the source signal can have a respectable voltage but a source resistance that is much greater than the load resistance.

Connecting the source directly to the load would result in **significant signal attenuation**. In such a case, one requires an amplifier with a high input resistance (much greater than the source resistance) and a low output resistance (much smaller than the load resistance) but with a modest voltage gain (or even unity gain).

Such an amplifier is referred to as a **buffer amplifier**.[18]



[18] a unity gain amplifier that copies a signal from one circuit to another while transforming its electrical impedance to provide a more ideal source (with a lower output impedance for a voltage buffer or a higher output impedance for a current buffer). This buffers the signal source in the first circuit against being affected by currents from the electrical load of the second circuit and may simply be called a buffer or follower when context is clear.

1.3.2 Cascaded Amplifiers

To meet given amplifier specifications, we often need to design the amplifier as a **cascade** of two (2) or more stages.

The stages are usually not identical but rather, each is designed to serve a specific purpose.

> For instance, to provide the overall amplifier with a large input resistance, the first stage is usually required to have a large input resistance.

Also, to equip the overall amplifier with a low output resistance, the final stage in the cascade is usually designed to have a low output resistance.

To illustrate the analysis and design of cascaded amplifiers, we consider a practical example.

A few comments need to be mentioned here.

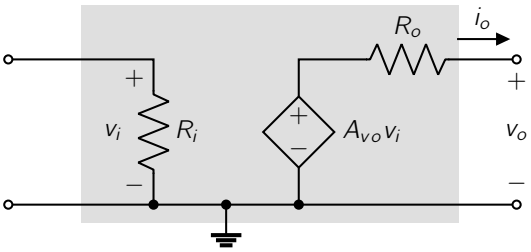
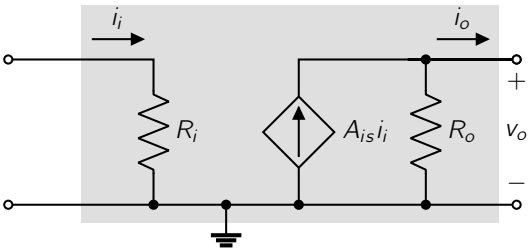
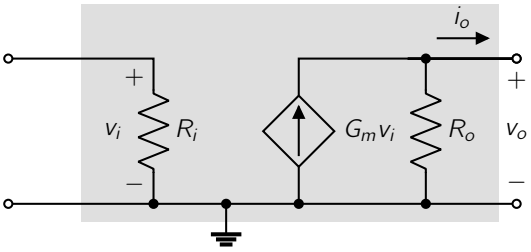
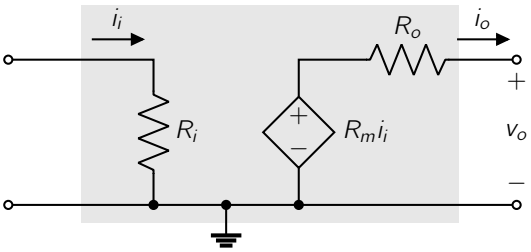
To avoid losing signal strength at the amplifier input where the signal is usually very small,

- the first stage is designed to have a **relatively large input resistance** ($1\,\text{M}\Omega$), which is much larger than the source resistance. The trade-off appears to be a moderate voltage gain ($10\,\text{V}\cdot\text{V}^{-1}$).
- The second stage does not need to have such a high input resistance. Rather, here we need to realize the bulk of the required voltage gain.
- The third and final, or output, stage is not asked to provide any voltage gain; rather, it functions as a **buffer amplifier**, providing a relatively large input resistance and a low output resistance, much lower than R_L.

  It is this stage that enables connecting the amplifier to the $100\,\Omega$ load.

These points can be made more concrete by solving the following exercises. In so doing, observe that in finding the gain of an amplifier stage in a cascade amplifier, the loading effect of the succeeding amplifier stage must be taken into account as we have done in the above example.

Figure 1.11
The four major amplifier types.

Circuit Model	Gain Parameter	Ideal
Voltage Amplifier 	**Open-Circuit Voltage Gain** $V_{vo} \equiv \left.\frac{v_o}{v_i}\right\|_{i_o=0}$ $\mathrm{V}\cdot\mathrm{V}^{-1}$	$R_i = \infty$ $R_o = 0$
Current Amplifier 	**Short-Circuit Voltage Gain** $A_{is} \equiv \left.\frac{i_o}{i_i}\right\|_{v_o=0}$ $\mathrm{A}\cdot\mathrm{A}^{-1}$	$R_i = 0$ $R_o = \infty$
Transconductance Amplifier 	**Short-Circuit Transconductance** $G_m \equiv \left.\frac{i_o}{v_i}\right\|_{v_o=0}$ $\mathrm{A}\cdot\mathrm{V}^{-1}$	$R_i = \infty$ $R_o = \infty$
Transresistance Amplifier 	**Open-Circuit Transresistance** $R_m \equiv \left.\frac{v_o}{i_i}\right\|_{i_o=0}$ $\mathrm{V}\cdot\mathrm{A}^{-1}$	$R_i = 0$ $R_o = 0$

1.3.3 Other Amplifier Types

In the design of an electronic system, the signal of interest[19] can be either a **voltage** or a **current**.

[19] whether at the system input, at an intermediate stage, or at the output.

For example, some transducers have very high output resistances and can be more appropriately modeled as current sources. Similarly, there are applications in which the output current rather than the voltage is of interest.

Therefore, although it is the most popular, the **voltage amplifier** considered above is just one of four possible amplifier types. The other three are the **current amplifier**, the **transconductance amplifier**, and the **transresistance amplifier**. **Fig.** 1.11 shows the four amplifier types, their circuit models, the definition of their gain parameters, and the ideal values of their input and output resistances.

1.3.4 Relationships between the Four Amplifier Models

Although for a given amplifier a particular one of the four models in **Fig.** 1.11 is most preferable, any of the four can be used to model any amplifier.

In fact, simple relationships can be derived to relate the parameters of the various models.

For instance, the open-circuit voltage gain A_{vo} can be related to the short-circuit current gain A_{is} as follows:

The open-circuit output voltage given by the voltage amplifier model of **Fig.** 1.11 is $A_{vo}v_i$. The current amplifier model in the same figure gives an open-circuit output voltage of $A_{is}i_iR_o$. Equating these two values and noting that $i_i = v_i/R_i$ gives

$$A_{vo} = A_{is}\left(\frac{R_o}{R_i}\right) \tag{1.8}$$

Similarly, we can show that

$$A_{vo} = G_mR_o \qquad \text{and} \qquad A_{vo} = \frac{R_m}{R_i} \tag{1.9}$$

The expressions in Eq. (**1.8**) and Eq. (**1.9**) can be used to relate any two of the gain parameters A_{vo}, A_{is}, G_m, and R_m.

1.3.5 Determining Input and Output Resistance

From the amplifier circuit models given in **Fig.** 1.11, we observe that the input resistance R_i of the amplifier can be determined by applying an input voltage v_i and measuring[20] the input current i_i; that is,

[20] or just plain calculating.

$$R_i = v_i/i_i$$

The output resistance is found as the ratio of the open-circuit output voltage to the short-circuit output current. Alternatively, the output resistance can be found by eliminating the input signal

source[21] and applying a voltage signal v_x to the output of the amplifier, as shown in **Fig.** 1.12. If we denote the current drawn from v_x into the output terminals as i_x,[22] then

$$R_o = v_x/i_x$$

[21] then i_i and v_i will both be zero.

[22] note that i_x is opposite in direction to i_o.

Although these techniques are conceptually correct, in actual practice more refined methods are employed in measuring R_i, and R_o.

1.3.6 Unilateral Models

The amplifier models considered above are **unilateral**.

This means signal flow is **unidirectional**, from input to output.

Most real amplifiers show some reverse transmission, which is usually undesirable but must nonetheless be modeled.[23]

Also, in later chapters, we will find it necessary in certain cases to augment the models of Table 1.1 to take into account the nonunilateral nature of some transistor amplifiers.

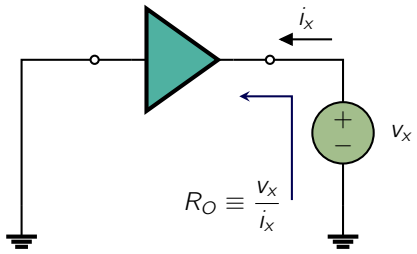


Figure 1.12 **Determining the output resistance.**

[23] We shall not pursue this point further at this time

Integrated Circuit — 1.4 Information

An IC, also known as a **microchip** or **chip**, is a compact assembly of electronic circuits formed from various electronic components[24] and their interconnections. These components are fabricated onto a thin, flat piece "chip" of semiconductor material, most commonly **silicon**. ICs are integral to a wide variety of electronic devices[25] performing functions such as data processing, control, and storage. They have transformed the field of electronics by enabling device miniaturisation, improving performance, and reducing cost.

Compared to assemblies built from discrete components, ICs are orders of magnitude smaller, faster, more energy-efficient, and less expensive, allowing for a very high transistor count.

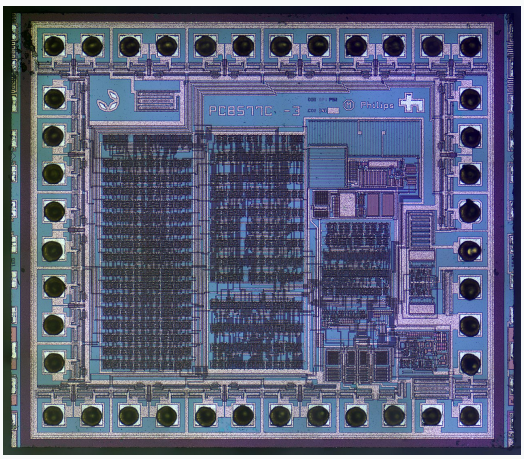


Figure 1.13: A microscope image of an integrated circuit die used to control LCDs. The pinouts are the dark circles surrounding the integrated circuit.

[24] such as transistors, resistors, and capacitors.

[25] including computers, smartphones, and televisions

Continues on next page →

← Continued from previous page

The IC's capability for mass production, its high reliability, and the standardised, modular approach of integrated circuit design facilitated rapid replacement of designs using discrete transistors. Today, ICs are present in virtually all electronic devices and have revolutionised modern technology.

> Products such as computer processors, microcontrollers, digital signal processors, and embedded chips in home appliances are foundational to contemporary society due to their small size, low cost, and versatility.

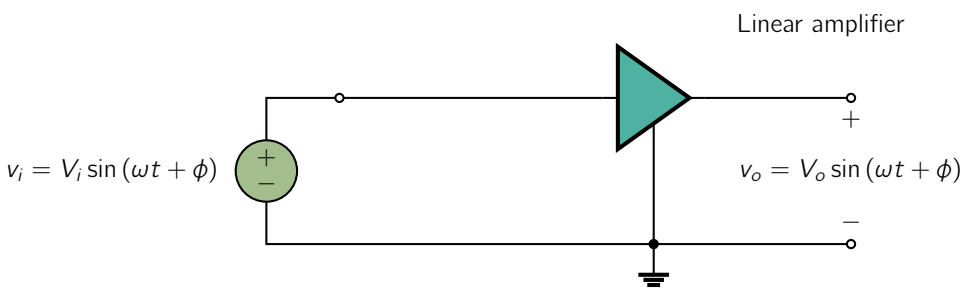


Figure 1.14
Measuring the frequency response of a linear amplifier: At the test frequency, the amplifier gain is characterized by its magnitude (V_o/V_i) and phase ϕ.

1.4 Frequency Response of Amplifiers

We now know from previous sections, that the input signal to an amplifier can always be expressed as the **sum of sinusoidal signals**. From this reasoning, it follows that an important characterisation of an amplifier is in terms of its response to input sinusoids of different frequencies.

> Such a characterization of amplifier performance is known as the **amplifier frequency response**.

1.4.1 Measuring the Amplifier Frequency Response

We shall introduce the subject of amplifier frequency response by showing how it can be measured. **Fig.** 1.14 depicts a linear voltage amplifier fed at its input with a sine-wave signal of amplitude V_i, and frequency ω.

As the figure indicates, the signal measured at the amplifier output also is sinusoidal with exactly the same frequency ω. This is an important point to note:

> Whenever a sine-wave signal is applied to a linear circuit, the resulting output is **sinusoidal with the same frequency as the input**.

In fact, the sine wave is the only signal that does not change shape as it passes through a linear circuit. Observe, however, that the output sinusoid will in general have a different amplitude and will be **shifted** in phase relative to the input.

The ratio of the amplitude of the output sinusoid (V_o) to the amplitude of the input sinusoid (V_i) is the magnitude of the amplifier gain, or transmission, at the test frequency ω. In addition, the angle ϕ is the phase of the amplifier transmission at the test frequency ω. If we denote the **amplifier transmission**, or **transfer function** as it is more commonly known, by $T(\omega)$, then:

$$|T(\omega)| = \frac{V_o}{V_i} \qquad \text{and} \qquad \angle T(\omega) = \phi$$

The response of the amplifier to a sinusoid of frequency ω is completely described by $|T(\omega)|$ and $\angle T(\omega)$. Now, to obtain the complete frequency response of the amplifier we simply change the

frequency of the input sinusoid and measure the new value for $|T|$ and $\angle T$.

> The end result will be a table and/or graph of gain magnitude $\left[T(\omega)|\right]$ versus frequency and a table and/or graph of phase angle $\left[\angle T(\omega)\right]$ versus frequency.

These two plots together constitute the frequency response of the amplifier;

- the first is known as the **magnitude** or **amplitude** response, and
- the second is the phase response.

> It is common to express the magnitude of transmission in dB and plot $20\log|T(\omega)|$ v. frequency.

1.4.2 Amplifier Bandwidth

Figure 1.21 shows the magnitude response of an amplifier.

It indicates that the gain is almost constant over a wide frequency range, roughly between ω_1 and ω_2. Signals whose frequencies are below ω_1 or above ω_2, will experience lower gain, with the gain decreasing as we move farther away from ω_1 and ω_2. The band of frequencies over which the gain of the amplifier is almost constant, to within a certain number of decibels (usually 3 dB), is called the **amplifier bandwidth**.

Normally the amplifier is designed so that its bandwidth coincides with the spectrum of the signals it is required to amplify.

> If this were not the case, the amplifier would **distort the frequency spectrum** of the input signal, with different components of the input signal being amplified by different amounts.

1.4.3 Evaluating the Frequency Response of Amplifiers

Above, we described the method used to measure the frequency response of an amplifier. We now briefly discuss the method for analytically obtaining an expression for the frequency response.

To evaluate the frequency response of an amplifier, one has to analyze the amplifier equivalent circuit model, taking into account all reactive components.[26] Circuit analysis proceeds in the usual fashion but with inductances and capacitances represented by their reactances.

An inductance L has a reactance or impedance $\mathbf{j}\omega L$, and a capacitance C has a reactance or impedance $1/\mathbf{j}\omega C$ or, equivalently, a susceptance or admittance $\mathbf{j}\omega C$. Therefore in a frequency-domain analysis we deal with impedances and/or admittances. The result of the analysis is the

[26] In the models considered in previous sections no reactive components were included. These were simplified models and cannot be used alone to predict the amplifier frequency response.

amplifier transfer function $T(\omega)$:

$$T(\omega) = \frac{V_o(\omega)}{V_i(\omega)}$$

where $V_i(\omega)$ and $V_o(\omega)$ denote the input and output signals, respectively. $T(\omega)$ is generally a complex function whose magnitude $|T(\omega)|$ gives the magnitude of transmission or the magnitude response of the amplifier. The phase of $T(\omega)$ gives the phase response of the amplifier.

In the analysis of a circuit to determine its frequency response, the algebraic manipulations can be considerably simplified by using the **complex frequency variable** s.

In terms of s, the impedance of an inductance L is sL and that of a capacitance C is $1/sC$. Replacing the reactive elements with their impedances and performing standard circuit analysis, we obtain the transfer function $T(s)$ as

$$T(s) \equiv \frac{V_o(s)}{V_i(s)}$$

Subsequently, we replace s by $\mathbf{j}\omega$ to determine the transfer function for **physical frequencies**, $T(\mathbf{j}\omega)$. Note that $T(\mathbf{j}\omega)$ is the same function we called $T(\omega)$ above;[27] the additional $\mathbf{j}$ is included to emphasize that $T(\mathbf{j}\omega)$ is obtained from $T(s)$ by replacing s with $\mathbf{j}\omega$.

[27] At this stage, we are using s simply as a shorthand for $\mathbf{j}\omega$.

1.4.4 Single-Time-Constant Networks

When analyzing amplifier circuits to determine their frequency response, the knowledge of the frequency-response characteristics of single-time-constant (STC) networks are always useful. An STC network is one that is composed of, or can be reduced to, one reactive component[28] and one resistance, which some examples are shown in **Fig.** 1.15.

[28] inductance or capacitance.

An STC network formed of an inductance L and a resistance R has a time constant $\tau = L/R$. The time constant τ of an STC network composed of a capacitance C and a resistance R is given by $\tau = CR$.

At this point we need in particular the frequency-response results; we will, in fact, briefly discuss this important topic now. Most STC networks can be classified into two (**2**) categories:[29]

[29] An important exception is the all-pass STC network

- low pass (LP), and

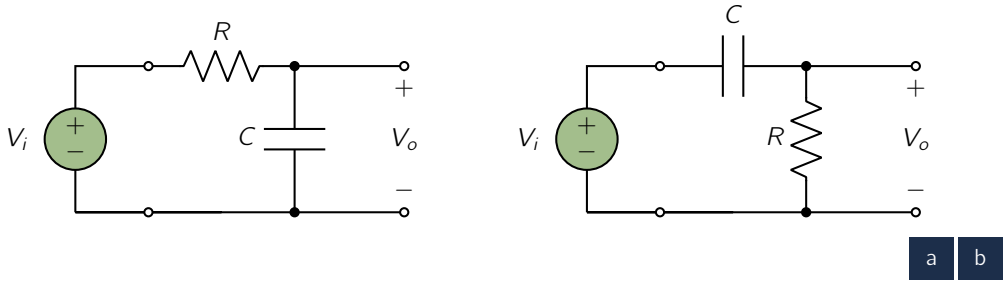


Figure 1.15
Two examples of STC networks: (a) a low-pass network and (b) a high-pass network.

- High-pass.

with each of the two categories displaying distinctly different signal responses.

> As an example, the STC network shown in **Fig.** 1.15a is of the **low-pass** type and that in **Fig.** 1.15b is of the **high-pass** type.

To see the reasoning behind this classification, observe that the transfer function of each of these two circuits can be expressed as a voltage-divider ratio, with the divider composed of a resistor and a capacitor. Now, recalling how the impedance of a capacitor varies with frequency ($Z = 1/\mathrm{j}\omega C$), it is easy to see that the transmission of the circuit in **Fig.** 1.15a will decrease with frequency and approach zero as ω approaches ∞.

Therefore the circuit of **Fig.** 1.15a acts as a **low-pass filter**;[30] it passes low-frequency, sine-wave inputs with little or no attenuation[31] and attenuates high-frequency input sinusoids. The circuit of **Fig.** 1.15b does the opposite. Its transmission is unity at $\omega = \infty$ and decreases as ω is reduced, reaching 0 for $\omega = 0$. The latter circuit, therefore, performs as a **high-pass filter**.

[30] A filter is a circuit that passes signals in a specified frequency band (the filter passband) and stops or severely attenuates (filters out) signals in another frequency band (the filter stopband).

[31] at $\omega = 0$, the transmission is unity.

Tbl. 1.1 provides a summary of the frequency-response results for STC networks of both types. Also, sketches of the magnitude and phase responses are given in Figs. 1.23 and 1.24. These frequency-response diagrams are known as **Bode plots**, and the 3-dB frequency (ω_0) is also known as the **corner frequency**, **break frequency**, or **pole frequency**.

It is important to develop a facility for the rapid determination of the time constant τ of an STC circuit. The process is very simple:

1. Set the independent voltage or current source to zero;
2. "grab hold" of the two terminals of the reactive element (capacitor C or inductor L); and
3. determine the equivalent resistance R that appears between these two terminals.

Table 1.1 Frequency Response of STC Networks.

Description	Low-Pass	High-Pass
Transfer Function $T(s)$	$\dfrac{K}{1+(s/\omega_0)}$	$\dfrac{Ks}{s+\omega_0}$
Transfer Function (for physical frequencies) $T(\mathrm{j}\omega)$	$\dfrac{K}{1+\mathrm{j}\,(\omega/\omega_0)}$	$\dfrac{K}{1-\mathrm{j}\,(\omega_0/\omega)}$
Magnitude Response $\lvert T(\mathrm{j}\omega)\rvert$	$\dfrac{\lvert K\rvert}{\sqrt{1+(\omega/\omega_0)^2}}$	$\dfrac{\lvert K\rvert}{\sqrt{1+(\omega_0/\omega)^2}}$
Phase Response $\angle T(\mathrm{j}\omega)$	$-\tan^{-1}(\omega/\omega_0)$	$-\tan^{-1}(\omega_0/\omega)$
Transmission at $\omega = 0$ (dc)	K	0
Transmission at $\omega =$ inf	0	K
3 dB Frequency	$\omega_0 = 1/\tau$; $\tau \equiv$ time constant (CR or L/R	

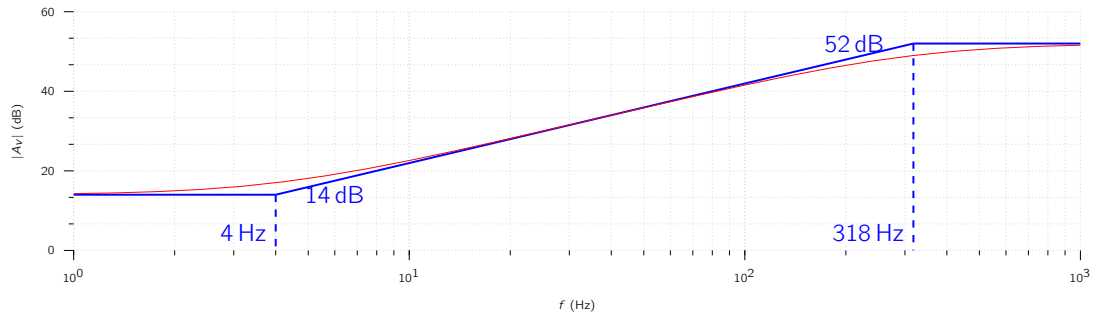


Figure 1.16

The time constant is then CR or L/R.

Bode Plots 1.5 Information

In the 1930s, while working at Bell Labs, Hendrik Bode devised a simple but accurate method for using linearized asymptotic responses to graph gain and phase shift against frequency on a logarithmic scale. Such gain and phase presentations, together called Bode plots, have enormous importance in the design and analysis of the frequency-dependent behavior of systems large and small.

1.4.5 Classification of Amplifiers Based on Frequency Response

Amplifiers can be classified based on the **shape of their magnitude-response curve**. Figure 1.26 shows typical frequency-response curves for various amplifier types. In Fig. 1.26(a) the gain remains constant over a wide frequency range, but falls off at low and high frequencies.

This type of frequency response is common in **audio amplifiers**.

Internal capacitances in the device[32] cause the fall-off of gain at high frequencies, just as C_i did in the circuit of Example 1.5.

[32] Such as that of a transistor

On the other hand, the falloff of gain at low frequencies is usually caused by coupling capacitors used to connect one amplifier stage to another, as indicated in Fig. 1.27. This practice is usually adopted to simplify the design process of the different stages. The coupling capacitors are usually chosen quite large[33] so their reactance (impedance) is small at the frequencies of interest. Nevertheless, at sufficiently low frequencies the reactance of a coupling capacitor will become large enough to cause part of the signal being coupled to appear as a voltage drop across the coupling capacitor, therefore not reaching the subsequent stage. Coupling capacitors will thus cause loss of gain at low frequencies and cause the gain to be zero at dc. This is not at all surprising, since from Fig. 1.27 we observe that the coupling capacitor, acting together with the input resistance of the subsequent stage, forms a high-pass STC circuit. It is the frequency response of this high-pass circuit that accounts for the shape of the amplifier frequency response in Fig. 1.26(a) at the low-frequency end.

[33] a fraction of a microfarad to a few tens of microfarads

There are many applications in which it is important that the amplifier maintain its gain at low frequencies down to dc. Furthermore, monolithic IC technology does not allow the fabrication of large coupling capacitors. Therefore IC amplifiers are usually designed as directly coupled or DC amplifiers.[34]

In a number of applications, such as in the design of radio and TV receivers, the need arises for an amplifier whose frequency response peaks around a certain frequency[35] and falls off on both sides of this frequency, as shown in Fig. 1.26(c). Amplifiers with such a response are called **tuned amplifiers**, **bandpass amplifiers**, or **bandpass filters**.

A tuned amplifier forms the heart of the front-end or tuner of a communication receiver; by adjusting its center frequency to coincide with the frequency of a desired communications channel (e.g., a radio station), the signal of this particular channel can be received while those of other channels are attenuated or filtered out.

[34] as opposed to capacitively coupled, or ac amplifiers.

[35] called the **center frequency**

Chapter 2

Operational Amplifiers

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Having learned the fundamental properties of basic amplifier concepts and terminology in Chapter 1, we now have the necessary tools to study a circuit building block of **universal importance**:



Figure 2.1 On of the first commercially available op-amps, the μA709.

> The **operational amplifier**, or simply **op-amp**.

Op amps have been in use for a long time, with their initial applications being primarily in the areas of **analog computation** and **sophisticated instrumentation**. Early op-amps were constructed from discrete components,[1] and their cost were comparatively high. In the mid 1960s is the first IC op-amp was produced. This was made up of a relatively large number of transistors and resistors all on the same silicon chip. Although its characteristics were poor[2] and its price was still quite high, its appearance signaled a new era in electronic circuit design.

Engineers started using op-amps in large quantities, which caused their price to significantly drop. They also demanded better-quality op-amps. Semiconductor manufacturers responded quickly, and within the span of a few years, high-quality op-amps became available at extremely low prices[3] from a large number of suppliers.

[1] vacuum tubes, and then transistors, and resistors

[2] by today's standards.

[3] It is more than possible to buy a dozen of IC op-amps for less than a euro.

One of the reasons for the popularity of the op-amp is its **versatility**. As we will shortly see in this chapter, almost anything can be done using op-amps. Equally important is the fact that the IC op-amp has characteristics which closely approach the assumed ideal. This implies that it is quite easy to design circuits using the IC op-amp. Also, op-amp circuits work at performance levels that are quite close to those predicted theoretically.

It is for this reason that we are studying op amps at this early stage.

As already implied, an IC op-amp is made up of a large number[4] of transistors together with resistors, and one capacitor connected in a rather complex circuit.[5]

[4] about 20.

The circuits inside the op-amp will **NOT** be discussed in this chapter.

We will rather treat the op-amp as a circuit building block and study its terminal characteristics and its applications. This approach is quite satisfactory in many op-amp applications.



[5] An example of an IC op-amp, LM358 Dual Op-Amp IC.

The Traitorous Eight — 2.1 Information

The traitorous eight[6] was a group of eight (**8**) employees who left **Shockley Semiconductor Laboratory** in 1957 to found **Fairchild Semiconductor**. William Shockley had in 1956 recruited a group of young Ph.D. graduates with the goal to develop and produce new semiconductor devices. While Shockley had received a Nobel Prize in Physics and was an experienced researcher and teacher, his management of the group was authoritarian and unpopular. This was accentuated by Shockley's research focus not proving fruitful. After the demand for Shockley to be replaced was rebuffed, the eight left to form their own company.

Shockley described their leaving as a "betrayal". The eight who left Shockley Semiconductor were Julius Blank, Victor Grinich, Jean Hoerni, Eugene Kleiner, Jay Last, Gordon Moore, Robert Noyce, and Sheldon Roberts. In August 1957, they reached an agreement with Sherman Fairchild, and on September 18, 1957, they formed Fairchild Semiconductor. The newly founded Fairchild Semiconductor soon grew into a leader in the semiconductor industry.

In 1960, it became an incubator of **Silicon Valley** and was directly or indirectly involved in the creation of dozens of corporations, including **Intel** and **AMD**. These many spin-off companies came to be known as "Fairchildren".



[6] The traitorous eight. From left to right: Gordon Moore, C. Sheldon Roberts, Eugene Kleiner, Robert Noyce, Victor Grinich, Julius Blank, Jean Hoerni and Jay Last (1960)

2.1 The Ideal Op Amp

It is always in the best interest for any engineer to study the simplest, i.e., the ideal, case of any product which will be the focus of this chapter.

2.1.1 Terminals of the Element

From a signal point of view the op-amp has three (**3**) terminals:

two (**2**) input terminals and one (**1**) output terminal.

Fig. 2.2 shows the symbol we shall use to represent the op-amp. Terminals 1 and 2 are input terminals, and terminal 3 is the output terminal. As explained in Chapter 1, amplifiers require DC power to operate.[7] Most IC op-amps require two (**2**) DC power supplies, as shown in **Fig.** 2.3.

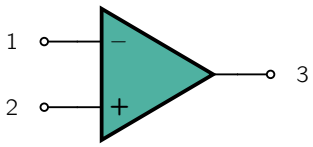


Figure 2.2
Circuit symbol for the standard op-amp.

Two terminals, 4 and 5, are brought out of the op-amp package and connected to a positive voltage V_{CC} and a negative voltage $-V_{EE}$, respectively. In Fig. 2(b) we explicitly show the two DC power supplies as batteries with a common ground. It is interesting to note that the reference grounding point in op-amp circuits is just the common terminal of the two power supplies.

[7] The power to amplify a given signal must come from somewhere.

> no terminal of the op-amp package is **physically connected to ground**.

In what follows we will not, for simplicity, explicitly show the op-amp power supplies.

> In addition to the three (**3**) signal terminals and the two power-supply terminals, an op-amp can have other terminals for specific purposes. These other terminals can include terminals for frequency compensation and terminals for offset pulling.

2.1.2 Function and Characteristics

We now consider the circuit function of the op-amp. The op-amp is designed to **sense the difference** between the voltage signals applied at its two (**2**) input terminals,[8] multiply this by a number A, and cause the resulting voltage $A(v_2 - v_1)$ to appear at output terminal 3. Thus $v_3 = A(v_2 - v_1)$. Here

[8] i.e., the quantity $v_2 - v_1$

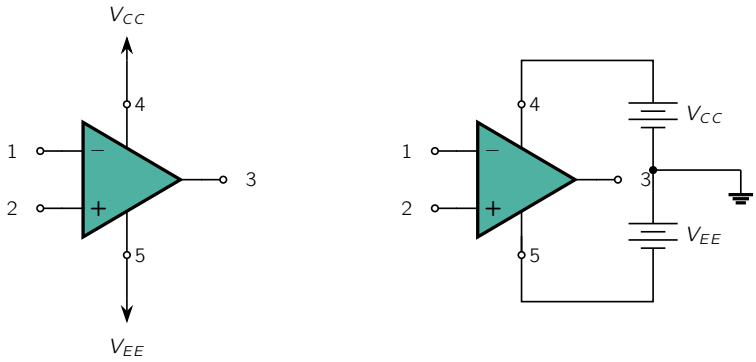


Figure 2.3
The op amp shown connected to dc power supplies (a) The short-hand representation where DC values are shown as arrows going out of the op-amp (b) The actual representation of how DC is fed to the op-amp.

it should be emphasized that when we talk about the voltage at a terminal we mean the voltage between that terminal and ground. Therefore v_1 means the voltage applied between terminal 1 and ground.

> The ideal op-amp is **NOT** supposed to draw any input current.

This means, the signal current into terminal 1 and the signal current into terminal 2 are both zero.

> In other words, the input impedance of an ideal op amp is supposed to be infinite.

How about the output terminal 3?

This terminal is supposed to act as the output terminal of an ideal voltage source. That is, the voltage between terminal 3 and ground will always be equal to $A(v_2 - v_1)$, independent of the current that may be drawn from terminal 3 into a load impedance.

> In other words, the output impedance of an ideal op amp is supposed to be zero.

Putting together all of the above, we arrive at the equivalent circuit model shown in **Fig.** 2.4. Note that the output is in phase with (as the same sign as) v_2 and is out of phase with (has the opposite sign of) v_1.

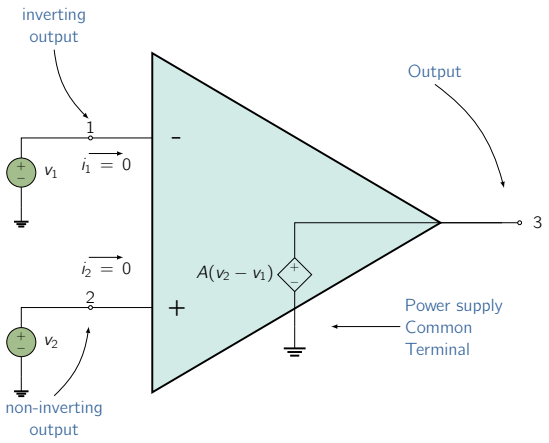


Figure 2.4 **Equivalent circuit of the ideal op amp.**

For this reason:

- input terminal 1 is called the **inverting input terminal** and is distinguished by a "−" sign,
- while input terminal 2 is called the **non-inverting input terminal** and is distinguished by a "+" sign.

As can be seen from **Fig.** 2.4, the op amp responds only to the *difference* signal $v_2 - v_1$ and hence ignores any signal common to both inputs. That is, if $v_1 = v_2 = 1$ V, then the output will (ideally) be zero. We call this property **common-mode rejection**, and we conclude an ideal op amp has zero common-mode gain or, equivalently, infinite common-mode rejection. We will have more to say about this point later. For the time being note that the op amp is a **differential-input**, **single-ended-output** amplifier, with the latter term referring to the fact that the output appears between terminal 3 and ground.

Furthermore, gain A is called the **differential gain**, for obvious reasons. Perhaps **NOT** so obvious is another name that we will attach to A:

the **open-loop gain**.

The reason for this name will become obvious later on when we "close the loop" around the op amp and define another gain, the closed-loop gain.

An important characteristic of op amps is that they are direct-coupled or dc amplifiers, where dc stands for direct-coupled.[9] The fact that op amps are direct-coupled devices will allow us to use them in many important applications.

[9] it could equally well stand for direct current, since a direct-coupled amplifier is one that amplifies signals whose frequency is as low as zero.

> Unfortunately, though, the direct-coupling property can cause some serious practical problems, as will be discussed in a later section.

How about bandwidth? The ideal op amp has a gain A which remains constant down to zero frequency and up to infinite frequency. That is, ideal op amps will amplify signals of any frequency with equal gain, and are thus said to have **infinite bandwidth**.

We have discussed all of the properties of the ideal op amp except for one, which in fact is the most important. This has to do with the value of A. The ideal op amp should have a gain A whose value is very large and ideally infinite. One may justifiably ask:

If the gain A is infinite, how are we going to use the op amp?

The answer is very simple:

In almost all applications the op amp will **NOT** be used alone in a so-called open-loop configuration. Rather, we will use other components to apply feedback to close the loop around the op amp. For future reference, lists the characteristics of the ideal op amp.

- Infinite input impedance
- Zero output impedance
- Zero common-mode gain or, equivalently, infinite common-mode rejection
- Infinite open-loop gain A
- Infinite bandwidth

2.1.3 Differential and Common-Mode Signals

The differential input signal v_{Id} is simply the **difference** between the two (2) given input signals v_1 and v_2, which we can write it as:

$$v_{1d} = v_2 - v_1 \tag{2.1}$$

The common-mode input signal v_{Im} is the **average** of the two (2) input signals v_1 and v_2; namely,

$$v_{Icm} = \frac{1}{2}(v_1 + v_2) \tag{2.2}$$

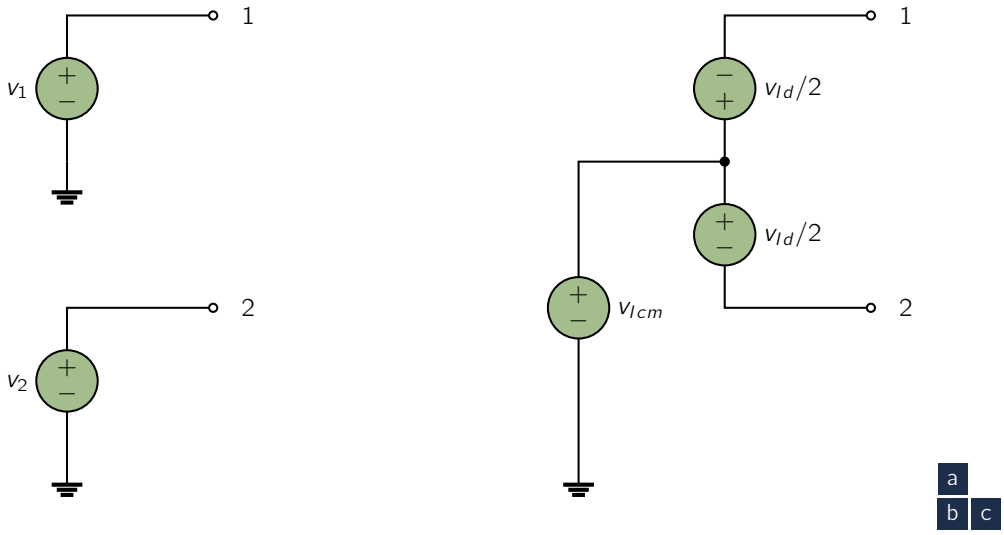


Figure 2.5
Representation of the signal sources (a) v_1, (b) v_2 and (c) in terms of their differential and common-mode components.

Both Eq. (**2.1**) and Eq. (**2.2**) can be used to express the input signals v_1 and v_2 in terms of their differential and common-mode components as: follows:

$$v_1 = v_{Im} - v_{Id}/2 \qquad \text{and} \qquad v_2 = v_{Im} + v_{Id}/2$$

These equations can in turn lead to the pictorial representation in **Fig.** 2.5.

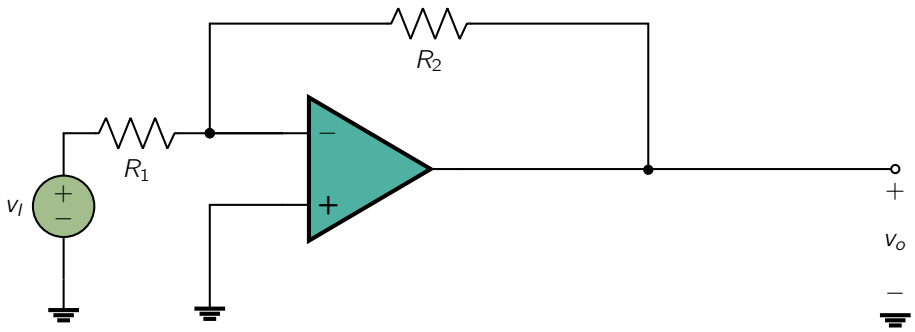


Figure 2.6
The inverting amplifier in closed-loop configuration.

2.2 The Inverting Amplifier

As mentioned previously, op amps are not used alone; rather, the op amp is connected to **passive components** in a feedback circuit. There are two (**2**) such basic circuit configurations employing an op amp and two (**2**) resistors:

1. inverting configuration,
2. non-inverting configuration.

Fig. 2.6 shows the inverting configuration. It consists of one op amp and two (**2**) resistors:

R_1 and R_2

Resistor R_2 is connected from the output terminal of the op amp (terminal 3) back to the **inverting** or **negative** input terminal (terminal 1). We speak of R_2 as applying **negative feedback**; if R_2 were connected between terminals 3 and 2 we would have called this **positive feedback**.

> Please observe that R_2 also closes the loop around the op amp.

In addition to adding R_2, we have grounded terminal 2 and connected a resistor R_1 between terminal 1 and an input signal source with a voltage v_t. The output of the overall circuit is taken at terminal 3.[10] Terminal 3 is, of course, a convenient point from which to take the output, since the impedance level there is ideally zero. Therefor, the voltage v_0 will not depend on the value of the current that might be supplied to a load impedance connected between terminal 3 and ground.

[10] i.e., between terminal 3 and ground.

2.2.1 The Closed-Loop Gain

We now wish to analyze the circuit in **Fig.** 2.6 to determine the closed-loop gain (G), which we defined here as:

$$G \equiv \frac{v_0}{v_0}$$

We will do so assuming the op amp to be **ideal**. Figure 2.6(a) shows the equivalent circuit, and the analysis proceeds as follows:

The gain of the op-amp (A) is very large.[11] If we assume the circuit is operating and producing a finite output voltage at terminal 3, then the voltage between the op-amp input terminals should be negligibly small and **ideally zero**. Specifically, if we call the output voltage v_0, then, by definition,

[11] ideally infinite.

$$v_2 - v_1 = \frac{v_0}{A} = 0$$

Through this equivalence, it follows that the voltage at the inverting input terminal (v_1) is given by $v_1 = v_2$. That is, because the gain (A) approaches infinity, the voltage v_1 approaches and ideally equals v_2.

> We speak of this as the two (2) input terminals **tracking each other in potential**.

We also speak of a **virtual short circuit** that exists between the two (**2**) input terminals. Here the word **virtual** should be emphasized.[12] A virtual short circuit means that whatever voltage is at 2 will automatically appear at 1 because of the infinite gain (A). However, terminal 2 happens to be connected to ground. Therefore $v_2 = 0$ and $v_1 = 0$. We speak of terminal 1 as being a **virtual ground**.[13] Now that we have determined v_1 we are in a position to apply Ohm's law and find the current i_1 through R_1 (see Fig. 2.6) as follows:

[12] One should **NOT** make the mistake of physically shorting terminals 1 and 2 together while analyzing a circuit.

[13] that is, having zero voltage but not physically connected to ground

$$i_1 = \frac{v_I - v_1}{R_1} = \frac{v_I - 0}{R_1} = \frac{v_I}{R_1}$$

Where will this current go?

It cannot go into the op amp, since the ideal op amp has an infinite input impedance and hence draws zero current. It follows that i_1 will have to flow through R_2, to the low-impedance terminal 3. We can then apply Ohm's law to R_2 and determine v_0.

That is,

$$\begin{aligned} v_o &= v_1 - i_1 R_2 \\ &= 0 - \frac{v_I}{R_1} R_2 \end{aligned} \qquad \text{therefore} \qquad \frac{v_o}{v_i} = -\frac{R_2}{R_1}$$

which is the required closed-loop gain. Figure 2.6(b) illustrates these steps and indicates by the circled numbers the order in which the analysis is performed. We therefore see the closed-loop gain is simply the ratio of the two (**2**) resistances R_2 and R_1. The minus sign means that the closed-loop amplifier provides signal inversion. Thus if $R_2/R_1 = 10$ and we apply at the input (v_1) a sine-wave signal of 1 V peak-to-peak, then the output vo will be a sine wave of 10 V peak-to-peak and phase-shifted 180ř with respect to the input sine wave. Because of the minus sign associated with the closed-loop gain, this configuration is called the inverting configuration.

The fact that the closed-loop gain depends entirely on external passive components[14] is very significant. It means that we can make the closed-loop gain as accurate as we want by selecting

[14] Here, these are resistors R_1 and $_2$

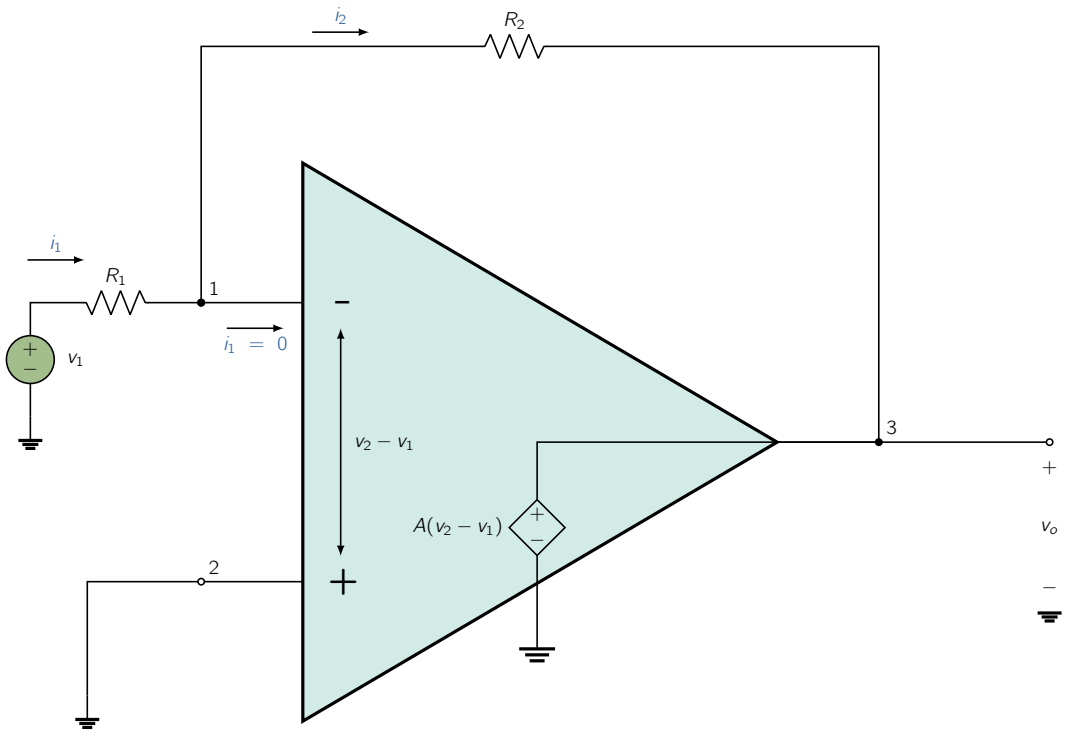


Figure 2.7
Analysis of the inverting configuration taking into account the finite open-loop gain of the op amp.

passive components of appropriate accuracy. It also means that the closed-loop gain is (ideally) independent of the op-amp gain. This is a dramatic illustration of negative feedback:

> We started out with an amplifier having very large gain A, and through applying negative feedback we have obtained a closed-loop gain R_2/R_1 that is much smaller than A but is stable and predictable.

That is, we are trading gain for accuracy.

2.2.2 Effect of Finite Open-Loop Gain

The points just made are more clearly illustrated by deriving an expression for the closed-loop gain under the assumption that the op-amp open-loop gain A is finite. If we denote the output voltage vo, then the voltage between the two input terminals of the op amp will be v_0/A. Since the positive input terminal is grounded, the voltage at the negative input terminal must be $-v_0/A$. The current i_1 through R_1 can now be found from

$$i_1 = \frac{v_I - (-v_O/A)}{R_1} = \frac{v_I + v_O/A}{R_1}$$

The infinite input impedance of the op amp forces the current i_1 to flow entirely through R_2. The

output voltage v_0 can thus be determined from

$$v_0 = -\frac{v_0}{4} - i_1 R_2 = -\frac{v_0}{A} - \left(\frac{v_1 + v_0/A}{R_1}\right) R_2$$

Collecting terms, the closed-loop gain G is found as:

$$G \equiv \frac{v_0}{v_t} = \frac{-R_2/R_1}{1 + (1 + R_2/R_1)/A} \tag{2.3}$$

We note that as A approaches ∞, G approaches the ideal value of $-R_2/R_1$. Also, from **Fig. ??** we see that as A approaches ∞, the voltage at the inverting input terminal approaches zero. This is the virtual-ground assumption we used in our earlier analysis when the op amp was assumed to be ideal. Finally, note that Eq. (**2.3**) in fact indicates that to minimize the dependence of the closed-loop gain G on the value of the open-loop gain A, we should make:

$$1 + \frac{R_2}{R} \ll A$$

2.2.3 Input and Output Resistances

Assuming an ideal op amp with infinite open-loop gain, the input resistance of the closed-loop inverting amplifier of Fig. 2.5 is simply equal to R_1. This can be seen from Fig. 2.6(b), where

$$R_i \equiv \frac{v_I}{i_1} = \frac{v_I}{v_I/R_1} = R_1$$

Now recall that in Section 1.5 we learned that the amplifier input resistance forms a voltage divider with the resistance of the source that feeds the amplifier. Thus, to avoid the loss of signal strength, voltage amplifiers are required to have high input resistance. In the case of the invert- ing op-amp configuration we are studying, to make R_i high we should select a high value for R_1. However, if the required gain R_2/R_1 is also high, then R_2 could become impractically large.[15] We may conclude that the inverting configuration suffers from a low input resistance.

[15] This is in the range of few MΩ.

Since the output of the inverting configuration is taken at the terminals of the ideal voltage source $A(v_2 - v_1)$ (see Fig. 2.6a), it follows that the output resistance of the closed-loop amplifier is zero.

2.2 Information

Application: The Weighted Summer

An important application of the **inverting configuration** is the weighted-summer circuit shown in Fig. 2.10. Here we have a resistance R_f, in the negative-feedback path, similar to before. But we have a number of input signals ($x_1, \cdots, x_n$) each applied to a corresponding resistor ($x_1, \cdots, x_n$), which are connected to the **inverting terminal** of the op-amp.

From our previous discussion, the ideal op amp will have a virtual ground appearing at its negative input terminal.

Continues on next page →

← Continued from previous page

Ohm's law then tells us that the currents ($x_1, \cdots, x_n$) are given by:

$$i_1 = \frac{v_1}{R_1},\ i_2 = \frac{v_2}{R_2},\ \ldots,\ i_n = \frac{v_n}{R_n}$$

All these currents sum together to produce the current i,

$$i = i_1 + i_2 + \cdots + i_n$$

which will be forced to flow through R_f.[16] The output voltage (v_o) may now be determined by another application of Ohm's law,

$$v_0 = 0 - iR_f = -iR_f$$

and therefore:

$$v_o = -\left(\frac{R_f}{R_1}v_1 + \frac{R_f}{R_2}v_2 + \cdots + \frac{R_f}{R_n}v_n\right)$$

That is, the output voltage is a weighted sum of the input signals $x_1, \cdots, x_n$. This circuit is therefore called a **weighted summer**. Note that each summing coefficient may be independently adjusted by adjusting the corresponding "feed-in" resistor (R_1 to R_n). This nice property, which greatly simplifies circuit adjustment, is a direct consequence of the virtual ground that exists at the inverting op-amp terminal. In the weighted summer of Fig. 2.10 all the summing coefficients must be of the same sign. The need occasionally arises for summing signals with opposite signs. Such a function can be implemented, however, using two op amps as shown in Fig. 2.11. Assuming ideal op amps, it can be easily shown that the output voltage is given by

$$v_1\left(\frac{R_a}{R_a}\right)\left(\frac{R_c}{R_a}\right) + v_2\left(\frac{R_a}{R_a}\right)\left(\frac{R_c}{R_a}\right) - v_3\left(\frac{R_c}{R_a}\right) - v_4\left(\frac{R_c}{R_a}\right)$$

Weighted summers are utilized in a variety of applications including in the design of audio systems, where they can be used in mixing signals originating from different musical instruments.

[16] since no current flows into the input terminals of an ideal op amp.

2.3 Non-Inverting Amplifier

The second closed-loop configuration we shall study is shown in **Fig.** 2.8. Here the input signal v_i is **applied directly** to the positive input terminal of the op-amp while one terminal of R_1 is connected to **ground**.

2.3.1 The Closed-Loop Gain

Analysis of the non-inverting circuit to determine its closed-loop gain (v_0/v_1) is illustrated in **Fig.** 2.8. Again the order of the steps in the analysis is indicated by circled numbers. Assuming the op-amp is **ideal** with **infinite gain**, a virtual short circuit exists between its two (**2**) input terminals.

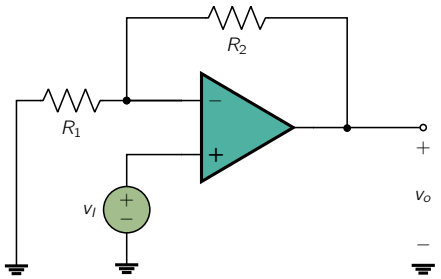


Figure 2.8: The noninverting configuration.

Therefore the difference input signal is:

$$v_{Id} = \frac{v_o}{A} = 0 \qquad \text{for} \qquad A = \infty$$

Therefore the voltage at the inverting input terminal will be equal to that at the non-inverting input terminal, which is the applied voltage v_I. The current through R_1 can then be determined as v_I/R_I. Because of the infinite input impedance of the op amp, this current will flow through R_2, as shown in **Fig. ??**.

Now the output voltage can be determined from

$$v_O = v_I + \left(\frac{v_I}{R_1}\right) R_2 \qquad \text{which gives} \qquad \frac{v_O}{v_I} = 1 + \frac{R_I}{R_2} \tag{2.4}$$

Further insight into the operation of the non-inverting configuration can be obtained by considering the following:

> Since the current into the op-amp inverting input is zero, the circuit composed of R_1 and R_2 acts in effect as a voltage divider feeding a fraction of the output voltage back to the inverting input terminal of the op amp;

that is,

$$v_1 = v_0 \left(\frac{R_1}{R_1 + R_2}\right)$$

Then the infinite op-amp gain and the resulting virtual short circuit between the two input terminals of the op amp forces this voltage to be equal to that applied at the positive input terminal; thus,

$$v_o \left(\frac{R_1}{R_1 + R_2}\right) = v_I$$

which produces the gain expression given in Eq. (**2.4**).

This is an appropriate point to reflect further on the action of the negative feedback present in the noninverting circuit of Fig. 2.12. Let vI increase. Such a change in vI will cause vId to increase, and vo will correspondingly increase as a result of the high[17] gain of the op amp. However, a fraction of the increase in vo will be fed back to the inverting input terminal of the op amp through the (R_1, R_2) voltage divider. The result of this feedback will be to counteract the increase in v_{Id}, driving v_{Id} back to zero, albeit at a higher value of v_o that corresponds to the increased value of v_t. This **degenerative** action of negative feedback gives it the alternative name **degenerative feedback**. Finally, note that the argument above applies equally well if v_I decreases.

[17] ideally infinite

2.3.2 Effect of Finite Open-Loop Gain

As we have done for the inverting configuration, we now consider the effect of the finite op-amp open-loop gain A on the gain of the noninverting configuration. Assuming the op amp to be ideal except for having a finite open-loop gain A, it can be shown that the closed-loop gain of the noninverting amplifier circuit of Fig. 2.12 is given by

$$G \equiv \frac{v_O}{v_I} = \frac{1 + (R_2/R_1)}{1 + \frac{1+(R_2/R_1)}{4}}$$

Observe that the denominator is identical to that for the case of the inverting configuration (Eq. 2.5). This is no coincidence; it is a result of the fact that both the inverting and the noninverting configurations have the same feedback loop, which can be readily seen if the input signal source is eliminated.[18] The numerators, however, are different, for the numerator gives the ideal or nominal closed-loop gain $(-R_2/R_1)$ for the inverting configuration, and $1 + R_2/R_1$ for the noninverting configuration). Finally, we note (with reassurance) that the gain expression in Eq. (2.11) reduces to the ideal value for $A = \infty$.

[18] i.e., short-circuited.

In fact, it approximates the ideal value for

$$A \gg 1 + \frac{R_2}{R_1}$$

This is the same condition as in the inverting configuration, except that here the quantity on the right-hand side is the nominal closed-loop gain. The expressions for the actual and ideal values of the closed-loop gain G in Eqs. (2.11) and (2.9), respectively, can be used to determine the percentage error in G resulting from the finite op-amp gain A as

$$\text{Percent gain error} = -\frac{1 + (R_2/R_1)}{A + 1 + (R_2/R_1)} \times 100$$

Thus, as an example, if an op amp with an open-loop gain of 1000 is used to design a non-inverting amplifier with a nominal closed-loop gain of 10, we would expect the closed-loop gain to be about 1% below the nominal value.

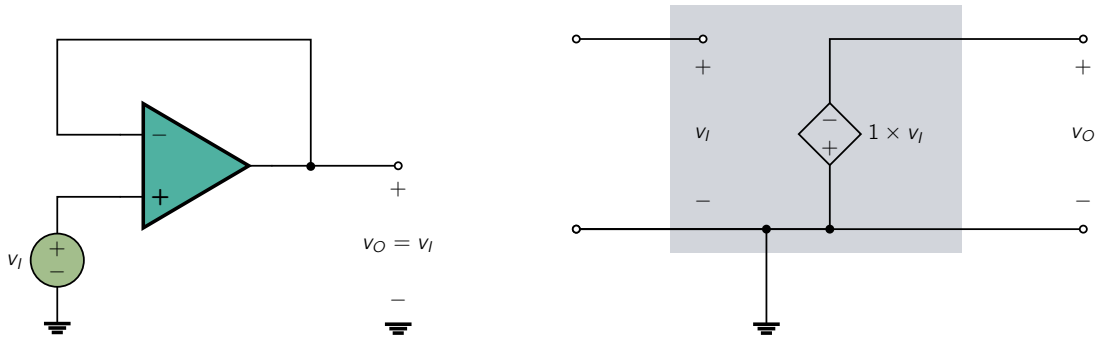


Figure 2.9: (a) The unity-gain buffer or follower amplifier. (b) Its equivalent circuit model.

2.3.3 Input and Output Resistance

The gain of the non-inverting configuration is positive and hence the name **noninverting**. The input impedance of this closed-loop amplifier is ideally infinite, since no current flows into the positive input terminal of the op amp. The output of the noninverting amplifier is taken at the terminals of the ideal voltage source $A(v_2 - v_1)$,[19] and thus the output resistance of the noninverting configuration is zero.

[19] see the op-amp equivalent circuit in Fig. 2.3

2.3.4 The Voltage Follower

The property of high input impedance is a very desirable feature of the non-inverting configuration. It enables using this circuit as a buffer amplifier to connect a source with a high impedance to a low-impedance load. We discussed the need for buffer amplifiers in Section 1.5. In many applications the buffer amplifier is not required to provide any voltage gain; rather, it is used mainly as an impedance transformer or a power amplifier. In such cases we may make $R_2 = 0$ and $R_1 = \infty$ to obtain the **unity-gain amplifier** shown in Fig. 2.14(a). This circuit is commonly referred to as a **voltage follower**, since the output "follows" the input. In the ideal case, $v_0 = v_I$, $R_{in} = \infty$, $R_{out} = 0$, and the follower has the equivalent circuit shown in Fig. 2.14(b).

Since in the voltage follower circuit the entire output is fed back to the inverting input, the circuit is said to have 100% negative feedback. The infinite gain of the op amp then acts to make $v_{Id} = 0$ and hence $v_0 = v_I$. Since the noninverting configuration has a gain greater than or equal to unity, depending on the choice of R_2/R_1, some prefer to call it "a follower with gain."

2.4 Difference Amplifiers

Having studied the two basic configurations of op-amp circuits together with some of their direct applications, we are now ready to consider a somewhat more involved but very important application. Specifically, we shall study the use of op amps to design difference.

A difference amplifier is one that responds to the **difference between the two signals** applied at its input and ideally rejects signals that are common to the two inputs. The representation of signals in terms of their differential and common-mode components was given in **Fig.** 2.5. Although ideally the difference amplifier will amplify only the differential input signal $v_{I,d}$ and reject completely the common-mode input signal $v_{I,cm}$, practical circuits will have an output voltage v_O given by

$$v_O = A_d v_{I,d} + A_{cm} v_{I,cm}$$

where A_d denotes the amplifier differential gain and A_{cm} denotes its common-mode gain (ideally zero). The efficacy of a differential amplifier is measured by the degree of its rejection of common-mode signals in preference to differential signals. This is usually quantified by a measure known as the **common-mode rejection ratio** (**CMRR**), defined as

$$\text{CMRR} = 20 \log \frac{|A_d|}{|A_{cm}|}$$

The need for difference amplifiers arises frequently in the design of electronic systems, especially those employed in instrumentation. As a common example, consider a transducer providing a small (e.g., 1 mV) signal between its two output terminals while each of the two wires leading from the transducer terminals to the measuring instrument may have a large interference signal (e.g., 1 V) relative to the circuit ground. The instrument front end obviously needs a difference amplifier.

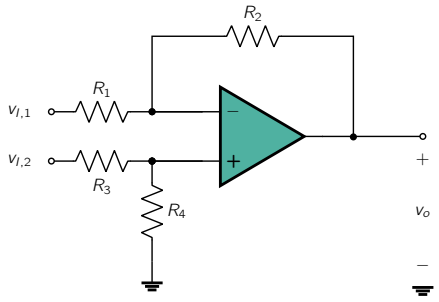
The need for a Difference Configuration 2.3 Information

Before we proceed any further we should address a question that the reader might have: The op amp is itself a difference amplifier; why not just use an op amp? The answer is that the very high,ideally infinite, gain of the op amp makes it impossible to use by itself. Rather, as we did before, we have to devise an appropriate feedback network to connect to the op amp to create a circuit whose closed-loop gain is finite, predictable, and stable.

2.4.1 A Single-Op-Amp Difference Amplifier

Our first attempt at designing a difference amplifier is motivated by the observation that the gain of the non-inverting amplifier configuration is positive, $\left(1 + R_2/R_1\right)$, while that of the inverting configuration is negative, $(-R_2/R_1)$. Combining the two configurations together is then a step in the right direction-namely, getting the difference between two input signals.

Figure 2.10
A difference amplifier.



Of course, we have to make the two gain magnitudes equal in order to reject common-mode signals. This, however, can be easily achieved by attenuating the positive input signal to reduce the gain of the positive path from $(1+R_2/R_1)$ to (R_2/R_1). The resulting circuit would then look like that shown in **Fig.** 2.10, where the attenuation in the positive input path is achieved by the voltage divider (R_3, R_4). The proper ratio of this voltage divider can be determined from

$$\frac{R_4}{R_4+R_3}\left(1+\frac{R_2}{R_1}\right)=\frac{R_2}{R_1}$$

which can be put in the form

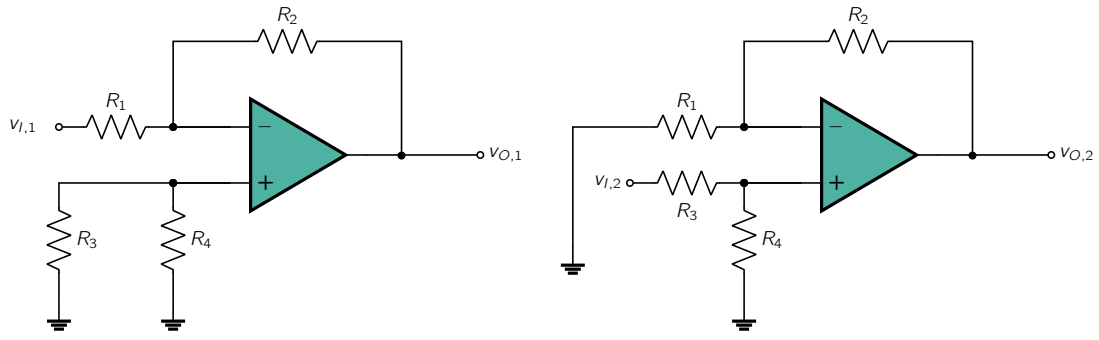
$$\frac{R_4}{R_4+R_3}=\frac{R_2}{R_2+R_1} \qquad \text{satisfied by} \qquad \frac{R_4}{R_2}=\frac{R_2}{R_1} \tag{2.5}$$

This completes our work. However, Let's step back and verify that the circuit in **Fig.** 2.10 with R_3 and R_4 selected according to Eq. (**2.5**) does in fact function as a difference amplifier. Specifically, we wish to determine the output voltage v_O in terms of v_{I1} and v_{I2}. Toward that end, we observe that the circuit is linear, and thus we can use superposition.

To apply superposition, we first reduce $v_{I,2}$ to zero-that is, ground the terminal to which v_{12} is applied-and then find the corresponding output voltage, which will be due entirely to v_{11}. We denote this output voltage v_{01}. Its value may be found from the circuit in Fig. 2.17(a), which we recognize as that of the inverting configuration. The existence of R_3 and R_4 does not affect the gain expression, since no current flows through either of them. Thus,

$$v_{01}=-\frac{R_2}{R_1}v_{I1}$$

Next, we reduce v_{I1} to zero and evaluate the corresponding output voltage v_{02}. The circuit will now take the form shown in Fig. 2.17(b), which we recognize as the noninverting configuration with an



(a) (b)

Figure 2.11: Application of superposition to the analysis of the circuit

additional voltage divider, made up of R_3 and R_4, connected to the input v_{12}. The output voltage v_{02} is therefore given by

$$v_{02} = v_{12}\frac{R_4}{R_2 + R_4}\left(1 + \frac{R_2}{R_1}\right) = \frac{R_2}{R_1}v_{12}$$

where we have utilized Eq. (**2.5**). The superposition principle tells us that the output voltage v_0 is equal to the sum of v_{01} and v_{02}. Thus we have

$$v_O = \frac{R_2}{R}(v_{I2} - v_{I1}) = \frac{R_2}{R}v_{Id}$$

Thus, as expected, the circuit acts as a difference amplifier with a differential gain A_d of

$$A_d = \frac{R_2}{R_1}$$

Of course this is predicated on the op amp being ideal and furthermore on the selection of R_3 and R_4 so that their ratio matches that of R_1 and R_2 (Eq. 2.15). To make this matching requirement a little easier to satisfy, we usually select

$$R_3 = R_1 \qquad \text{and} \qquad R_4 = R_2$$

Let's next consider the circuit with only a common-mode signal applied at the input, as shown in Fig. 2.18. The figure also shows some of the analysis steps. Thus,

$$\begin{aligned} i_1 &= \frac{1}{R_1}\left[v_{Icm} - \frac{R_4}{R_4 + R_3}v_{Icm}\right] \\ &= v_{Icm}\frac{R_3}{R_4 + R_2}\frac{1}{R_1} \end{aligned}$$

The output voltage can now be found from

$$v_O = \frac{R_4}{R_4 + R_3}v_{Icm} - i_2R_2$$

Substituting $i_2 = i_1$ and for i_1 from Eq. (2.18),

$$\begin{aligned} v_O &= \frac{R_4}{R_4 + R_3}v_{Icm} - \frac{R_2}{R_1}\frac{R_3}{R_4 + R_3}v_{Icm} \\ &= \frac{R_4}{R_4 + R_2}\left(1 - \frac{R_2}{R_1}\frac{R_3}{R_4}\right)v_{Icm} \end{aligned}$$

Thus,

$$A_{cm} \equiv \frac{v_O}{v_{cm}} = \left(\frac{R_4}{R_4 + R_3}\right)\left(1 - \frac{R_2}{R_1}\frac{R_3}{R_4}\right)$$

For the design with the resistor ratios selected according to Eq. (2.15), we obtain

$$A_{cm} = 0$$

as expected. Note, however, that any mismatch in the resistance ratios can make A_{cm} nonzero, and hence CMRR finite.

In addition to rejecting common-mode signals, a difference amplifier is usually required to have a high input resistance. To find the input resistance between the two input terminals (i.e., the resistance seen by v_{Id}), called the **differential input resistance** R_{Id}, consider **Fig.** 2.12. Here we have assumed that the resistors are selected so that

$$R_3 = R_1 \qquad \text{and} \qquad R_4 = R_2$$

Now

$$R_{id} \equiv \frac{v_{Id}}{i_I}$$

Since the two input terminals of the op amp track each other in potential, we may write a loop equation and obtain

$$v_{Id} = R_1 i_I + 0 + R_1 i_I$$

Thus,

$$R_{id} = 2R_1$$

Note that if the amplifier is required to have a large differential gain (R_2/R_1), then R_1 of necessity will be relatively small and the input resistance will be correspondingly low, a drawback of this circuit.

Another drawback of the circuit is that it is not easy to vary the differential gain of the amplifier. Both of these drawbacks are overcome in the instrumentation amplifier discussed next.

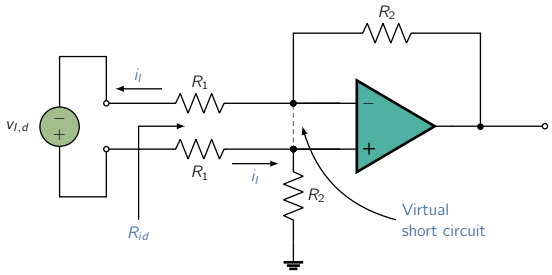


Figure 2.12 **Finding the input resis-tance of the difference amplifier for thecase R3= R1 and R4= R2.**

2.4.2 A Versatile Circuit: The Instrumentation Amplifier

The low-input-resistance problem of the difference amplifier of Fig. 2.16 can be solved by using voltage followers to buffer the two input terminals; that is, a voltage follower of the type in Fig. 2.14 is connected between each input terminal and the corresponding input terminal of the difference amplifier. However, if we are going to use two additional op amps, we should ask the question: Can we get more from them than just impedance buffering? An obvious answer would be that we should try to get some voltage gain. It is especially interesting that we can achieve this without compromising the high input resistance simply by using followers with gain rather than unity-gain followers. Achieving some or indeed the bulk of the required gain in this new first stage of the differential amplifier eases the burden on the difference amplifier in the second stage, leaving it to its main task of implementing the differencing function and thus rejecting common-mode signals.

The resulting circuit is shown in Fig. 2.20(a). It consists of two stages in cascade. The first stage is formed by op amps A_1 and A_2 and their associated resistors, and the second stage is the by-now-familiar difference amplifier formed by op amp A_3 and its four associated resistors. Observe that as we set out to do, each of A_1 and A_2 is connected in the noninverting configu- ration and thus

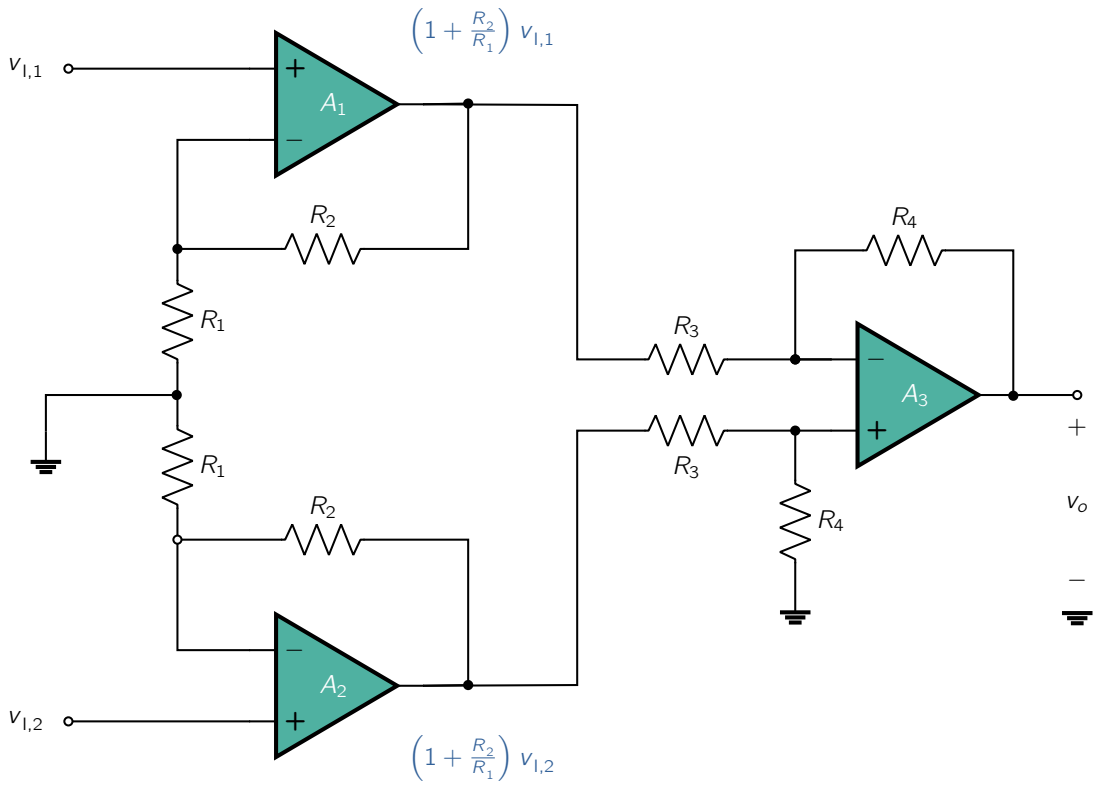


Figure 2.13
A popular circuit for an instrumentation amplifier. (a) Initial approach to the circuit. (b) The circuit in (a) with the connection between node X and ground removed and the two resistors R1 and R1 lumped together. This simple wiring change dramatically improves performance.

realizes a gain of $(1 + R_2/R_1)$. It follows that each of v_{I1} and v_{I2} is amplified by this factor, and the resulting amplified signals appear at the outputs of A_1 and A_2, respectively. The difference amplifier in the second stage operates on the difference signal $(1 + R_2/R_1)(v_{12} - v_{11}) = (1 + R_2/R_1)v_{1d}$ and provides at its output

$$v_O = \frac{R_4}{R_3}\left(1 + \frac{R_2}{R_1}\right)v_{Id}$$

Thus the differential gain realized is

$$A_d = \left(\frac{R_4}{R_3}\right)\left(1 + \frac{R_2}{R_1}\right)$$

The common-mode gain will be zero because of the differencing action of the second-stage amplifier

The circuit in Fig. 2.20(a) has the advantage of very high (ideally infinite) input resistance and high differential gain. Also, provided A_1 and A_2 and their corresponding resistors are matched, the two signal paths are symmetric-a definite advantage in the design of a differential amplifier. The circuit, however, has three major disadvantages:

1. The input common-mode signal v_{Im} is amplified in the first stage by a gain equal to that experienced by the differential signal v_{Id}. This is a very serious issue, for it could result in the signals at the outputs of A_1 and A_3 being of such large magnitudes that the op amps saturate (more on op-amp saturation in Section 2.8). But even if the op amps do not saturate, the difference amplifier of the second stage will now have to deal with much larger common-mode signals, with the result that the CMRR of the overall amplifier will inevitably be reduced. 2. The two amplifier channels in

the first stage have to be perfectly matched, otherwise a spurious signal may appear between their two outputs. Such a signal would get amplified by the difference amplifier in the second stage. 3. To vary the differential gain A_d, two resistors have to be varied simultaneously, say the two resistors labeled R_1. At each gain setting the two resistors have to be perfectly matched: a difficult task.

All three problems can be solved with a very simple wiring change: Simply disconnect the node between the two resistors labeled R_1, node X, from ground. The circuit with this small but functionally profound change is redrawn in Fig. 2.20(b), where we have lumped the two resistors (R_1 and R_1) together into a single resistor ($2R_1$).

Analysis of the circuit in Fig. 2.20(b), assuming ideal op amps, is straightforward, as is illustrated in Fig. 2.20(c). The key point is that the virtual short circuits at the inputs of op amps A_1 and A_2 cause the input voltages v_{I1} and v_{I2} to appear at the two terminals of resistor ($2R_1$). Thus the differential input voltage $v_{12} - v_{11} \equiv v_{1d}$ appears across $2R_1$ and causes a current $i = v_{Id}/2R_1$ to flow through $2R_1$ and the two resistors labeled R_2. This current in turn produces a voltage difference between the output terminals of A_1 and A_2 given by

$$v_{O2} - v_{O1} = \left(1 + \frac{2R_2}{2R_1}\right) v_{Id}$$

The difference amplifier formed by op amp A_3 and its associated resistors senses the voltage difference ($v_{02} - v_{01}$) and provides a proportional output voltage v_0:

$$\begin{aligned} v_0 &= \frac{R_4}{R_1}(v_{02} - v_{01}) \\ &= \frac{R_4}{R_2}\left(1 + \frac{R_2}{R_1}\right) v_{Id} \end{aligned}$$

Thus the overall differential voltage-gain is given by

$$A_d \equiv \frac{v_O}{v_H} = \frac{R_4}{R_2}\left(1 + \frac{R_2}{R_1}\right)$$

Observe that proper differential operation does not depend on the matching of the two resistors labeled R_2. Indeed, if one of the two is of different value, say R_2', the expression for A_d becomes

$$A_d = \frac{R_4}{R_2}\left(1 + \frac{R_2 + R_2'}{2R_1}\right)$$

Consider next what happens when the two input terminals are connected together to a common-mode input voltage v_{Im}. It is easy to see that an equal voltage appears at the negative input terminals of A_1 and A_2, causing the current through $2R_1$ to be zero. Thus there will be no current flowing in the R_2 resistors, and the voltages at the output terminals of A_1 and A_2 will be equal to the input (i.e., v_{Icm}). Thus the first stage no longer amplifies v_{Icm}; it simply propagates v_{Icm} to its two output terminals, where they are subtracted to produce a zero common-mode output by A_3. The difference amplifier in the second stage, however, now has a much improved situation at its input: The difference signal has been amplified by $(1 + R_2/R_1)$ while the common-mode voltage remained unchanged.

Finally, we observe from the expression in Eq. (2.22) that the gain can be varied by changing only one resistor, $2R_1$. We conclude that this is an excellent differential amplifier circuit and is widely

employed as an instrumentation amplifier, that is, as the input amplifier used in a variety of electronic instruments.

2.5 Integrators and Differentiators

The applications we have studied up to now have **only** utilised resistors in the op-amp feedback path and in connecting the signal source to the circuit, that is, in the feed-in path. As a result, circuit operation has been[20] **independent of frequency**. By allowing the use of both capacitors along with resistors in the feedback and feed-in paths of op-amp circuits, we can extend the use of op-amps and create wide range of useful and exciting applications.

[20] ideally

We begin our study of op-amp RC circuits by considering two (**2**) basic applications:

signal **integrators** and **differentiators**.

2.5.1 The Inverting Configuration with General Impedances

To start, let us consider the **inverting closed-loop configuration** with impedances as $Z_1(s)$ and $Z_2(s)$ replacing the previously used resistors R_1 and R_2, respectively.

The resulting circuit is shown in **Fig.** 2.14 and, for an ideal op amp, has the closed-loop gain or, more appropriately, the closed-loop transfer function

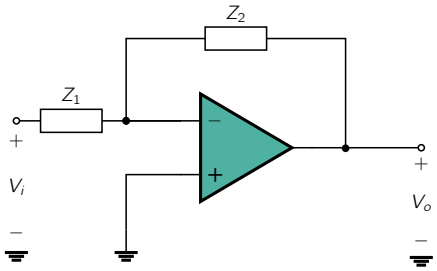


Figure 2.14 **The inverting configuration with general impedances in the feedback and the feed-in paths.**

$$\frac{V_o(s)}{V_i(s)} = -\frac{Z_2(s)}{Z_1(s)} \qquad (2.6)$$

As explained in the previous chapter, replacing s by $\mathbf{j}\omega$ provides the transfer function for physical frequencies ω:

the transmission magnitude and phase for a sinusoidal input signal of frequency (ω).

2.5.2 The Inverting Integrator

By placing a capacitor in the feedback path[21] and a resistor at the input, in place of Z_1, we obtain the circuit as seen in Fig. 2.24(a). We will now see that this circuit realizes the mathematical operation of integration.

[21] i.e., in place of Z_2 in **Fig.** 2.14

Let the input be a time-varying function of $v_1(t)$. The virtual ground at the inverting op-amp input causes $v_1(t)$ to appear in effect across R, and therefore the current $i_1(t)$ will be $v_1(t)/R$. This current flows through the capacitor C, causing **charge to accumulate** on C. If we assume that the circuit begins operation at time $t = 0$, then at an arbitrary time t, the current $i_1(t)$ will have deposited on C a charge equal to $\int_0^t i_1(t)dt$.

Therefore the capacitor voltage $v_c(t)$ will change by:

$$\frac{1}{C}\int_0^t i_1(t)\ \mathrm{d}t.$$

If the initial voltage on C (at $t = 0$) is denoted V_C, then

$$v_c(t) = V_c + \frac{1}{C}\int_0^t i_1(t)dt$$

Now the output voltage $v_o(t) = -v_C(t)$; thus,

$$v_O(t) = -\frac{1}{CR}\int_0^t v_I(t)dt - V_C$$

Therefore the circuit provides an output voltage which is **proportional to the time integral of the input**, with V_C being the initial condition of integration and CR the integrator time constant.

As expected, there is a negative sign attached to the output voltage, and thus this integrator circuit is said to be an **inverting integrator**.

It is also known as a Miller integrator[22] after an early worker in this field.



[22] Noted American electrical engineer, best known for discovering the Miller effect and inventing fundamental circuits for quartz crystal oscillators.

The operation of the integrator circuit can be described alternatively in the frequency domain by substituting $Z_1(s) = R$ and $Z_2(s) = 1/sC$ in Eq. (**2.6**) to obtain the transfer function:

$$\frac{V_o(s)}{V(s)} = -\frac{1}{sCR}$$

For physical frequencies, $s = \mathbf{j}\omega$ and

$$\frac{V_o(\mathbf{j}\omega)}{V_i(\mathbf{j}\omega)} = -\frac{1}{\mathbf{j}\omega CR}$$

Thus the integrator transfer function has magnitude

$$\left|\frac{V_o}{V}\right| = \frac{1}{\omega CR} \qquad \text{and phase} \qquad \phi = +\angle 90 \tag{2.7}$$

The Bode plot for the integrator magnitude response can be obtained by noting from Eq. (**2.7**) that as ω doubles[23] the magnitude is halved (decreased by 6 dB). Therefore the Bode plot is a straight line of slope -6 dB/octave (or, equivalently, -20 dB/ decade). This line (shown in **Fig.** 2.15) intercepts the 0-dB line at the frequency that makes $|V_a/V_i| = 1$, which from Eq. (**2.7**) is:

[23] increases by an octave.

$$\omega_{int} = \frac{1}{CR}$$

The frequency ω_{int} is known as the **integrator frequency** and is simply the inverse of the integrator time constant.

Comparison of the frequency response of the integrator to that of an STC low-pass network indicates that the integrator behaves as a low-pass filter with a corner frequency of zero. Observe also that

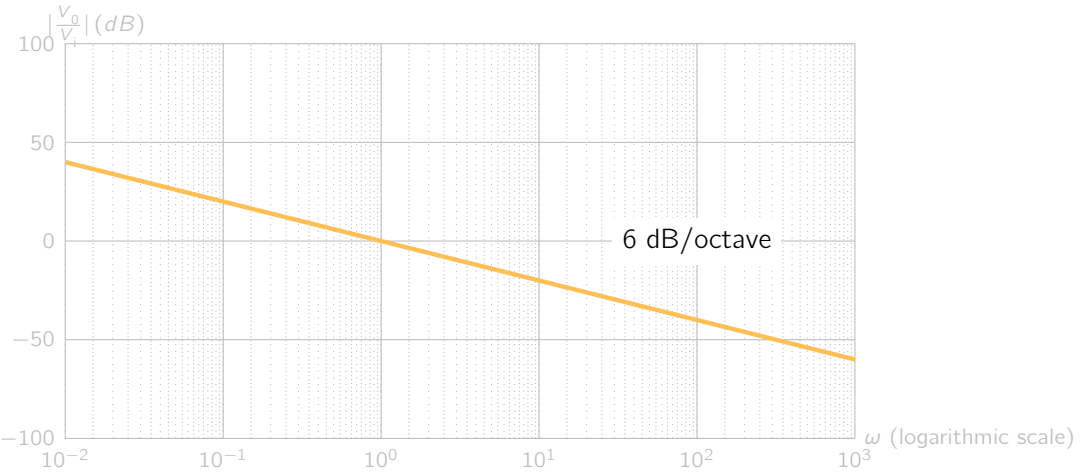


Figure 2.15: Frequency response of the integrator.

at $\omega = 0$, the magnitude of the integrator transfer function is infinite. This indicates that at dc the op amp is operating with an open loop. This should also be obvious from the integrator circuit itself. Reference to Fig. 2.24(a) shows that the feedback element is a capacitor, and thus at dc, where the capacitor behaves as an open circuit, there is no negative feedback! This is a very significant observation and one that indicates a source of problems with the integrator circuit: Any tiny dc component in the input signal will theoretically produce an infinite output. Of course, no infinite output voltage results in practice; rather, the output of the amplifier saturates at a voltage close to the op-amp positive or negative power supply (L_+ or L_-), depending on the polarity of the input dc signal.

The dc problem of the integrator circuit can be alleviated by connecting a resistor R_F across the integrator capacitor C, as shown in **Fig.** 2.16, and therefore the gain at dc will be $-R_F/R$ rather than infinite. Such a resistor provides a DC feedback path. Unfortunately, however, the integration is no longer ideal, and the lower the value of R_F, the less ideal the integrator circuit becomes.

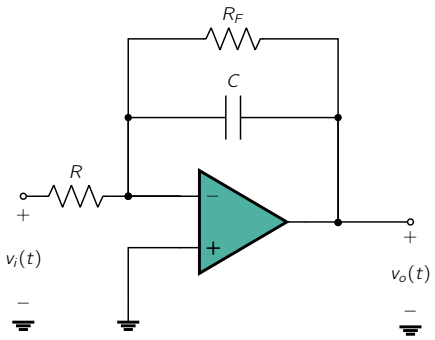


Figure 2.16
The Miller integrator with a large resistance RF connected in parallel with C in order to provide negative feedback and hence finite gain at dc.

> This is because R_F causes the frequency of the integrator pole to move from its ideal location at $\omega = 0$ to one determined by the corner frequency of the STC network (R_F, C).

Specifically, the integrator transfer function becomes

$$\frac{V_o(s)}{V_i(s)} = -\frac{R_F/R}{1 + sCR_F}$$

as opposed to the ideal function of $-1/sCR$. The lower the value we select for RF, the higher the corner frequency $(1/CRF)$ will be and the more nonideal the integrator becomes. Therefore selecting a value for R_F presents the designer with a trade-off between dc performance and signal performance. The effect of R_F on integrator performance is investigated further in

The preceding example hints at an important application of integrators, namely, their use in providing triangular waveforms in response to square-wave inputs.

2.5.3 The Op-Amp Differentiator

Interchanging the location of the capacitor and the resistor of the integrator circuit results in the circuit in Fig. 2.27(a), which performs the mathematical function of differentiation. To see how this comes about, let the input be the time-varying function $vI(t)$, and note that the virtual ground at the inverting input terminal of the op amp causes $vI(t)$ to appear in effect across the capacitor C. Thus the current through C will be $C(dvI/dt)$, and this current flows through the feedback resistor R providing at the op-amp output a voltage $v_o(t)$,

$$v_o(t) = -CR\frac{dv_i(t)}{dt}$$

The frequency-domain transfer function of the differentiator circuit can be found by substituting in Eq. (2.24), $Z_1(s) = 1/sC$ and $Z_2(s) = R$ to obtain

$$\frac{V_o(s)}{V.(s)} = -sCR$$

which for physical frequencies $s = j\omega$ yields

$$\frac{V_o(j\omega)}{V_i(j\omega)} = -j\omega CR$$

Thus the transfer function has magnitude

$$\left|\frac{V_o}{V}\right| = \omega CR$$

and phase

$$\phi = -90^\circ$$

The Bode plot of the magnitude response can be found from Eq. (2.34) by noting that for an octave increase in ω, the magnitude doubles (increases by 6 dB). Thus the plot is simply a straight line of slope +6 dB/octave (or, equivalently, +20 dB/decade) intersecting the 0-dB line (where $|V_o/V_i| = 1$) at $\omega = 1/CR$, where CR is the **differentiator time constant** [see Fig. 2.27(b)].

The frequency response of the differentiator can be thought of as the response of an STC high-pass filter with a corner frequency at infinity (refer to Fig. 1.24). Finally, we should note that the very nature of a differentiator circuit causes it to be a "noise magnifier." This is due to the spike introduced at the output every time there is a sharp change in $v1(t)$; such a change could be interference coupled electromagnetically ("picked up") from adjacent signal sources. For this reason

and because they suffer from stability problems (Chapter 11), differentiator circuits are generally avoided in practice. When the circuit of Fig. 2.27(a) is used, it is usually necessary to connect a small-valued resistor in series with the capacitor. This modification, unfortunately, turns the circuit into a nonideal differentiator.

2.6 DC Imperfections

Up to now we have considered the op-amp to be **ideal**.

> The only exception has been a brief discussion of the effect of the op-amp finite gain (A) both the closed-loop gain of the **inverting** and **non-inverting** configurations.

While for many applications the assumption of an ideal op amp is **NOT** a bad practice on itself, a circuit designer needs to be thoroughly familiar with the characteristics of practical op-amps and the effects of such characteristics on the performance of op-amp circuits.[24]

[24] Only then will the designer be able to use the op amp intelligently, especially if the application at hand is not a straightforward one.

The non-ideal properties of op amps will, of course, limit the operation range of the circuits we have analysed previously. Now, we will consider some of the important non-ideal properties of the op amp. We do this by treating one non-ideal case at a time, beginning with the dc problems to which op-amps are susceptible.

2.6.1 Offset Voltage

As op-amps are direct-coupled devices with large gains at dc, they are prone to dc problems. The first such problem is the dc offset voltage.

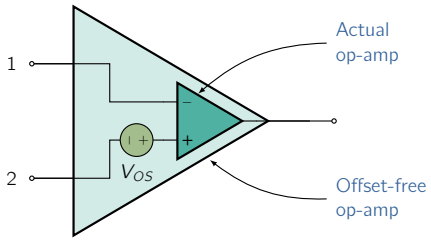


Figure 2.17 **Circuit model for an op amp with input offset voltage V_{OS}.**

To understand this problem consider the following **conceptual** experiment:

> If the two (**2**) input terminals of the op amp are tied together and connected to ground (GND), we will see, despite the fact that $v_{Id} = 0$, a there will be a finite dc voltage at the output.

In fact, if the op-amp has a high dc gain, the output will be at either the **positive** or **negative** saturation level. The op-amp output can be brought back to its ideal value of 0 V by connecting a DC voltage source of appropriate polarity and magnitude between the two (**2**) input terminals of the op amp. This external source **balances out** the input offset voltage of the op amp.

> The input offset voltage (V_{OS}) must be of equal magnitude and of opposite polarity to the voltage we applied externally.

The input offset voltage arises as a result of the unavoidable mismatches present in the input differential stage inside the op-amp. Here, our concern is to investigate the effect of V_{OS} on the operation of closed-loop op-amp circuits.[25]

[25] From a practical perspective, general-purpose op amps exhibit V_{OS} in the range of 1 mV to 5 mV

The value of V_{OS} depends on temperature. The op-amp data sheets generally specify typical and

maximum values for V_{OS} at room temperature as well as the temperature coefficient of V_{OS}, usually in $\mu \cdot V \cdot {}^{\circ}C^{-1}$. They do not, however, specify the polarity of V_{OS} because the component mismatches that give rise to V_{OS} are obviously not known a priori; different units of the same op-amp type may exhibit either a positive or a negative V_{OS}.

To analyze the effect of V_{OS} on the operation of op-amp circuits, we need a circuit model for the op amp with input offset voltage. Such a model is shown in **Fig.** 2.17. It consists of a de source of value V_{OS} placed in series with the positive input lead of an offset-free op amp. The justification for this model follows from the description above.

Analysis of op-amp circuits to determine the effect of the op-amp V_{OS} on their performance is straightforward: The input voltage signal source is short-circuited and the op amp is replaced with the model of **Fig.** 2.17.[26] Following this procedure, we find that both the inverting and the noninverting amplifier configurations result in the same circuit, that shown in Fig. 2.29, from which the output dc voltage due to V_{OS} is found to be

$$V_O = V_{OS}\left[1 + \frac{R_2}{R_1}\right]$$

This output dc voltage can have a large magnitude.

> As an example, a non-inverting amplifier with a closed-loop gain of 1000, when constructed from an op-amp with a 5 mV input offset voltage, will have a DC output voltage of +5 V or -5 V[27] rather than the ideal value of 0 V.

[26] Eliminating the input signal, done to simplify matters, is based on the principle of superposition.

[27] The output polarity of course depends on the polarity of V_{OS}.

Now, when an input signal is applied to the amplifier, the corresponding signal output will be superimposed on the 5 V DC. Obviously then, the allowable signal swing at the output will be reduced.

> Even worse, if the signal to be amplified is DC, we would not know whether the output is due to V_{OS} or to the signal.

Some op amps are provided with two (2) additional terminals to which a specified circuit can be connected to trim to zero the output DC voltage due to V_{OS}. **Fig.** 2.18 shows such an arrangement that is typically used with general-purpose op amps. A potentiometer is connected between the offset-nulling terminals with the wiper of the potentiometer connected to the op-amp negative supply. Moving the potentiometer wiper introduces an imbalance that counteracts the asymmetry present in the internal op-amp circuitry and that gives rise to V_{OS}. It should be noted, however, that even though the dc output offset can be trimmed to zero, the problem remains of the variation (or drift) of V_{OS} with temperature.

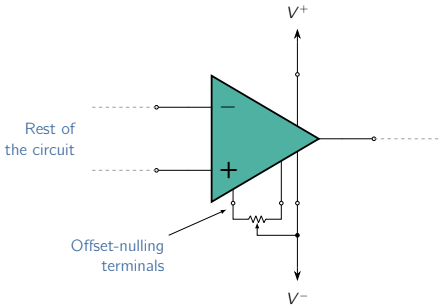


Figure 2.18 **The output dc offset voltage of an op amp can be trimmed to zero by connecting a potentiometer to the two offset-nulling terminals. The wiper of the potentiometer is connected to the negative supply of the op amp.**

One way to overcome the dc offset problem is by **capacitively coupling** the amplifier. This, however, will be possible only in applications where the closed-loop amplifier is not required to amplify dc or

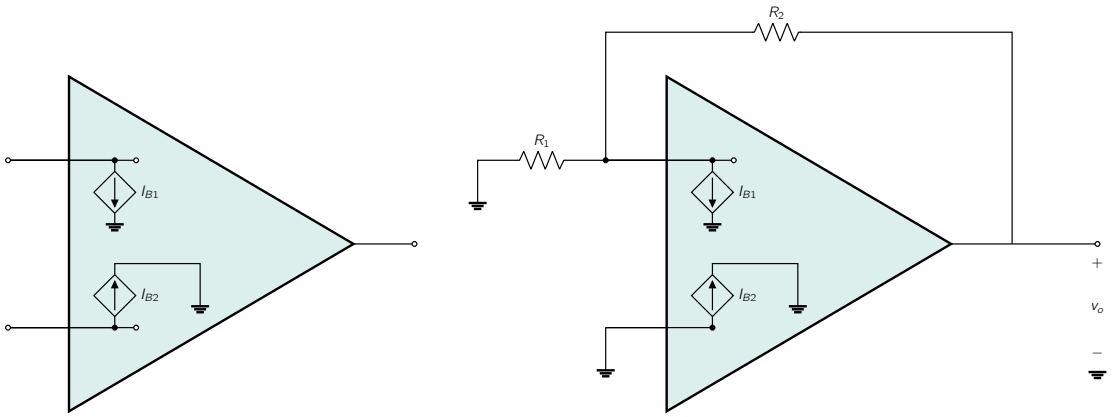


Figure 2.19: (a) The op-amp input bias currents represented by two current sources IB1 and IB2 (b) Analysis of the closed-loop amplifier, taking into account the input bias currents.

very-low-frequency signals. Figure 2.31(a) shows a capacitively coupled amplifier. Because of its infinite impedance at dc, the coupling capacitor will cause the gain to be zero at dc. As a result, the equivalent circuit for determining the dc output voltage resulting from the op-amp input offset voltage V_{OS} will be that shown in Fig. 2.31(b).

Therefore V_{OS} sees in effect a unity-gain voltage follower, and the dc output voltage V_0 will be equal to V_{OS} rather than $V_{OS}(1 + R_2/R_1)$, which is the case without the coupling capacitor. As far as input signals are concerned, the coupling capacitor C forms together with R_1 an STC high-pass circuit with a corner frequency of $\omega_0 = 1/CR_1$. Thus the gain of the capacitively coupled amplifier will fall off at the low-frequency end [from a magnitude of $(1 + R_2/R_1)$ at high frequencies] and will be 3 dB down at ω_0.

2.6.2 Input Bias and Offset Currents

The second DC problem encountered in op amps is illustrated in **Fig.** 2.19a. For the op amp to operate, its two input terminals have to be supplied with dc currents, termed the input bias currents.[28] In **Fig.** 2.19a these two currents are represented by two current sources, I_{B1} and I_{B2}, connected to the two input terminals. It should be emphasized that the input bias currents are **independent** of the fact that a real op amp has finite (though large) input resistance (not shown in **Fig.** 2.19a).

[28] This is the case for op amps constructed using bipolar junction transistors (BJTs). Those using MOSFETs in the first (input) stage do not draw an appreciable input bias current; nevertheless, the input terminals should have continuous dc paths to ground. More on this in later chapters.

> The op-amp manufacturer usually specifies the average value of I_{B1} and I_{B2} as well as their expected difference. The average value I_B is called the **input bias current**,

$$I_B = \frac{I_{B1} + I_{B2}}{2}$$

and the difference is called the **input offset current** and is given by

$$I_{OS} = |I_{B1} - I_{B2}|$$

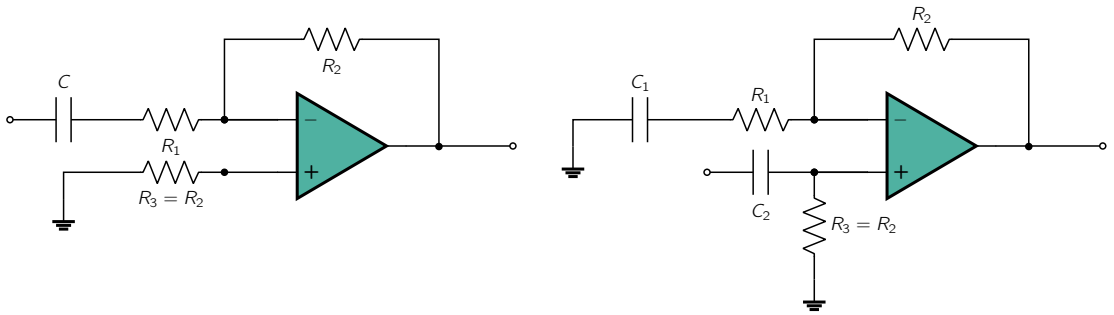


Figure 2.20: (a)In an ac-coupled amplifier the dc resistance seen by the inverting terminal is R2 ; hence R3 is chosen equal to R2. (b) Illustrating the need for a continuous dc path for each of the op-amp input terminals. Specifically, note that the amplifier will not work without resistor R3.

Typical values for general-purpose op amps that use bipolar transistors are $IB = 100$ nA and $I_{OS} = 10$ nA.

We now wish to find the dc output voltage of the closed-loop amplifier due to the input bias currents. To do this we ground the signal source and obtain the circuit shown in **Fig.** 2.19b for both the inverting and noninverting configurations. As shown in **Fig.** 2.19b, the output dc voltage is given by

$$V_0 = I_{B1}R_2 \simeq I_B R_2 \tag{2.8}$$

This obviously places an upper limit on the value of R_2. Fortunately, however, a technique exists for reducing the value of the output dc voltage due to the input bias currents. The method consists of introducing a resistance R_3 in series with the noninverting input lead, From a signal point of view, R_3 has a negligible effect.[29] The appropriate value for R_3 can be determined by analyzing the circuit in Fig. 2.34, where analysis details are shown, and the output voltage is given by

[29] ideally no effect

$$V_\alpha = -I_\alpha R_3 + R_2(I_{\alpha 1} - I_{\alpha 2}R_3/R_1)$$

Consider first the case $I_{B1} = I_{B2} = I_B$, which results in

$$V_o = I_R[R_2 - R_3(1 + R_2/R_1)]$$

Thus we can reduce V_0 to zero by selecting R_3 such that

$$R_3 = \frac{R_2}{1 + R_2/R_1} = \frac{R_1 R_2}{R_1 + R_2}$$

That is, R_3 should be made equal to the parallel equivalent of R_1 and R_2. Having selected R_3 as above, let us evaluate the effect of a finite offset current I_{OS}. Let $I_{B1} = I_B + I_{OS}/2$ and $I_{B2} = I_B - I_{OS}/2$, and substitute in Eq. (2.38). The result is

$$V_o = I_{os}R_2$$

which is usually about an order of magnitude smaller than the value obtained without R_3 (Eq. (**2.8**)). We can therefore conclude that to minimize the effect of the input bias currents, one should place in the positive lead a resistance equal to the equivalant dc resistance seen by the inverting terminal.

We emphasize the word *dc* in the last statement; note that if the amplifier is ac-coupled, we should select $R_3 = R_2$, as shown in Fig. 2.35.

While we are on the subject of ac-coupled amplifiers, we should note that one must always provide a continuous dc path between each of the input terminals of the op amp and ground. This is the case no matter how small I_B is. For this reason the ac-coupled noninverting amplifier of Fig. 2.36 will **not** work without the resistance R_3 to ground.

Unfortunately, including R_3 lowers considerably the input resistance of the closed-loop amplifier.

2.6.3 Effects of Offset of Inverting Integrator

Our discussion of the inverting integrator circuit mentioned the susceptibility of this circuit to saturation in the presence of small dc voltages or currents and therefore we have to consider the effect of the op-amp DC offsets on its operation. As will be seen, these effects can be quite dramatic.

To see the effect of the input dc offset voltage V_{OS}, consider the integrator circuit in Fig. 2.37, where for simplicity we have short-circuited the input signal source. Analysis of the circuit is straightforward and is shown in Fig. 2.37. Assuming for simplicity that at time $t = 0$ the voltage across the capacitor is zero, the output voltage as a function of time is given by

$$v_O = V_{OS} + \frac{V_{OS}}{CR}t$$

Therefore v_O increases linearly with time until the op amp **saturates**. As should be expected, the dc input offset current I_{OS} produces a similar problem. Figure 2.38 illustrates the situation. Observe that we have added a resistance R in the op-amp positive-input lead in order to keep the input bias current IB from flowing through C. Nevertheless, the offset current I_{OS} will flow through C and cause v_O to ramp linearly with time until the op amp saturates.

As mentioned previously on the study of integrator, the dc problem of the integrator circuit can be alleviated by connecting a resistor R_F across the integrator capacitor C, as shown in Fig. 2.25. Such a resistor provides a dc path through which the dc currents (V_{OS}/R) and I_{OS} can flow,[30] with the result that v_0 will now have a dc component $[V_{OS}(1 + R_F/R) + I_{OS}R_F]$ instead of rising linearly. To keep the dc offset at the output small, one would select a low value for RF.

[30] assuming a resistance equal to $R \parallel R_F$ is connected in the positive op-amp lead

Unfortunately, however, the lower the value of R_F, the less ideal the integrator circuit becomes.

2.7 Finite Open-Loop Gain and Bandwidth

2.7.1 Frequency Dependence of the Open-Loop Gain

The differential open-loop gain (A) of an op amp is **NOT infinite**. It is **finite** and decreases with frequency. Figure 2.39 shows a plot for $|A|$, with the numbers typical of some commercially available general-purpose op amps.

> although the gain is quite high at dc and low frequencies, it starts to fall off at a rather low frequency (10 Hz in our example).

The uniform -20-dB/decade gain rolloff shown is typical of **internally compensated** op amps. These are units that have a network[31] included within the same IC chip whose function is to cause the op-amp gain to have the single-time-constant (STC) low-pass response shown. This process of modifying the open-loop gain is termed **frequency compensation**, and its purpose is to ensure that op-amp circuits will be stable (as opposed to oscillatory).

[31] usually a single capacitor

By analogy to the response of low-pass STC circuits, the gain $A(s)$ of an internally compensated op amp may be expressed as

$$A(s) = \frac{A_0}{1 + s/\omega_b} \tag{2.9}$$

which for physical frequencies, $s = \mathbf{j}\omega$, becomes the following:

$$A(\mathbf{j}\omega) = \frac{A_0}{1 + \mathbf{j}\omega/\omega_b}$$

where A_0 denotes the dc gain and ω_b is the 3-dB frequency.[32] For the example shown in Fig. 2.39, $A_0 = 10^5$ and $\omega_b = 2\pi \times 10$ rad/s. For frequencies $\omega \gg \omega_b$ (about 10 times and higher) Eq. (2.43) may be approximated by

[32] corner frequency or "break" frequency

$$A(\mathbf{j}\omega) \simeq \frac{A_0\omega_b}{\mathbf{j}\omega} \qquad \text{therefore} \qquad |A(\mathbf{j}\omega)| = \frac{A_0\omega_b}{\omega} \tag{2.10}$$

from which it can be seen that the gain |A| reaches unity (0 dB) at a frequency denoted by ω and given by

$$\omega_t = A_0\omega_b \tag{2.11}$$

Substituting in Eq. (**2.10**) gives:

$$A(\mathbf{j}\omega) \simeq \frac{\omega_t}{i\omega} \tag{2.12}$$

The frequency $f_t = \omega_t/2\pi$ is usually specified on the data sheets of commercially available op amps and is known as the **unity-gain bandwidth**. Also note that for $\omega \gg \omega_b$ the open-loop gain in Eq. (**2.9**) becomes:

$$A(s) \simeq \frac{\omega_t}{s}$$

The gain magnitude can be obtained from Eq. (**2.12**) as

$$|A(\mathbf{j}\omega)| \simeq \frac{\omega_t}{\omega} = \frac{f_t}{f}$$

Therefore if f_t is known (10⁶ Hz in our example), one can easily determine the magnitude of the op-amp gain at a given frequency f. Furthermore, observe that this relationship correlates with the Bode plot in Fig. 2.39. Specifically, for $f \gg f_b$, doubling f (an octave increase) results in halving the gain (a 6-dB reduction). Similarly, increasing f by a factor of 10 (a decade increase) results in reducing $|A|$ by a factor of 10 (20 dB).

As a matter of practical importance, we note that the production spread in the value of f_t between op-amp units of the same type is usually much smaller than that observed for A_0 and fb. For this reason ft is preferred as a specification parameter. Finally, it should be mentioned that an op amp having this uniform -6-dB/octave (or equivalently -20-dB/decade) gain rolloff is said to have a **single-pole model**. Also, since this single pole **dominates** the amplifier frequency response, it is called a **dominant pole**. For more on poles (and zeros), the reader may wish to consult Appendix F.

2.7.2 Frequency Response of Closed-Loop Amplifiers

We next consider the effect of limited op-amp gain and bandwidth on the closed-loop transfer functions of the two (**2**) basic configurations:

- the inverting circuit of Fig. 2.5, and the
- noninverting circuit of Fig. 2.12.

The closed-loop gain of the inverting amplifier, assuming a finite op-amp open-loop gain A, was derived in Eq. (**2.3**), which we repeat here as:

$$\frac{V_o}{V} = \frac{-R_2/R_1}{1 + (1 + R_2/R_1)/A}$$

Substituting for A from Eq. (**2.9**) and using Eq. (**2.11**) gives

$$\frac{V_o(s)}{V_i(s)} = \frac{-R_2/R_1}{1 + \frac{1}{A_o}\left(1 + \frac{R_2}{R}\right) + \frac{s}{\omega l(1+R_2/R_1)}}$$

For $A_0 \gg 1 + R_2/R_1$, which is usually the case,

$$\frac{V_o(s)}{V_i(s)} \simeq \frac{-R_2/R_1}{1 + \frac{s}{\omega l(1+R/lR_1)}}$$

which is of the same form as that for a low-pass STC network (see Table 1.2, page 36). Thus the inverting amplifier has an STC low-pass response with a dc gain of magnitude equal to R_2/R_1. The closed-loop gain rolls off at a uniform -20-dB/decade slope with a corner frequency (3-dB frequency) given by

$$\omega_{3\text{dB}} = \frac{\omega_t}{1 + R_2/R_1}$$

Similarly, analysis of the noninverting amplifier of Fig. 2.12, assuming a finite open-loop gain A, yields the closed-loop transfer function

$$\frac{V_o}{V_i} = \frac{1 + R_2/R_1}{1 + (1 + R_2/R_1)/A}$$

Substituting for A from Eq. (2.42) and making the approximation $A_0 \gg 1 + R_2/R_1$ results in

$$\frac{V_o(s)}{V_i(s)} \simeq \frac{1 + R_2/R_1}{1 + \frac{s}{\omega/(1+R_2/R_1)}}$$

Thus the noninverting amplifier has an STC low-pass response with a dc gain of $(1 + R_2/R_1)$ and a 3-dB frequency given also by Eq. (2.53).

The table in Example 2.6 above clearly illustrates the trade-off between gain and bandwidth: For a given op amp, the lower the closed-loop gain required, the wider the bandwidth achieved. Indeed, the noninverting configuration exhibits a constant **gain-bandwidth product** equal to f_t of the op amp. An interpretation of these results in terms of feedback theory will be given in Chapter 11.

2.8 Large-Signal Operation of Op Amps

In our final section for studying op-amps, we will look at the limitations on the performance of op-amp circuits when large output signals are present.

2.8.1 Output Voltage Saturation

Similar to all other amplifiers, op-amps **operate linearly** over a limited range of output voltages. Specifically, the op-amp output saturates in the manner shown discussed in the previous chapter with L_+ and L_- within 1 V or so of the positive and negative power supplies, respectively.

Therefore, an op-amp operating from ± 15 V supplies will saturate when the output voltage reaches about $+13$ V in the positive direction and -13 V in the negative direction. For this particular op amp the **rated** output voltage is said to be ± 13 V.

> To avoid clipping off the peaks of the output waveform, and the resulting waveform distortion, the input signal must be kept correspondingly small.

2.8.2 Output Current Limits

Another limitation on the operation of op-amps is their output current is limited to a specified maximum.

> For instance, the popular 741 op amp is specified to have a maximum output current[33] of ± 20 mA.

Therefore, in designing closed-loop circuits utilizing the 741, the designer[34] has to ensure that under no condition will the op-amp be required to supply an output current, in either direction, exceeding ± 20 mA.

This, of course, has to include both the current in the feedback circuit as well as the current supplied to a load resistor. If the circuit requires a larger current, the op-amp output voltage will saturate at the level corresponding to the maximum allowed output current.

[33] This is the maximum amount of current the op amp can source or sink without damage. Exceeding this limit can damage the device.

[34] That is you.

2.8.3 Slew Rate

Another phenomenon which can cause non-linear distortion when large output signals are present is **slew-rate limiting**. The name refers to the fact that there is a specific **maximum rate of change** possible at the output of a real op-amp. This maximum is known as the **slew rate** (SR) of the

op-amp and is defined as:[35]

$$SR = \left.\frac{d[^{t}_{v_o]}}{dt}\right|_{max}$$

and is usually specified on the op-amp data sheet in units of $V \cdot \mu s^{-1}$. It follows, if the input signal applied to an op-amp circuit is such that it demands an output response that is faster than the specified value of SR, the op-amp will not comply.

[35] Slew derives from a nautical term, and you can find many 1800s references to slewing a cannon on its trunnions (that would be a vertical movement). The earliest recorded usage (according to the OED) is from Falconer's Marine dictionary (1769):

> Rather, its output will change at the maximum possible rate, which is equal to its SR.

As an example, consider an op amp connected in the unity-gain voltage-follower configuration shown in Fig. 2.43(a), and let the input signal be the step voltage shown in Fig. 2.43(b). The output of the op amp will not be able to rise instantaneously to the ideal value V; rather, the output will be the linear ramp of slope equal to SR, shown in Fig. 2.43(c). The amplifier is then said to be **slewing**, and its output is **slew-rate limited**.

To understand the origin of the slew-rate phenomenon, we need to know about the internal circuit of the op-amp. For the time being, however, it is sufficient to know about the phenomenon and to note that it is distinct from the finite op-amp bandwidth that limits the frequency response of the closed-loop amplifiers, studied previously.

The limited bandwidth is a linear phenomenon and does not result in a change in the shape of an input sinusoid; that is, it does not lead to nonlinear distortion. The slew-rate limitation, on the other hand, can cause nonlinear distortion to an input sinusoidal signal when its frequency and amplitude are such that the corresponding ideal output would require vo to change at a rate greater than SR. This is the origin of another related op-amp specification, its full-power bandwidth, to be explained later.

Before leaving the example in Fig. 2.43, however, we should point out that if the step input voltage V is sufficiently small, the output can be the exponentially rising ramp shown in Fig. 2.43(d). Such an output would be expected from the follower if the only limitation on its dynamic performance were the finite op-amp bandwidth. Specifically, the transfer function of the follower can be found by substituting $R_1 = \infty$ and $R_2 = 0$ in Eq. (2.55) to obtain

$$\frac{V_o}{V_i} = \frac{1}{1 + s/\omega_i}$$

which is a low-pass STC response with a time constant $1/\omega_t$. Its step response would therefore be (see Appendix E)

$$v_o(t) = V(1 - e^{-\omega_t t})$$

The initial slope of this exponentially rising function is $(\omega_t V)$. Thus, as long as V is sufficiently small so that $\omega_t V \leqslant SR$, the output will be as in Fig. 2.43(d).

2.8.4 Full-Power Bandwidth

Op-amp slew-rate limiting can cause non-linear distortion in sinusoidal waveforms. Consider once more the unity-gain follower with a sine-wave input given by:

$$v_I = V_i[][h] \sin \omega t$$

The rate of change of this waveform is given as:

$$\frac{\mathrm{d}[}{\mathrm{d}v}[I]][t] = \omega \hat{V}_\mathrm{i} \cos \omega t$$

with a maximum value of $\omega \hat{V}_i$. This maximum occurs at the zero crossings of the input sinusoid. Now if ωV_i exceeds the slew rate of the op amp, the output waveform will be distorted in the manner shown in Fig. 2.44. Observe that the output cannot keep up with the large rate of change of the sinusoid at its zero crossings, and the op amp **slews**.

The op-amp data sheets usually specify a frequency f_M called the **full-power bandwidth**. It is the frequency at which an output sinusoid with amplitude equal to the rated output voltage of the op-amp begins to show distortion due to slew-rate limiting. If we denote the rated output voltage V_max, then f_M is related to SR as follows:

$$\omega_\mathrm{M} V_\mathrm{o,max} = \mathrm{SR} \qquad \text{and} \qquad f_\mathrm{M} = \frac{\mathrm{SR}}{2\pi V_\mathrm{o,max}}$$

Based on the above given equation, it should be apparent the output sinusoids of amplitudes smaller than V_max will show slew-rate distortion at frequencies higher than ω_M.

In fact, at a frequency ω higher than ω_M, the maximum amplitude of the undistorted output sinusoid is given by

$$V_\mathrm{o} = V_\mathrm{o,max} = \left(\frac{\omega_\mathrm{M}}{\omega}\right)$$

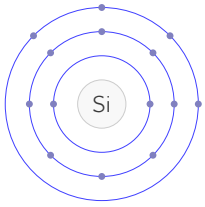
Chapter 3

Semiconductors

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In this chapter, we will have a gentle introduction the properties and physics of **semiconductors**. The objective is to provide a basis for understanding the physical operation of **diodes** and **transistors** to enable their effective use in the design of circuits. While many concepts studied in this chapter apply to semiconductor materials in general, our focus will be heavily biased towards **silicon**, as it is the **primary** material used in the vast majority of microelectronic circuits.[1]



[1] The atomic structure and the free electrons present in a neutral Silicon atom.

This chapter will begins with a study of the crystal structure of semiconductors and introduces the two (**2**) types of charge carriers available for current conduction:

1. electrons, and
2. holes.

Unique among Many — 3.1 Information

The most significant property of semiconductors is their **conductivity can be varied over a very wide range** through the introduction of controlled amounts of **impurity atoms** into the semiconductor crystal in a process called **doping**.

Now we shall start to look at the building blocks which make up a semiconductor.

3.1 Semiconductor Physics

Semiconductors play a vital role in how modern electronics are made and therefore is in our best interest to get a good overview on them. To that end we shall study two (2) types: Intrinsic, and Doped.



[2]**Silicon** is an element, with symbol **Si** and atomic number 14. It is a hard, brittle crystalline solid with a blue-grey metallic lustre, and is a non-metal and semiconductor. It is a member of group 14 in the periodic table: carbon is above it; and *germanium*, *tin*, *lead*, and *flerovium* are below it. It is relatively unreactive. Silicon is widely regarded as the predominant semiconductor material due to its versatile applications in various electrical devices such as transistors, solar cells, IC, and others.

3.1.1 Intrinsic Semiconductors

As their name gives it away, **semiconductors** are materials whose conductivity lies between a of conductors, such as copper, and insulators, such as glass. From an element standpoint, there are two (2) kinds of semiconductors:

Single-element Semiconductors

such as germanium and silicon,[2] which are in group IV in the periodic table.

Compound Semiconductors

such as gallium-arsenide, which are formed by combining elements from groups III and V or groups II and VI.

Compound semiconductors are useful in special electronic circuit applications as well as in applications involving light, such as Light-Emitting Diode (LED)s. Of the two (2) elemental semiconductors, germanium[3] was used in the fabrication of the first transistors, between late 1940s, early 1950s. It was quickly supplanted, however, with *silicon*, on which today's IC technology is based.



[3]Germanium is a chemical element, with symbol **Ge** and atomic number 32. It is lustrous, hard-brittle, grayish-white and similar in appearance to silicon. It is a metalloid or a nonmetal in the carbon group that is chemically similar to silicon. Like silicon, germanium naturally reacts and forms complexes with oxygen in nature. Elemental germanium is used as a semiconductor in transistors and various other electronic devices.

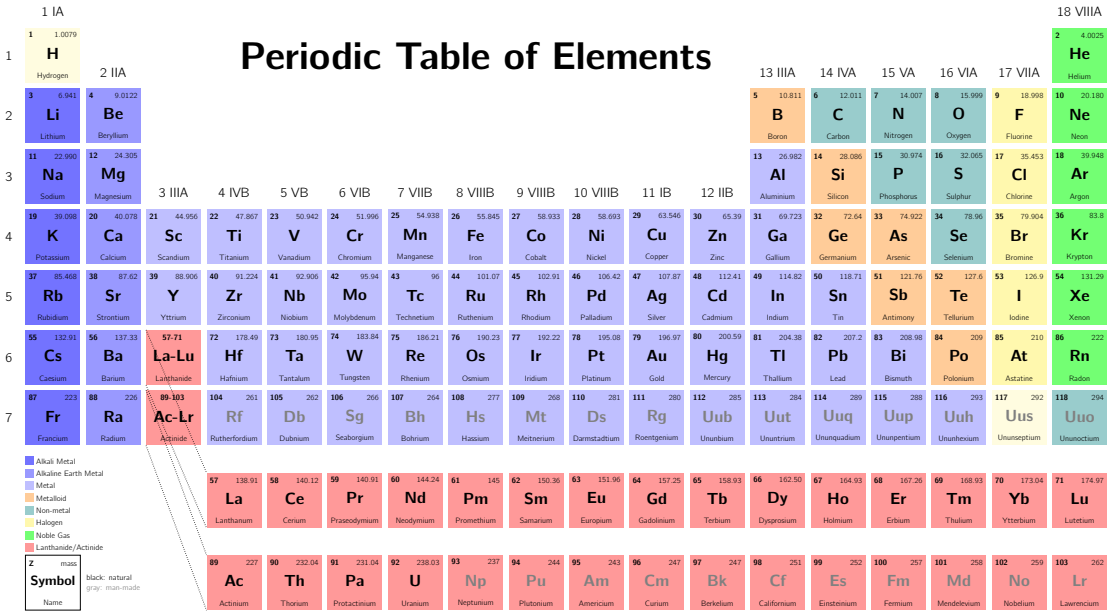


Figure 3.1 The periodic table of elements.

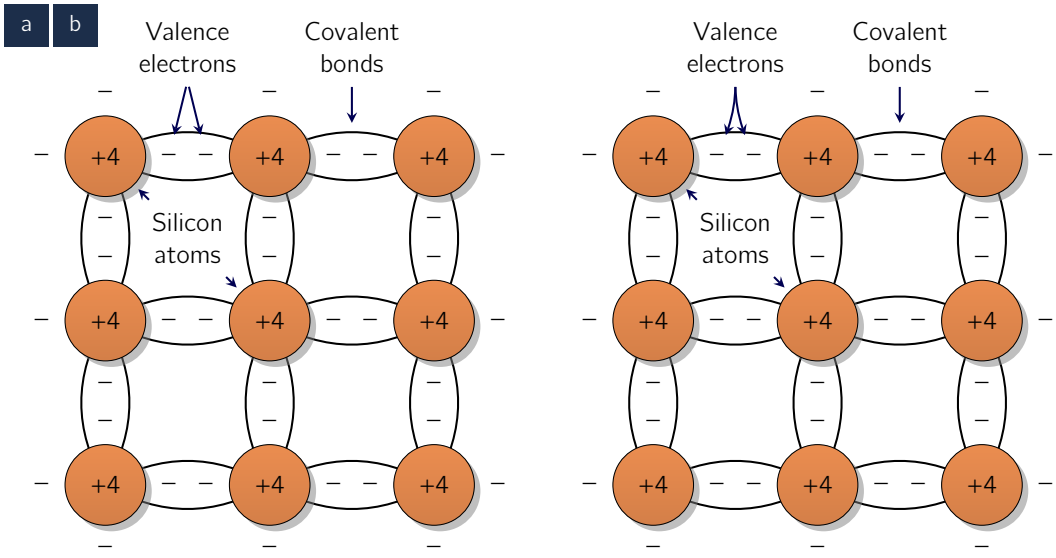


Figure 3.2
(a) Two-dimensional representation of the silicon crystal. The circles represent the inner core of silicon atoms, with +4 indicating its positive charge of +4q, which is neutralized by the charge of the four valence electrons. Please observe how the covalent bonds are formed by sharing of the valence electrons. At 0 K, all bonds are intact and no free electrons are available for current conduction (b) At room temperature, some of the covalent bonds are broken by thermal generation. Each broken bond gives rise to a free electron and a hole, both of which become available for current conduction.

For this reason, we will deal mostly with **silicon** devices throughout this lecture.

A silicon atom has four (**4**) valence electrons, and requires another four (**4**) to complete its outermost shell. This is achieved by sharing one of its valence electrons with each of its four neighboring atoms. Each pair of shared electrons forms a **covalent bond**. The result is a crystal of pure or intrinsic silicon with a regular lattice structure, where the atoms are held in their position by the covalent bonds. **Fig.** 3.2a shows a two-dimensional representation of such a structure.

At sufficiently low temperatures,[4] all the covalent bonds are intact and no electrons are available to conduct electric current.

[4] close to, if not, absolute zero (0 K.)

Too Cold to Move — 3.2 Information

At such low temperatures, the intrinsic silicon crystal behaves as an **insulator**. At room temperature, sufficient thermal energy exists to break some of the covalent bonds, a process known as **thermal generation**.

As shown in **Fig.** 3.2b, when a covalent bond is broken, an electron is freed. The free electron can wander away from its parent atom, and it becomes available to conduct electric current if an electric field (**E**) is applied to the crystal. As the electron leaves its parent atom, it leaves behind a net

Valence Electrons
Valence electrons are the electrons in the outermost shell of an atom. They control how elements react, form chemical bonds, and share or give away electrons.

Covalent Bond
A chemical bond that involves the sharing of electrons to form electron pairs between atoms. These electron pairs are known as shared pairs or bonding pairs. The stable balance of attractive and repulsive forces between atoms, when they share electrons, is known as covalent bonding

positive charge which is equal to the magnitude of the electron charge.

Therefore, an electron from a neighbouring atom could be attracted to this positive charge, and leaves its parent atom. This action fills up the "hole" which existed in the ionized atom but creates a new hole in the other atom.

This process may repeat itself, with the result that we effectively have a positively charged carrier, or **hole**, moving through the silicon crystal structure and being available to conduct electric current. The charge of a hole is equal in magnitude to the charge of an electron. We can therefore see that, as temperature increases, more covalent bonds are broken and electron-hole pairs are generated.

The increase in free electrons and holes results in an **increase in the conductivity of silicon**.

Thermal generation results in free electrons and holes in equal numbers and hence equal concentrations, where concentration refers to the number of charge carriers per unit volume (cm^3). The free electrons and holes move randomly through the silicon crystal structure, and in the process some electrons may fill some of the holes. This process, called **recombination**, results in the disappearance of free electrons and holes. The recombination rate is proportional to the number of free electrons and holes, which in turn is determined by the thermal generation rate.

The latter is a strong function of **temperature**.

In thermal equilibrium, the recombination rate is equal to the generation rate, and we can conclude that the concentration of free electrons (n) is equal to the concentration of holes (p),

$$n = p = n_i$$

where n_i denotes the number of free electrons and holes in a unit volume (cm^3) of intrinsic silicon at a given temperature. Results from semiconductor physics gives n_i as

$$n_i = BT^{3/2} \exp\left(E_g/2kT\right) \tag{3.1}$$

where B is a material-dependent parameter which is $7{,}3 \times 10^{15}$ for silicon, T is the temperature in K; E_g, a parameter known as the **bandgap energy**, is 1.12 electron volt[5] (eV) for silicon; and k is Boltzmann's constant (8.62×10^{-5} eV/K).

The bandgap energy (E_g) is the minimum energy required to break a covalent bond and generate an electron-hole pair and is defined as an energy range in a solid where no electronic states exist.

Carriers in a Semiconductor — 3.1 Exercise

Calculate the value of n_i, for silicon at room temperature ($T \simeq 300$ K).

Continues on next page →

[5] also written electron-volt and electron volt, is the measure of an amount of kinetic energy gained by a single electron accelerating through an electric potential difference of one volt in vacuum. When used as a unit of energy, the numerical value of $1\,e \cdot V$ in joules (J) is equal to the numerical value of the charge of an electron in coulombs (C).

Under the 2019 revision of the SI, this sets $1\,e \cdot V$ equal to the exact value $1{,}602.176.634 \times 10^{19}$

← Continued from previous page

Substituting the values in Eq. (3.1) gives the following:

$$n_i = 7.3 \times 10^{15}(300)^{3/2}e^{-1.12/(2\times 8.62\times 10^5 \times 300)}$$
$$= 1.5 \times 10^{10}$$

Although this number seems large, to place it into context note that silicon has 5×10^{22} atoms/cm. Thus at room temperature only one in about 5×10^{12} atoms is ionized and contributing a free electron and a hole.

Finally, it is useful for future purposes to express the product of the hole and free-electron concentration as

$$pn = n_i^2 \tag{3.2}$$

where for silicon at room temperature, $n_i \approx 1{,}5 \times 10^{10}\,\text{cm}^{-3}$. As will be seen shortly, this relationship extends to extrinsic or doped silicon as well.

Liquid Crystal Display — 3.3 Information

The existence of liquid crystals whose color could be changed by means of an external heat source was first reported in 1888 by an Austrian botanical physiologist. The LC idea lay dormant until the late 1940s, however. Subsequent developments in the field of solid-state electronics provided the technology to harness the technique in display media, with the first LCDs being demonstrated by RCA beginning in 1962. Today, LCDs are an essential component in every mobile device as the interface to the world of electronics within. At the other end of the scale, large LCDs are used in flat-panel TVs, and very large LCDs are appearing as "dynamic" wallpaper in museum display settings.

3.1.2 Doped Semiconductors

The intrinsic silicon crystal described previously has **equal concentrations of free electrons and holes**, generated by thermal generation. These concentrations are far too small for silicon to conduct measurable current at room temperature. In addition, the carrier concentrations and hence the conductivity are strong functions of temperature, which is **NOT** a desirable property in an electronic device. Fortunately, a method was developed to change the carrier concentration in a semiconductor crystal substantially and in a precisely controlled manner.

This process is known as **doping**, and the resulting silicon is referred to as **doped silicon**.

Doping requires introducing **impurity atoms** into the silicon crystal in sufficient numbers to significantly increase the concentration of either free electrons or holes but with little or no change in the crystal properties of silicon.

- To increase the concentration of free electrons (n) silicon is doped with an element with a valence of 5, such as *phosphorus*. The resulting doped silicon is then said to be of n-type.[6]
- To increase the concentration of holes (p) silicon is doped with an element having a valence of 3, such as *boron*, and the resulting doped silicon is said to be of p-type.

[6] This means, again, that the material has more free moving electrons than there are holes.

Figure 3.3 shows a silicon crystal doped with phosphorus impurity. The dopant (phosphorus) atoms replace some of the silicon atoms in the crystal structure. Since the phosphorus atom has five (**5**) electrons in its outer shell, four of these electrons form covalent bonds with the neighboring atoms, and the fifth electron becomes a free electron. Therefore each phosphorus atom donates a free electron to the silicon crystal, and the phosphorus impurity is called a **donor**.

No Such Thing as Free Holes — 3.4 Information

It should be clear, though, that no holes are generated by this process. The net positive charge associated with the phosphorus atom is a bound charge that does not move through the crystal.

If the concentration of donor atoms is N_D, where N_D is usually much greater than n_i, the concentration of free electrons in the n-type silicon will be:

$$n_n \approx N_D \tag{3.3}$$

where the subscript n denotes n-type silicon. Therefore, n_n is determined by the doping concentration and not by temperature. This is not the case, however, for the hole concentration. All the holes in the n-type silicon are those generated by thermal ionization. Their concentration p_n can be found by noting that the relationship in Eq. (**3.2**) applies equally well for doped silicon, provided thermal equilibrium is achieved. Therefore for n-type silicon:

$$p_n n_n = n_i^2$$

Substituting for n_n from Eq. (**3.3**), we obtain for p_n:

$$p_n \simeq \frac{n_i^2}{N_n}$$

Therefore p_n will have the same dependence on temperature as that of n_i^2. Finally, in n-type silicon the concentration of free electrons n_n will be much larger than that of holes. Hence electrons are said to be the majority charge carriers and holes the minority charge carriers in n-type silicon.

To obtain p-type silicon in which holes are the majority charge carriers, a trivalent impurity such as boron is used. Figure 3.4 shows a silicon crystal doped with *boron*.

The *boron* atoms **replace some of the silicon atoms** in the silicon crystal structure.

Since each boron atom has three electrons in its outer shell, it accepts an electron from a neighbouring atom, thus forming covalent bonds. The result is a hole in the neighboring atom and a bound negative charge at the **acceptor** (boron) atom. It follows that each acceptor atom provides a hole. If the acceptor doping concentration is N_A, where $N_A \gg n_i$, the hole concentration becomes

$$p_p \simeq N_A \tag{3.4}$$

where the subscript p denotes p-type silicon. Therefore, here the majority carriers are holes and their concentration is determined by N_A. The concentration of minority electrons can be found by using the relationship $p_p n_p = n_i^2$ and substituting for p_p from Eq. (**3.4**):

$$n_p \simeq \frac{n_i^2}{N_A}$$

Therefore, the concentration of the minority electrons will have the same temperature dependence as that of n_i^2.

It should be emphasized that a piece of *n*-type or *p*-type silicon is **electrically neutral**. The charge of the majority free carriers[7] are neutralized by the bound charges associated with the impurity atoms.

[7] electrons in the *n*-type and holes in the *p*-type silicon.

Electrons and Holes — 3.2 Exercise

Consider an *n*-type silicon for which the dopant concentration $N_D = 1{,}0 \times 10^{18}\,\text{cm}^{-3}$. Find the electron and hole concentrations at $T = 300\,\text{K}$.

The concentration of the majority electrons is

$$n_n \simeq N_D = 1{,}0 \times 10^{18}\,\text{cm}^{-3}$$

The concentration of the minority holes is

$$p_n \simeq \frac{n_i^2}{N_n}$$

In Example 3.1 we found that at $T = 300\,\text{K}$, $n_i = 1{,}5 \times 10^{10}\,\text{cm}^{-3}$. Therefore,

$$p_n = \frac{(1.5 \times 10^{10})^2}{10^{17}} = 2{,}25 \times 10^{3}\,\text{cm}^{-3}$$

Observe that $n_n \gg n_i$ and that n_n is vastly higher than p_n.

3.2 Current Flow in Semiconductors

There are two (2) fundamentally different mechanisms for the movement of charge carriers and therefore for current flow in semiconductors:

- drift, and
- diffusion.

Let's have a look at both these options in more detail and understand their effect on semiconductors.

3.2.1 Drift Current

When an electrical field (**E**) is established in a semiconductor crystal, holes are accelerated in the direction of **E**, and free electrons are accelerated in the direction **opposite** to that of **E**. This situation is illustrated in **Fig.** 3.3. The holes acquire a velocity $v_{\text{p-drift}}$ given by:

$$v_{\text{p-drift}} = \mu_p E \tag{3.5}$$

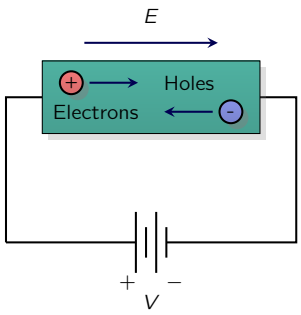


Figure 3.3 **An electric field** E **established in a bar of silicon causes the holes to drift in the direction of** E **and the free electrons to drift in the opposite direction. Both the hole and electron drift currents are in the direction of** E.

where μ_n is a constant called the **hole mobility** where it represents the degree of ease by which holes move through the silicon crystal in response to the electrical field **E**.

Given the velocity has the units of $\text{cm} \cdot \text{s}^{-1}$ and **E** has the units of $\text{V} \cdot \text{m}^{-1}$, we see from Eq. (**3.5**), the mobility (μ_p) needs to have the units of $\text{cm}^2 \cdot \text{V}^{-1} \cdot \text{s}^{-1}$.

For intrinsic[8] silicon:

$$\mu_p = 480\,\text{cm}^2 \cdot \text{V}^{-1} \cdot \text{s}^{-1}.$$

The free electrons acquire a drift velocity ($v_{\text{n-drift}}$) given by:

$$v_{n\text{-drift}} = -\mu_n \mathbf{E} \tag{3.6}$$

where the result is negative as the electrons move in the direction **opposite** to **E**. Here μ_n is the **electron mobility**[9]. Note that μ_n is about 2.5 times μ_n, signifying that electrons move with much greater ease through the silicon crystal than do holes.

Let's now return to the single-crystal silicon bar shown in **Fig.** 3.3. Let the concentration of holes be p and that of free electrons n. We wish to calculate the current component due to the flow of holes.

[8] chemically pure silicon with no added impurities, meaning its electrical conductivity comes only from thermally generated electrons and holes.

[9] for intrinsic silicon is about $1{,}350 \times 10^3\,\text{cm}^2 \cdot \text{V}^1 \cdot \text{s}^1$

Consider a plane perpendicular to the x-direction. In one second, the hole charge that crosses that plane will be $(Aqpv_{\text{p-drift}})$ coulombs, where A is the cross-sectional area of the silicon bar and q is the magnitude of electron charge.

This then must be the hole component of the drift current flowing through the bar,

$$I_p = Aqpv_{\text{p-drift}}$$

Substituting for $v_{\text{p-drift}}$ from Eq. (**3.5**), we obtain

$$I_p = Aqp\mu_p\mathbf{E}$$

We are usually interested in the current density (J_p), which is the current per unit cross-sectional area $(\text{A} \cdot \text{m}^{-2})$,

$$J_p = \frac{I_p}{A} = qp\mu_p\mathbf{E} \tag{3.7}$$

The current component due to the drift of free electrons can be found in a similar manner.

Directions Matter 3.5 Information

Note that electrons drifting from right to left result in a current component from left to right. This is because of the convention of taking the direction of current flow as the direction of flow of positive charge and opposite to the direction of flow of negative charge.

Therefore,

$$I_n = -Aqnv_{\text{n-drift}}$$

Substituting for $v_{\text{n-drift}}$ from Eq. (**3.6**), we obtain the current density $J_n = I_n/A$ as

$$J_n = qn\mu_n\mathbf{E} \tag{3.8}$$

The total drift current density can now be found by summing J_p and J_n from Eq. (**3.7**) and Eq. (**3.8**),

$$J = J_n + J_n = q\left(p\mu_n + n\mu_n\right)\mathbf{E}$$

This relationship can be written as

$$J = \sigma E \qquad \text{or} \qquad J = E/\rho \tag{3.9}$$

where the conductivity (σ) is given as:

$$\sigma = q\left(p\mu_p + n\mu_n\right)$$

and the resistivity (ρ) is given by:

$$\rho \equiv \frac{1}{\sigma} = \frac{1}{q\left(p\mu_p + n\mu_n\right)}$$

Please observe Eq. (**3.9**) is a form of *Ohm's law* and can be written alternately as

$$\rho = \frac{E}{I}$$

Therefore the units of ρ are obtained from:

$$\frac{\mathrm{V}\cdot\mathrm{cm}^{-1}}{\mathrm{A}\cdot\mathrm{cm}^{-2}} = \Omega\cdot\mathrm{cm}$$

3.2.2 Diffusion Current

Carrier diffusion occurs when the density of charge carriers in a piece of semiconductor is **NOT** uniform.

> For example, if by some mechanism the concentration of, say, holes, is made higher in one part of a piece of silicon than in another, then holes will diffuse from the region of high concentration to the region of low concentration.[10]

[10] Such a diffusion process is like similar if one drops a few ink drops in a water-filled tank.

The diffusion of charge carriers gives rise to a net flow of charge, or **diffusion current**.

As an example, consider the bar of silicon shown in **Fig.** 3.4a: By some unspecified process, we have arranged to inject holes into its left side. This continuous hole injection gives rise to and maintains a hole **concentration profile** such as that shown in **Fig.** 3.4b. This profile in turn causes holes to diffuse from left to right along the silicon bar, resulting in a hole current in the x direction. The magnitude of the current at any point is proportional to the slope of the concentration profile, or the **concentration gradient**, at that point,

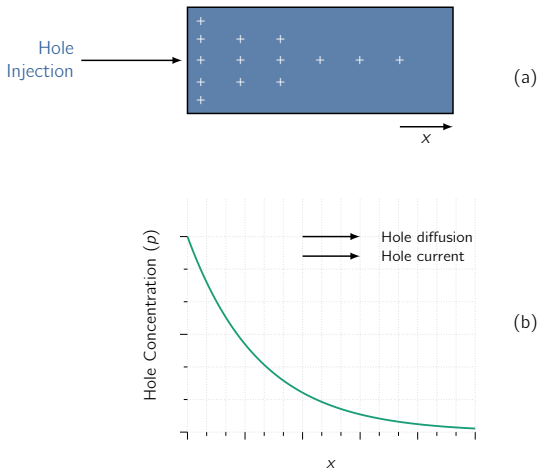
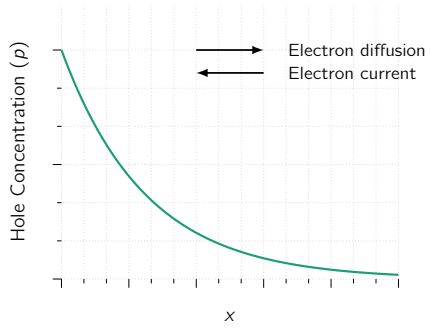


Figure 3.4 **A bar of silicon (a) into which holes are injected, thus creating the hole concentration profile along the x axis, shown in (b). The holes diffuse in the positive direction of x and give rise to a hole diffusion current in the same direction. Note that we are not showing the circuit to which the silicon bar is connected.**

$$J_p = -qD_p\frac{\mathrm{d}p(x)}{\mathrm{d}x}$$

where J_p is the hole-current density ($\mathrm{A}\cdot\mathrm{cm}^{-2}$), q is the magnitude of electron charge, D_p is a constant called the **diffusion constant** or diffusivity of holes; and $p(x)$ is the hole concentration at point x.

> The gradient $\mathrm{d}p/\mathrm{d}x$ is negative, resulting in a **positive current** in the x-direction.



In the case of electron diffusion resulting from an electron concentration gradient, as shown in **Fig.** 3.5, a similar relationship applies, giving the electron-current density,

$$J_n = qD_n\frac{\mathrm{d}n(x)}{\mathrm{d}x}$$

where D_n is the diffusion constant or diffusivity of electrons. Please observe that a negative (dn/dx) gives

Figure 3.5 **If the electron concentration profile shown is established in a bar of silicon, electrons diffuse in the x direction, giving rise to an electron diffusion current in the negative-x direction.**

rise to a negative current, a result of the convention that the positive direction of current is taken to be that of the flow of positive charge.[11]

[11] and opposite to that of the flow of negative charge.

> For holes and electrons diffusing in intrinsic silicon, typical values for the diffusion constants are $D_p = 12\,\text{cm}^2 \cdot \text{s}^{-1}$ and $D_n = 35\,\text{cm}^2 \cdot \text{s}^{-1}$.

At this point the reader is probably wondering where the diffusion current in the silicon bar in Fig. 3.6(*a*) goes. A good question, as we are not showing how the right-side end of the bar is connected to the rest of the circuit. We will address this and related questions in detail in our discussion of the *pn* junction in later sections.

Relationship between D and μ — 3.6 Information

A simple yet powerful relationship ties the **diffusion constant** with the **mobility**:

$$\frac{D_n}{\mu_n} = \frac{D_p}{\mu_p} = V_T \qquad \text{where} \qquad V_T = \frac{kT}{q}.$$

The parameter V_T is known as the **thermal voltage**. At room temperature, $T \simeq 300\,\text{K}$ and $V_T = 25{,}9\,\text{mV}$.

This relationship is known as the **Einstein relationship**.[12]

[12] The Einstein relation connects a particle's diffusion coefficient to its mobility, linking random thermal motion with how easily it moves through a medium under a force.

3.3 The PN Junction

Having learned important semiconductor concepts, we are now ready to consider our first practical semiconductor structure:

the *pn* junction.

As previously mentioned, the *pn* junction implements the diode and plays the dominant role in the structure and operation of the Bipolar Junction Transistor (BJT).

Understanding *pn* junctions is very important to the study of the MOSFET operation.

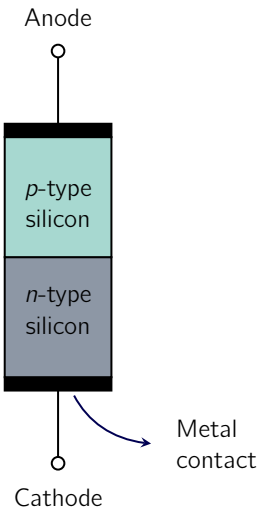


Figure 3.6 **Simplified physical structure of the pn junction. As the pn junction implements the junction diode, its terminals are labeled anode and cathode.**

3.3.1 Implementation

Fig. 3.6 shows a simplified physical structure of the *pn* junction. It consists of a *p*-type semiconductor[13] brought into close contact with an *n*-type semiconductor material, which is also silicon.

[13] e.g., silicon

In actual practice, both *p* and *n* regions are part of the same silicon crystal. That is, the *pn* junction is formed within a single silicon crystal by creating regions of different dopings (*p* and *n* regions).

As indicated in **Fig.** 3.6, external wire connections are made to the *p* and *n* regions through metal (aluminum) contacts. If the *pn* junction is used as a diode, these constitute the diode terminals and are therefore labeled **anode** and **cathode** in keeping with diode terminology.[14]

[14] This terminology in fact is a carryover from that used with vacuum-tube technology, which was the technology for making diodes and other electronic devices until the invention of the transistor in 1947. This event ushered in the era of solid-state electronics, which changed not only electronics, communications, and computers but indeed the world!

3.3.2 Open Circuit Operation

Fig. 3.7 shows a *pn* junction under open-circuit conditions-that is, the external terminals are left open. The "+" signs in the *p*-type material denote the **majority holes**. The charge of these holes is neutralized by an equal amount of bound negative charge associated with the acceptor atoms.

For simplicity, these bound charges are not shown in the diagram. Also not shown are the minority electrons generated in the *p*-type material by thermal ionization. In the *n*-type material the majority electrons are indicated by "-" signs. Here also, the bound positive charge, which neutralizes the charge of the majority electrons, is not shown in order to keep the diagram simple. The *n*-type material also contains minority holes generated by thermal ionization but not shown in the diagram.

Diffustion Current

Because the concentration of holes is high in the *p* region and low in the *n* region, holes diffuse across the junction from the *p* side to the *n* side. Similarly, electrons diffuse across the junction from the *n* side to the *p* side.

These two current components add together to form the diffusion current I_p, whose direction is from the p side to the n side, as indicated in **Fig.** 3.7.

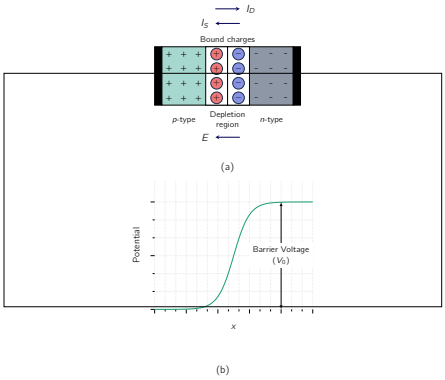


Figure 3.7
(a) The pn junction with no applied voltage (open-circuited terminals). (b) The potential distribution along an axis perpendicular to the junction.

Depletion Region

The holes that diffuse across the junction into the *n* region quickly recombine with some of the majority electrons present there and therefore disappear from the scene.

This recombination process results also in the disappearance of some free electrons from the *n*-type material. Therefore some of the bound positive charge will no longer be neutralized by free electrons, and this charge is said to have been **uncovered**. Given the recombination takes place close to the junction, there will be a region close to the junction that is **depleted of free electrons** and contains uncovered bound positive charge, as indicated in Fig. 3.9.

The electrons that diffuse across the junction into the *p* region quickly recombine with some of the majority holes there, and thus disappear from the scene. This results also in the disappearance of some majority holes, causing some of the bound negative charge to be uncovered (i.e., no longer neutralized by holes). Thus, in the *p* material close to the junction, there will be a region **depleted of holes** and containing uncovered bound negative charge, as indicated in **Fig.** 3.7.

From the above it follows that a **carrier-depletion region** will exist on both sides of the junction, with the *n* side of this region positively charged and the *p* side negatively charged. This carrier-depletion

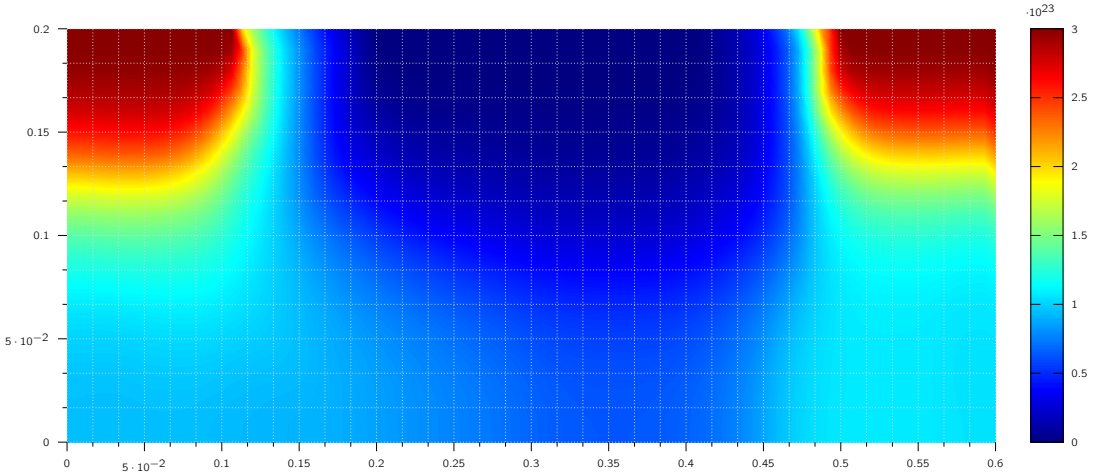


Figure 3.8
The electron density of a MESFET. Please observe the central point where the environment is void of electrons.

region-or, simply, depletion region-is also called the space-charge **region**. The charges on both sides of the depletion region cause an electric field E to be established across the region in the direction indicated in **Fig.** 3.7. Hence a potential difference results across the depletion region, with the n side at a positive voltage relative to the p side, as shown in **Fig.** 3.7b. Thus the resulting electric field opposes the diffusion of holes into the n region and electrons into the p region. In fact, the voltage drop across the depletion region acts as a **barrier** that has to be overcome for holes to diffuse into the n region and electrons to diffuse into the p region. The larger the barrier voltage, the smaller the number of carriers that will be able to overcome the barrier, and hence the lower the magnitude of diffusion current. Thus it is the appearance of the barrier voltage V_0 that limits the carrier diffusion process. It follows that the diffusion current ID depends strongly on the voltage drop $V0$ across the depletion region.

Drift Current and Equilibrium

In addition to the current component I_p due to majority-carrier diffusion, a component due to minority-carrier drift exists across the junction. Specifically, some of the thermally generated holes in the n material move toward the junction and reach the edge of the depletion region. There, they experience the electric field in the depletion region, which sweeps them across that region into the p side. Similarly, some of the minority thermally generated electrons in the p material move to the edge of the depletion region and get swept by the electric field in the depletion region across that region into the n side. These two current components-electrons moved by drift from p to n and holes moved by drift from n to p-add together to form the drift current I_s, whose direction is from the n side to the p side of the junction, as indicated in Fig. 3.9. Since the current I_s is carried by thermally generated minority carriers, its value is strongly dependent on temperature; however, it is independent of the value of the depletion-layer voltage V_0. This is due to the fact that the drift current is determined by the number of minority carriers that make it to the edge of the depletion region; any minority carriers that manage to get to the edge of the depletion region will be swept across by E irrespective of the value of E or, correspondingly, of V_0.

Under open-circuit conditions (**Fig.** 3.7) no external current exists; thus the two opposite currents across the junction must be equal in magnitude:

$$I_D = I_S$$

This equilibrium condition[15] is maintained by the barrier voltage (V_0). Therefore, if for some reason I_D exceeds I_S, then more bound charge will be uncovered on both sides of the junction, the depletion layer will widen, and the voltage across it (V_0) will increase. This in turn causes I_D to decrease until equilibrium is achieved with $I_D = I_S$.

On the other hand, if I_S exceeds I_D, then the amount of uncovered charge will decrease, the depletion layer will narrow, and the voltage across it (V_0) will decrease. This causes I_D to increase until equilibrium is achieved with $I_D = I_S$.

[15] In fact, in equilibrum the equality of drift and diffusion currents applies not just to the total currents but also to their individual components. That is, the hole drift current must equal the hole diffusion current and, similarly, the electron drift current must equal the electron diffusion current.

The Junction Built-in Voltage

With no external voltage applied, the barrier voltage (V_0) across the *pn* junction can be shown to be given by:

$$V_0 = V_T \ln\left(\frac{N_A N_D}{n_i^2}\right)$$

where N_A and N_D are the doping concentrations of the *p*-side and *n*-side of the junction, respectively. Therefore V_0 depends both on doping concentrations and on temperature.

It is known as the **junction built-in voltage**.

Typically, for silicon at room temperature, V_0 is in the range of 0,6 V to 0,9 V.

When the *pn* junction terminals are left open-circuited, the voltage measured between them will be zero. That is, the voltage V_0 across the depletion region does **NOT** appear between the junction terminals. This is because of the contact voltages existing at the metal-semiconductor junctions at the terminals, which counter and exactly balance the barrier voltage.

If this were not the case, we would have been able to draw energy from the isolated *pn* junction, which would clearly **violate the principle of conservation of energy**.

Width of and Charge Stored in the Depletion Region

Figure 3.10 provides further illustration of the situation occuring in the *pn* junction when the junction is in equilibrium. In Fig. 3.10(a) we see a junction in which $N_A > N_D$, a typical situation in practice. This is borne out by the carrier concentration on both sides of the junction, as shown in Fig. 3.10(b).

We have denoted the minority-carrier concentrations in both sides by n_{p0} and p_{n0}, with the additional subscript "0" signifying equilibrium.[16]

[16] i.e., before external voltages are applied, as will be seen in the next section.

Observe that the depletion region extends in both the *p* and *n* materials and **equal amounts of charge exist** on both sides.[17] However, since usually unequal dopings N_A and N_D are used, as in the case illustrated in Fig. 3.10, the width of the depletion layer will **NOT** be the same on the two (**2**) sides. Rather, to uncover the same amount of charge, the depletion layer will extend deeper into the more lightly doped material. Specifically, if we denote the width of the depletion region in the *p* side by x_n and in the *n* side by x_n, we can express the magnitude of the charge of the junction as

[17] (Q and Q in Fig. 3.10c)

$$\text{on the } n \text{ side} \quad |Q_+| = qAx_nN_D \quad \text{and on the } p \text{ side} \quad |Q_-| = qAx_pN_A$$

where A is the cross-sectional area of the junction in the plane perpendicular to the page. The charge equality condition can now be written as $qAx_nN_D = qAx_nN_A$ which can be rearranged to give us:

$$\frac{x_n}{r} = \frac{N_A}{N_B}$$

In actual practice, it is usual for one side of the junction to be much more heavily doped than the other, with the result that the depletion region exists almost entirely on one side (the lightly doped side).

The width W of the depletion layer can be shown to be given by

$$W = x_{\mathrm{n}} + x_{\mathrm{p}} = \sqrt{\frac{2\epsilon_s}{q}\left(\frac{1}{N_{\mathrm{A}}} + \frac{1}{N_{\mathrm{D}}}\right)V_0}$$

where ϵ_s is the electrical permittivity of silicon = $11.7\epsilon_0 = 11.7 \times 8.85 \times 10^{-14}$ F/cm = 1.04×10^{-12} F/cm. Typically W is in the range 0.1 μm to 1 μm. Eqs. (3.25) and (3.26) can be used to obtain x_n and x_p in terms of W as:

$$x_{\mathrm{n}} = W\frac{N_{\mathrm{A}}}{N_{\mathrm{A}} + N_{\mathrm{D}}} \qquad \text{and} \qquad x_{\mathrm{p}} = W\frac{N_{\mathrm{D}}}{N_{\mathrm{A}} + N_{\mathrm{D}}} \tag{**3.10**}$$

The charge stored on either side of the depletion region can be expressed in terms of W by utilizing Eqs. (3.23) and Eq. (**3.10**) to obtain:

$$Q_I = |Q_+| = |Q_-|$$
$$Q_J = Aq\left(\frac{N_A N_D}{N_A + N_D}\right)W$$

Finally, we can substitute for W from Eq. (3.26) to obtain

$$Q_J = A\sqrt{2\epsilon_s q\left(\frac{N_A N_D}{N_A + N_D}\right)V_0}$$

These expressions for QJ will prove useful in subsequent sections.

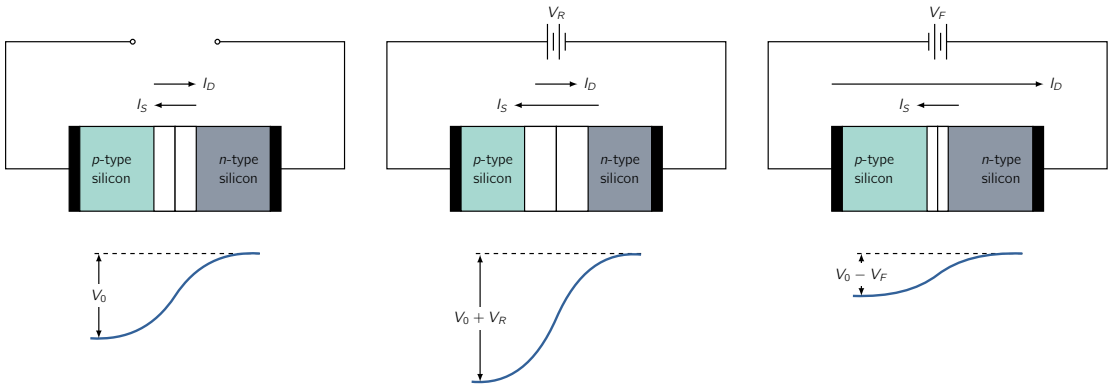


Figure 3.9

3.4 PN Junction with Applied Voltage

Now that we had a look at the open-circuited *pn* junction in detail, we are now ready to apply a DC voltage between its two (**2**) terminals to find its electrical conduction properties. If the voltage is applied so that the *p* side is made more positive than the *n* side, it is referred to as a **forward-bias** voltage.[18] Conversely, if our applied dc voltage is such that it makes the *n* side more positive than the *p* side, it is said to be a reverse-bias voltage. As will be seen, the *pn* junction exhibits vastly different conduction properties in its forward and reverse directions. Our plan is as follows. We begin by a simple qualitative description in Section 3.5.1 and then consider an analytical description of the $i - v$ characteristic of the junction in Section 3.5.2.

[18] we take the term **bias** to refer simply to the application of a dc voltage.

3.4.1 Qualitative Description of Junction Operation

Fig. 3.9 shows the *pn* junction under three different conditions: (a) the open-circuit or equilibrium condition studied in the previous section; (b) the reverse-bias condition, where a DC voltage V_R is applied; and (c) the forward-bias condition, where a DC voltage V_F is applied.

Observe that in the open-circuit case, a barrier voltage V_0 develops, making **n** more positive than p, and limiting the diffusion current Ip to a value exactly equal to the drift current Is, thus resulting in a zero current at the junction terminals, as should be the case, since the terminals are open-circuited. Also, as mentioned previously, the barrier voltage V_0, though it establishes the current equilibrium across the junction, does **not** in fact appear between the junction terminals.

Consider now the reverse-bias case in (b). The externally applied reverse-bias voltage V_R is in the direction to add to the barrier voltage, and it does, thus increasing the effective barrier voltage to $(V_0 + V_R)$ as shown. This reduces the number of holes that diffuse into the *n* region and the number of electrons that diffuse into the *p* region. The end result is that the diffusion current ID is dramatically reduced. As will be seen shortly, a reverse-bias voltage of a volt or so is sufficient to

cause $I_p \simeq 0$, and the current across the junction and through the external circuit will be equal to I_s. Recalling that I_s is the current due to the drift across the depletion region of the thermally generated minority carriers, we expect I_s to be very small and to be strongly dependent on temperature. We will show this to be the case very shortly. We thus conclude that in the reverse direction, the *pn* junction conducts a very small and almost-constant current equal to I_s.

Before leaving the reverse-bias case, observe that the increase in barrier voltage will be accompanied by a corresponding increase in the stored uncovered charge on both sides of the depletion region. This in turn means a wider depletion region, needed to uncover the additional charge required to support the larger barrier voltage $(V_0 + V_R)$. Analytically, these results can be obtained easily by a simple extension of the results of the equilibrium case. Thus the width of the depletion region can be obtained by replacing V_0 in Eq. (3.26) by $(V_0 + V_0)$,

$$W = x_n + x_p = \sqrt{\frac{2\epsilon_s}{q}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)(V_0 + V_R)}$$

and the magnitude of the charge stored on either side of the depletion region can be determined by replacing V_0 in Eq. (3.30) by $(V_0 + V_R)$,

$$Q_J = A\sqrt{2\epsilon_s q\left(\frac{N_A N_D}{N_A + N_D}\right)(V_0 + V_R)} \tag{3.11}$$

We next consider the forward-bias case shown in Fig. 3.11(c). Here the applied voltage V_F is in the direction that subtracts from the built-in voltage V_0, resulting in a reduced barrier voltage $(V_0 - V_F)$ across the depletion region. This reduced barrier voltage will be accompanied by reduced depletion-region charge and correspondingly narrower depletion-region width W. Most importantly, the lowering of the barrier voltage will enable more holes to diffuse from p to n and more electrons to diffuse from n to p. Thus the diffusion current Ip increases substantially and, as will be seen shortly, can become many orders of magnitude larger than the drift current I_s. The current I in the external circuit is of course the difference between I_p and I_s, $I = I_p - I_s$ and it flows in the forward direction of the junction, from p to n. We thus conclude that the **pn** junction can conduct a substantial current in the forward-bias region and that current is mostly a diffusion current whose value is determined by the forward-bias voltage V_F.

Substituting for $p_n(x)$ from Eq. (3.35) gives

$$J_p(x) = q\left(\frac{D_p}{L_p}\right)p_{n0}\left(e^{V/V_T} - 1\right)e^{-(x-x_n)/L_p}$$

As expected, $J_p(x)$ is highest at $x = x_n$,

$$J_p(x_n) = q\left(\frac{D_p}{L_n}\right)p_{n0}\left(e^{V/V_T} - 1\right) \tag{3.12}$$

and decays exponentially for $x > x_n$, as the minority holes recombine with the majority electrons. This recombination, however, means that the majority electrons will have to be replenished by a current that injects electrons from the external circuit into the n region of the junction. This latter current component has the same direction as the hole current (because electrons moving from right

to left give rise to current in the direction from left to right). It follows that as $J_p(x)$ decreases, the electron current component increases by exactly the same amount, making the total current in the n material constant at the value given by Eq. (3.37).

An exactly parallel development can be applied to the electrons that are injected from the n to the p region, resulting in an electron diffusion current given by a simple adaptation of Eq. (3.37),

$$J_n(-x_p) = q\left(\frac{D_n}{L_n}\right) n_{p0}\left(e^{V/V_T} - 1\right)$$

Now, although the currents in Eqs. (3.37) and (3.38) are found at the two edges of the depletion region, their values do not change in the depletion region. Thus we can drop the location descriptors (x_n), $(-x_p)$, add the two current densities, and multiply by the junction area A to obtain the total current I as $I = A(J_p + J_n)$ $I = Aq\left(\frac{D_p}{L_n}p_{n0} + \frac{D_n}{L_n}n_{p0}\right)\left(e^{V/V_T} - 1\right)$ Substituting for $p_{n0} = n_i^2/N_D$ and for $n_{p0} = n_i^2/N_A$ gives

$$I = Aqn_i^2\left(\frac{D_p}{L_nN_D} + \frac{D_n}{L_nN_A}\right)\left(e^{V/V_T} - 1\right) \tag{3.13}$$

From this equation we note that for a negative V (reverse bias) with a magnitude of a few times V_T (25.9 mV), the exponential term becomes essentially zero, and the current across the junction becomes negative and constant. From our qualitative description in Section 3.5.1, we know that this current must be I_s. Thus,

$$I = I_s(e^{V/V_T} - 1) \tag{3.14}$$

where

$$I_S = Aqn_i^2\left(\frac{D_p}{L_pN_p} + \frac{D_n}{L_nN_A}\right)$$

Figure 3.13 shows the $I - V$ characteristic of the pn junction (Eq. 3.40). Observe that in the reverse direction the current saturates at a value equal to $-I_s$. For this reason, I_s is given the name **saturation current**. From Eq. (3.41) we see that I_s is directly proportional to the cross-sectional area A of the junction. Thus, another name for I_s, one we prefer to use in this book, is the junction scale current. Typical values for I_s, for junctions of various areas, range from 10^{-18} A to 10^{-12} A.

Besides being proportional to the junction area A, the expression for I_s in Eq. (3.41) indicates that I_s is proportional to n_i^2, which is a very strong function of temperature (see Eq. 3.2).

3.4.2 Reverse Breakdown

The description of the operation of the pn junction in the reverse direction, and the $I-V$ relationship of the junction in Eq. (**3.14**), indicate that at a reverse-bias voltage $(-V)$, with the condition of $V \gg Vr$, when applied will result in reverse current that flows across the junction is **approximately equal** to I_S and therefore is very small. However, as the magnitude of the reverse-bias voltage V is increased, a value is reached at which a very large reverse current flows as shown in Fig. 3.14.

Observe that as V reaches the value $V[Z]$, the dramatic increase in reverse current is accompanied by a very small increase in the reverse voltage; that is, the reverse voltage across the junction

remains very close to the value V_Z. The phenomenon that occurs at $V = V_Z$ is known as **junction breakdown**.

> It is **NOT** a destructive phenomenon. That is, the *pn* junction can be repeatedly operated in the breakdown region without a permanent effect on its characteristics.

This, however, is predicated on the assumption that the magnitude of the reverse-breakdown current is limited by the external circuit to a **safe** value. The "safe" value is one that results in the limitation of the power dissipated in the junction to a safe, allowable level.

There are two (**2**) possible mechanisms for *pn* junction breakdown:

- the zener effect,[19] and
- avalanche effect.

[19] Named after an early worker in the area. Note that the subscript Z in V_Z denotes zener. We will use V_Z to denote the breakdown voltage whether the breakdown mechanism is the zener effect or the avalanche effect.

If a **pn** junction breaks down with a breakdown voltage $V_Z < 5$ V, the breakdown mechanism is usually the zener effect. Avalanche breakdown occurs when V_Z is greater than approximately 7 V. For junctions that break down between 5 V and 7 V, the breakdown mechanism can be either the zener or the avalanche effect or a combination of the two.

Zener breakdown occurs when the electric field in the depletion layer increases to the point of breaking covalent bonds and generating electron-hole pairs. The electrons generated in this way will be swept by the electric field into the *n* side and the holes into the *p* side. Therefore these electrons and holes constitute a reverse current across the junction. Once the zener effect starts, a large number of carriers can be generated, with a negligible increase in the junction voltage. Therefore the reverse current in the breakdown region will be large and its value must be determined by the external circuit, while the reverse voltage appearing between the diode terminals will remain close to the specified breakdown voltage V_Z.

The other breakdown mechanism, **avalanche breakdown**, occurs when the minority carriers that cross the depletion region under the influence of the electric field gain sufficient kinetic energy to be able to break covalent bonds in atoms with which they collide. The carriers liberated by this process may have sufficiently high energy to be able to cause other carriers to be liberated in another ionizing collision. This process keeps repeating in the fashion of an avalanche, with the result that many carriers are created that are able to support any value of reverse current, as determined by the external circuit, with a negligible change in the voltage drop across the junction.

As will be seen when we study Diodes in the next chapter, some **pn** junction diodes are fabricated to operate specifically in the breakdown region, where use is made of the nearly constant voltage V_Z.

3.5 Capacitive Effects in the pn Junction

There are two (2) charge-storage mechanisms in a given *pn* junction.

- One is associated with the charge stored in the depletion region, and
- the other is associated with the minority-carrier charge stored in the *n* and *p* materials as a result of the concentration profiles established by carrier injection.

> While the 1st is easier to see when the *pn* junction is reverse biased, the 2nd is in effect only when the junction is **forward biased**.

3.5.1 Depletion or Junction Capacitance

When a *pn* junction is **reverse biased** with a voltage V_R, the charge stored on either side of the depletion region is given by Eq. (**3.11**),

$$Q_J = A\sqrt{2\epsilon_s q\frac{N_A N_D}{N_A + N_D}}\,(V_0 + V_R)$$

Therefore, for a given *pn* junction,

$$Q_J = \alpha\sqrt{V_0 + V_R} \qquad \text{where} \qquad \alpha = A\sqrt{2\epsilon_s q\frac{N_A N_D}{N_A + N_D}} \tag{3.15}$$

The non-linear relation between Q_J and V_R, is shown in Fig. 3.15.

> This non-linear relationship makes it difficult to define a capacitance which accounts for the need to change Q_J whenever V_R is changed.

We can, however, assume the junction is operating at a point such as Q, as indicated in Fig. 3.15, and define a capacitance C_j which relates the change in the charge Q_J to a change in the voltage V_R,

$$C_j = \left.\frac{dQ_j}{dV_R}\right|_{V_R = V_Q} \tag{3.16}$$

> This **incremental-capacitance approach** turns out to be quite useful in electronic circuit design.

Using Eq. (**3.16**) together with Eq. (**3.15**) gives us:

$$C_j = \frac{\alpha}{2\sqrt{V_0 + V_R}} \tag{3.17}$$

The value of C_j at zero reverse bias can be obtained from Eq. (**3.17**) as

$$C_{j0} = \frac{\alpha}{2\sqrt{V_0}} \tag{3.18}$$

which enables us to express C_j as:

$$C_j = \frac{C_{j0}}{\sqrt{1 + \frac{V_R}{V_0}}} \tag{3.19}$$

where C_{j0} is given by Eq. (**3.18**) or alternatively if we substitute for α from Eq. (**3.15**) by

$$C_{j0} = A\sqrt{\left(\frac{\epsilon_s q}{2}\right)\left(\frac{N_A N_D}{N_A + N_D}\right)\left(\frac{1}{V_0}\right)}$$

Before leaving the subject of depletion-region or junction capacitance we point out that in the *pn* junction we have been studying, the doping concentration is made to change abruptly at the junction boundary. Such a junction is known as an **abrupt junction**. There is another type of pn junction in which the carrier concentration is made to change gradually from one side of the junction to the other. To allow for such a **graded junction**, the formula for the junction capacitance (Eq. (**3.19**)) can be written in the more general form

$$C_j = \frac{C_{j0}}{\left(1 + \frac{V_R}{V}\right)^m}$$

where m is a constant called the **grading coefficient**, whose value ranges from 1/3 to 1/2 depending on the manner in which the concentration changes from the p to the n side.

3.5.2 Diffusion Capacitance

Consider a forward-biased *pn* junction.

In steady state, minority-carrier distributions in the p and n materials are established, as shown in Fig. 3.12. Therefore, a certain amount of excess minority-carrier charge is stored in each of the p and n bulk regions.[20] If the terminal voltage (V) changes, this charge will have to change before a new steady state is achieved.

[20] outside the depletion region.

> This **charge-storage phenomenon** gives rise to another capacitive effect, distinctly different from that due to charge storage in the depletion region.

To calculate the excess minority-carrier charge, refer to Fig. 3.12. The excess hole charge stored in the n region can be found from the shaded area under the exponential as follows:[21]

$$\begin{aligned} Q_n &= Aq \times \text{shaded area under the } p_n(x) \text{ curve} \\ &= Aq[p_n(x_n) - p_{n0}]L_n \end{aligned}$$

[21] Recall that, from calculus, the area under an exponential curve Ae^{xB} is equal to AB.

Substituting for $p_n(x_n)$ from Eq. (3.33) and using Eq. (**3.12**) enables us to express Q_p as

$$Q_p = \frac{L_p^2}{D} I_p$$

The factor (L_p^2/D_p) relating Q_p to I_p is a useful device parameter that has the dimension of time (s) and is denoted τ_p

$$\tau_p = \frac{L_p^2}{D_p} \qquad \text{therefore} \qquad Q_p = \tau_p I_p \tag{3.20}$$

The time constant τ_p is known as the excess **minority-carrier (hole) lifetime**.

> It is the **average time** it takes for a hole injected into the n region to recombine with a majority electron.

This definition of τ_p implies that the entire charge Q_p disappears and has to be replenished every τ_p seconds. The current that accomplishes the replenishing is $I_p = Q_p/\tau_p$. This is an alternate derivation for Eq. (**3.20**).

A relationship similar to that in Eq. (**3.20**) can be developed for the electron charge stored in the p region,

$$Q_n = \tau_n I_n$$

where τ_n is the electron lifetime in the p region. The total excess minority-carrier charge can be obtained by adding together Q_p and Q_n,

$$Q = \tau_p I_p + \tau_n I_n$$

This charge can be expressed in terms of the diode current $I = I_p + I_n$ as

$$Q = \tau_r I$$

where τ_T is called the **mean transit time** of the junction. Obviously, τ_T is related to τ_p and τ_n. Furthermore, for most practical devices, one side of the junction is much more heavily doped than the other.

> For example, if $N_A \gg N_D$, one can show that $I_p \gg I_n$, $I \simeq I_p$, $Q_p \gg Q_n$, $Q \simeq Q_p$, and thus $\tau_T \simeq \tau_p$.

For small changes around a bias point, we can define an incremental diffusion capacitance C_d as:

$$C_d = \frac{\mathrm{d}Q}{\mathrm{d}V} \qquad \text{and therefore} \qquad C_d = \left(\frac{\tau_T}{V_T}\right) I$$

where I is the forward-bias current.

As can be seen C_d is directly proportional to the forward current (I) and therefore is negligibly small when the diode is reverse biased. In addition, to keep C_d small, the transit time (τ_t) **must** be made small, an important requirement for a *pn* junction intended for high-speed or high-frequency operation.

References



Part II

Semiconductor Devices

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The number of transistors on an integrated circuit will double in about 18 months.

(Gordon Moore,co-founder of Fairchild Semiconductor and Intel)

Chapter 4

Diodes

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In Chapter 2, we dealt **almost** entirely with **linear circuits**;

> Any non-linear behavior, such as **amplifier output saturation**, was treated as a problem to be solved by the circuit designer.

However, there are numerous signal-processing functions which can only be implemented by **non-linear circuits**.

> For example, the generation of DC voltages from the AC power supply, and the generation of signals of various waveforms.[1]

[1] e.g., sinusoids, square waves, pulses.

Also, digital **logic** and **memory circuits** constitute a special class of non-linear circuits. The simplest and most fundamental non-linear circuit element is the **diode**. Similar to a resistor, the diode has two (**2**) terminals. However, unlike the resistor, which has a linear relationship[2] between the current flowing through it and the voltage appearing across it, the diode has a **non-linear** $i - v$ characteristic.

[2] i.e., the relation between voltage and current is proportional.

Chapter Structure This chapter is concerned with the study of diodes. To understand the key idea of the diode function, we begin with a **fictitious element**, the ideal diode.[3]

[3] Similar to what we did with the op-amp.

We then introduce the silicon junction diode, explain its terminal characteristics, and provide techniques for the analysis of diode circuits. The latter task involves the important subject of *device modeling*. Our study of modeling the diode characteristics will lay the foundation for our study of modeling *transistor* operation in the following three chapters.

Of the many applications of diodes, their use in the design of rectifiers[4] is the most common. Therefore we shall study **rectifier circuits** in some detail and briefly look at a number of other diode applications.

[4] which convert ac to dc.

The junction diode is nothing more than the *pn* junction we studied in Chapter 3, and most of this chapter is concerned with the study of silicon *pn*-junction diodes.

In the last section, however, we briefly consider some specialized diode types, including the **photodiode** and the LED.

4.1 Information

A History of the Diode

The diode is one of the oldest and most important electronic devices, although it is not as famous as the transistor. Used in all sorts of electronic systems, the diode functions as a **one-way valve** for electric current.[5] There are two (**2**) basic types of diodes.

- One is an electron tube similar to the **triode**.
- The other type uses semiconductors, like the **transistor**.

Both were invented early in the 20th century.



Figure 4.1
John Fleming's valve or diode (1904) allowed electrical current to flow in one direction only.

[5] It only allows current to flow in one direction.

The first diode was a modified light bulb. *Thomas Edison* discovered that including an extra electrode in a light bulb and connecting it to the positive side of a battery resulted in a current flowing from the filament through the empty space. He was not sure what to do with this discovery, and moved on to other projects. *Joseph J. Thomson* announced the discovery of the electron in April 1897 and explained the *Edison effect* where current travels just one way through a vacuum tube.

Others found another use for this device. In the early 1900s, for example, English engineer *John Ambrose Fleming* used this one-way electrical "valve", to convert radio waves into a flow of current that could be measured by a galvanometer. The *Fleming valve* is remembered as the first true electronic device. It came into use for radio transmission and soon became the basis of *Lee De Forest*'s Audion electron tube, which he invented in 1906.

Also around 1906, American engineer *Greenleaf W. Pickard* invented a new type of diode. Pickard based his design on the earlier discovery that electricity can flow in only one direction through certain types of mineral crystals, such as **silicon**. By placing a silicon crystal between a metal base and a carefully placed fine wire, Pickard created a valve that could also be used to detect radio waves.

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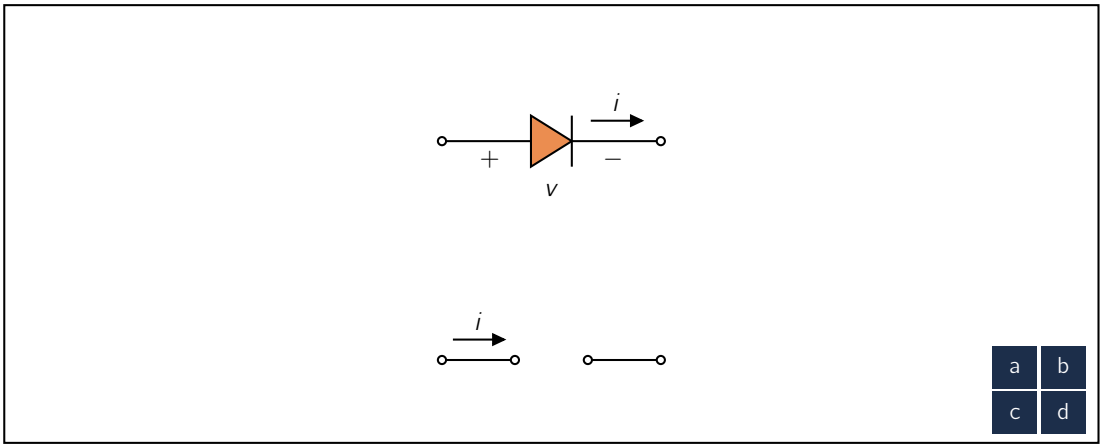


Figure 4.2
The ideal diode: (a) diode circuit symbol; (b) iv characteristic; (c) equivalent circuit in the reverse direction; (d) equivalent circuit in the forward direction.

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This type of "cat's whisker" diode[6] became more popular after American H. C. Dunwoody patented a version of it that used a material called **carborundum**.

[6] so-named because of the fine wire used in it.

4.1 The Ideal Diode

As usual it is worth studying the ideal case to get a good grasp of its fundamentals before tackling the imperfections caused by manufacturing, physical limitations, etc. We will start with understanding probably its most important property.

4.1.1 i-v Characteristics

The ideal diode may be considered to be the most important non-linear circuit element. It is a **two-terminal device** having the circuit symbol of Fig. 4.1(a) and the $i - v$ characteristic shown in Fig. 4.1(b). The terminal characteristic of the ideal diode can be interpreted as follows:

> If a negative voltage[7] is applied to the diode, no current flows and the diode behaves as an open circuit (Fig. 4.1c).

[7] relative to the reference direction indicated in Fig. 4.1a

Diodes operated in this mode are said to be **reverse biased**, or operated in the reverse direction.

> An ideal diode has zero current when operated in the reverse direction and is said to be cut off, or simply off.

On the other hand, if a positive current[8] is applied to the ideal diode, **zero voltage drop** appears across the diode. In other words, the ideal diode behaves as a short circuit in the forward direction (Fig. 4.1d); it passes any current with zero voltage drop.

[8] relative to the reference direction indicated in Fig. 4.1(a).

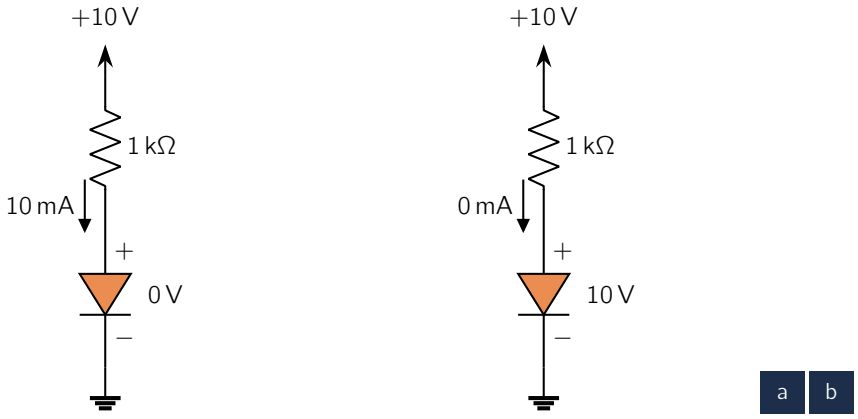


Figure 4.3
The two modes of operation of ideal diodes and the use of an external circuit to limit (a) the forward current and (b) the reverse voltage.

A forward-biased diode is said to be **turned on**, or simply on.

From the above description it should be noted that the external circuit must be designed to limit the forward current through a **conducting diode**, and the reverse voltage across a **cutoff diode**, to predetermined values.

A Study of Two Diodes **Fig.** 4.3 shows two diode circuits illustrating this point. In the circuit of **Fig.** 4.3a the diode is obviously conducting. Therefore its voltage drop will be zero, and the current through it will be determined by the +10 V supply and the 1 kΩ resistor as 10 mA. The diode in the circuit of **Fig.** 4.3b is obviously cut off, and thus its current will be zero, which in turn means that the entire 10 V supply will appear as reverse bias across the diode.

The positive terminal of the diode is called the **anode** and the negative terminal the **cathode**.[9] The $i - v$ characteristic of the ideal diode, conducting in one direction and not in the other, should explain the choice of its arrow-like circuit symbol.

[9] This terminology is a carryover from the days of vacuum-tube diodes.

As should be evident from the preceding description, the $i - v$ characteristic of the ideal diode is **highly non-linear**. Although it consists of two straight-line segments, they are at **perpendicular** to one another.

A nonlinear curve that consists of straight-line segments is said to be **piecewise linear**.

If a device having a piecewise-linear characteristic is used in a particular application in such a way that the signal across its terminals swings along only one of the linear segments, then the device can be considered a linear circuit element as far as that particular circuit application is concerned.

On the other hand, if signals swing past one or more of the break points in the characteristic, linear analysis is no longer possible.

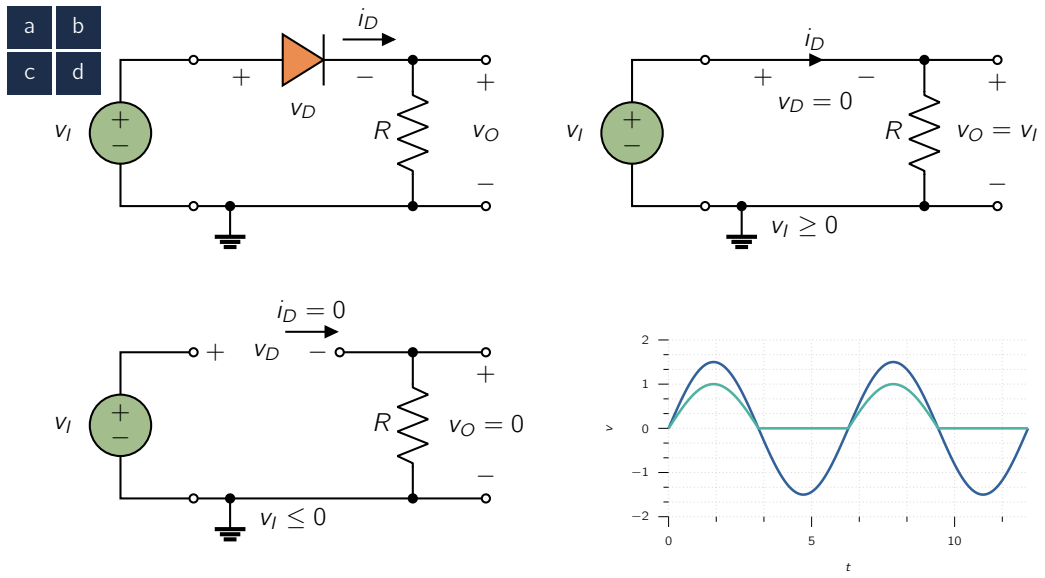


Figure 4.4
A rectifier circuit using a diode (a) Rectifier circuit (b) Equivalent circuit when $v_I \geq 0$ (c) Equivalent circuit when $v_I \leq 0$ (d) Output waveforms, with the rectified wave shown with lower amplitude for easy identification.

4.1.2 Application: The Rectificer

A simple application of the diode, one that makes use of its severely non-linear $i - v$ curve, is the **rectifier circuit** shown in **Fig.** 4.4a. The circuit consists of the series connection of a diode (D) and a resistor (R).

Let the input voltage v_I be the sinusoid shown in **Fig.** 4.4d, and assume the diode to be **ideal**.

During the positive half-cycles of the input sinusoid, the positive v_I will cause current to flow through the diode in its **forward direction**. It follows that the diode voltage v_D will be very small-ideally zero. Therefore, the circuit will have the equivalent shown in **Fig.** 4.4b, and the output voltage v_O will be equal to the input voltage v_I .

On the other hand, during the negative half-cycles of v_I , the diode will not conduct. Therefore, the circuit will have the equivalent shown in **Fig.** 4.4c, and v_O will be zero. This in turn with generate an output voltage with the waveform shown in **Fig.** 4.4d.

> While v_I alternates in polarity and has a zero average value, v_O is **unidirectional** and has a finite average value or a DC component.

Therefore the circuit of **Fig.** 4.4a rectifies the signal and therefore is called a **rectifier**. It can also be used to generate DC from AC.[10]

[10] We will study rectifier circuits in a later section of this chapter.

4.1.3 Application: Logic Gates

Diodes together with resistors can be used to implement **digital logic** functions.[11] **Fig.** 4.5 shows two (**2**) diode logic gates.



[11] An example of a diode based logic gate in IC form

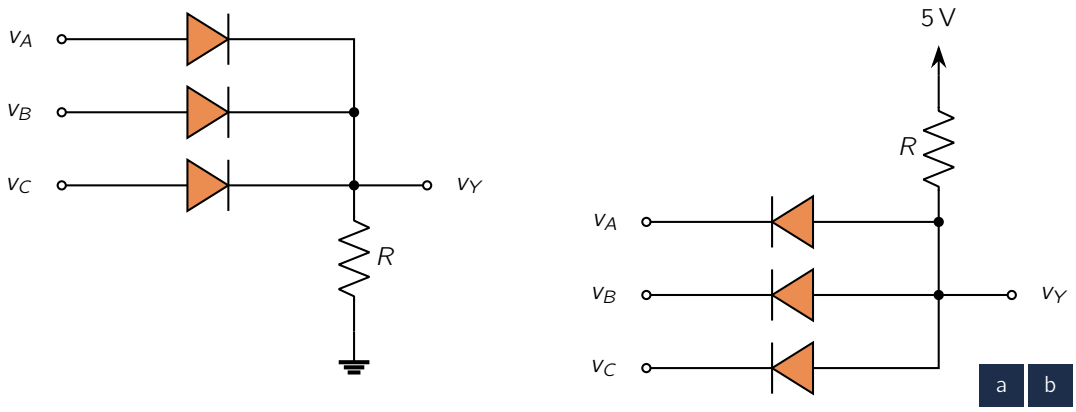


Figure 4.5
Diode logic gates: (a) OR gate; (b) AND gate (in a positive-logic system).

To see how these circuits function, consider a positive-logic system in which voltage values close to 0 V correspond to logic 0, or **low**, and voltage values close to +5 V correspond to logic 1, or **high**.

The circuit in **Fig.** 4.5a has three (**3**) inputs: v_A, v_B, and v_C. It is easy to see that diodes connected to +5 V inputs will conduct, therefore clamping the output v_Y to a value equal to +5 V. This positive voltage at the output will keep the diodes whose inputs are low (around 0 V) cut off. Therefore the output will be high if one or more of the inputs are high. The circuit therefore implements the logic **OR** function, which in Boolean notation is expressed as:

$$Y = A + B + C$$

Similarly, the circuit of **Fig.** 4.5b implements the logic **AND** function,

$$Y = A \cdot B \cdot C$$

Ideal Diodes in Circuit — 4.1 Exercise

ex:ideal-diodes-in-circuit

Assuming the diodes to be ideal, find the values of I and V in the circuits of **Fig.** 4.6.

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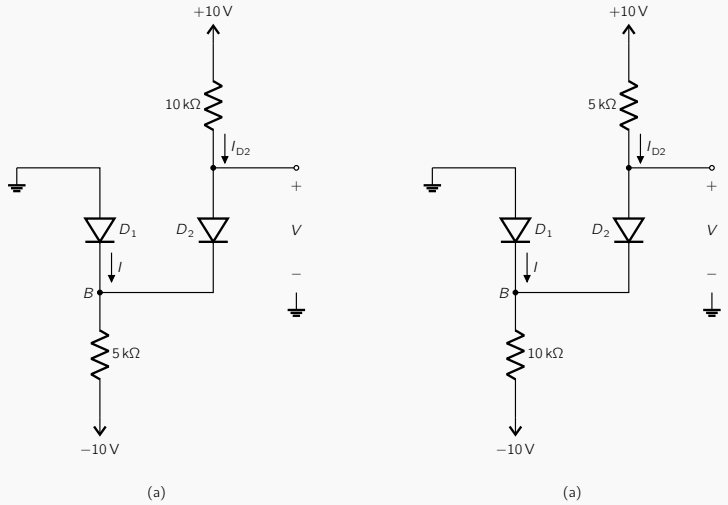


Figure 4.6

4.2 Terminal Characteristics of Junction Diodes

The most common implementation of the diode utilizes a *pn* junction. We have studied the physics of the *pn* junction and derived its $i - v$ characteristic in sufficient detail in Chapter 3. The *pn* junction being used to implement the diode function should come as no surprise:

> The *pn* junction can conduct **substantial current in the forward direction** and almost **no current in the reverse direction**.

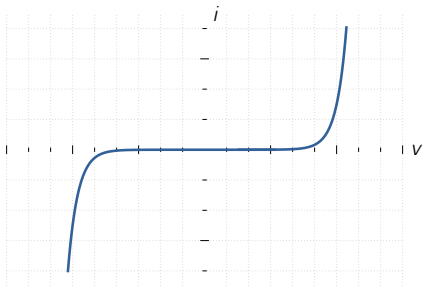


Figure 4.7
The $i - v$ characteristic of a silicon junction diode.

In this section we study the $i - v$ characteristic of the *pn* junction diode in detail to prepare ourselves for diode circuit applications.

Fig. 4.7 shows the $i - v$ characteristic of a silicon junction diode. The same characteristic is shown in Fig. 4.8 with some scales expanded and others compressed to reveal details. Note that the scale changes have resulted in the apparent discontinuity at the origin. As indicated, the characteristic curve consists of three distinct regions:

- The forward-bias region, determined by $v > 0$,
- The reverse-bias region, determined by $v < 0$,
- The breakdown region, determined by $v < -V_{ZK}$

We will have a more closer look at the three (**3**) regions of operation in the following sections.

[13] Known as **saturation current**, or **scale current**, more accurately the **reverse saturation current**, is the part of the reverse current in a semiconductor diode caused by *diffusion* of *minority carriers* from the neutral regions to the **depletion region**. This current is almost independent of the reverse voltage [1].

4.2.1 The Forward-Bias Region

The forward-bias region[12] of operation is entered when the terminal voltage (v) is **positive**. In the forward region the $i - v$ relationship is closely approximated by

$$i = I_s \left(\exp\left(v/V_T \right) - 1 \right) \tag{4.1}$$

In this aforementioned equation I_S is a constant[13] for a given diode at a given temperature.

A formula for I_S in terms of the diode's physical parameters and temperature was given in **??**.

For anyone who doesn't read the margin comments, the current I_S is usually called the **saturation current**.[14] Another name for I_S, and one that we will occasionally use, is the **scale current**. This

[12] or simply forward region.

[14] For reasons that will become apparent shortly.

name arises from the fact that I_S being directly proportional to the cross-sectional area of the diode.

Therefore doubling of the junction area results in a diode with double the value of I_S and, as the diode equation indicates, doubles the value of current i for a given forward voltage v. For "small-signal" diodes, which are small-size diodes intended for low-power applications, I_S is on the order of $1{,}0 \times 10^{-14}$ A.

> The value of I_S is, however, a very strong function of temperature. As a rule of thumb, I_S doubles in value for every 5 °C rise in temperature.

The voltage V_T in Eq. (**4.1**) is a constant called the **thermal voltage** and is given by:

$$V_T = \frac{k_B T}{q} \tag{4.2}$$

where

- k_B is the *Boltzmann*'s constant[15] ($1{,}38 \times 10^{-23}\,\mathrm{J \cdot K^{-1}}$),
- T the absolute temperature in kelvin (K),
- q is the magnitude of electronic charge ($q = 1{,}6 \times 10^{-19}$ C),

If we were to substitute k_B, with proper units, to Eq. (**4.2**), we can derive the following:

$$V_T = 0.0862T \text{ mV}$$

Therefore, at room temperature (20 °C) the value of V_T is around 25,3 mV.

> For quick circuit analysis we shall use $V_T = 25$ mV at room temperature.[16]

For appreciable current i in the **forward direction**, more specifically for $i > I_S$, Eq. (**4.1**) can be approximated by the exponential relationship

$$i \approx I_S \exp\left(v/V_T\right) \qquad \text{or in logaritmic form} \qquad v = V_T \ln\left(\frac{i}{I_S}\right) \tag{4.3}$$

where, of course, ln denotes the natural (base e) logarithm. The exponential relationship of the current (i) to the voltage (v) holds over many decades of current.[17] This is quite a remarkable property of junction diodes, one that is also found in BJT and that has been exploited in many interesting applications.

Let us consider the forward $i - v$ relationship in Eq. (**4.3**) and evaluate the current I_1 corresponding to a diode voltage (V_1):

$$I_1 = I_S \exp\left(V_1/V_T\right) \qquad \text{and for } V_2 \text{ and } I_2 \qquad I_2 = I_S \exp\left(V_2/V_T\right)$$

These two (**2**) equations can be combined to produce

$$\frac{I_2}{I_1} = \exp\left(\left(V_2 - V_1\right)/V_T\right) \quad \text{or} \quad V_2 - V_1 = V_T \ln\frac{I_2}{I_1} \quad \text{or in base-10} \quad V_2 - V_1 = 2.3 V_T \log\frac{I_2}{I_1}$$



[15] **Ludwig Eduard Boltzmann** *(1844 - 1906)*

was an Austrian mathematician and theoretical physicist. His greatest achievements were the development of **statistical mechanics** and the statistical explanation of the 2nd law of thermodynamics. In 1877, he provided the current definition of entropy, $S = k_B \ln \Omega$, where Ω is the number of microstates whose energy equals the system's energy, interpreted as a measure of the statistical disorder of a system. *Max Planck* named the constant k_B the Boltzmann constant.

[16] A slightly higher ambient temperature (25 C or so) is usually assumed for electronic equipment operating inside a cabinet. At this temperature, $V_T \approx 25{,}8$ mV

[17] a span of as many as seven decades, i.e., a factor of $1{,}0 \times 10^8$, can be found

This equation simply states that for a decade[18] change in current, the diode voltage drop changes by $2.3V_T$, which is approximately 60 mV. This also suggests that the diode $i - v$ relationship is most conveniently plotted on semi-log paper.[19]

A glance at the *i-v* characteristic in the forward region (Fig. 4.8) reveals that the current is negligibly small for v smaller than about 0,5 V. This value is usually referred to as the **cut-in voltage**.

This apparent threshold in the characteristic is simply a **consequence of the exponential relationship**.

Another consequence of this relationship is the rapid increase of i. Therefore, for a **fully conducting** diode, the voltage drop lies in a narrow range, approximately 0,6 V to 0,8 V. This gives rise to a simple model for the diode where it is assumed that a conducting diode has approximately a 0,7 V drop across it.

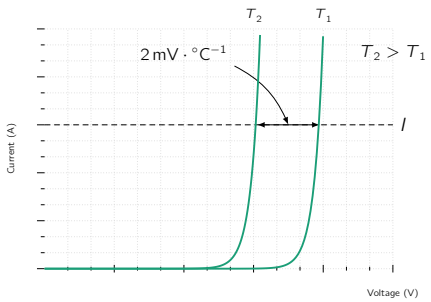


Figure 4.8 **Temperature dependence of the diode forward characteristic. At a constant current, the voltage drop decreases by approximately 2 mV for every 1 C increase in temperature.**

Diodes with different current ratings[20] will exhibit the 0,7 V drop at different currents.

> For example, a small-signal diode may be considered to have a 0,7 V drop at $i = 1$ mA, while a higher-power diode may have a 0,7 V drop at $i = 1$ A.

Given both I_S and V_T are functions of temperature, the forward *i-v* characteristic varies with temperature, as illustrated in **Fig.** 4.8. At a given constant diode current, the voltage drop across the diode decreases by approximately 2 mV for every 1 °C increase in temperature.

Thermometers 4.2 Information

The change in diode voltage with temperature has been exploited in the design of electronic thermometers.

4.2.2 Reverse Bias Region

The reverse-bias region of operation is entered when the diode voltage v is made **negative**. Eq. (**4.1**) predicts that if v is negative and a few times larger than V_T (25 mV) in magnitude, the exponential term becomes **negligibly small compared to unity**, and the diode current becomes:

$$i \approx -I_S$$

That is, the current in the reverse direction is constant and equal to I_S.

This constancy is the reason behind the term saturation current.

Real diodes exhibit reverse currents that, though quite small, are much larger than I_S.

[18] which means in a factor of 10.

[19] Using the vertical, linear axis for v and the horizontal, log axis for i, one obtains a straight line with a slope of $60\,mV \cdot decade^{1}$ of current.

[20] different areas and correspondingly different I_S.

For example, a small-signal diode whose I_S is on the order of 1×10^{-14} A to 1×10^{-15} A could show a reverse current on the order of 1 nA.

The reverse current also increases somewhat with the increase in magnitude of the reverse voltage.[21]

A large part of the reverse current is due to **leakage effects**. These leakage currents are proportional to the junction area, just as I_S is. Their dependence on temperature, however, is different from that of I_S. Therefore, whereas I_S doubles for every 5 °C rise in temperature, the corresponding rule of thumb for the temperature dependence of the reverse current is that it doubles for every 10 °C rise in temperature.

[21] Because of the very small magnitude of the current, these details are not clearly evident on the diode $i - v$ characteristic of Fig. 4.8.

4.2.3 The Breakdown Region

The 3rd distinct region of diode operation is the **breakdown region**, which can be easily identified on the diode *i-v* characteristic in Fig. 4.8. The breakdown region is entered when the magnitude of the reverse voltage exceeds a threshold value that is specific to the particular diode, called the **breakdown voltage**. This is the voltage at the "knee" of the *i-v* curve in Fig. 4.8 and is denoted V_{ZK}, where the subscript Z stands for **zener** (see Section 3.5.3) and K denotes **knee**.

As can be seen from Fig. 4.8, in the breakdown region the reverse current increases rapidly, with the associated increase in voltage drop being very small.

Diode breakdown is normally **NOT** destructive, provided the power dissipated in the diode is limited by external circuitry to a "safe" level.This safe value is normally specified on the device data sheets.

It therefore is necessary to limit the reverse current in the breakdown region to a value consistent with the permissible power dissipation.

The diode *i-v* characteristic in breakdown being almost a vertical line enables it to be used in voltage regulation applications

4.3 Modelling the Diode Forward Characteristic

Having studied the diode terminal characteristics we are now ready to consider the analysis of circuits using **forward-conducting diodes**. **Fig.** 4.9 shows such a circuit. It consists of a DC source V_{DD}, a resistor R, and a diode. We wish to analyze this circuit to determine the diode voltage V_D and current I_D. To aid in our analysis, we need to represent the diode with a **model**.

There are a variety of diode models, of which we now know two:

- the ideal-diode model, and the
- exponential model.

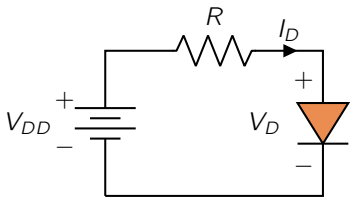


Figure 4.9
A simple circuit used to illustrate the analysis of circuits in which the diode is forward conducting.

In the following discussion we shall assess the suitability of these two models in various analysis situations. Also, we shall develop and comment on other models.

This material, besides being useful in the analysis and design of diode circuits, establishes a foundation for the modeling of transistor operation that we will study in the next three chapters.

4.3.1 The Exponential Model

The most accurate description of the diode operation in the forward region is provided by the exponential model. Unfortunately, however, its **severely non-linear** nature makes this model highly difficult to use.

To get a better understanding, let's analyze the circuit in **Fig.** 4.9 using the exponential diode model.

Assuming V_{DD} is greater than 0,5 V or so, the diode current will be much greater than I_S, and we can represent the diode *i-v* characteristic by the exponential relationship, resulting in:

$$I_D = I_S \exp\left(V_D/V_T\right) \tag{4.4}$$

The other equation that governs circuit operation is obtained by writing a *Kirchhoff* loop equation, resulting in

$$I_D = \frac{V_{DD} - V_D}{R} \tag{4.5}$$

Assuming that the diode parameter I_S is known, Eq. (**4.4**) and Eq. (**4.5**) are two equations in the two unknown quantities I_D and V_D. Two alternative ways for obtaining the solution are **graphical analysis** and **iterative analysis**.

4.3.2 Graphical Analysis Using the Exponential Model

Graphical analysis is performed by plotting the relationships of Eqs. (4.6) and (4.7) on the $i - v$ plane. The solution can then be obtained as the coordinates of the point of intersection of the two graphs. A sketch of the graphical construction is shown in Fig. 4.11. The curve represents the exponential diode equation (Eq. 4.6), and the straight line represents Eq. (4.7). Such a straight line is known as the load line, a name that will become more meaningful in later chapters. The load line intersects the diode curve at point Q, which represents the operating point of the circuit. Its coordinates give the values of Ip and Vp.

Graphical analysis aids in the visualization of circuit operation. However, the effort involved in performing such an analysis, particularly for complex circuits, is too great to be justified in practice.

4.3.3 The Need for Rapid Analysis

The iterative analysis procedure utilized in the example above is simple and yields accurate results after two or three iterations. Nevertheless, there are situations in which the effort and time required are still greater than can be justified. Specifically, if one is doing a pencil-and-paper design of a relatively complex circuit, rapid circuit analysis is a necessity.

Through quick analysis, the designer is able to evaluate various possibilities before deciding on a suitable circuit design. To speed up the analysis process, one must be content with less precise results. This, however, is seldom a problem, because the more accurate analysis can be postponed until a final or almost-final design is obtained. Accurate analysis of the almost-final design can be performed with the aid of a computer circuit-analysis program such as SPICE (see Appendix B and the website). The results of such an analysis can then be used to further refine or "fine-tune" the design.

To speed up the analysis process, we must find a simpler model for the diode forward characteristic.

4.3.4 The Constant-Voltage-Drop Model

The simplest, and arguably, the most widely used diode model is the **constant-voltage-drop** model.

> This model is based on the observation that a forward-conducting diode has a voltage drop which varies in a relatively narrow range, say, 0,6 V to 0,8 V.

The model assumes this voltage to be constant at a value, say, 0,7 V. This development is illustrated in Fig. 4.12.

The constant-voltage-drop model is the one most frequently employed in the initial phases of analysis and design. This is especially true if at these stages one does not have detailed information about

the diode characteristics, which is often the case.

Finally, note that if we employ the constant-voltage-drop model to solve the problem in Example 4.4, we obtain:

$$V_D = 0{,}7\,\text{V} \qquad \text{and} \qquad I_D = \frac{V_{DD} - 0.7}{R} = \frac{5 - 0.7}{1} = 4{,}3\,\text{mA}$$

which are not very different from the values obtained before with the more elaborate exponential model.

4.3.5 The Ideal-Diode Model

In applications involving voltages significantly greater than the diode voltage drop,[22] we may *neglect the diode voltage drop* altogether while calculating the diode current. The result is the ideal-diode model, which we studied in Section 4.1.

[22] In this case these values range between 0,6 V to 0,8 V.

> the greatest utility of the **ideal-diode model** is in determining **which diodes are on and which are off** in a multi-diode circuit.

4.3.6 The Small-Signal Model

Consider the situation in **Fig.** 4.10a, where a DC voltage (V_{DD}) establishes a DC current (I_D) through the series combination of a resistance R and a diode D. The resulting diode voltage is denoted V_D. As mentioned previously, values of I_D and V_D can both be obtained by solving the circuit using the diode exponential characteristic or, much more quickly, approximate values can be found using the diode constant-voltage-drop model.

Next, consider the situation of V_{DD} undergoing a small change (ΔV_{DD}), as shown in **Fig.** 4.10b. As indicated, the current I_D changes by an increment ΔI_D, and the diode voltage V_D changes by an increment ΔV_D.

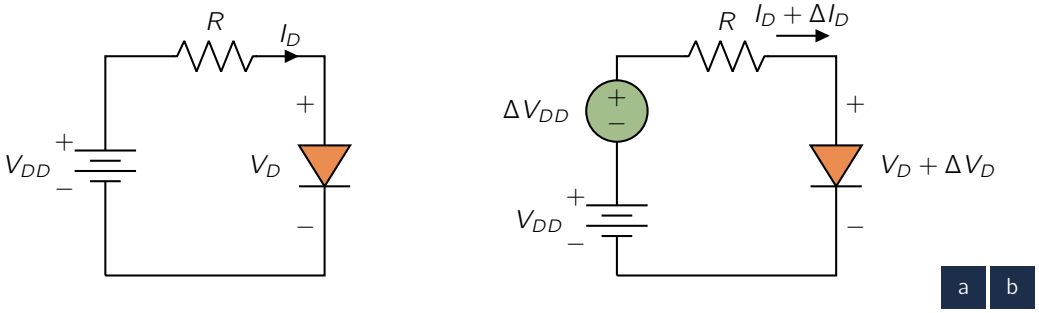


Figure 4.10
(a) A simple diode circuit (b) the situation when V_{DD} changes by ΔV_{DD}.

We wish to find a quick way to determine the values of these incremental changes. Toward that end, we develop a "small-signal" model for the diode.

Here the word **signal** emphasizes that in general, ΔV_{DD} can be a time-varying quantity. The qualifier "small" indicates that this diode model applies only when ΔV_D is kept sufficiently small, with **sufficiently** to be quantified shortly.

To develop the diode small-signal model, have a look at Fig. 4.14. We express the voltage across the diode as the sum of the DC voltage V_D and the time-varying signal $v_d(t)$,

$$v_D(t) = V_D + v_d(t) \tag{4.6}$$

Correspondingly, the total instantaneous diode current ($i_D(t)$) will then be;

$$i_D(t) = I_S e^{\,v_D \,/\, v_T}$$

Substituting for v_D from Eq. (**4.6**) gives:

$$i_D(t) = I_s e^{(V_D+v_d)/V_T} \qquad \text{rewritten as} \qquad i_D(t) = I_s e^{V_D/V_T} e^{v_d/V_T} \tag{4.7}$$

In the absence of the signal $v_d(t)$, the diode voltage is equal to V_D, and the diode current is I_D, given by

$$I_D = I_S e^{V_D/V_T}$$

Thus, $i_D(t)$ in Eq. (**4.7**) can be expressed as

$$i_p(t) = I_p e^{v_d/V_T} \tag{4.8}$$

Now if the amplitude of the signal $v_d(t)$ is kept sufficiently small such that

$$\frac{v_d}{V_T} \ll 1 \tag{4.9}$$

then we may expand the exponential of Eq. (**4.8**) in a series and truncate the series after the first two (**2**) terms to obtain the approximate expression:

$$i_D(t) \simeq I_D \left(1 + \frac{v_d}{V_x}\right) \tag{4.10}$$

This is the **small-signal approximation**. It is valid for signals whose amplitudes are smaller than about 5 mV.[23] From Eq. (**4.10**) we have:

$$i_D(t) = I_D + \frac{I_D}{V_T} v_d$$

Thus, superimposed on the DC current I_D, we have a signal current component directly proportional to the signal voltage v_d. That is,

$$i_p = I_p + i_d \qquad \text{where} \qquad i_d = \frac{I_D}{V_r} v_d$$

The quantity relating the signal current i_d to the signal voltage v_d has the dimensions of conductance (℧) and is called the **diode small-signal conductance**. The inverse of this parameter is the **diode small-signal resistance**, or **incremental resistance**, r_d,

$$r_d = \frac{V_T}{I_D}$$

[23] Have a look at Eq. (**4.9**), and recall that $V_T = 25\,\text{mV}$.

Note that the value of r_d is inversely proportional to the bias current I_p.

Additional insight into the small-signal approximation and the small-signal diode model can be obtained by considering again the graphical construction in Fig. 4.14. Here the diode is seen to be operating at a dc bias point Q characterized by the dc voltage Vp and the corresponding dc current I_p. Superimposed on V_p we have a signal $v_d(t)$, assumed (arbitrarily) to have a triangular waveform.

It is easy to see that using the small-signal approximation is equivalent to assuming that the signal amplitude is sufficiently small such that the excursion along the $i - v$ curve is limited to a short almost-linear segment. The slope of this segment, which is equal to the slope of the tangent to the $i - v$ curve at the operating point Q, is equal to the small-signal conductance. The reader is encouraged to prove that the slope of the **i-v** curve at $i = I_D$ is equal to I_D/V_T, which is $1/r_d$; that is,

$$r_d = 1/\left[\frac{\partial i_D}{\partial v_p}\right]_{i=1}$$

From the preceding we conclude that superimposed on the quantities V_D and I_D that define the dc bias point, or **quiescent point**, of the diode will be the small-signal quantities $v_d(t)$ and $i_a(t)$, which are related by the diode small-signal resistance r_a evaluated at the bias point (Eq. 4.18). Thus the small-signal analysis can be performed separately from the dc bias analysis, a great convenience that results from the linearization of the diode characteristics inherent in the small-signal approximation. Specifically, after the dc analysis is performed, the small-signal equivalent circuit is obtained by eliminating all dc sources (i.e., short-circuiting dc voltage sources and open-circuiting dc current sources) and replacing the diode by its small-signal resistance. Thus, for the circuit in Fig. 4.13(b), the dc analysis is obtained by using the circuit in Fig. 4.13(a), while the incremental quantities ΔI_p and ΔV_p can be determined by using the small-signal equivalent circuit shown in Fig. 4.15. The following example should further illustrate the application of the small-signal model.

From the above we see that for a diode circuit that involves both dc and signal quantities, a small-signal equivalent circuit can be obtained by eliminating the dc sources and replacing each diode with its small-signal resistance r_d. Such a circuit is linear and can be solved using linear circuit analysis. Finally, we note that while r_d models the small-signal operation of the diode at low frequencies, its dynamic operation is modeled by the capacitances C_j and C_d, which we

studied in Section 3.6 and which also are small-signal parameters. A complete model of the diode includes C_i and C_j in parallel with r_j.

4.3.7 Use of the Diode Forward Drop in Voltage Regulation

A further application of the diode small-signal model is found in a popular diode application, namely, the use of diodes to create a regulated voltage. A voltage regulator is a circuit whose purpose is to provide a constant dc voltage between its output terminals. The output voltage is required to remain as constant as possible in spite of (a) changes in the load current drawn from the regulator output terminal and (b) changes in the dc power-supply voltage that feeds the regulator circuit.

Since the forward-voltage drop of the diode remains almost constant at approximately 0.7 V while the current through it varies by relatively large amounts, a forward-biased diode can make a simple voltage regulator. For instance, we have seen in Example 4.5 that while the 10-V dc supply voltage had a ripple of 2 V peak-to-peak (a $\pm 10\%$) variation), the corresponding ripple in the diode voltage was only about ± 2.7 mV (a $\pm 0.4\%$) variation). Regulated voltages greater than 0.7 V can be obtained by connecting a number of diodes in series. For example, the use of three forward-biased diodes in series provides a voltage of about 2 V. One such circuit is investigated in the following example, which utilizes the diode small-signal model to quantify the efficacy of the voltage regulator that is realized.

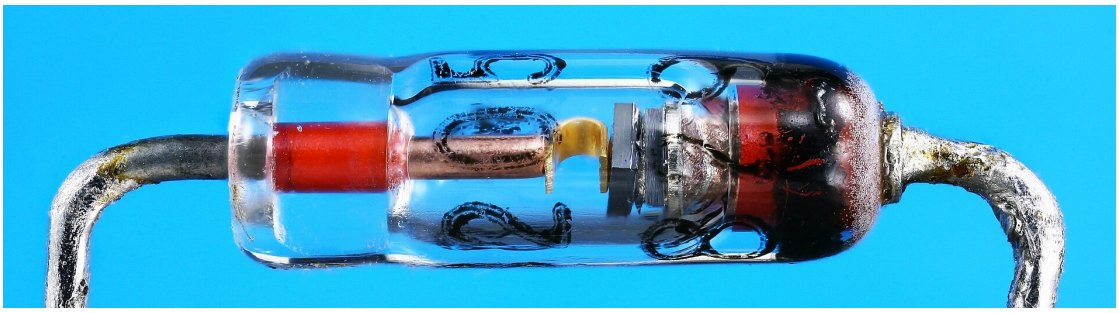


Figure 4.11
A clouse look at the 1N829 Zener Diode 6,2 V 500mW ±5 % Through Hole DO-35 (DO-204AH).

4.4 Zener Diodes

A Zener diode is a heavily doped semiconductor device designed to safely operate in **reverse breakdown**. Unlike standard diodes, when its specific breakdown threshold is reached, it conducts current in reverse while maintaining a constant voltage across its terminals which makes it ideal for voltage regulation and circuit protection. In this section we will have a look at how these diodes function and then study their usage in circuits.

4.4.1 Operating in the Reverse Breakdown Region

The very steep *i-v* curve which the diode exhibits in the breakdown region (Fig. 4.8) and the almost-constant voltage drop that this indicates suggest diodes operating in the breakdown region can be used in the design of voltage regulators.

From previous experience, we know voltage regulators are circuits which provide a constant DC output voltage in the face of changes in their load current and in the system power-supply voltage. This in fact turns out to be an important application of diodes operating in the reverse breakdown region, and special diodes are manufactured to operate specifically in the breakdown region.

> Such diodes are called breakdown diodes or, more commonly, **zener diodes**.

Figure 4.18 shows the circuit symbol of the zener diode. In normal applications of zener diodes, current flows into the cathode, and the cathode is positive with respect to the anode. Thus I_z and V_z in Fig. 4.18 have positive values.

4.4.2 Specifying and Modeling the Zener Diode

Figure 4.19 shows details of the diode *i-v* characteristic in the breakdown region. We observe that for currents greater than the knee current I_{ZK},[24] the *i-v* characteristic is almost a straight line. The manufacturer usually specifies the voltage across the zener diode V_z at a specified test current, I_{ZT}. We have indicated these parameters in Fig. 4.19 as the coordinates of the point labeled Q.

[24] specified on the data sheet of the zener diode.

Therefore a 6,8 V zener diode will exhibit a 6,8 V drop at a specified test current of, say, 10 mA. As the current through the zener deviates from I_{ZT}, the voltage across it will change, though only slightly. Figure 4.19 shows that corresponding to current change ΔI the zener voltage changes by ΔV, which is related to ΔI by:

$$\Delta V = r_z \Delta I$$

where r_z is the inverse of the slope of the **almost-linear** *i-v* curve at point Q. Resistance r_z is the **incremental resistance** of the zener diode at operating point Q. It is also known as the **dynamic resistance** of the zener, and its value is specified on the device data sheet.

Typically, r_z is in the range of a few ohms to a few tens of ohms. Obviously, the lower the value of r_z is, the more constant the zener voltage remains as its current varies, and therefore the more ideal its performance becomes in the design of voltage regulators. In this regard, we observe from Fig. 4.19 that while r_z remains low and almost constant over a wide range of current, its value increases considerably in the vicinity of the knee.

> Therefore, as a general design guideline, one should avoid operating the zener in this low-current region.

Zener diodes are fabricated with voltages V_z in the range of a few volts to a few hundred volts. In addition to specifying V_z (at a particular current I_{ZT}), r_z, and I_{ZK}, the manufacturer also specifies the maximum power that the device can safely dissipate. Thus a 0.5-W, 6.8-V zener diode can operate safely at currents up to a maximum of about 70 mA.

The almost-linear *i-v* characteristic of the zener diode suggests the device can be modeled as indicated in Fig. 4.20. Here V_{Z0} denotes the point at which the straight line of slope $1/r_z$ intersects the voltage axis (refer to Fig. 4.19). Although V_{z0} is shown in Fig. 4.19 to be slightly different from the knee voltage V_{ZK}, in practice their values are almost equal.

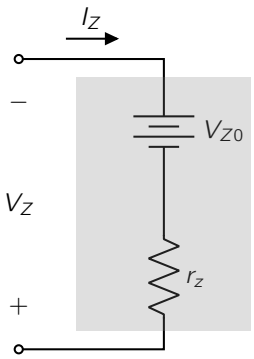


Figure 4.12
Model for the zener diode

The equivalent circuit model of Fig. 4.20 can be analytically described by

$$V_Z = V_{Z0} + r_z I_Z$$

and it applies for $I_Z > I_{ZK}$ and, obviously, $V_Z > V_{Z0}$.

4.4.3 Temperature Effects

The dependence of the zener voltage (V_z) on temperature is specified in terms of its **temperature** coefficient TC, or **temco** as it is commonly known, which is usually expressed in $\text{mV} \cdot {}^{\circ}\text{C}^{-1}$.

The value of TC depends on the zener voltage, and for a given diode the TC varies with the operating current. Zener diodes whose V_Z are lower than about 5 V exhibit a **negative** TC, whereas zeners with higher voltages exhibit a **positive** TC.

The TC of a zener diode with a V_Z of about 5 V can be made zero by operating the diode at a specified current. Another commonly used technique for obtaining a reference voltage with low temperature coefficient is to connect a zener diode with a positive temperature coefficient of about $2\,\text{mV} \cdot {}^{\circ}\text{C}^{-1}$ in series with a forward-conducting diode. Since the forward-conducting diode has a voltage drop of $\approx 0{,}7\,\text{V}$ and a TC of about - $2\,\text{mV} \cdot {}^{\circ}\text{C}^{-1}$, the series combination will provide a voltage of $(V_Z + 0.7)$ with a TC of about zero.

The Current Status of Zener Diodes 4.3 Information

Though simple and useful, zener diodes have lost a great deal of their popularity in recent years. They have been virtually replaced in voltage-regulator design by specially designed ICs which perform the voltage-regulation function much more effectively and with greater flexibility than zener diodes.

4.5 Limiting and Clamping Circuits

In this section, we will look at the additional non-linear circuit applications of diodes.

4.5.1 Limiter Circuits

Figure 4.30 shows the general transfer characteristic of a limiter circuit. As indicated, for inputs in a certain range, $L_N \leqslant v_1 \leqslant L_+/K$, the limiter acts as a linear circuit, providing an output proportional to the input, $v_O = Kv_I$.

> Although in general K can be greater than 1, the circuits discussed in this section have $K \leqslant 1$ and are known as **passive limiters**.

If vI exceeds the upper threshold $(L+/K)$, the output voltage is **limited** or clamped to the upper limiting level L_+. On the other hand, if v_i is reduced below the lower limiting threshold $(L_\perp/K)$, the output voltage v_0 is limited to the lower limiting level L_-.

The general transfer characteristic of Fig. 4.30 describes a **double limiter**-that is, a limiter that works on both the positive and negative peaks of an input waveform. Single limiters, of course, exist. Finally, note that if an input waveform such as that shown in Fig. 4.31 is fed to a double limiter, its two peaks will be **clipped off**. Limiters therefore are sometimes referred to as clippers. The limiter whose characteristics are depicted in Fig. 4.30 is described as a hard limiter. **Soft limiting** is characterized by smoother transitions between the linear region and the saturation regions and a slope greater than zero in the saturation regions, as illustrated in Fig. 4.32. Depending on the application, either hard or soft limiting may be preferred. Limiters find application in a variety of signal-processing systems. One of their simplest applications is in limiting the voltage between the two input terminals of an op amp to a value lower than the breakdown voltage of the transistors that make up the input stage of the op-amp circuit.

Diodes can be combined with resistors to provide simple realizations of the limiter function. A number of examples are depicted in Fig. 4.33. In each part of the figure both the circuit and its transfer characteristic are given. The transfer characteristics are obtained using the constant-voltage-drop (V_p = 0.7 V) diode model but assuming a smooth transition between the linear and saturation regions of the transfer characteristic. The circuit in Fig. 4.33(a) is that of the half-wave rectifier except that here the output is taken across the diode. For $v_t < 0.5$ V, the diode is cut off, no current flows, and the voltage drop across R is zero; thus $v_0 = v_t$. As v_t exceeds 0.5 V, the diode turns on, eventually limiting vo to one diode drop (0.7 V). The circuit of Fig. 4.33(b) is similar to that in Fig. 4.33(a) except that the diode is reversed. Double limiting can be implemented by placing two diodes of opposite polarity in paral- lel, as shown in Fig. 4.33(c). Here the linear region of the characteristic is obtained for $-0.5V \leqslant v_1 \leqslant 0.5$ V. For this range of v_1, both diodes are off and $v_0 = v_1$. As v_1 exceeds 0.5 V, D_1 turns on and eventually limits v_0 to +0.7 V. Similarly, as v_t

goes more negative than -0.5 V, D_2 turns on and eventually limits v_0 to -0.7 V. The thresholds and saturation levels of diode limiters can be controlled by using strings of diodes and/or by connecting a dc voltage in series with the diode(s). The latter idea is illustrated in Fig. 4.33(d). Finally, rather than strings of diodes, we may use two zener diodes in series, as shown in Fig. 4.33(e). In this circuit, limiting occurs in the positive direction at a voltage of V_{Z2} + 0.7, where 0.7 V represents the voltage drop across zener diode Z_1 when conducting in the **forward** direction. For negative inputs, Z_1 acts as a zener, while Z_2 conducts in the forward direction. It should be mentioned that pairs of zener diodes connected in series are available commercially for applications of this type under the name **double-anode** zener. More flexible limiter circuits are possible if op amps are combined with diodes and resistors. Examples of such circuits are discussed in Chapter 18.

4.5.2 The Clamped Capacitor or DC Restorer

If in the basic peak-rectifier circuit, the output is taken across the diode rather than across the capacitor, an interesting circuit with important applications results.

The circuit, called a **DC restorer**, is shown in Fig. 4.34 fed with a square wave where the SPICE simulation code is presented below.

spice

```
1
2  VIN 0   1   0 AC 0 PULSE -6 10 0 1.e-8 1.e-8 1e-3 2e-3
3  C1  2   1   1e-6
4  D1  0   2   DCLAMP
5
6  .MODEL DCLAMP D (IS = 1e-14)
7
8  .TRAN 1e-4 8e-3 0 1e-5
9
```

Because of the polarity in which the diode is connected, the capacitor will charge to a voltage v_c with the polarity indicated in Fig. 4.34 and equal to the magnitude of the most negative peak of the input signal. Subsequently, the diode turns off and the capacitor retains its voltage indefinitely. If, for instance, the input square wave has the arbitrary levels -6 V and +4 V, then v_c will be equal to 6 V. Now, since the output voltage vo is given by $v_o = v_I + v_C$ it follows that the output waveform will be identical to that of the input, except that it is shifted upward by v_c volts. In our example the output will thus be a square wave with levels of 0 V and +10 V. Another way of visualizing the operation of the circuit in Fig. 4.34 is to note that because the diode is connected across the output with the polarity shown, it prevents the output voltage from going below 0 V (by conducting and charging up the capacitor, thus causing the output to rise to 0 V), but this connection will not constrain the positive excursion of v_0. The output waveform will therefore have its lowest peak **clamped** to 0 V, which is why the circuit is called a clamped capacitor. It should be obvious that reversing the diode polarity will provide an output waveform whose highest peak is clamped to 0 V. In either case, the output waveform will have a finite average value or dc component. This dc component is entirely unrelated to the average value of the input waveform. As an application, consider a

pulse signal being transmitted through a capacitively coupled or ac-coupled system. The capacitive coupling will cause the pulse train to lose whatever dc component it originally had. Feeding the resulting pulse waveform to a clamping circuit provides it with a well-determined dc component, a process known as **dc** restoration. This is why the circuit is also called a **dc** restorer. Restoring dc is useful because the dc component or average value of a pulse waveform is an effective measure of its **duty cycle**.⁵ The duty cycle of a pulse waveform can be modulated (in a process called **pulsewidth modulation**) and made to carry information. In such a system, detection or demodulation could be achieved simply by feeding the received pulse waveform to a dc restorer and then using a simple RC low-pass filter to separate the average of the output waveform from the superimposed pulses. When a load resistance R is connected across the diode in a clamping circuit, as shown in Fig. 4.35, the situation changes significantly. While the output is above ground, a current must flow in R. Since at this time the diode is off, this current obviously comes from the capacitor, thus causing the capacitor to discharge and the output voltage to fall. This is shown in Fig. 4.35 for a square-wave input. During the interval t_0 to t_1, the output voltage falls exponentially with time constant CR. At t_1 the input decreases by V_a volts, and the output attempts to follow. This causes the diode to conduct heavily and to quickly charge the capacitor. At the end of the interval t_1 to t_2, the output voltage would normally be a few tenths of a volt negative (e.g., -0.5 V). Then, as the input rises by V_a volts (at t_2), the output follows, and the cycle repeats itself. In the steady state the charge lost by the capacitor during the interval t_0 to t_1 is recovered during the interval t_1 to t_2. This charge equilibrium enables us to calculate the average diode current as well as the details of the output waveform.

4.5.3 The Voltage Doubler

Fig. 4.13 shows a circuit composed of two (**2**) sections in cascade:

a clamped capacitor formed by C_1 and D_1, and a peak rectifier formed by D_2 and C_2.

When excited by a sinusoid of amplitude V_p, the clamping section provides the voltage waveform v_{D1} shown, **assuming ideal diodes**, in Fig. 4.36(b).

> While the positive peaks are clamped to 0 V, the negative peak reaches $-2V_p$.

In response to this waveform, the peak-detector section provides across capacitor C_2 a DC voltage equal to the negative peak of v_{D1}, that is, $-2V_p$. Because the output voltage is double the input peak, the circuit is known as a **voltage doubler**.

> The technique can be extended to provide output dc voltages that are higher multiples of V_p.

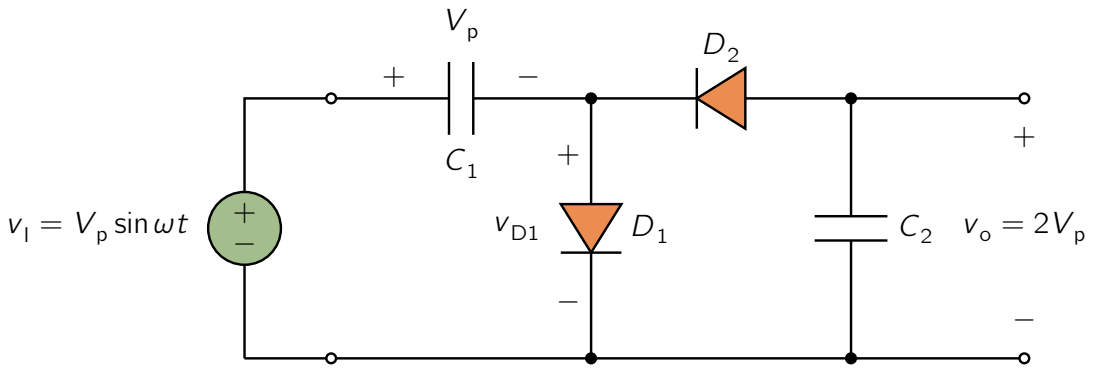


Figure 4.13: The voltage doubler.

4.6 Special Diode Types

In this section, we discuss briefly some important special types of diodes which are relevant in industry and in research.

4.6.1 The Schottky-Barrier Diode (SBD)



Figure 4.14
Diagram used to describe the schottky barrier diode.

The Schottky-barrier diode (SBD) is formed by bringing metal into contact with a moderately doped *n*-type semiconductor material. The resulting metal-semiconductor junction behaves like a diode, conducting current in one direction[25] and acting as an open circuit in the other, and is known as the **Schottky-barrier diode** or simply the **Schottky diode**.

[25] from the metal anode to the semiconductor cathode

In fact, the *i-v* characteristic of the SBD is remarkably similar to that of a *pn*-junction diode, with two (2) important exceptions:

1. In the SBD, current is conducted by majority carriers (electrons). Thus the SBD does not exhibit the minority-carrier charge-storage effects found in forward-biased pn junctions. As a result, Schottky diodes can be switched from on to off, and vice versa, much faster than is possible with **pn**-junction diodes.

2. The forward voltage drop of a conducting SBD is lower than that of a *pn*-junction diode. For example, an SBD made of silicon exhibits a forward voltage drop of 0.3 V to 0.5 V, compared to the 0.6 V to 0.8 V found in silicon pn-junction diodes. SBDs can also be made of gallium arsenide (GaAs) and, in fact, play an important role in the design of GaAs circuits.⁶ Gallium-arsenide SBDs exhibit forward voltage drops of about 0.7 V.

Apart from GaAs circuits, Schottky diodes find application in the design of a special form of bipolar-transistor logic circuits, known as Schottky-TTL, where TTL stands for transistor-transistor logic.

Use of Metal Contacts 4.4 Information

Before leaving the subject of Schottky-barrier diodes, it is important to note that not every metal-semiconductor contact is a diode. In fact, metal is commonly deposited on the semiconductor surface in order to make terminals for the semiconductor devices and to connect different devices in an integrated-circuit chip. Such metal-semiconductor contacts are known as **ohmic contacts** to distinguish them from the rectifying contacts that result in SBDs. Ohmic contacts are usually made by depositing metal on very heavily doped (and thus low-resistivity) semiconductor regions. (Recall that SBDs use moderately doped material.)

4.6.2 Varactors

In the **Semiconductor** chapter, we have learned that reverse-biased *pn*-junctions exhibit a charge-storage effect which is modeled with the depletion-layer or junction capacitance C_j. As Eq. (**3.13**) indicates, C_j is a function of the reverse-bias voltage V_R. This dependence turns out to be useful in a number of applications, such as the automatic tuning of radio receivers. Special diodes are therefore fabricated to be used as voltage-variable capacitors known as **varactors**.[26]

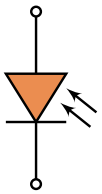


Figure 4.15 Diagram used to describe the varactor, or varicap.

[26] This is also known as **varicap** in some markets.

These devices are optimized to make the capacitance a strong function of voltage by arranging that the grading coefficient m is 3 or 4.

4.6.3 Photodiodes



If a reverse-biased *pn*-junction is illuminated-that is, exposed to incident light-the photons impacting the junction cause covalent bonds to break, and thus electron-hole pairs are generated in the depletion layer. The electric field in the depletion region then sweeps the liberated electrons to the n side and the holes to the p side, giving rise to a reverse current across the junction. This current, known as photocurrent, is proportional to the intensity of the incident light. Such a diode, called a photodiode, can be used to convert light signals into electrical signals.

Figure 4.16 Diagram used to describe the photodiode.

Photodiodes are usually fabricated using a compound semiconductor[27] such as gallium arsenide (GaAs). The photodiode is an important component of a growing family of circuits known as **optoelectronics** or **photonics**. As the name implies, such circuits utilise an optimum combination of electronics and optics for signal processing, storage, and transmission. Usually, electronics is the preferred means for signal processing, whereas optics is most suited for transmission and storage.

> Examples include fiber-optic transmission of telephone and television signals and the use of optical storage in CD-ROM computer discs [2].

Optical transmission provides very wide bandwidths and low signal attenuation. Optical storage allows vast amounts of data to be stored reliably in a small space.

> Without reverse bias, the illuminated photodiode functions as a solar cell. Usually fabricated from low-cost silicon, a solar cell converts light to electrical energy.

[27] An elemental semiconductor, such as silicon, uses an element from column IV of the periodic table, whereas a compound semiconductor uses a combination of elements from columns III and V or II and VI.

For example, GaAs is formed of gallium (column III) and arsenic (column V) and is thus known as a III-V compound.

4.6.4 Light Emitting Diodes

The light-emitting diode (LED) performs the inverse of the function of the photodiode;

> it converts a forward current into light.

If were to remember, in a forward-biased *pn*-junction, minority carriers are injected across the junction and diffuse into the *p* and *n* regions. The diffusing minority carriers then recombine with the majority carriers. Such recombination can be made to give rise to light emission. This can be done by fabricating the *pn*-junction using a semiconductor of the type known as **direct-bandgap** materials. GaAs belongs to this group and can thus be used to fabricate light-emitting diodes.

Direct and Indirect Band Gap Semiconductors 4.5 Information

The band gap represents the **minimum energy difference** between the top of the valence band and the bottom of the conduction band, However, the top of the valence band and the bottom of the conduction band are **NOT** generally at the same value of the electron momentum.

1. In a direct band gap semiconductor, the top of the valence band and the bottom of the conduction band occur at the same value of momentum, as in the schematic below.

2. In an indirect band gap semiconductor, the maximum energy of the valence band occurs at a different value of momentum to the minimum in the conduction band energy.

The difference between the two is most important in optical devices [3].

The light emitted by an LED is proportional to the number of recombination which take place, which in turn is proportional to the forward current in the diode.

LEDs are very popular devices. They find application in the design of numerous types of displays, including the displays of laboratory instruments such as digital voltmeters. They can be made to produce light in a variety of colors. Furthermore, LEDs can be designed so as to produce coherent light with a very narrow bandwidth. The resulting device is a **laser diode**. Laser diodes find application in optical communication systems and in DVD players, among other things.

Combining an LED with a photodiode in the same package results in a device known as an **optoisolator**. The LED converts an electrical signal applied to the optoisolator into light, which the photodiode detects and converts back to an electrical signal at the output of the optoisolator. Use of the optoisolator provides complete electrical isolation between the electrical circuit that is connected to the isolator's input and the circuit that is connected to its output.

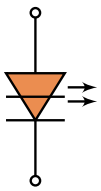


Figure 4.17
Diagram used to describe the laser diode.

Such isolation can be useful in reducing the effect of electrical interference on signal transmission within a system, and thus optoisolators are frequently employed in the design of digital systems. They can also be used in the design of medical instruments to reduce the risk of electrical shock to patients.

the optical coupling between an LED and a photodiode need **NOT** be accomplished inside a small package. Indeed, it can be implemented over a long distance using an optical fiber, as is done in fiber-optic communication links.

From Indication to Illumination 4.6 Information

Light-emitting diodes (LEDs), which once served only as low-powered status indicators, are now lighting our way! Increasingly, automotive lighting uses LEDs; increasingly, too, LED bulbs of higher and higher power are replacing both incandescent and fluorescent lighting in homes and offices. Incandescent bulbs are only 5% efficient in the conversion of electricity into light-the other 95% is dissipated as heat. The light conversion efficiency of LEDs, however, is 60%. Moreover, LEDs last 25 times longer (25,000 hours) than incandescent bulbs and 3 times longer than fluorescents.

References

1. Dorf, R. C. *The electrical engineering handbook* (CRC press, 1997).
2. Kieschnick, K, Zimmermann, H, Seegebrecht, P & Pleß, H. *Integrated photodiodes for DVD and CD-ROM applications* in *30th European Solid-State Device Research Conference* (2000), 252–255.
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Chapter 5

MOS Field Effect Transistors

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5.4 Body Effect and Other Topics 147

Having studied the simple *pn*-junction diode, which is simplest two-terminal semiconductor device, we now turn our attention to **three-terminal semiconductor devices**.

Three-terminal devices are far more useful compared to its two-terminal counterparts as they can be used in a multitude of applications, ranging from **signal amplification** to **digital logic and memory**. The basic principle involved is the use of the voltage between two (2) terminals to control the current flowing in the third terminal. In this way a three-terminal device can be used to realise a controlled source.[1] Also, in the extreme, the control signal can be used to cause the current in the third terminal to change from zero to a large value, thus allowing the device to act as a switch.[2]

[1] which as we have is the basis for amplifier design.

[2] the switch is the basis for the realization of the **logic inverter**, the basic element of digital circuits.

There are two (2) major types of three-terminal semiconductor devices:

the Metal-Oxide Semiconductor Field-Effect Transistor (MOSFET), and the BJT.

> While each of the two (2) transistor types offers unique features and areas of application, the MOSFET has become by far the most widely used electronic device, especially in the design of IC.

Compared to BJTs, MOSFETs can be made quite small,[3] and their manufacturing process is *relatively* simple. In addition, their operation requires comparatively little power. Furthermore, circuit designers have found clever ways to implement digital and analog functions utilizing MOSFETs almost exclusively.[4] All of these properties have made it possible to pack large numbers of MOSFETs, as many as **4 billion**, on a single IC chip to implement very sophisticated, Very Large System

[3] requiring a small area on the silicon IC chip.

[4] i.e., with very few or no resistors.

Integration (BJT) digital circuits such as those for memory and microprocessors.

Analog circuits such as amplifiers and filters can also be implemented in Metal-Oxide Semiconductor (MOS) technology, albeit in smaller, less-dense chips. Also, both analog and digital functions are increasingly being implemented on the same IC chip, in what is known as **mixed-signal design**.[5]

[5] These circuits are mostly used on cell phones, telecommunications, portable electronics, and automobiles with electronics and digital sensors.

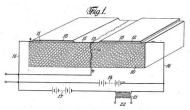
Chapter Structure The objective of studying MOSFETs is to develop a good understanding of the circuit element:

its physical structure and operation, terminal characteristics, and DC circuit applications.

Although discrete MOS transistors exist, and the material studied in this chapter will enable us to design discrete MOS circuits, our study of the MOSFET is strongly influenced by its usage in IC design.

The First Field-Effect Devices 5.1 Information

In 1925 a patent[6] for solid-state electric-field-controlled conductor was filed in Canada by *Julius E. Lilienfeld*, a physicist at the University of Leipzig, Germany. Other patent refinements followed in the United States in 1926 and 1928. Regrettably, no research papers were published. Consequently, in 1934 *Oskar Heil*, a German physicist working at the University of Cambridge, U.K., filed a patent on a similar idea. But all these early concepts of electric-field control of a semiconducting path languished because suitable technology was not available.



[6] an image from the patent showcasing its behaviour.

The invention of the BJT in 1947 at Bell Telephone Laboratories resulted in the speedy development of bipolar devices, a circumstance that further delayed the development of field-effect transistors. Although the field-effect device was described in a paper by *William Shockley* in 1952, it was not until 1960 that a patent on an insulated-gate field-effect device, the MOSFET, was filed by *Dawon Kahng* and *Martin Atalla*, also at Bell Labs.

Clearly, the idea of field-effect control for amplification and switching has changed the world. With IC chips today containing billions of MOS devices, it dominates the electronic world.

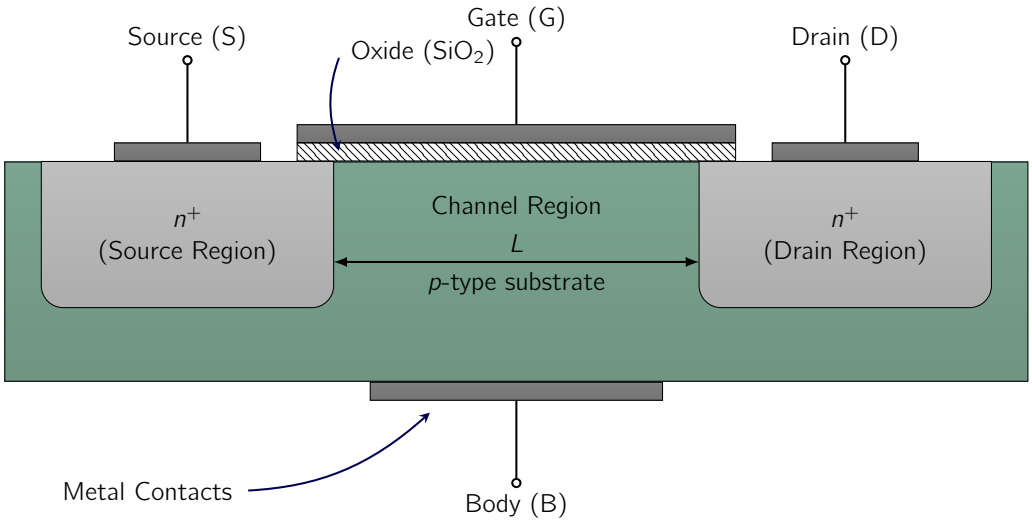


Figure 5.1
The physical structure of the enhancement-type NMOS transistor in cross-sectional view with a depth of W. As an example the value are typically $L = 0{,}03\,\mu\cdot$m to $1\,\mu\cdot$m, $W = 0{,}05\,\mu\cdot$m to $100\,\mu\cdot$m, and the thickness of the oxide layer (t_{ox}) is in the range of 1 to 10 nm.

5.1 Device Structure and Physical Operation

The enhancement-type MOSFET is **the most widely used field-effect transistor**.

> Except for the last section of this chapter, we will focus only on the enhancement-type MOSFET.

We will start in this section by learning about its structure and physical operation. This will lead to the *i-v* characteristics of the device, studied in the later section.

5.1.1 Device Structure

Fig. 5.1 shows the physical structure of the *n*-channel enhancement-type MOSFET.[7] The transistor is fabricated on a *p*-type substrate, which is a single-crystal silicon wafer that provides physical support for the device.[8] Two heavily doped *n*-type regions, indicated in **Fig.** 5.1 as the n^+ source and the n^+ drain regions,[9] are created in the substrate. A thin layer of silicon dioxide (SiO_2) of thickness t_{ox},[10] which is an excellent electrical insulator, is grown on the surface of the substrate, covering the area between the source (**S**) and drain (**D**) regions. Metal is deposited on top of the oxide layer to form the **gate electrode** of the device. Metal contacts are also made to the source region, the drain region, and the substrate, also known as the body.[11]

Therefore four (**4**) terminals are brought out:

1. the gate terminal (**G**),
2. the source terminal (**S**),

[7] The meaning of the names "enhancement" and "*n*-channel" will become apparent shortly.

[8] and for the entire circuit in the case of an IC.

[9] The notation *n* indicates heavily doped *n*-type silicon. Conversely, *n* is used to denote lightly doped.

[10] this value is typically 1 nm to 10 nm.

[11] in actual ICs, contact to the body is made at a location on the top of the device.

3. the drain terminal (**D**), and
4. the substrate or body terminal (**D**).

At this point it should be clear the name of the device MOSFET is derived from its **physical structure**. The name, however, has become a general one and is used also for Field-Effect Transistor (FET)s that do **NOT** use metal for the gate electrode. In fact, most modern MOSFETs are fabricated using a process known as **silicon-gate technology**, in which a certain type of silicon,[12] called *polysilicon*, is used to form the gate electrode.

[12] Polysilicon (polycrystalline silicon) is an **ultra-pure** (often 99.9999999% pure), high-tech material made of many tiny silicon crystals. It is the main raw material used to build solar panels and computer microchips.

Our description of MOSFET operation and characteristics applies **irrespective of the type of gate electrode**.

Another name for the MOSFET is the **insulated-gate FET** or **IGFET**. This name also arises from the physical structure of the device, emphasizing the fact the gate electrode is electrically insulated from the device body.[13] It is this insulation which causes the current in the gate terminal to be extremely small with values on the order of 10^{-15} A.

[13] This is achieved by the use of the oxide layer.

Looking at it closely, the substrate forms *pn* junctions with the **source** and **drain** regions. In normal operation these *pn* junctions are kept reverse biased at all times. Since, as we shall see shortly, the drain will always be at a positive voltage relative to the source, the two (2) *pn* junctions can be effectively cut off by simply connecting the substrate terminal to the source terminal. We shall assume this to be the case in the following description of MOSFET operation.

Therefore, here, the substrate will be considered as having **no effect** on device operation, and the MOSFET will be treated as a three (3)-terminal device, with the terminals being

the gate (**G**), the source (**S**), and the drain (**D**).

The Body Diode — 5.2 Information

Within a MOSFET, an inherent structure known as the **body diode** exists. This diode forms between the source and the body (or substrate) of the transistor. While it is **NOT** the primary path for current flow in normal operation, the body diode can become significant under certain conditions, particularly when the MOSFET is off or switching states.

It will be shown that a voltage applied to the gate controls current flow between source and drain. This current will flow in the longitudinal direction from drain to source in the region labeled **channel region**.

This region has a length (L) and a width (W), two (2) important parameters of the MOSFET.

Typically, L is in the range of 0.03 μm to 1 μm, and W is in the range of 0.05 μm to 100 μm. A last point worth mentioning is that the MOSFET is a **symmetrical device** where its source and drain can be interchanged with no change in device characteristics.

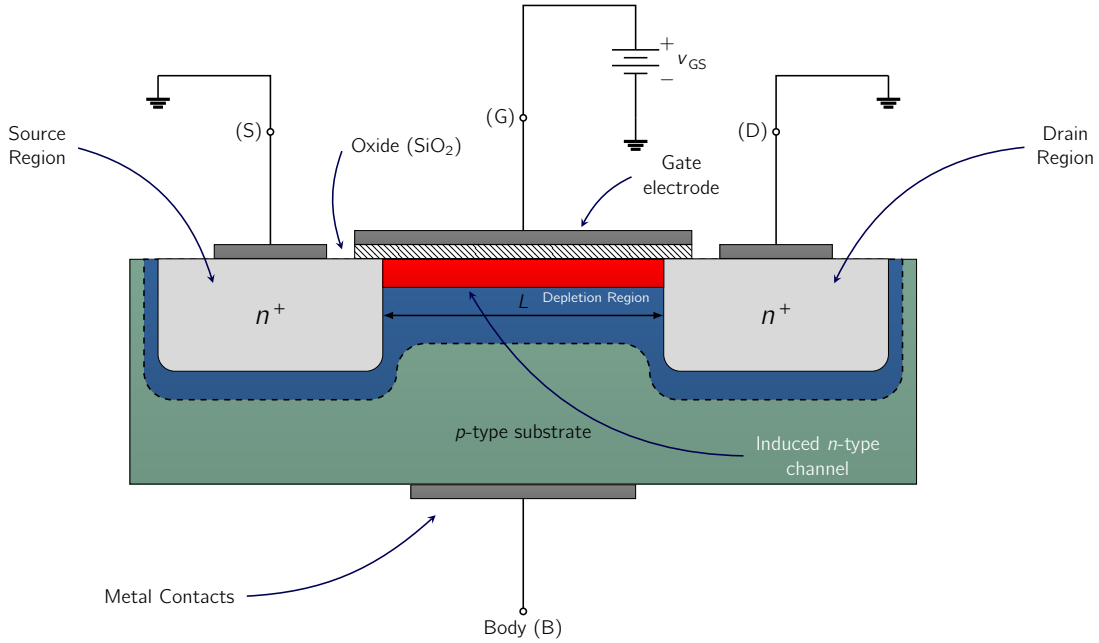


Figure 5.2: The enhancement-type NMOS transistor with a positive voltage applied to the gate. An n channel is induced at the top of the substrate beneath the gate.

5.1.2 Operation with Zero Gate Voltage

With zero voltage applied to the gate, two back-to-back diodes exist in series between drain and source. One diode is formed by the *pn* junction between the n^+ drain region and the *p*-type substrate, and the other diode is formed by the *pn* junction between the *p*-type substrate and the n^+ source region. These back-to-back diodes prevent current conduction from drain to source when a voltage v_{DS} is applied.

> The path between drain and source has a very high resistance, of the order of $1{,}0 \times 10^{13}\,\Omega$.

5.1.3 Creating a Channel for Current Flow

Consider next the situation depicted in **Fig.** 5.2.

Here we have grounded the source and the drain and applied a positive voltage to the gate. Since the source is grounded, the gate voltage appears in effect between gate and source and therefore is denoted v_{GS}. The positive voltage on the gate causes, in the first instance, the free holes, which are positively charged, to be repelled from the region of the substrate under the gate, called the channel region. These holes are pushed downward into the substrate, leaving behind a **carrier-depletion region**. The depletion region is populated by the bound negative charge associated with the acceptor atoms. These charges are "uncovered" because the neutralising holes have been pushed downward into the substrate.

As well, the positive gate voltage attracts electrons from the n^+ source and drain regions[14] into the channel region. When a sufficient number of electrons accumulate near the surface of the substrate under the gate, an *n*-region is in effect created, connecting the source and drain regions, as indicated in **Fig.** 5.2. Now if a voltage is applied between drain and source, current flows through this induced *n*-region, carried by the mobile electrons. The induced *n*-region therefore forms a channel for current flow from drain to source and is aptly called so. Correspondingly, the MOSFET of **Fig.** 5.2 is called an *n*-channel MOSFET or, alternatively, an NMOS transistor.

[14] where they are in abundance.

Note that an *n*-channel MOSFET is formed in a *p*-type substrate: The channel is created by inverting the substrate surface from *p*-type to *n*-type. Hence the induced channel is also called an **inversion layer**.

The value of v_{GS} at which a sufficient number of mobile electrons accumulate in the channel region to form a conducting channel is called the **threshold voltage** and is denoted as[15] V_t. Obviously, V_t for an *n*-channel FET is positive. The value of V_t is controlled during device fabrication and typically lies in the range of 0,3 V to 1,0 V.

[15] Some texts use V_T to denote the threshold voltage. We use V_t to avoid confusion with the thermal voltage V_T.

The gate and the channel region of the MOSFET form a parallel-plate capacitor, with the oxide layer acting as the capacitor *dielectric*. The positive gate voltage causes positive charge to accumulate on the top plate of the capacitor.[16] The corresponding negative charge on the bottom plate is formed by the electrons in the *induced channel*. An electric field ($|\mathbf{E}|$) therefore develops in the vertical direction. It is this field which controls the amount of charge in the channel, and thus it determines the channel conductivity and, in turn, the current which will flow through the channel when a voltage v_{DS} is applied.

[16] The electrode which is on the gate.

> This is the origin of the name "field-effect transistor" (FET).

The voltage across this parallel-plate capacitor, which is, the **voltage across the oxide**, **must** exceed V_t for a channel to form. When $v_{DS} = 0$, as in **Fig.** 5.2, the voltage at every point along the channel is zero, and the voltage across the oxide[17] is uniform and equal to v_{GS}. The excess of v_{GS} over V_t is termed the **effective voltage** or the **overdrive voltage** and is the quantity that determines the charge in the channel.

[17] i.e., between the gate and the points along the channel.

For our lecture, we shall denote $(v_{GS} - V_t)$ by v_{OV},

$$v_{GS} - V_t \equiv v_{ov}$$

We can express the magnitude of the electron charge ($|Q|$) in the channel by:

$$|Q| = C_{ox}(WL)\, v_{OV} \qquad [C] \qquad (5.1)$$

where C_{ox}, called the **oxide capacitance**, is the capacitance of the parallel-plate capacitor per unit gate area (in units of $F/m2$), W is the width of the channel, and L is the length of the channel. The oxide capacitance C_{ax} is given by

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$$

where ϵ_{ox} is the permittivity of the silicon dioxide,

$$\epsilon_{\infty} = 3.9\epsilon_0 = 3.98{,}854 \times 10^{-12} = 3{,}45 \times 10^{-11}$$

The oxide thickness (t_{ox}) is determined by the **process technology** used to fabricate the MOSFET.

As an example, for a process with $t_{ox} = 4\,\text{nm}$,

$$C_{ox} = \frac{3.45 \times 10^{-11}}{4 \times 10^{-9}} = 8.6 \times 10^{-3}\ \text{F/m}^2$$

It is much more convenient to express C_{ox} per micron squared. For our example, this gives us $8{,}6\,\text{fF} \cdot \mu\text{m}^{-2}$, where fF denotes femtofarad (10⁻¹⁵ F). For a MOSFET fabricated in this technology with a channel length $L = 0.18\,\mu\text{m}$ and a channel width $W = 0.72\,\mu\text{m}$, the total capacitance between gate and channel is

$$C = C_{ox}WL = 8.6 \times 0.18 \times 0.72 = 1{,}1\,\text{fF}$$

Finally, note from Eq. (**5.1**) that as v_{ov} is increased, the magnitude of the channel charge increases proportionately. Sometimes this is depicted as an increase in the depth of the channel. That is, the larger the overdrive voltage, the deeper the channel.

5.1.4 Applying a Small Drain-to-Source Voltage

Now that we have induced a channel, let's now apply a **positive voltage** between drain and source (v_{DS}), as shown in Fig. 5.3. We first consider the case where v_{DS} is small.[18] The voltage v_{DS} causes a current i_D to flow through the induced n-channel. Current is carried by free electrons traveling from source to drain.[19]

[18] In this case we are considering levels of 50 mV.

[19] hence the names source and drain.

By convention, the direction of current flow is opposite to that of the flow of negative charge.

Therefore the current in the channel, i_D, will be from drain to source, as indicated in Fig. 5.3.

We now wish to calculate the value of i_D. Toward that end, we first note that because v_{DS} is small, we can continue to assume that the voltage between the gate and various points along the channel remains **approximately constant** and equal to the value at the source end, i.e., v_{GS}. Therefore, the effective voltage between the gate and the various points along the channel remains equal to v_{OV}, and the channel charge Q is still given by Eq. (**5.1**). Of particular interest

in calculating the current i_D is the charge per unit channel length, which can be found from Eq. (**5.1**) as:

$$\frac{|Q|}{\text{unit channel length}} = C_{ox}Wv_{ov} \tag{5.2}$$

The voltage v_{DS} establishes an electric field E across the length of the channel,

$$|E| = \frac{v_{DS}}{L}$$

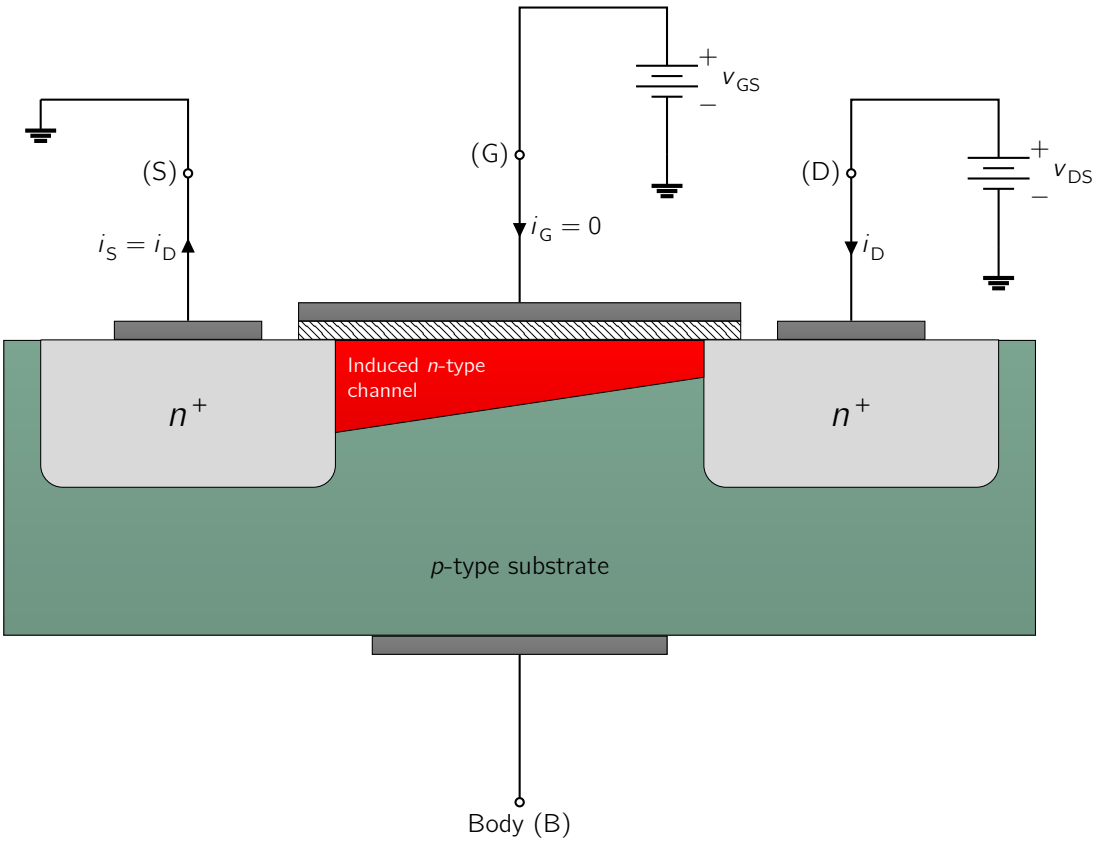


Figure 5.3

This electric field in turn causes the channel **electrons to drift toward the drain** with a velocity given by:

$$\text{Electron Drift Velocity} = \mu_n |E| = \mu_n \frac{v_{DS}}{L} \tag{5.3}$$

where μ_n is the mobility of the electrons at the surface of the channel.[20] The value of i_D can now be found by multiplying the charge per unit channel length Eq. (**5.2**) by the electron drift velocity Eq. (**5.3**),

[20] It is a physical parameter whose value depends on the **fabrication process technology**.

$$i_D = \left[(\mu_n C_{ox}) \left(\frac{W}{L} \right) v_{OV} \right] v_{DS}$$

Therefore, for small v_{DS}, the channel behaves as a linear resistance whose value is controlled by the overdrive voltage v_{ov}, which in turn is determined by v_{GS}.

$$i_D = \left[(\mu_n C_{ox}) \left(\frac{W}{L} \right) (v_{GS} - V_t) \right] v_{DS}$$

The conductance g_{DS} of the channel can be found from Eq. (5.7) or (5.8) as

$$g_{DS} = (\mu_n C_{ox}) \left(\frac{W}{L} \right) v_{OV} \quad \text{or} \quad g_{DS} = (\mu_n C_{ox}) \left(\frac{W}{L} \right) (v_{GS} - V[t]) \tag{5.4}$$

Observe that the **conductance** is determined by the product of three (**3**) factors:

$(\mu_n C_{ox})$, (W/L), and v_{ov} (or equivalently, $v_{GS} - V_t$).

To gain insight into MOSFET operation, we consider each of the three factors in turn.

The first factor, $(\mu_n C_{ox})$, is determined by the process technology used to fabricate the MOSFET. It is the product of the electron mobility (μ_n) and the oxide capacitance (C_{ox}). It makes physical sense for the channel conductance to be proportional to each of μ_n and C_{ox} and therefore to their product, which is termed the **process transconductance parameter** and given the symbol k'_n, where the subscript n denotes n channel,

$$k'_n = \mu_n C_{ox}$$

It can be shown that with μ_n having the dimensions of meters squared per volt-second $(m^2 \cdot V^{-1} \cdot s^{-1})$ and C_{ox} having the dimensions of farads per meter squared $(F \cdot m^{-2})$, the dimensions of k'_n are amperes per volt squared $(A \cdot V^{-2})$.

The second factor in the expression for the conductance g_{DS} in Eq. (**5.4**) is the transistor **aspect ratio** (W/L).

> The channel conductance is proportional to the channel width (W) and inversely proportional to the channel length (L) should make perfect physical sense.

The (W/L) ratio is obviously a dimensionless quantity that is determined by the device designer. Indeed, the values of W and L can be selected by the device designer to give the device the *i-v* characteristics desired. For a given fabrication process, however, there is a minimum channel length, L_{min}. In fact, the minimum channel length that is possible with a given fabrication process is used to characterize the process and is being continually reduced as technology advances.

> For example, in 2014, the state-of-the-art in commercially available MOS technology was a 32 nm process, meaning that for this process the minimum channel length possible was 32 nm.

Finally, we should note that the oxide thickness t_{ox} scales down with L_{min}. Therefore, for a 0.13-μm technology, t_{ox} is 2.7 nm, but for the currently popular 65-nm technology, t_{ox} is about 2.2 nm.

The product of the process transconductance parameter (k'_n) and the transistor aspect ratio (W/L) is the MOSFET transconductance parameter (k_n):

$$k_n = k'_n(W/L) \qquad \text{or} \qquad k_n = (\mu_n C_{ox})(W/L)$$

The MOSFET parameter k_n has the dimensions of $A \cdot V^{-2}$.

The third term in the expression of the channel conductance g_{DS} is the overdrive voltage v_{ov}. This is hardly surprising, given v_{ov} directly determines the magnitude of electron charge in the channel. As will be seen, v_{0V} is a very important circuit-design parameter.

> Throughout this lecture, we will use v_{ov} and $v_{GS} - V_t$ interchangeably.

We conclude this subsection by noting that with v_{DS} kept small, the MOSFET behaves as a linear resistance r_{DS} whose value is controlled by the gate voltage v_{GS},

$$r_{DS} = \frac{1}{g_{\mathrm{DS}}}$$
$$r_{DS} = \frac{1}{(\mu_{\mathrm{n}} C_{\mathrm{ox}})\,(W/L)\,v_{OV}}$$
$$r_{\mathrm{DS}} = \frac{1}{(\mu_{\mathrm{n}} C_{\mathrm{ox}})\,(W/L)\,(v_{gs} - V_{\mathrm{t}})}$$

The operation of the MOSFET as a voltage-controlled resistance is further illustrated in Fig. 5.4, which is a sketch of i_{D} versus v_{DS} for various values of vGS. Observe that the resistance is infinite for $v_{GS} \leqslant V_t$, and decreases as v_{GS} is increased above V_t. It is interesting to note that although v_{GS} is used as the parameter for the set of graphs in Fig. 5.4, the graphs in fact depend only on v_{ov} (and, of course, k_n).

The description above indicates that for the MOSFET to conduct, a channel has to be **induced**. Then, increasing v_{GS} above the threshold voltage V_t enhances the channel, hence the names enhancement-mode operation and enhancement-type MOSFET. Finally, we note that the current that leaves the source terminal (i_{s}) is equal to the current that enters the drain terminal (i_{D}), and the gate current $i_{\mathrm{G}} = 0$.

5.1.5 Operation as v_{DS} is Increased

We next consider the situation as v_{DS} is increased. For this purpose, let v_{GS} be held constant at a value greater than V_{t};

that is, let the MOSFET be operated at a constant overdrive voltage V_{OV}.

Please refer to Fig. 5.5 for a visual aid, and note that v_{DS} appears as a voltage drop across the length of the channel. That is, as we travel along the channel from source (S) to drain (D), the voltage[21] increases from zero to v_{DS}. Therefore the voltage between the gate and points along the channel decreases from $v_{GS} = V_{\mathrm{t}} + V_{OV}$ at the source end to $v_{GD} = v_{GS} v_{DS} = V_{\mathrm{t}} + V_{OV} v_{DS}$ at the drain end. Given the channel depth depends on this voltage, and specifically on the amount by which this voltage exceeds V_{t}, we find that the channel is **no longer of uniform depth** but rather, the channel will take the tapered shape shown in Fig. 5.5, being deepest at the source end[22] and shallowest at the drain end6 (where the depth is proportional to VOVvDS). This point is further illustrated in Fig. 5.6.

[21] measured relative to the source.

[22] where the depth is proportional to V_{OV}.

As v_{DS} is increased, the channel becomes more tapered and its resistance increases correspondingly. Therefore, the $i_{\mathrm{D}} v_{DS}$ curve does not continue as a straight line but bends as shown in Fig. 5.7. The equation describing this portion of the iDvDS curve can be easily derived by utilizing the information in Fig. 5.6. Specifically, please note that the charge in the tapered channel is proportional to the channel cross-sectional area shown in Fig. 5.6(b). This area in turn can be easily seen as proportional to 1 2 [VOV +(VOVvDS)] or VOV 1 2 vDS . Thus, the relationship between iD and vDS can be

found by replacing VOV in Eq. (5.7) by VOV 1 2 vDS ,

$$i_D = k_n' \left(\frac{W}{L}\right) \left(V_{OV} - \frac{1}{2} v_{DS}\right) v_{DS}$$

This relationship describes the semiparabolic portion of the iDvDS curve in Fig. 5.7. It applies to the entire segment down to vDS = 0. Specifically, note that as vDS is reduced, we can neglect 1 2 vDS relative to VOV in the factor in parentheses, and the expression reduces to that in Eq. (5.7). The latter of course is an approximation and applies only for small vDS (i.e., near the origin). There is another useful interpretation of the expression in Eq. (5.14). From Fig. 5.6(a) we see that the average voltage along the channel is 1 2 vDS . Thus, the average voltage that gives rise to channel charge and hence to iD is no longer VOV but VOV 1 2 vDS , which is indeed the factor that appears in Eq. (5.14). Finally, we note that Eq. (5.14) is frequently written in the alternate form

$$i_D = k_n' \left(\frac{W}{L}\right) \left(V_{OV} v_{DS} - \frac{1}{2} v_{DS}^2\right)$$

Furthermore, for an arbitrary value of VOV , we can replace VOV by (vGSVt) and rewrite Eq. (5.15) as

$$i_D = k_n' \left(\frac{W}{L}\right) \left[(v_{GS} - V_t)\, v_{DS} - \frac{1}{2} v_{DS}^2\right]$$

5.1.6 Channel Pinch-Off and Current Saturation

The above description of operation assumed that even though the channel became tapered, it still had a finite (nonzero) depth at the drain end. This in turn is achieved by keeping vDS sufficiently small that the voltage between the gate and the drain, vGD, exceeds Vt . This is indeed the situation shown in Fig. 5.6(a). Note that for this situation to obtain, vDS must not exceed VOV , for as vDS = VOV , vGD = Vt , and the channel depth at the drain end reduces to zero. Figure 5.8 shows vDS reaching VOV and vGD correspondingly reaching Vt . The zero depth of the channel at the drain end gives rise to the term channel pinch-off. Increasing vDS beyond this value (i.e., vDS >VOV) has no effect on the channel shape and charge, and the current through the channel remains constant at the value reached for vDS =VOV . The drain current thus saturates at the value found by substituting vDS =VOV in Eq. (5.14),

$$i_D = \frac{1}{2} k_n' \left(\frac{W}{L}\right) V_{OV}^2$$

The MOSFET is then said to have entered the **saturation region**.[23] The voltage v_{DS} at which saturation occurs is denoted V_{DSsat},

[23] or, equivalently, the saturation mode of operation.

$$V_{DSsat} = V_{OV} = V_{GS} - V_t$$

It should be noted that channel pinch-off does **NOT** mean channel blockage:

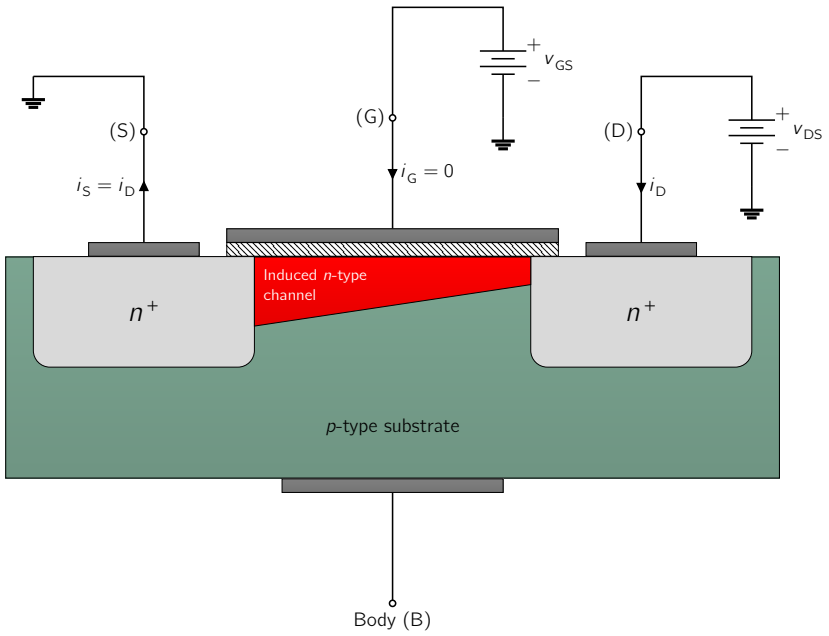


Figure 5.4: Operation of the enhancement NMOS transistor as vDS is increased. The induced channel acquires a tapered shape, and its resistance increases as vDS is increased. Here, vGS is kept constant at a value > Vt ; vGS = Vt +VOV.

Current continues to flow through the pinched-off channel, and the electrons that reach the drain end of the channel are accelerated through the depletion region which exists there and into the drain terminal.

Any increase in v_{DS} above V_{DSsat} appears as a voltage drop across the depletion region. Thus, both the current through the channel and the voltage drop across it remain constant in saturation.

The saturation portion of the $ip - vDS$ curve is, as expected, a horizontal straight line, as indicated in Fig. 5.7. Also indicated in Fig. 5.7 is the name of the region of operation obtained with a continuous (non-pinched-off) channel, the **triode region**, a carryover from the days of vacuum-tube devices, whose operation a FET resembles.

Finally, we note that the expression for i_D in saturation can be generalized by replacing the constant overdrive voltage V_{ov} by a variable one, v_{ov}:

$$i_D = \frac{1}{2}k'_n\left(\frac{W}{L}\right)v_{OV}^2$$

Also, v_{ov} can be replaced by $(v_{GS} - V_t)$ to obtain the alternate expression for saturation- mode i_D,

$$i_D = \frac{1}{2}k'_n\left(\frac{W}{L}\right)(v_{GS} - V_t)^2$$

5.1.7 The p-Channel MOSFET

Figure 5.9(a) shows a cross-sectional view of a *p*-channel enhancement-type MOSFET. The structure is similar to that of the NMOS device except that here the substrate is *n* type and the source and the drain regions are p^+ type:

> that is, all semiconductor regions are reversed in polarity relative to their counterparts in the NMOS case.

The PMOS and NMOS transistors are said to be **complementary** devices.

To induce a channel for current flow between source and drain, a **negative voltage** is applied to the gate, that is, between gate and source, as indicated in Fig. 5.9(b). By increasing the magnitude of the negative v_{GS} beyond the magnitude of the threshold voltage V_{tp}, which by convention is negative, a *p* channel is established as shown in Fig. 5.9(b). This condition can be described as

$$v_{GS} \leqslant V_{tp}$$

or, to avoid dealing with negative signs,

$$|v_{GS}| \geqslant |V_{tp}|$$

Now, to cause a current *ip* to flow in the p channel, a negative voltage *vDS* is applied to the drain.[24] The current i_D is carried by holes and flows through the channel from source to drain. As we have done for the NMOS transistor, we define the process transconductance parameter for the PMOS device as

$$k'_p = \mu_p C_{ox}$$

where μ_p is the mobility of the holes in the induced p channel. Typically, $\mu_p = 0.25\mu_n$ to $0.5\ \mu_n$ and is process-technology dependent. The transistor transconductance parameter k_p is obtained by multiplying k'_p by the aspect ratio W/L,

$$k_p = k'_p (W/L)$$

[24] If a positive voltage is applied to the drain, the pn junction between the drain region and the substrate will become forward biased, and the device will no longer operate as a MOSFET. Proper MOSFET operation is predicated on the pn junctions between the source and drain regions and the substrate being always reverse biased.

The remainder of the description of the physical operation of the *p*-channel MOSFET follows that for the NMOS device, except of course for the sign reversals of all voltages. We will present the complete current-voltage characteristics of both NMOS and PMOS transistors in the next section.

PMOS technology originally dominated MOS IC manufacturing, and the original microprocessors utilized PMOS transistors. As the technological difficulties of fabricating NMOS transistors were solved, NMOS completely supplanted PMOS. The main reason for this change is that electron mobility μ_n is higher by a factor of 2 to 4 than the hole mobility μ_n, resulting in NMOS transistors having greater gains and speeds of operation than PMOS devices. Subsequently, a technology was developed that permits the fabrication of both NMOS and PMOS transistors on the same chip. Appropriately called **complementary MOS**, or **CMOS**, this technology is currently the dominant electronics technology.

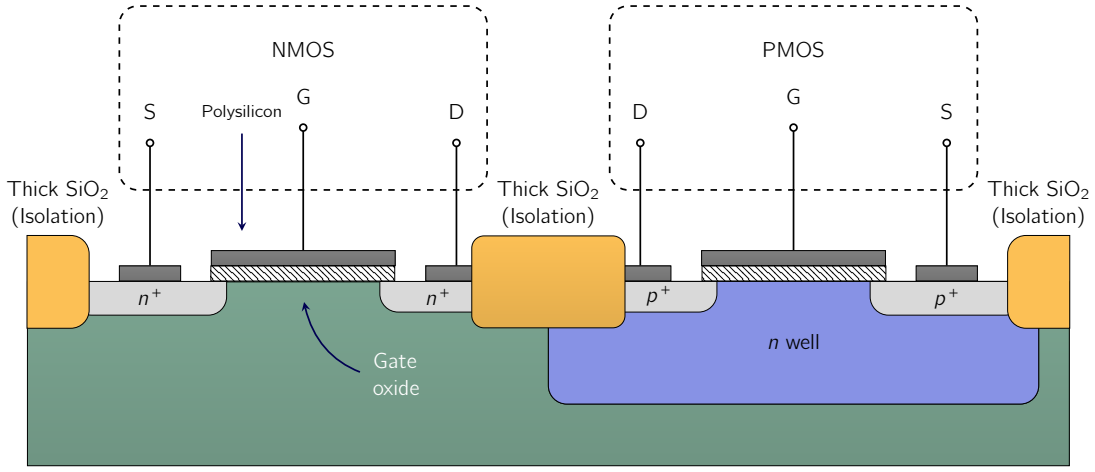


Figure 5.5

5.1.8 Complementary MOS or CMOS

As the name implies, complementary MOS technology uses MOS transistors of both polarities. Although CMOS circuits are somewhat more difficult to fabricate than NMOS, the availability of complementary devices makes possible many powerful circuit configurations. Indeed, at the present time CMOS is the most widely used of all the IC technologies. This statement applies to both analog and digital circuits. CMOS technology has virtually replaced designs based on NMOS transistors alone. Furthermore, by 2014 CMOS technology had taken over many applications that just a few years earlier were possible only with bipolar devices.

Fig. 5.5 shows a cross section of a CMOS chip illustrating how the PMOS and NMOS transistors are fabricated. Observe that while the NMOS transistor is implemented directly in the *p*-type substrate, the PMOS transistor is fabricated in a specially created *n* region, known as an *n* well. The two devices are isolated from each other by a thick region of oxide that functions as an insulator. Not shown on the diagram are the connections made to the *p*-type body and to the *n* well. The latter connection serves as the body terminal for the PMOS transistor.

5.1.9 Operating the MOS Transistor in the Subthreshold Region

The above description of the n-channel MOSFET operation implies that for vGS < Vt , no current flows and the device is cut off. This is not entirely true, for it has been found that for values of vGS smaller than but close to Vt , a small drain current flows. In this subthreshold region of operation, the drain current is exponentially related to vGS , much like the iC-vBE relationship of a BJT, as will be shown in the next chapter. Although in most applications the MOS transistor is operated with vGS >Vt , there are special, but a growing number of, applications that make use of subthreshold operation. In Chapter 14, we will briefly consider subthreshold operation.

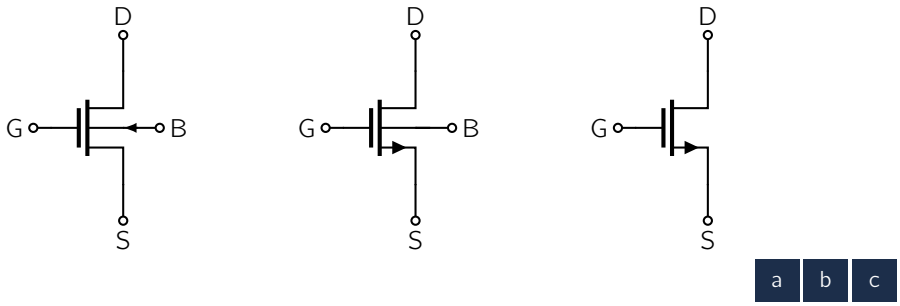


Figure 5.6
(a) Circuit symbol for the n-channel enhancement-type MOSFET. (b) Modified circuit symbol with an arrowhead on the source terminal to distinguish it from the drain and to indicate device polarity (i.e., n channel). (c) Simplified circuit symbol to be used when the source is connected to the body or when the effect of the body on device operation is unimportant.

5.2 Current-Voltage Characteristics

Building on the physical foundation established in the previous section for the operation of the enhancement MOS transistor, in this section we present its complete current-voltage characteristics. These characteristics can be measured at dc or at low frequencies and thus are called static characteristics. The dynamic effects that limit the operation of the MOSFET at high frequencies and high switching speeds will be discussed in Chapter 10.

5.2.1 Circuit Symbol

Fig. 5.9a shows the circuit symbol for the *n*-channel enhancement-type MOSFET. Please observe that the spacing between the two (2) vertical lines that represent the gate and the channel indicates the fact that the gate electrode is insulated from the body of the device. The polarity of the **p**-type substrate (body) and the *n* channel is indicated by the arrowhead on the line representing the body (B). This arrowhead also indicates the polarity of the transistor, namely, that it is an *n*-channel device.

> Although the MOSFET is a symmetrical device, it is often useful in circuit design to designate one terminal as the source and the other as the drain[25]

[25] without having to write S and D beside the terminals.

This objective is achieved in the modified circuit symbol shown in **Fig.** 5.9b. Here an arrowhead is placed on the source terminal, thus distinguishing it from the drain terminal. The arrowhead points in the normal direction of current flow and thus indicates the polarity of the device (i.e., *n*-channel). Observe that in the modified symbol, there is no need to show the arrowhead on the body line. Although the circuit symbol of **Fig.** 5.9b clearly distinguishes the source from the drain, in practice it is the polarity of the voltage impressed across the device that determines source and drain;

> The drain is always positive relative to the source in an *n*-channel FET.

In applications where the source is connected to the body of the device, a further simplification of

the circuit symbol is possible, as indicated in **Fig.** 5.9c. This symbol is also used in applications when the effect of the body on circuit operation is **NOT** important, as will be seen later.

5.2.2 The $i_D - v_{DS}$ Characteristics

Table 5.1 provides a compilation of the conditions and the formulas for the operation of the NMOS transistor in each of the three possible regions:

the cutoff region, the triode region, and the saturation region.

The first two (**2**) are useful if the MOSFET is to be utilized as a switch. On the other hand, if the MOSFET is to be used to design an amplifier, it must be operated in the saturation region. At the top of Table 5.1 we show a circuit consisting of an NMOS transistor and two dc supplies providing v_{GS} and v_{DS}. This conceptual circuit can be used to measure the $i_D - v_{DS}$ characteristic curves of the NMOS transistor. Each curve is measured by setting v_{GS} to a desired constant value, varying v_{DS}, and measuring the corresponding i_D. Two of these characteristic curves are shown in the accompanying diagram: one for $v_{GS} < V_m$ and the other for $v_{GS} = V_m + v_{OV}$. (Note that we now use V_m to denote the threshold voltage of the NMOS transistor, to distinguish it from that of the PMOS transistor, denoted V_ν.) As Table 5.1 shows, the boundary between the triode region and the saturation region is determined by whether v_{DS} is less or greater than the overdrive voltage v_{OV} at which the transistor is operating. An equivalent way to check for the region of operation is to examine the relative values of the drain and gate voltages. To operate in the triode region, the gate voltage must exceed the drain voltage by at least V_m volts, which ensures that the channel remains continuous (not pinched off). On the other hand, to operate in saturation, the channel must be pinched off at the drain end; pinch-off is achieved here by keeping vD higher than $vG - Vm$, that is, not allowing v_D to fall below v_G by more than V_m volts. The graphical construction of Fig. 5.12 should serve to remind the reader of these conditions. A set of $i_p - v_{DS}$ characteristics for the NMOS transistor is shown in Fig. 5.13. Observe that each graph is obtained by setting v_{GS} above V_{tn} by a specific value of overdrive voltage, denoted V_{OV1}, V_{OV2}, V_{OV3}, and V_{OV4}. This in turn is the value of v_{DS} at which the corresponding graph saturates, and the value of the resulting saturation current is directly determined by the value of v_{0V}, namely, $\frac{1}{2}k_nV_{0V1}^2$, $\frac{1}{2}k_nV_{0V2}^2$... The reader is advised to commit to memory both the structure of these graphs and the coordinates of the saturation points.

spice

```
RC 4 3 1k
M1 3 2 0 0 NFET
V1 1 0 dc 12
V2 4 0 dc 10

.MODEL NFET NMOS LEVEL=3          PHI=0.600000    TOX=2.1200E-08
                + XJ=0.200000U    TPG=1           VTO=0.7860       DELTA=6.9670E-01
                + LD=1.6470E-07   KP=9.6379E-05   UO=591.7         THETA=8.1220E-02
                + RSH=8.5450E+01  GAMMA=0.5863    NSUB=1.6160E+16
                + NFS=5.0000E+12  VMAX=2.0820E+05 ETA=7.0660E-02
```

Continues on next page →

← Continued from previous page

```spice
                + KAPPA=1.3960E-01 CGDO=4.0241E-10 CGSO=4.0241E-10
                + CGBO=3.6144E-10  CJ=3.8541E-04   MJ=1.1854
                + CJSW=1.3940E-10  MJSW=0.125195   PB=0.800000

.dc V2 0 10 0.1 V1 1.5 3.5 0.5
```

Finally, please observe the boundary between the triode and the saturation regions, that is, the locus of the saturation points, is a *parabolic curve* described by

$$i_{\mathrm{D}} = \frac{1}{2}k'_{\mathrm{n}}\left(\frac{W}{L}\right)v_{\mathrm{DS}}^2$$

5.2.3 The $i_D - v_{GS}$ Characteristic

When the MOSFET is used in the design of an *amplifier*, it is operated in the **saturation region**. As Fig. 5.13 indicates, in saturation the drain current (i_D) is constant, determined by v_{GS}[26] and is **independent** of v_{DS}. That is, the MOSFET operates as a **constant-current source** where the value of the current is determined by v_{GS}. In effect, then, the MOSFET operates as a **voltage-controlled current source** with the control relationship described by

$$i_D = \frac{1}{2}k'_n\left(\frac{W}{L}\right)(v_{GS} - V_{tn})^2 \qquad \text{or in } v_{ov} \qquad i_D = \frac{1}{2}k'_n\left(\frac{W}{L}\right)v_{OV}^2 \tag{5.5}$$

[26] We can also use v_{OV} for this as well.

> This is the relationship which underlies the application of the MOSFET as an **amplifier**, a **non-linear** relationship that should be of concern to those interested in **designing linear amplifiers**.

Figure 5.14 shows the $i_D - v_{GS}$ characteristic of an NMOS transistor operating in saturation.[27]

[27] Note that if we are interested in a plot of i_D versus v_{OV}, we simply shift the origin to the point $v_{GS} = V_{tn}$.

The view of the MOSFET in the **saturation region** as a voltage-controlled current source is illustrated by the equivalent-circuit representation shown in Fig. 5.15. For reasons that will become apparent shortly, the circuit in Fig. 5.15 is known as a **large-signal equivalent** circuit.

> Here the current source is **ideal**, with an infinite output resistance representing the independence, in saturation, of i_D from v_{DS}.

This, of course, has been assumed in the idealized model of device operation utilized thus far. We are about to rectify an important shortcoming of this model.

5.2.4 Finite Output Resistance in Saturation

The Eq. (**5.5**) and the corresponding large-signal equivalent circuit in Fig. 5.15, as well as the graphs in Fig. 5.13, indicate that in saturation, i_D is **independent** of v_{DS}. Therefore, a change Δv_{DS} in the drain-to-source voltage causes a zero change in i_D, which implies that the incremental resistance looking into the drain of a saturated MOSFET is infinite.

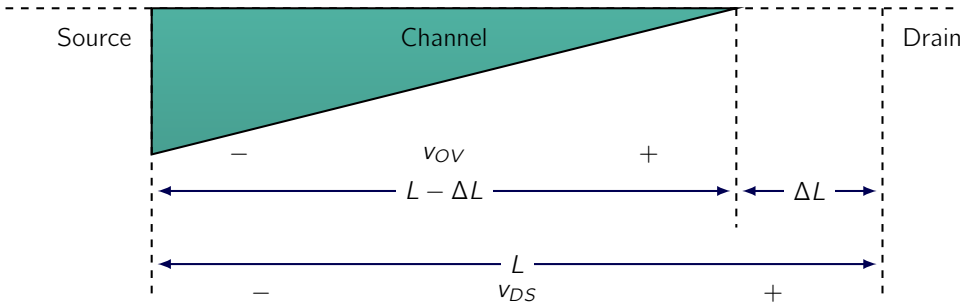


Figure 5.7 Increasing v_{DS} beyond v_{DSsat} causes the channel pinch-off point to move slightly away from the drain, therefore reducing the effective channel length (by L).

This, however, is an *idealization* based on the premise that once the channel is pinched off at the drain end, further increases in v_{DS} have no effect on the channel's shape.

> However, in practice, increasing v_{DS} beyond v_{OV} does affect the channel somewhat.

Specifically, as v_{DS} is increased, the channel pinch-off point is moved slightly **away** from the drain, toward the source. This is illustrated in **Fig.** 5.7, from which we note that the voltage across the channel remains constant at v_{OV}, and the additional voltage applied to the drain appears as a voltage drop across the narrow depletion region between the end of the channel and the drain region.

This voltage accelerates the electrons that reach the drain end of the channel and sweeps them across the depletion region into the drain. Note, however, that[28] the channel length is in effect reduced, from L to $L - \Delta L$, a phenomenon known as **channel-length modulation**.

[28] with depletion-layer widening

Now, since i_D is inversely proportional to the channel length which was given in Eq. (**5.5**), and i_D increases with v_{DS}.

This effect can be accounted for in the expression for i_D by including a factor $1 + \lambda (v_{DS} - v_{OV})$ or, for simplicity, $(1 + \lambda v_{DS})$,

$$i_D = \frac{1}{2} k'_n \left(\frac{W}{L}\right) (v_{GS} - V_{tn})^2 (1 + \lambda v_{DS}) \qquad (\mathbf{5.6})$$

Here λ is a device parameter having the units of reciprocal volts (V^{-1}).

> λ depends both on the **process technology** used to fabricate the device and on the **channel length** L that the circuit designer selects.

Specifically, the value of λ is much larger for newer submicron technologies than for older technologies. This makes intuitive sense:

> Newer technologies have very short channels, and are therefore much more greatly impacted by the **channel-length modulation effect**.

Also, for a given process technology, λ is inversely proportional to L.

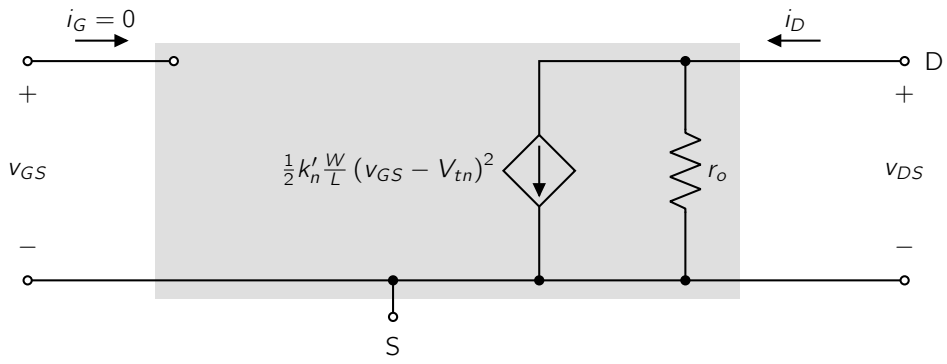


Figure 5.8 Large-signal, equivalent-circuit model of the n-channel MOSFET in saturation, incorporating the output resistance r_o. The output resistance models the linear dependence of i_D on v_{DS} and is given by Eq. (5.7).

A typical set of $i_D - v_{DS}$ characteristics showing the effect of channel-length modulation is displayed in Fig. 5.17. The observed linear dependence of i_D on v_{DS} in the saturation region is represented in Eq. (**5.6**) by the factor $(1 + \lambda v_{DS})$. From Fig. 5.17 we observe that when the straight-line $i_D - v_{DS}$ characteristics are extrapolated, they intercept the v_{DS} axis at the point, $v_{DS} = -V_A$, where V_A is a positive voltage. Eq. (**5.6**), however, indicates that $i_D = 0$ at $v_{DS} = -1/\lambda$. It follows that

$$V_A = \frac{1}{\lambda}$$

and therefore V_A is a device parameter with the dimensions of V. For a given process, V_A is proportional to the channel length L that the designer selects for a MOSFET.

We can isolate the dependence of V_A on L by expressing it as

$$V_A = V_A' L$$

where $V_A' L$ is entirely **process-technology dependent**, with the dimensions of volts per micron. Generally, V_A' falls in the range of 5 V/μm to 50 V/μm.

> The voltage, V_A, is usually referred to as the Early voltage, after *J. M. Early*, who discovered a similar phenomenon for the BJT.

Eq. (**5.6**) indicates that when channel-length modulation is taken into account, the saturation values of i_D depend on v_{DS}. Therefore, for a given v_{GS}, a change Δv_{DS} gives a corresponding change Δi_D in the drain current i_D. It follows that the output resistance of the current source representing i_D in saturation is no longer infinite.

Defining the output resistance r_o as:[29]

$$r_o \equiv \left[\frac{\partial i_D}{\partial v_{DS}}\right]^{-1} \qquad \text{where} \qquad v_{GS} \text{ is kept constant.}$$

and using Eq. (**5.6**) results in

$$r_o = \left[\lambda \frac{k_n'}{2}\frac{W}{L}(V_{GS} - V_m)^2\right]^{-1}$$

which can be written as

$$r_o = \frac{1}{\lambda I_D} \qquad \text{or} \qquad r_o = \frac{V_A}{I_D'} \tag{5.7}$$

[29] We use r_o to denote the output resistance in saturation, and r_{DS} to denote the drain-to-source resistance in the triode region, for small v_{DS}.

where I'_D is the drain current *without* channel-length modulation taken into account; that is,

$$I'_D = \frac{1}{2}k'_n\frac{W}{l}(V_{GS} - V_m)^2$$

Thus the output resistance is **inversely proportional** to the drain current.[30]

Finally, we see in **Fig.** 5.8 the large-signal, equivalent-circuit model incorporating r_o.

[30] In applying Eq. (**5.7**) we will usually drop the prime on I_D and simply use $r_o = V_A/I_D$ where I_D is the drain current without channel-length modulation.

5.2.5 Characteristics of the p-Channel MOSFET

The circuit symbol for the *p*-channel enhancement-type MOSFET is shown in **Fig.** 5.9a. **Fig.** 5.9b shows a modified circuit symbol in which an arrowhead pointing in the normal direction of current flow is included on the source terminal. For the case where the source is connected to the substrate, the simplified symbol of **Fig.** 5.9c is usually used.

The regions of operation of the PMOS transistor and the corresponding conditions and expression for i_D are shown in Table 5.2. Observe that the equations are written in a way that emphasizes physical intuition and avoids the confusion of negative signs. Thus while V_{tp} is by convention negative, we use $|V_\nu|\rangle$, and the voltages v_{SG} and v_{SD} are positive. Also, in all of our circuit diagrams we will always draw *p*-channel devices with their sources on top so that current flows from top to bottom. Finally, we note that PMOS devices also suffer from the channel-length modulation effect. This can be taken into account by including a factor $(1 + |\lambda|v_{SD})$ in the saturation-region expression for i_D as follows

$$i_D = \frac{1}{2}k'_p\left(\frac{W}{L}\right)\left(v_{SG} - |V_{ip}|\right)^2(1 + |\lambda|v_{SD})$$

or equivalently

$$i_D = \frac{1}{2}k'_p\left(\frac{W}{L}\right)\left(v_{SG} - |V_{ip}|\right)^2\left(1 + \frac{v_{SD}}{|V_A|}\right)$$

where λ and V_A (the Early voltage for the PMOS transistor) are by convention negative quantities, hence we use $|\lambda|$ and $|V_A|$. Finally, we should note that for a given CMOS fabrication process λ_n and $|\lambda_p|$ are generally not equal, and similarly for V_{An} and $|V_{Ap}|$.

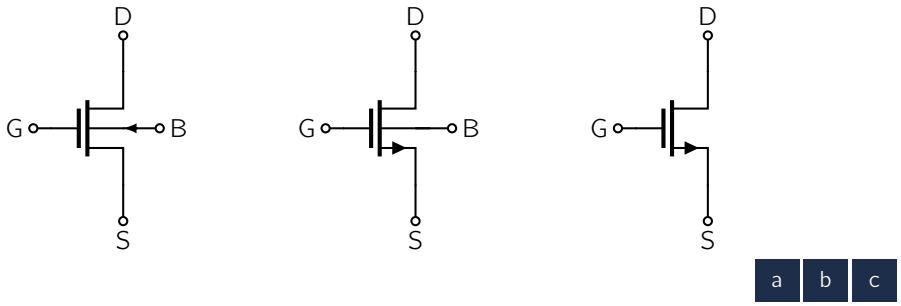


Figure 5.9
(a) Circuit symbol for the p-channel enhancement-type MOSFET. (b) Modified symbol with an arrowhead on the source lead. (c) Simplified circuit symbol for the case where the source is connected to the body.

To recap, to turn a PMOS transistor on, the gate voltage has to be made lower than that of the source by at least $|V_{v_p}|$. To operate in the triode region, the drain voltage has to exceed that of the gate by at least $|V_{v_p}|$; otherwise, the PMOS operates in saturation. Finally, Fig. 5.20 provides a pictorial representation of these operating conditions.

5.3 MOSFET Circuits at DC

Having studied the *i-v* characteristics of MOSFETs, we now consider circuits in which only DC voltages and currents are of concern. Specifically, we shall present a series of design and analysis examples of MOSFET circuits at operating with DC. The objective is to instill a familiarity with the device and the ability to perform MOSFET circuit analysis both rapidly and effectively.

In the following examples, to keep matters simple and focus our attention on the essence of MOSFET circuit operation, we will generally neglect channel-length modulation;

that is, we will assume $\lambda = 0$.

We will find it convenient to work in terms of the overdrive voltage:

$V_{ov} = V_{GS} - V_m$ for NMOS and $|V_{ov}| = V_{SG} - |V_m|$ for PMOS.

5.3 Information

Moore's Law

A half-century ago, Gordon Moore, who would go on to become a cofounder first of Fairchild Semiconductor and then of Intel, presented a startling idea in **HIS LAW:** the issue of Electronics Magazine dated April 19, 1965. Moore, who had a doctorate in chemistry, had projected the potential growth of the IC industry based on five points spanning a seven-year period from 1958 to 1965. The conclusion he reached-that the number of transistors per chip had been increasing and would continue to increase by a factor of 2 every two years or so-was destined to propel progress in ICs over the succeeding decades into the twenty-first century. Doubling of the number of transistors was predicted on the basis of another prediction: the continuing shrinkage of transistor dimensions. In early recognition of the importance of this prediction, Carver Mead, a pioneer in very large scale integration (VLSI), soon began to refer to this prediction as "Moore's law." (See Chapter 15, Section 15.1, for the implications of Moore's law).

5.4 Body Effect and Other Topics

Before we close of MOSFETs, let's briefly consider a number of important though secondary issues.

5.4.1 The Role of the Substrate - The Body Effect

In numerous applications the source terminal (S) is connected to the substrate (or body) terminal (B), which results in the *pn* junction between the substrate and the induced channel (review Fig. 5.5) having a constant zero (cutoff) bias. In such a case the substrate does not play any role in circuit operation and its existence can be ignored altogether.

In ICs, however, the substrate is usually common to many MOS transistors. In order to maintain the cutoff condition for all the substrate-to-channel junctions, the substrate is usually connected to the most negative power supply in an NMOS circuit (the most positive in a PMOS circuit). The resulting reverse-bias voltage between source and body (V_{SR} in an n-channel device) will have an effect on device operation. To appreciate this fact, consider an NMOS transistor and let its substrate be made negative relative to the source. The reverse-bias voltage will widen the depletion region (refer to Fig. 5.2). This in turn reduces the channel depth. To return the channel to its former state, v_{GS} has to be increased. The effect of V_{SB} on the channel can be most conveniently represented as a change in the threshold voltage V_t. Specifically, it has been shown that increasing the reverse substrate bias voltage V_{SB} results in an increase in V_t according to the relationship

$$V_t = V_{t0} + \gamma\left[\sqrt{2\phi_f + V_{SB}} - \sqrt{2\phi_f}\right]$$

where V_{t0} is the threshold voltage for $V_{SB} = 0$; ϕ_f is a physical parameter with $(2\phi_f)$ typically 0.6 V; γ is a fabrication-process parameter given by

$$\gamma = \frac{\sqrt{2qN_A\epsilon_s}}{C_{\infty}}$$

where q is the magnitude of the electron charge (1.6×10^{-19} C), N_A is the doping concentration of the *p*-type substrate, and ϵ_s is the permittivity of silicon ($11.7\epsilon_0 = 11.7 \times 8.854 \times 10^{-14} = 1.04 \times 10^{-12}$ F/cm). The parameter γ has the dimension of $\sqrt{V}$ and is typically 0.4 V^{1/2}. Finally, note that Eq. (5.30) applies equally well for p-channel devices with V_{SB} replaced by the reverse bias of the substrate, V_{BS} (or, alternatively, replace V_{SB} by $|V_{SB}|$) and note that γ is negative. Also, in evaluating γ, N_A must be replaced with N_D, the doping concentration of the n well in which the PMOS is formed. For p-channel devices, $2\phi_f$ is typically 0.75 V, and γ is typically -0.5 V^{1/2}.

Equation (5.30) indicates that an incremental change in V_{SB} gives rise to an incremental change in V_t, which in turn results in an incremental change in iD even though v_{GS} might have been kept constant. It follows that the body voltage controls i_D; thus the body acts as another gate for the MOSFET, a phenomenon known as the **body effect**. Here we note that the parameter γ is known as the **body-effect parameter**.

5.4.2 Temperature Effects

Both V_t and k' are **temperature sensitive**. The magnitude of V_t decreases by about $2\,\text{mV}\cdot{}^\circ\text{C}^{-1}$ rise in temperature. This decrease in $|V_t|$ gives rise to a corresponding increase in drain current as temperature is increased. However, because k' decreases with temperature and its effect is a dominant one, the overall observed effect of a temperature increase is a **decrease** in drain current.[31]

[31] This very interesting result is put to use in applying the MOSFET in power circuits.

5.4.3 Breakdown and Input Protection

As the voltage on the drain is increased, a value is reached at which the *pn*-junction between the drain region and substrate suffers avalanche breakdown. This breakdown usually occurs at voltages of 20 V to 150 V and results in a somewhat rapid increase in current.

This is known as a **weak avalanche**.

Another breakdown effect that occurs at lower voltages[32] in modern devices is called **punch-through**. It occurs in devices with relatively short channels when the drain voltage is increased to the point that the depletion region surrounding the drain region extends through the channel to the source. The drain current then increases rapidly.

[32] This occurs usually around 20 V.

Normally, punch-through does not result in permanent damage to the device.

Yet another kind of breakdown occurs when the gate-to-source voltage exceeds about 30 V. This is the breakdown of the gate oxide and results in **permanent damage** to the device.

Although 30 V may seem high, it must be remembered that the MOSFET has a very high input resistance and a very small input capacitance, and thus small amounts of static charge accumulating on the gate capacitor can cause its breakdown voltage to be exceeded.

To prevent the accumulation of static charge on the gate capacitor of a MOSFET, gate-protection devices are usually included at the input terminals of MOS integrated circuits. The protection mechanism invariably makes use of **clamping diodes**.

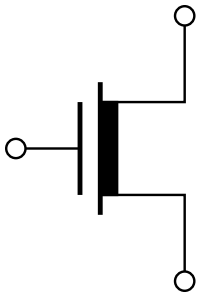
5.4.4 Velocity Saturation

At high longitudinal electric fields, the drift velocity of charge carriers in the channel reaches an upper limit.[33] This effect, which in modern very-short-channel devices can occur for v_{DS} lower than 1 V, is called **velocity saturation**. It can be shown that when velocity saturation occurs, the current i_D will no longer be related to v_{GS} by the square-law relationship. Rather, i_D becomes linearly dependent on v_{GS} and the transconductance g_m becomes constant and independent of v_{GS}.

[33] This is approximately $10^7\,\text{cm}\cdot\text{s}^1$ for electrons and holes in silicon.

5.4.5 Depletion-Type MOSFET

We will finish of MOSFET with a brief discussion of another type of MOSFET, called the depletion-type MOSFET.[34]



[34] Diagram of an *n*-type depletion MOSFET.

Its structure is similar to that of the enhancement-type MOSFET with one significant difference:

> The depletion MOSFET has a physically implanted channel.

Therefore an *n*-channel depletion-type MOSFET has an *n*-type silicon region connecting the n^+ source and the n^+ drain regions at the top of the *p*-type substrate. Therefore if a voltage v_{DS} is applied between drain and source, a current i_D flows for $v_{GS} = 0$.

> In other words, there is no need to induce a channel, unlike the case of the enhancement MOSFET.

The channel depth and therefore its conductivity can be controlled by v_{GS} in exactly the same manner as in the enhancement-type device whereby applying a positive v_{GS} enhances the channel by attracting more electrons into it. Here, however, we also can apply a negative v_{GS}, which causes electrons to be repelled from the channel, which makes the channel **shallower** and causes its conductivity to decrease. The negative v_{GS} is said to **deplete** the channel of its charge carriers, and this mode of operation[35] is called **depletion mode**. As the magnitude of v_{GS} is increased in the negative direction, a value is reached at which the channel is completely depleted of charge carriers and i_D is reduced to zero even though *vps* may be still applied. This negative value of v_{GS} is the threshold voltage of the *n*-channel depletion-type MOSFET.

[35] negative v_{GS}.

The description above suggests that a depletion-type MOSFET can be operated in the enhancement mode by applying a positive v_{GS} and in the depletion mode by applying a negative v_{GS}.

One final point worth mentioning is the $i_D - v_{DS}$ characteristics are similar to those for the enhancement-type MOSFET except for the negative V_{tn}.

> Note that the device symbol denotes the existing channel via the shaded area next to the vertical line.

Depletion-type MOSFETs can be fabricated on the same IC chip as enhancement-type devices, resulting in circuits with improved characteristics, as will be shown in a later chapter. The depletion-type MOSFET, however, is a specialty device and is not commonly used.

References

Chapter 6

Bipolar Junction Transistors

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In this chapter, we will study the other major three-terminal device:

the BJT.

Similar but Different — 6.1 Information

The presentation of the material in this chapter parallels but does **NOT** depend on the MOSFET we studied previously and therefore if needed, the BJT can be studied before the MOSFET.

Three-terminal devices are far more useful than two-terminal ones,[1] as they can be used in a multitude of applications,

ranging from signal amplification to the design of digital logic and memory circuits.

The basic principle involved is the use of the voltage between two (2) terminals to **control the current** flowing in the **third terminal**.

In this way, a three-terminal device can be used to realise a **controlled source**, which is the basis for amplifier design. Also, in the extreme, the control signal can be used to cause the current in the third terminal to change from zero to a large value, allowing the device to act as a **switch**.[2]

The invention of the BJT in 1948 at the *Bell Telephone Laboratories* ushered in the era of **solid-state**

[1] such as the diodes we have covered previously.

[2] The switch is the basis for the realization of the logic inverter, the basic element of digital circuits.

circuits [1, 2]. The result was not just the replacement of *vacuum tubes* by transistors in radios and television sets but the eruption of an electronics revolution that led to major changes in the way we work, play, and indeed, live. The invention of the transistor also eventually led to the dominance of information technology and the emergence of the knowledge-based economy.[3]



[3] One of the first commercial product to use a transistor was *Regency TR-1*, a transistor radio, introduced in 1954 [3].

The BJT enjoyed nearly three decades as the device of choice in the design of both discrete and integrated circuits. Although the MOSFET had been known very early on, it was not until the 1970s and 1980s that it became a serious competitor to the BJT. By 2011, the MOSFET was undoubtedly the most widely used electronic device, and Complimentary MOS (CMOS) technology the technology of choice in the design of IC [4]. Nevertheless, the BJT remains a significant device that excels in certain applications.

The BJT remains popular in **discrete-circuit design**, where it is used together with other discrete components such as resistors and capacitors to implement circuits that are assembled on Printed-Circuit Board (PCB)s.

> There is a very wide selection of BJT types that fit nearly every conceivable application.

As well, the BJT is still the preferred device in some very demanding analogue and digital IC applications. This is especially true in very-high-frequency and high-speed circuits.[4]

[4] In particular, a very-high-speed digital logic-circuit family based on bipolar transistors, namely, **emitter-coupled logic**, is still in use.

Finally, bipolar transistors can be combined with MOSFETs to create innovative circuits that take advantage of the high-input-impedance and low-power operation of MOSFETs and the very-high-frequency operation and high-current-driving capability of bipolar transistors. The resulting technology is known as BiCMOS, and it is finding increasingly larger areas of application.

In this chapter, we shall start with a description of the physical operation of the BJT. While it is relatively simple, this physical description provides considerable insight regarding the performance of the transistor as a circuit element. We then quickly move from describing current flow in terms of electrons and holes to a study of the transistor terminal characteristics. Circuit models for transistor operation in different modes will be developed and utilized in the analysis and design of transistor circuits.

6.1 Physical Implementation

As with every other device we've discussed previously, let's look at how BJT functions.

6.1.1 Simplified Structure and Operation Modes

Fig. 6.1a shows a simplified structure for the BJT, where the it consists of three (**3**) semiconductor regions:

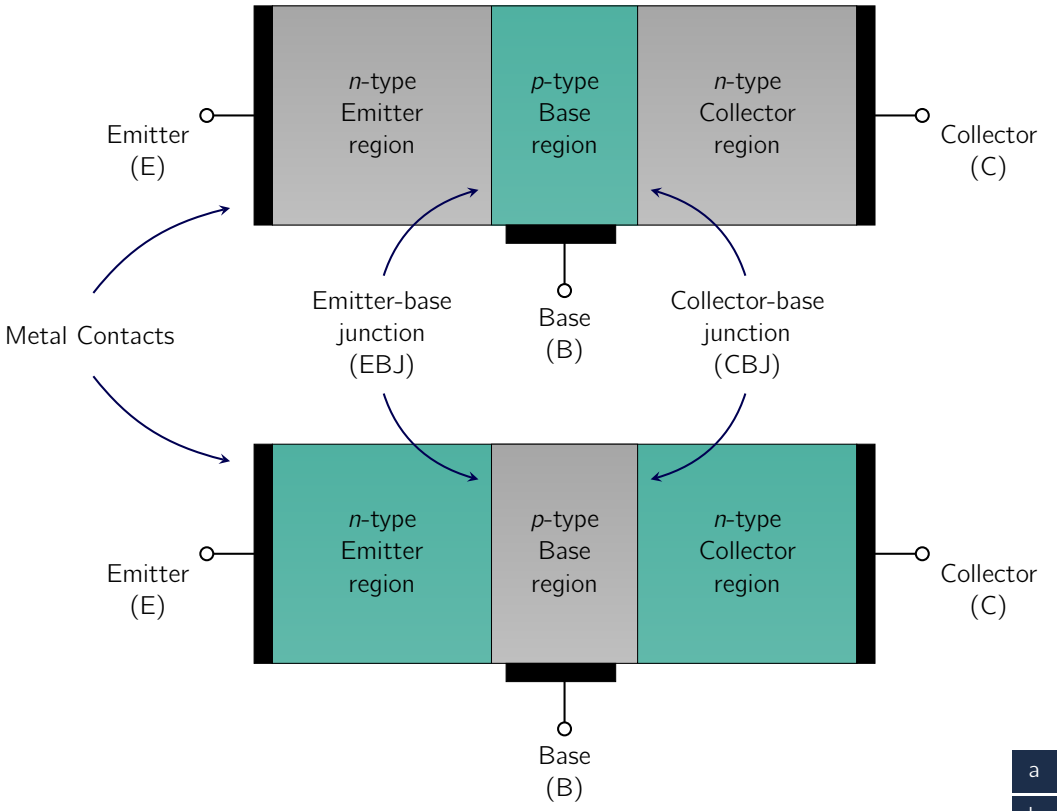


Figure 6.1
(a) A simplified structure of the npn transistor. (b) A simplified structure of the pnp transistor.

- the **emitter** region (n type),
- the **base** region (p type), and
- the **collector** region (n type).

Such a transistor is called an *npn* transistor. Another transistor, a dual of the *npn* as shown in **Fig.** 6.1b, has a p-type emitter, an n-type base, and a p-type collector, and is appropriately called a *pnp* transistor.

A terminal is connected to each of the three semiconductor regions of the transistor, with the terminals labeled **emitter** (E), **base** (B), and **collector** (C).

The transistor consists of two (2) *pn* junctions, the **emitter-base junction** (EBJ) and the collector-base junction (CBJ). Depending on the bias condition[5] of each of these junctions, different modes of operation of the BJT are obtained, as shown in **Tbl.** 6.1.

[5] which can either be **forward** or **reverse**.

The **active mode** is the one used if the transistor is to operate as an amplifier. Switching applications (e.g., logic circuits) utilize both the **cutoff mode** and the **saturation** mode.

In **cutoff mode** no current flows as both junctions are reverse biased.

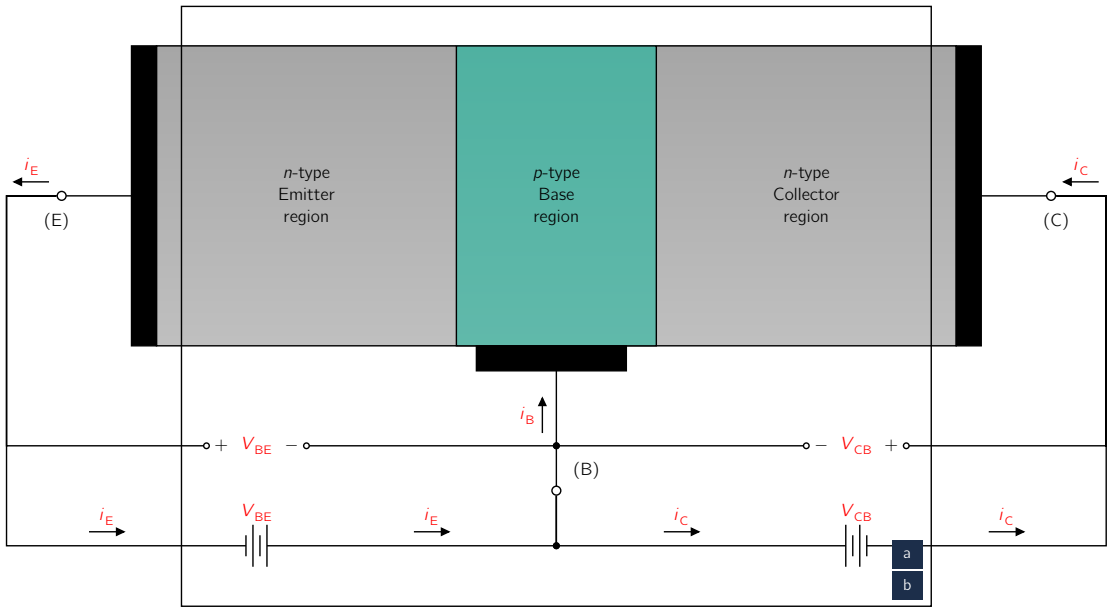


Figure 6.2 Current flow in an npn transistor biased to operate in the active mode. (Reverse current components due to drift of thermally generated minority carriers are not shown.)

As we will see shortly, charge carriers of both polarities;

electrons and holes,

participate in the current-conduction process in a BJT, which is the reason for the name bipolar.[6]

[6] This should be contrasted with the situation in the MOSFET, where current is conducted by charge carriers of one type only: electrons in *n*-channel devices or holes in *p*-channel devices. In earlier days, some referred to FETs as **unipolar devices**.

6.1.2 Operation of the npn Transistor in the Active Mode

Of the three (**3**) modes of operation of the BJT, the active mode is probably the most important and relevant to our use case. Therefore, it is in our best interest to begin our study of the BJT by considering its physical operation in the active mode.[7] This situation is illustrated in **Fig.** 6.2 for the *npn* transistor. Two external voltage sources[8] are used to establish the **required bias conditions** for active-mode operation. The base-emitter voltage (V_{BE}) causes the *p*-type base to be higher in potential than the *n*-type emitter, thus **forward biasing the emitter-base junction**. The collector-base voltage V_{CB} causes the *n*-type collector to be at a higher potential than the *p*-type base, thus reverse biasing the collector-base junction.

[7] To get the most out of this chapter, please refresh your knowledge on semiconductors, which we have covered in Chapter 3.

[8] which are shown in the figure as batteries.

Current Flow

Table 6.1 The operation modes of a bipolar-junction transistor.

Mode	EBJ	CBJ
Cutoff	Reverse	Reverse
Active	Forward	Reverse
Saturation	Forward	Forward

The forward bias on the emitter-base junction will cause current to flow across this junction. Current will consist of two (2) primary components:

1. electrons injected from the emitter into the base, and
2. holes injected from the base into the emitter.

As will become apparent shortly, it is highly desirable to have the first component[9] be much larger than the second component .[10] This can be accomplished by **fabricating the device with a heavily doped emitter and a lightly doped base**.

[9] electrons from emitter to base.

[10] holes from base to emitter.

> This makes the device to have a high density of electrons in the emitter and a low density of holes in the base.

The current that flows across the emitter-base junction will constitute the emitter current i_E, as indicated in **Fig.** 6.2. The direction of i_E is **out of** the emitter lead, which, following the usual conventions, is in the direction of the positive-charge flow (hole current) and opposite to the direction of the negative-charge flow (electron current), with the emitter current i_E being equal to the sum of these two (2) components.

> However, since the electron component is much larger than the hole component, the emitter current will be dominated by the electron component.

From our study in Section 3.4 of the current flow across a forward-biased *pn*-junction, we know that the magnitude of both the electron component and the hole component of i_E will be proportional to

$$\exp\left(\frac{v_{BE}}{V_T}\right)$$

where v_{BE} is the forward voltage across the base-emitter junction and V_T is the thermal voltage, which is approximately 25 mV at room temperature.

Let's now focus our attention on the first current component, namely, that carried by electrons injected from the emitter into the base. These electrons will be minority carriers in the *p*-type base region. Because their concentration will be highest at the emitter side of the base, the injected electrons will *diffuse through* the base region toward the collector. In their journey across the base, some of the electrons will combine with holes, which are majority carriers in the base.

However, since the base is usually very thin and, as mentioned earlier, lightly doped, the proportion of electrons that are "lost" through this recombination process will be quite small. Thus, most of the diffusing electrons will reach the boundary of the collector-base depletion region. Because the collector is more positive than the base,[11] these successful electrons will be swept across the CBJ depletion region into the collector. They will thus get collected and constitute the collector current i_C.

[11] by the reverse-bias voltage v_{CB}.

Collector Current

From the aforementioned statements, we see the collector current (i_C) is carried by the electrons that reach the collector region. Its direction will be opposite to that of the flow of electrons, and thus into the collector terminal. Its magnitude will be proportional to $exp\left(v_{BE}/V_T\right)$, therefore we can derive the following relation:

$$i_C = I_S \exp\left(\frac{v_{BE}}{V_T}\right)$$

where the constant of proportionality (I_S), as in the case of the diode, is called the **saturation current** and is a transistor parameter.[12] An important observation to make here is that i_C is independent of the value of v_{CB}.

[12] We will have more to say about I_S shortly.

> As long as the collector is positive with respect to the base, the electrons that reach the collector side of the base region will be swept into the collector and will register as collector current.

The Base Current

Reference to **Fig.** 6.2 shows that the base current i_B is composed of two components.

1. The first component i_{B1} is due to the holes injected from the base region into the emitter region. This current component is proportional to e^{v_{BE}/V_T}.
2. The second component of base current, i_{B2}, is due to holes that have to be supplied by the external circuit in order to replace the holes lost from the base through the recombination process.

Because i_{B2} is proportional to the number of electrons injected into the base, it also will be proportional to e^{v_{BE}/V_T}. Thus the total base current, $i_B = i_{B1} + i_{B2}$, will be proportional to e^{v_{BE}/V_T}, and can be expressed as a fraction of the collector current i_c as follows:

$$i_B = \frac{i_C}{\beta} = \left(\frac{I_S}{\beta}\right) e^{v_{BE}/V_T} \quad [A] \quad (\mathbf{6.1})$$

where β is a transistor parameter.

For modern **npn** transistors, β is in the range 50 to 200, but it can be as high as 1000 for special devices. For reasons that will become clear later, the parameter β is called the **common-emitter current gain**. The above description indicates that the value of β is highly influenced by two factors:

- the width of the base region W, and
- the relative dopings of the base region and the emitter region, N_A/N_p.

To obtain a high β[13] the base should be thin, by making W small, and lightly doped and the emitter heavily doped, by making N_A/N_D small.

[13] which is highly desirable since β represents a gain parameter.

> For modern integrated circuit fabrication technologies, W is in the **nanometer range**.

Emitter Current

Since the current that enters a transistor must leave it, it can be seen from **Fig.** 6.2 that the emitter current i_E is equal to the sum of the collector current i_C and the base current i_B; that is,

$$i_E = i_C + i_B \tag{6.2}$$

Use of Eq. (**6.1**) and Eq. (**6.2**) gives

$$i_E = \frac{\beta + 1}{\beta} i_C \qquad \text{where} \qquad i_E = \frac{\beta + 1}{\beta} I_S e^{v_{BE}/V_T} \tag{6.3}$$

Alternatively, we can express Eq. (**6.3**) in the form

$$i_C = \alpha i_E \qquad \text{where} \qquad \alpha = \frac{\beta}{\beta + 1} \tag{6.4}$$

Thus the emitter current in Eq. (**6.3**) can be written

$$i_E = (\frac{I_S}{\alpha}) e^{v_{BE}/V_T}$$

Finally, we can use Eq. (**6.4**) to express β in terms of α, that is,

$$\beta = \frac{\alpha}{1 - \alpha} \tag{6.5}$$

It can be seen from Eq. (**6.4**) that α is a constant for a particular transistor that is less than but very close to unity.

For instance, if $\beta = 100$, then $\alpha \simeq 0.99$.

Eq. (**6.5**) reveals an important fact:

> Small changes in α correspond to very large changes in β.

This mathematical observation manifests itself physically, with the result that transistors of the same type may have widely different values of β. For reasons that will become apparent later, α is called the **common-base current gain**.

Minority Carrier Distribution

Our understanding of the physical operation of the BJT can be enhanced by considering the distribution of minority charge carriers in the base and the emitter. Figure 6.4 shows the profiles of the concentration of electrons in the base and holes in the emitter of an *npn* transistor operating in the **active mode**. Observe that since the doping concentration in the emitter, N_D, is much higher than the doping concentration in the base, N_A, the concentration of electrons injected from emitter to base, $n_p(0)$, is much higher than the concentration of holes injected from the base to the emitter, $p_n(0)$.

Both quantities are proportional to e^{v_{BE}/V_T}, thus

$$n_p(0) = n_{p0} e^{v_{BE}/V_T} \tag{6.6}$$

where n_{n0} is the thermal-equilibrium value of the minority-carrier, i.e., electron, concentration in the base region. Next, observe that because the base is very thin, the concentration of excess electrons decays almost linearly.[14] Furthermore, the reverse bias on the collector-base junction causes the concentration of excess electrons at the collector side of the base to be zero.[15]

[14] As opposed to the usual exponential decay, as observed for the excess holes in the emitter region.

[15] Recall that electrons that reach that point are swept into the collector.

The tapered minority-carrier concentration profile (Fig. 6.4) causes the electrons injected into the base to diffuse through the base region toward the collector. This electron diffusion current I_n is directly proportional to the slope of the straight-line concentration profile,

$$I_n = A_E q D_n \frac{dn_p(x)}{dx} = A_E q D_n \left(-\frac{n_p(0)}{W}\right) \tag{6.7}$$

where A_E is the cross-sectional area of the base-emitter junction,[16] q is the magnitude of the electron charge, D_n is the electron diffusivity in the base, and W is the effective width of the base.

[16] in the direction perpendicular to the page.

Please observe that the **negative slope** of the minority-carrier concentration results in a negative current I_n across the base. That is, I_n flows from right to left, in the negative direction of x, which corresponds to the usual convention, namely, opposite to the direction of electron flow.

The recombination in the base region, though slight, causes the excess minority-carrier concentration profile to deviate from a straight line and take the slightly concave shape indicated by the broken line in Fig. 6.4. The slope of the concentration profile at the EBJ is slightly higher than that at the CBJ, with the difference accounting for the small number of electrons lost in the base region through recombination.

Finally, we have the collector current $i_C = I_n$, which will give a negative value for i_C, indicating that i_C flows in the negative direction of the x axis.[17] Since we will take this to be the positive direction of i_C, we can drop the negative sign in Eq. (**6.7**). Doing this and substituting for $n_p(0)$ from Eq. (**6.6**), we can therefore express the collector current i_c as

[17] i.e., from right to left.

$$i_C = I_S e^{v_{BE}/V_T}$$

where the saturation current I_S is given by

$$I_S = A_E q D_n n_{p0}/W$$

Substituting $n_{p0} = n_i^2/N_A$, where n_i is the intrinsic carrier density and N_A is the doping concentration in the base, we can express I_S as

$$I_S = \frac{A_E q D_n n_i^2}{N_A W}$$

The saturation current I_S is inversely proportional to the base width W and is directly proportional to the area of the EBJ.

Typically I_S is in the range of 1×10^{-12} A to 1×10^{-18} A, depending on the size of the device. Because I_S is proportional to n_i^2, it is a strong function of temperature, approximately doubling for every 5řC rise in temperature.

> For the dependence of n_i^2 on **temperature**, refer to Eq. (**3.1**).

Since I_S is directly proportional to the junction area, it will also be referred to as the **scale current**. Two transistors that are identical except that one has an EBJ area, say, twice that of the other will have saturation currents with that same ratio (i.e., 2).

Therefore for the same value of v_{BE} the larger device will have a collector current twice that in the smaller device.

> This concept is frequently employed in **integrated-circuit design**.

Recapitulation and Equivalent-Circuit Models

We have presented a first-order model for the operation of the *npn* transistor in the **active mode**.

Basically, the forward-bias voltage v_{BE} causes an exponentially related current i_C to flow in the collector terminal. The collector current i_C is independent of the value of the collector voltage as long as the collector-base junction remains reverse biased; that is, $v_{CB} \geqslant 0$. Therefore in the **active mode** the collector terminal behaves as an ideal constant-current source where the value of the current is determined by v_{BE}. The base current i_B is a factor $1/\beta$ of the collector current, and the emitter current is equal to the sum of the collector and base currents. Since i_B is much smaller than i_C (i.e., $\beta \gg 1$), $i_E \simeq i_C$. More precisely, the collector current is a fraction α of the emitter current, with α smaller than, but close to, unity.

This first-order model of transistor operation in the active mode can be represented by the equivalent circuit shown in Fig. 6.5(a). Here, diode DE has a scale current ISE equal to (I_s/α) and thus provides a current i_E related to v_{BE} according to Eq. (6.9). The current of the controlled source, which is equal to the collector current, is controlled by v_{RF} according to the exponential relationship indicated, a restatement of Eq. (6.1). This model is in essence a nonlinear voltage-controlled current source. It can be converted to the current-controlled current-source model shown in Fig. 6.5(b) by expressing the current of the controlled source as αi_E. Note that this model is also nonlinear because of the exponential relationship of the current i_E through diode D_E and the voltage v_{BE}. From this model we observe that if the transistor is used as a two-port network with the input port between E and B and the output port between C and B (i.e., with B as a common terminal), then the current gain observed is equal to α. Thus α is called the common-base current gain.

Two other equivalent-circuit models, shown in Fig. 6.5(c) and (d), may be used to represent the operation of the BJT. The model of Fig. 6.5(c) is essentially a voltage-controlled current source. However, here diode DB conducts the base current and thus its current scale factor is I_s/β, resulting in the $i_B - v_{BE}$ relationship given in Eq. (6.3). By simply expressing the collector current as βi_R we obtain the current-controlled current-source model shown in Fig. 6.5(d). From this latter model we observe that if the transistor is used as a two-port network with the input port between B and E and the output port between C and E (i.e., with E as the common terminal), then the current gain observed is equal to β. Thus β is called the common-emitter current gain.

Figure 6.5 Large-signal equivalent-circuit models of the **npn** BJT operating in the forward active

mode.

Finally, we note that the models in Fig. 6.5 apply for any positive value of v_{BE}. That is, unlike the models we will be discussing in Chapter 7, here there is no limitation on the size of v_{BE}, and thus these models are referred to as **large-signal models**.

6.1 Exercise

An npn Transistor

An *npn* transistor having $I_S = 10^{-15}$ A and $\beta = 100$ is connected as follows:

The emitter is **grounded**, the base is fed with a constant-current source supplying a DC current of 10 μA, and the collector is connected to a 5 V DC supply via a resistance R_C of 3 kΩ. Assuming the transistor is operating in the active mode, find V_{BE} and V_{CE}.

Use these values to verify active-mode operation. Replace the current source with a resistance connected from the base to the 5 V dc supply.

What resistance value is needed to result in the same operating conditions?

If the transistor is operating in the active mode, it can be represented by one of the four possible equivalent-circuit models. Because the emitter is grounded, either the model in Fig. 6.5(c) or that in Fig. 6.5(d) would be suitable.

Given we know the base current I_B, the model of Fig. 6.5(d) is the most suitable.

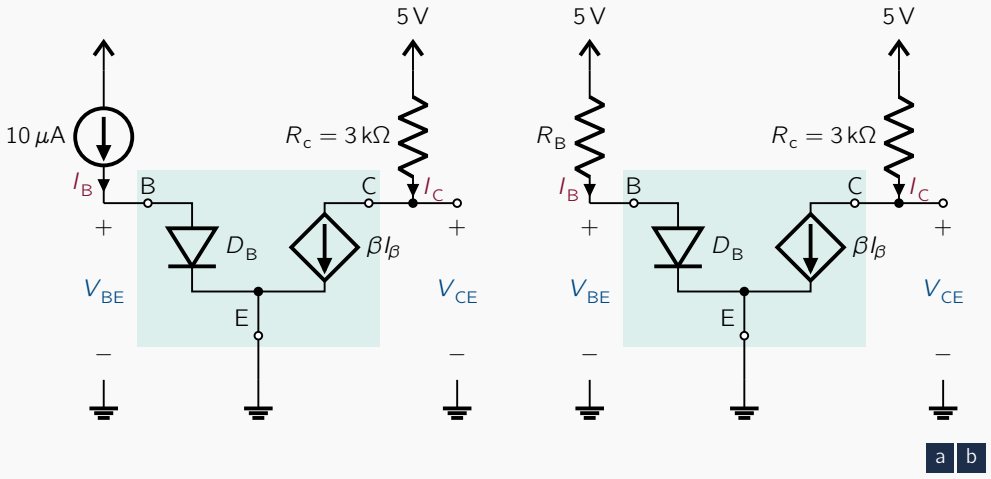


Figure 6.3 Circuits for the question "An npn Transistor". (a) The circuit with a constants current source attached (b) The constant current source is replaced with a resistance (R_B) connected to a 5 V.

Fig. 6.3 shows the circuit as described with the transistor represented by the model of Fig. 6.5(*d*). We can determine V_{BE} from the exponential characteristic of D_B as follows:

$$V_{BE} = V_T \ln \frac{I_B}{I_S/\beta} = 25 \ln \left(\frac{10 \times 10^{-6}}{10^{-17}} \right)$$

$$= 690\,\text{mV} = 0.69V$$

Continues on next page →

← Continued from previous page

Next we determine the value of V_{CE} from

$$V_{CE} = V_{CC} - R_C I_C \qquad \text{where} \qquad I_C = \beta I_B = 100 \times 10 \times 10^{-6} = 10^{-3} A = 1mA$$

Therefore,

$$V_{CF} = 5 - 3 \times 1 = +2V$$

Since V_C at +2 V is higher than V_B at 0,69 V, the transistor is indeed operating in the active mode. Now, replacing the 10-μA current source with a resistance R_B connected from the base to the 5-V dc supply V_{cc}, as in Fig. 6.6(b), the value of R_B must be

$$R_B = \frac{V_{CC} - V_{BE}}{I_B} = \frac{5 - 0.69}{10\mu A} = 431k \quad \blacksquare$$

6.1.3 Structure of Actual Transistor

Fig. 6.4 shows a more realistic, but still simplified, cross section of an *npn*-type BJT. Please observe that, the collector virtually **surrounds** the emitter region, thus making it difficult for the electrons injected into the thin base to escape being collected. In this way, the resulting α is close to unity[18] and β is large.

[18] i.e., $\alpha \approx 1$.

In addition, observe the device is **NOT** symmetrical, and therefore the emitter and collector cannot be interchanged.[19] The structure in **Fig.** 6.4 indicates also that the CBJ has a much larger area than the EBJ.

[19] If the emitter and collector are reversed-that is, the CBJ is forward biased and the EBJ is reverse biased-the device operates in a mode called the "reverse-active mode." The resulting values of α and β, denoted α_R and β_R (with R denoting reverse), are much lower than the values of α and β, respectively, obtained in the "forward"-active mode discussed above. Hence, the reverse-active mode has no practical application. The MOSFET, on the other hand, having a perfectly symmetrical device, can operate equally well with its drain and source terminals interchanged.

Therefore, the CB diode D_C has a saturation current I_{SC} that is much larger than the saturation current of the EB diode D_E.

Typically, I_{SC} is 10 to 100 times larger than I_{SE}, recalling that

$$I_{SE} = I_S/\alpha \simeq I_S$$

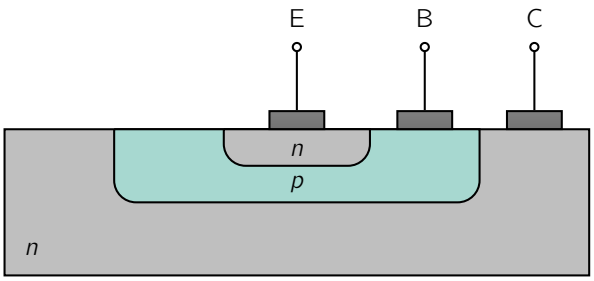


Figure 6.4 Cross-sectional view of an npn BJT.

6.1.4 Operation in the Saturation Mode

As previously mentioned, for BJT to operate in the **active mode**, the CBJ must be **reverse biased**. Till now, we have stated this condition for the *npn* transistor as $v_{CB} \geqslant 0$. However, we know that a *pn*-junction does **NOT** effectively become forward biased until the forward voltage across it exceeds approximately 0,4 V. It follows that one can maintain active-mode operation of an *npn* transistor for negative v_{CB} down to approximately -0,4 V. This is illustrated in Fig. 6.8, which is a sketch of i_C versus v_{CB} for an *npn* transistor operated with a constant emitter current I_E. As expected, i_C is independent of v_{CB} in the active mode, a situation that extends for v_{CB} going negative to approximately 0,4 V.

> Below this value of v_{CB}, the CBJ begins to conduct sufficiently that the transistor leaves the active mode and **enters the saturation mode** of operation, where i_C decreases.

To see why i_C decreases in saturation, we can construct a model for the saturated *npn* transistor as follows.

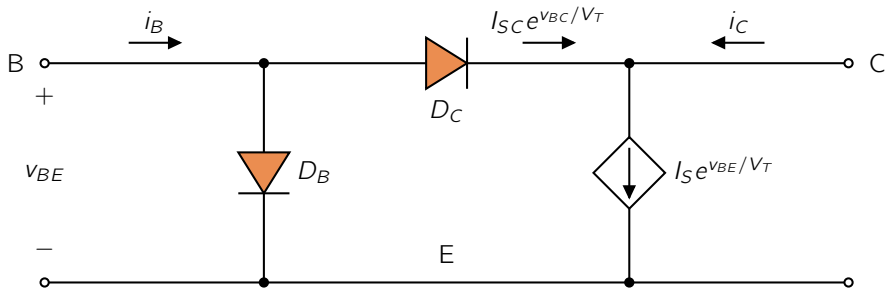


Figure 6.5 Modeling the operation of an npn transistor in saturation by augmenting the model of Fig. 6.5(c) with a forward-conducting diode DC . Note that the current through DC increases iB and reduces iC.

We augment the model of Fig. 6.5(c) with the forward-conducting CBJ diode DC , as shown in Fig. 6.9. Observe that the current iBC will subtract from the controlled-source current, resulting in the reduced collector current iC given by

$$i_C = I_S \exp\left(v_{BE}/V_T\right) - I_{SC} \exp\left(v_{BC}/V_T\right)$$

where ISC is the saturation current for DC and is related to IS by the ratio of the areas of the CBJ and the EBJ. The second term in Eq. (6.14) will play an increasing role as v BC exceeds 0.4 V or so, causing iC to decrease and eventually reach zero. Figure 6.9 also indicates that in saturation the base current will increase to the value iB = (IS /)ev BE /VT + ISC evBC /V T (6.15) Equations (6.14) and (6.15) can be combined to obtain the ratio iC /iB for a saturated transistor. We observe that this ratio will be lower than the value of . Furthermore, the ratio will decrease as v BC is increased and the transistor is driven deeper into saturation. Because iC /i

of a saturated transistor can be set to any desired value lower than β by adjusting v_{BC}, this ratio is known as **forced** β and denoted β_{forced},

$$\beta_{\text{forced}} = \left.\frac{i_C}{i_R}\right|_{\text{exturation}} \leqslant \beta$$

As we will see later, in analysing a circuit we can determine whether BJT is in saturation mode by either of the following two (2) tests:

1. Is the CBJ forward biased by more than 0,4 V?
2. Is the ratio i_C/i_B lower than β?

The collector-to-emitter voltage v_{CE} of a saturated transistor can be found from Fig. 6.9 as the difference between the forward-bias voltages of the EBJ and the CBJ,

$$V_{CEsat} = V_{BE} - V_{BC}$$

We know the CBJ has a much larger area than the EBJ. This property translates to V_{BC} being smaller than V_{BE} by 0,1 V to 0,3 V.

Therefore we can state:

$$V_{CESat} \simeq 0{,}1\,\text{V to } 0{,}3\,\text{V}.$$

Typically we assume a transistor at the edge of saturation has $V_{CESat} = 0{,}3\,\text{V}$, whereas a transistor deep in saturation has $V_{CESat} = 0{,}2\,\text{V}$.

6.1.5 The pnp Transistor

The *pnp* transistor operates in a similar manner to that of the *npn* device described previously where Figure 6.10 shows a *pnp* transistor biased to operate in the active mode.

Here the voltage V_{EB} causes the *p*-type emitter to be higher in potential than the *n*-type base, therefore **forward biasing** the emitter-base junction. The collector-base junction is **reverse biased** by the voltage V_{BC}, which keeps the *p*-type collector lower in potential than the *p*-type base.

Unlike the *npn* transistor, however, current in the *pnp* device is mainly conducted by holes injected from the emitter into the base as a result of the forward-bias voltage V_{EB}. Since the component of emitter current contributed by electrons injected from base to emitter is kept small by using a lightly doped base, most of the emitter current will be due to holes. The electrons injected from base to emitter give rise to the first component of base current, i_{B1}. Also, a number of the holes injected into the base will recombine with the majority carriers in the base (electrons) and will thus be lost. The disappearing base electrons will have to be replaced from the external circuit, giving rise to the second component of base current, i_{B2}. The holes that succeed in reaching the boundary of the depletion region of the collector-base junction will be attracted by the negative voltage on the collector. Thus these holes will be swept across the depletion region into the collector and appear as collector current.

It can easily be seen from the above description that the *i-v* relationship of the *pnp* transistor will be identical to that of the *npn* transistor except that v_{BE} has to be replaced by v_{EB}.

Also, the large-signal, active-mode operation of the *pnp* transistor can be modeled by any of four equivalent circuits similar to those for the npn transistor in Fig. 6.5. Two of these four circuits are shown in Fig. 6.11.

Finally, we note that the *pnp* transistor can operate in the saturation mode in a manner analogous to that described for the *npn* device.

Figure 6.11 Two large-signal models for the **pnp** transistor operating in the active mode.

6.2 Information

The Invention of BJT

The first working transistor was demonstrated at the Bell Labs in late 1947 by John Bardeen and Walter Brattain, who were part of a team led by *William Shockley*. Made of germanium, the device became known as a point-contact transistor and operated on the field-effect principle. Within a few weeks, however, Shockley wrote a complete description of the bipolar junction transistor (BJT) and filed for a U.S. patent with the title "Circuit Element Utilizing Semiconductor Material." BJTs dominated the electronics world from the early 1950s to the mid-1970s, when MOSFETs took over the leading position. In 1956, Shockley, Bardeen, and Brattain shared the Nobel Prize in Physics for the discovery of the transistor effect.

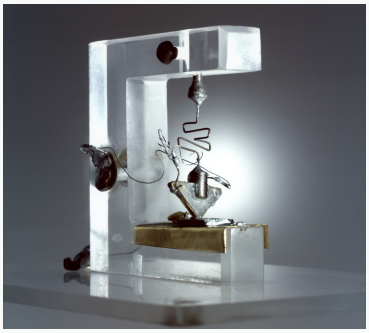
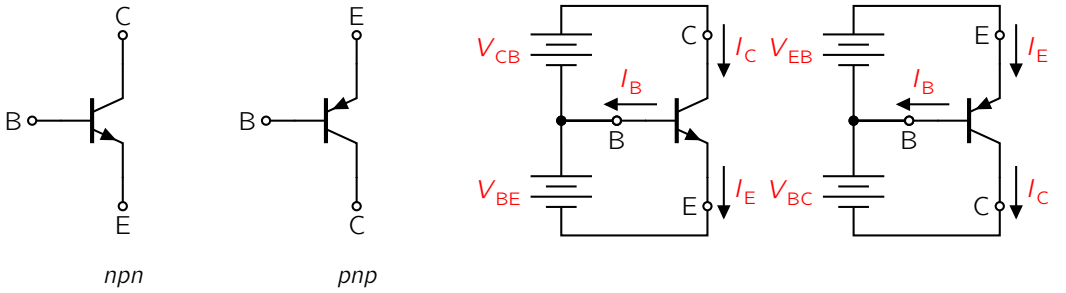


Figure 6.6
A replica of the first transistor made from germanium [5].

Figure 6.7
x



a b c d

6.2 The i-v Characteristics

6.2.1 Symbols and Conventions

The physical structure used up to now explain transistor operation is rather cumbersome for use in drawing the schematic of a multitransistor circuit. Fortunately, a very descriptive and convenient circuit symbol exists for the BJT. Figure 6.12(a) shows the symbol for the **npn** transistor; the **pnp** symbol is given in Fig. 6.12(b). In both symbols the emitter is distinguished by an arrowhead. This distinction is important because, as we have seen in the last section, practical BJTs are not symmetric devices.

The polarity of the device-**npn** or **pnp**-is indicated by the **direction of the arrowhead on the emitter**. This arrowhead points in the direction of normal current flow in the emitter, which is also the forward direction of the base-emitter junction. Since we have adopted a drawing convention by which currents flow from top to bottom, we will always draw pnp transistors in the manner shown in Fig. 6.12(b) (i.e., with their emitters on top). Figure 6.13 shows **npn** and **pnp** transistors connected to dc sources so as to operate in the active mode. Figure 6.13 also indicates the reference and actual directions of current flow throughout the transistor. Our convention will be to take the reference direction to coincide

with the normal direction of current flow. Hence, normally, we should not encounter a negative value for iE , iB, or iC. The convenience of the circuit-drawing convention that we have adopted should be obvious from Fig. 6.13. Note that currents flow from top to bottom and that voltages are higher at the top and lower at the bottom. The arrowhead on the emitter also implies the polarity of the emitter-base voltage that should be applied in order to forward bias the emitter-base junction. Just a glance at the circuit symbol of the pnp transistor, for example, indicates that we should make the emitter higher in voltage than the base (by vEB) in order to cause current to flow into the emitter (downward). Note that the symbol vEB means the voltage by which the emitter (E) is higher than the base (B). Thus for a pnp transistor operating in the active mode vEB is positive, while in an

npn transistor vBE is positive. From the discussion of Section 6.1 it follows that an npn transistor whose EBJ is forward biased (usually, VBE 0.7 V) will operate in the active mode as long as the collector voltage does not fall below that of the base by more than approximately 0.4 V. Otherwise, the transistor leaves the active mode and enters the saturation region of operation.5 In a parallel manner, the pnp transistor will operate in the active mode if the EBJ is forward biased (usually, VEB 0.7 V) and the collector voltage is not allowed to rise above that of the base by more than 0.4 V or so. Otherwise, the CBJ becomes forward biased, and the pnp transistor enters the saturation region of operation.

Collector Base Reverse Current 6.3 Information

In our study of current flow in transistors we ignored the small reverse currents carried by thermally generated minority carriers. While such currents can be safely neglected in modern transistors, the reverse current across the collector-base junction deserves some mention. This current, denoted as I_{CBO}, is the reverse current flowing from collector to base with the emitter open-circuited.[20] This current is usually in the nA range, a value that is many times higher than its theoretically predicted value. As with the diode reverse current, I_{CBO} contains a substantial leakage component, and its value is dependent on v_{CB}. I_{CBO} depends strongly on temperature, approximately doubling for every 10řC rise.⁶ ⁶The temperature coefficient of I_{CBO} is different from that of I_s because I_{CBO} contains a substantial leakage component.

[20] This is why we use the subscript O.

6.2.2 Graphical Representation of Transistor Characteristics

It is sometimes useful to describe the transistor $i - v$ characteristics graphically. Figure 6.16 shows the $i_c - v_{BE}$ characteristic, which is the exponential relationship $i_C = I_s e^{v_{BE}/V_T}$ which is identical to the diode *i*-**v** relationship. The $i_E - v_{BE}$ and $i_B - v_{BE}$ characteristics are also exponential but with different scale currents: I_s/α for i_E, and I_s/β for i_B. Since the constant of the exponential characteristic, $1/V_T$, is quite high ($\simeq 40$), the curve rises very sharply. For v_{BE} smaller than about 0.5 V, the current is negligibly small. Also, over most of the normal current range v_{BE} lies in the range of 0.6 V to 0.8 V. In performing rapid first-order dc calculations, we normally will assume that $V_{BE} \simeq 0.7$ V, which is similar to the approach used in the analysis of diode circuits (Chapter 4). For a **pnp** transistor, the $i_c - v_{EB}$ characteristic will look identical to that of Fig. 6.16 with v_{BE} replaced with v_{EB}.

Figure 6.16 The $i_c - v_{BE}$ characteristic for an **npn** **Figure 6.17** Effect of temperature on the $i_c - v_{BE}$

As in silicon diodes, the voltage across the emitter-base junction decreases by about 2 mV for each rise of 1řC in temperature, provided the junction is operating at a constant current. Figure 6.17 illustrates this temperature dependence by depicting $i_c - v_{BE}$ curves for an **npn** transistor at three different temperatures.

6.2.3 Dependence of ic on the Collector Voltage - The early Effect

When operated in the active region, practical BJTs show some dependence of the collector current on the collector voltage, with the result that, unlike the graph shown in Fig. 6.8, their $i_C - v_{CB}$ characteristics are not perfectly horizontal straight lines. To see this dependence more

Figure 6.18 (a) Conceptual circuit for measuring the $i_c - v_{CE}$ characteristics of the BJT. (b) The $i_c - v_{CE}$

clearly, consider the conceptual circuit shown in Fig. 6.18(a). The transistor is connected in the common-emitter configuration; that is, here the emitter serves as a common terminal between the input and output ports. The voltage V_{BE} can be set to any desired value by adjusting the dc source connected between base and emitter. At each value of V_{BE}, the corresponding $i_c - v_{CE}$ characteristic curve can be measured point by point by varying the dc source connected between collector and emitter and measuring the corresponding collector current. The result is the family of $i_c - v_{CE}$ characteristic curves shown in Fig. 6.18(b) and known as common-emitter characteristics.

At low values of v_{CE} (lower than about 0.3 V), as the collector voltage goes below that of the base by more than 0.4 V, the collector-base junction becomes forward biased and the transistor leaves the active mode and enters the saturation mode. Shortly, we shall look at the details of the i_c-v_{CE} curves in the saturation region. At this time, however, we wish to examine the characteristic curves in the active region in detail. We observe that the characteristic curves, though still straight lines, have finite slope. In fact, when extrapolated, the characteristic lines meet at a point on the negative v_{CE} axis, at $v_{CE} = -V_A$. The voltage V_A, a positive number, is a parameter for the particular BJT, with typical values in the range of 10 V to 100 V. As noted earlier, it is called the **Early voltage**, after J. M. Early, the engineering scientist who first studied this phenomenon.

At a given value of v_{BE}, increasing v_{CE} increases the reverse-bias voltage on the collector-base junction, and thus increases the width of the depletion region of this junction (refer to Fig. 6.4). This in turn results in a decrease in the **effective base width** W. Recalling that I_s is inversely proportional to W (Eq. 6.13), we see that I_s will increase and that i_c increases proportionally. This is the Early effect. For obvious reasons, it is also known as the base-width modulation effect.⁸ βRecall that the MOSFET's counterpart is the channel-length modulation effect. These two effects are remarkably similar and have been assigned the same name, Early effect.

The linear dependence of i_c on v_{CE} can be explicitly accounted for by assuming that I_s remains constant and including the factor $(1 + v_{CE}/V_A)$ in the equation for i_C as follows:

$$i_C = I_S e^{v_{BE}/V_T}\left(1 + \frac{v_{CE}}{V_A}\right)$$

The nonzero slope of the $i_c - v_{CE}$ straight lines indicates that the **output resistance** looking into the collector is not infinite. Rather, it is finite and defined by

$$r_o \equiv \left|\frac{\partial i_C}{\partial v_{CE}}\bigg|_{v_{pc}=\text{constant}}\right|^{-1}$$

Using Eq. (6.18) we can show that

$$r_o = \frac{V_A + V_{CE}}{I_C}$$

where I_c and V_{CE} are the coordinates of the point at which the BJT is operating on the particular $i_c - v_{CE}$ curve (i.e., the curve obtained for v_{BE} equal to constant value V_{BE} at which Eq. (6.19) is evaluated). Alternatively, we can write

$$r_o = \frac{V_A}{I'}$$

where I'_c is the value of the collector current with the Early effect neglected; that is,

$$I'_C = I_s e^{V_{BE}/V_T}$$

It is rarely necessary to include the dependence of i_c on v_{CE} in dc bias design and analysis that is performed by hand. Such an effect, however, can be easily included in the SPICE simulation of circuit operation, which is frequently used to "fine-tune" pencil-and-paper analysis or design. The finite output resistance ro can have a significant effect on the gain of transistor amplifiers. This is particularly the case in integrated-circuit amplifiers, as will be shown in Chapter 8. Fortunately, there are many situations in which r_a can be included relatively easily in pencil-and-paper analysis. The output resistance ro can be included in the circuit model of the transistor.⁹ This is illustrated in Fig. 6.19, where we show the two large-signal circuit models of a

Figure 6.19 Large-signal, equivalent-circuit models of an npn BJT operating in the active mode in the common-emitter configuration with the output resistance r_a included.

common-emitter **npn** transistor operating in the active mode, those in Fig 6.5(c) and (d), with the resistance re connected between the collector and the emitter terminals.

6.2.4 An Alternative Form of the Common-Emitter Characteristics

An alternative way of expressing the transistor common-emitter characteristics is illustrated in Fig. 6.20. Here the base current i_B rather than the base-emitter voltage v_{BE} is used as a parameter. That is, each $i_c - v_{CE}$ curve is measured with the base fed with a constant current IB. The resulting characteristics, shown in Fig. 6.20(b), look similar to those in Fig. 6.18. Figure 6.20(c) shows an expanded view of the characteristics in the saturation region.

The Common-Emitter Current Gain β In the active region of the characteristics shown in Fig. 6.20(b) we have identified a particular point Q. Note that this operating point for the transistor is characterized by a base current I_B, a collector current I_C, and a collector-emitter voltage V_{CE}. The ratio I_C/I_B is the transistor β. However, there is another way to measure β: change the base current by an increment Δi_R and measure the resulting increment Δi_C, while keeping V_{CE} constant. This is illustrated in Fig. 6.20(b). The ratio $\Delta i_C/\Delta i_B$ should, according to our study thus far, yield an identical value for β. It turns out, however, that the latter value of β (called **incremental**, or ac, β) is a little different from the dc β (i.e., I_c/I_B). Such a distinction, however, is too subtle for our needs in this book. We shall use β to denote both dc and incremental values.¹⁰

The Saturation Voltage V_{CEsat} and Saturation Resistance R_{CEsat} Refer next to the expanded view of the common-emitter characteristics in the saturation region shown in Fig. 6.20(c). The "bunching together" of the curves in the saturation region implies that the incremental β is lower there than in the active region. A possible operating point in the saturation region is that labeled X. It is characterized by a base current IB, a collector current I_{Csat}, and a collector-emitter voltage V_{CEsat}. From our previous discussion of saturation, recall that $I_{\text{Cast}} = \beta_{\text{forced}} I_B$, where $\beta_{\text{forced}} < \beta$. ¹⁰Manufacturers of bipolar transistors use h_{FE} to denote the dc value of β and h_{fe} to denote the incremental β. These symbols come from the h-parameter description of two-port networks (see Appendix C), with the subscript $F(f)$ denoting forward and $E(e)$ denoting common emitter.

Figure 6.20 Common-emitter characteristics. (a) Basic CE circuit; note that in (b) the horizontal scale is expanded around the origin to show the saturation region in some detail. A much greater expansion of the saturation region is shown in (c). The i_c-v_{CE} curves in saturation are rather steep, indicating that the saturated transistor exhibits a low collector-to-emitter resistance R_{CEsat},

$$R_{CEsat} \equiv \left.\frac{\partial v_{CE}}{\partial i_C}\right|_{\substack{i_B=I_B \\ i_C=I_{Csa}}}$$

Typically, R_{CEsat} ranges from a few ohms to a few tens of ohms.

Figure 6.21 A simplified equivalent-circuit model of the saturated transistor. That the collector-to-emitter resistance of a saturated BJT is small should have been anticipated from the fact that between C and E we now have two forward-conducting diodes in series¹¹ (see also Fig. 6.9). A simple model for the saturated BJT is shown in Fig. 6.21. Here V_{BE} is assumed constant (approximately 0.7 V) and V_{CE} also is assumed constant, $V_{CEsat} \simeq 0.2$ V. That is, we have neglected the small saturation resistance R_{CEsat} for the sake of making the model simple for hand calculations.

Determining Base Voltage 6.2 Exercise

ex:determining-base-voltage

For the circuit below, we need to determine the value of the voltage V_{BB} that results in the transistor operating:

Continues on next page →

6.2 Exercise

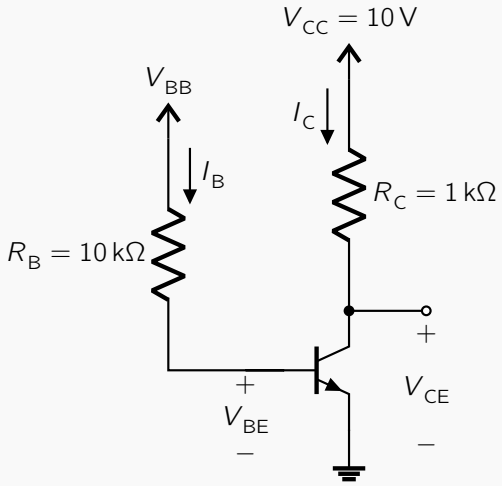


Figure 6.8: Circuit diagram for the question "Determining Base Voltage".

a. in the active mode with $V_{CE} = 5$ V

b. at the edge of saturation

c. deep in saturation with $\beta_{forced} = 10$

To keep things simple, assume that V_{BE} remains constant at 0.7 V. The transistor β is specified to be 50.

The solution is as follows:

a. To operate in the active mode with $V_{CE} = 5$ V,

$$I_c = \frac{V_{cc} - V_{CE}}{R_c} = \frac{10 - 5}{1\ k\Omega} = 5mA$$

$$I_B = \frac{I_C}{\beta} = \frac{5}{50} = 0.1\ mA$$

Now the required value of V_{BB} can be found as follows:

$$V_{BB} = I_B R_B + V_{BE} = 0.1 \times 10 + 0.7 = 1.7V$$

b. Operation at the edge of saturation is obtained with $V_{CE} = 0.3$ V. Thus

$$I_c = \frac{10 - 0.3}{1} = 9.7\ mA$$

Since, at the edge of saturation, I_c and I_B are still related by β,

$$I_B = \frac{9.7}{50} = 0.194\ mA$$

The required value of V_{BB} can be determined as

$$V_{BB} = 0.194 \times 10 + 0.7 = 2.64V$$

---

c. To operate deep in saturation,

$$V_{CF} = V_{CFsat} \simeq 0.2V$$

Thus,

$$I_c = \frac{10 - 0.2}{1} = 9.8\ mA$$

← Continued from previous page

We then use the value of forced β to determine the required value of I_B as

$$I_B = \frac{I_C}{\beta_{\text{forced}}} = \frac{9.8}{10} = 0.98 \text{ mA}$$

and the required V_{BB} can now be found as

$$V_{BB} = 0.98 \times 10 + 0.7 = 10.5V$$

Please observe that once the transistor is in saturation, increasing V_{BB} and thus I_B results in negligible change in I_c since V_{CEsat} will change only slightly. Thus I_c is said to **saturate**, which is the origin of the name "saturation mode of operation".

6.3 BJT Circuits at DC

Like we did with the MOSFET in the previous chapter, we are now ready to consider the analysis of BJT circuits to which only DC voltages are applied. In the following examples we will use the simple model in which $|V_{BE}|$ of a conducting transistor is 0,7 V and $|V_{CE}|$ of a saturated transistor is 0,2 V, and we will neglect the **Early effect**.

Better models can, of course, be used to obtain more accurate results. This, however, is usually achieved at the expense of speed of analysis.

> The additional complexity could impede the circuit designer's ability to gain insight regarding circuit behavior.

Accurate results using elaborate models can be obtained using circuit simulation with SPICE which is almost always done in the final stages of a design and certainly before circuit fabrication.[21] As we will see, in analyzing a circuit the first question that one must answer is:

In which mode is the transistor operating?

In some cases, the answer will be obvious. For instance, a quick check of the terminal voltages will indicate whether the transistor is cut off or conducting. If it is conducting, we have to determine whether it is operating in the active mode or in saturation. In some cases, however, this may not be obvious. Needless to say, as we gain practice and experience in transistor circuit analysis and design, the answer will be apparent in a much larger proportion of problems. The answer, however, can always be determined by utilizing the following procedure.

Assume that the transistor is operating in the **active mode** and, using the active-mode model in Table 6.3, proceed to determine the various voltages and currents that correspond. Then check for consistency of the results with the assumption of active-mode operation; that is, is V_{CB} of an *npn* transistor greater than -0,4 V (or V_{CB} of a *pnp* transistor lower than 0,4 V)? If the answer is yes, then our task is complete. If the answer is no, assume saturation-mode operation and, using the saturation-mode model in Table 6.3, proceed to determine currents and voltages

Table 6.3 Simplified Models for the Operation of the BJT in DC Circuits

and then check for consistency of the results with the assumption of **saturation-mode** operation.

Here the test is usually to compute the ratio I_C/I_B and to verify that it is lower than the transistor β (i.e., $\beta_{\text{forced}} < \beta$). Since β for a given transistor type varies over a wide range,[22] one must use the lowest specified β for this test. Finally, note that the order of these two assumptions can be reversed.

> **A Note on Units** — Except when otherwise specified, in this LectureBook we use a consistent set of units, namely, volts (V), milliamps (mA), and kilohms ($k\Omega$).

[21] Computer simulation, however, is **NOT** a substitute for quick pencil-and-paper circuit analysis, an essential ability that aspiring circuit designers must master.

[22] That is, if one buys BJTs of a certain part number, the manufacturer guarantees only that their values of β fall within a certain range, say 50 to 150.

6.4 Transistor Breakdown and Temperature Effects

We will finish this chapter with a brief discussion of two (2) important non-ideal effects in the BJT:

- voltage breakdown, and
- the dependence of β on I_c and temperature.

6.4.1 Transistor Breakdown

The maximum voltages which can be applied to a BJT are limited by the EBJ and CBJ breakdown effects that follow the **avalanche multiplication mechanism** described in the **Semiconductors** chapter.

Consider first the common-base configuration (Fig. 6.32(a)). The $i_c - v_{CB}$ characteristics in Fig. 6.32(b) indicate that for $i_E = 0$ (i.e., with the emitter open-circuited) the collector-base junction breaks down at a voltage denoted by BV_{CBO}. For $i_E > 0$, breakdown occurs at voltages smaller than BV_{CBO}. Typically, for discrete BJTs, BV_{CBO} is greater than 50 V.

Figure 6.32 The BJT common-base characteristics including the transistor breakdown region.

Figure 6.33 The BJT common-emitter characteristics including the breakdown region.

Next consider the common-emitter characteristics of Fig. 6.33, which show breakdown occurring at a voltage BV_{CEO}. Here, although breakdown is still of the avalanche type, the effects on the characteristics are more complex than in the common-base configuration. We will not explain these in detail; it is sufficient to point out that typically BV_{CEO} is about half BV_{CBO}. On transistor data sheets, BV_{CEO} is sometimes referred to as the sustaining voltage LV_{CEO}.

Breakdown of the CBJ in either the common-base or common-emitter configuration is **NOT** destructive as long as the power dissipation in the device is kept within safe limits.[23] This, however, is **NOT** the case with the breakdown of the emitter-base junction. The EBJ breaks down in an avalanche manner at a voltage BV_{EBO} much smaller than BV_{CBO}. Typically, BV_{EBO} is in the range of 6 V to 8 V, and the **breakdown is destructive** in the sense that the β of the transistor is **permanently reduced**.

[23] This is similar to operating the diode in the breakdown region. As long as heat-dissipation is handled correctly, the device can safely be used.

This does not prevent use of the EBJ as a zener diode to generate reference voltages in IC design. In such applications one is not concerned with the β-degradation effect.

Dependence of β on I_C and Temperature 6.4 Information

Throughout this chapter we've assumed the transistor common-emitter DC current gain, β or h_{FE}, is constant for a given transistor. In reality however, β depends on the DC current at which the transistor is operating, and there is a current range over which β is highest. Normally, one arranges to operate the transistor at a current within this range.

The fact that β increases with temperature can lead to serious problems in transistors that operate at large power levels

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Part III

Appendix

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The end of a melody is not its goal: but nonetheless, had the melody not reached its end it would not have reached its goal either.

A parable.

(Friedrich Nietzsche)

Appendix A

SI Unit

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A.1 An Introduction to Scientific Quantification 177

The International Bureau of Weights and Measures (BIPM) was established by Article 1 of the Metre Convention, which was signed on 20 May 1875. It is charged with providing the basis for a **single**, **coherent system of measurements** to be used throughout the world and it operates under the authority of the *International Committee of Weights and Measures* (CIPM).

A.1 An Introduction to Scientific Quantification

Before we dive deep into the topic of standardized units, let's have a look at the history of its development.

A.1.1 A Historical Background

The decimal metric system, dating from the time of the French Revolution, was based in 1799 on the metre and the kilogram. Under the terms of the Metre Convention, new international prototypes of the metre and kilogram were manufactured and formally adopted by the first General Conference on Weights and Measures (CGPM) in 1889. In 1960 the 11th CGPM formally defined and established the International System of Units (SI). Since then the SI has been periodically updated to take account of advances in science and the need for measurements in new domains. The last major revision was adopted by the 26th CGPM (2018), which decided that the SI would be based on the **fixed numerical values of a set of seven defining constants** from which the definitions of the

seven base units of the SI would be deduced.

A.1.2 Base Units

The SI is the system of units in which:

- Unperturbed ground state hyperfine transition frequency of the caesium 133 atom $\Delta\nu_{Cs}$ is $9{,}192{,}631{,}770 \times 10^{9}$ Hz,
- Speed of light in vacuum c is $2{,}997{,}924{,}58 \times 10^{8}\,\mathrm{m}\cdot\mathrm{s}^{-1}$,
- Planck constant h is $6{,}626{,}070{,}15 \times 10^{-34}\,\mathrm{J}\cdot\mathrm{s}^{-1}$,
- Elementary charge e is $1{,}602{,}176{,}634 \times 10^{-19}$ C,
- Boltzmann constant k is $1{,}380{,}649 \times 10^{-23}\,\mathrm{J}\cdot\mathrm{K}^{-1}$,
- Avogadro constant N_A is $6{,}022{,}140{,}76 \times 10^{23}\,\mathrm{mol}^{-1}$,
- Luminous efficacy of monochromatic radiation of frequency $5{,}40 \times 10^{14}$ Hz, K_{cd} is $683\,\mathrm{lm}\cdot\mathrm{W}^{-1}$,

where the hertz, joule, coulomb, lumen, and watt, with unit symbols Hz, J, C, lm, and W, respectively, are related to the units second, metre, kilogram, ampere, kelvin, mole, and candela, with unit symbols s, m, kg, Å, K, mol, and cd, respectively, according to:

$$\mathrm{Hz} = \mathrm{s}^{-1},\ \mathrm{J} = \mathrm{kg}\cdot\mathrm{m}^2\cdot\mathrm{s}^{-2},\ \mathrm{C} = \mathrm{A}\cdot\mathrm{s},\ \mathrm{lm} = \mathrm{cd}\cdot\mathrm{m}^2\cdot\mathrm{m}^{-2} = \mathrm{cd}\cdot\mathrm{sr},\ \text{and}\ \mathrm{W} = \mathrm{kg}\cdot\mathrm{m}^2\cdot\mathrm{s}^{-3}$$

These definitions specify the **exact numerical value** of each constant when its value is expressed in the corresponding SI unit. By fixing the exact numerical value the unit becomes defined, since the product of the numerical value and the unit has to equal the value of the constant, which is invariant.

The defining constants have been chosen such that, when taken together, their units cover all of the units of the SI. In general, there is no one-to-one correspondence between the defining constants and the SI base units, except for the caesium frequency $\Delta\nu_{Cs}$ and the Avogadro constant N_A.

Any SI unit is a product of powers of these seven constants and a dimensionless factor.

As an example, using $\mathrm{Hz} = \mathrm{s}^{-1}$, one metre can be derived from the speed of light c and caesium frequency $\Delta\nu_{Cs}$:

$$1\,\mathrm{m} = \left(\frac{c}{2{,}997{,}924{,}58 \times 10^{8}}\right)\mathrm{s} = \frac{9{,}192{,}631{,}770 \times 10^{9}}{2{,}997{,}924{,}58 \times 10^{8}}\frac{c}{\Delta\nu_{Cs}} \approx 30{,}663.319\frac{c}{\Delta\nu_{Cs}}.$$

Table A.1
The seven base units of SI.

Quantity	SI Unit
Time	The **second** (s) is the SI unit of **time**. It is defined by taking the fixed numerical value of the time caesium frequency $\Delta\nu_{Cs}$, the unperturbed ground-state hyperfine transition frequency of Caesium-133, to be $9{,}192{.}631{.}770 \times 10^9$ when expressed in the unit Hz, which is equal to s^{-1}.
Length	The **metre** (m), is the SI unit of length. It is defined by taking the fixed numerical value of length the speed of light in vacuum c to be $2{,}997{.}924{.}58 \times 10^8$ m when expressed in the unit $m \cdot s^{-1}$, where the second is defined in terms of $\Delta\nu_{Cs}$.
Mass	The **kilogram** (kg), is the SI unit of mass. It is defined by taking the fixed numerical value of the Planck constant h to be $6,62607015 \times 10^{-34}$ when expressed in the unit $J \cdot s$, which is equal to mass $kg \cdot m^2 \cdot s^{-1}$, where the metre and the second are defined in terms of c and $\Delta\nu_{Cs}$.
Electric Current	The **ampere** (A), is the SI unit of electric current. It is defined by taking the fixed numerical electric current value of the elementary charge e to be $1{,}602\ 176\ 634 \times 10^{-19}$ when expressed in the unit C, which is equal to $A \cdot s$, where the second is defined in terms of $\Delta\nu_{Cs}$.
Thermodynamic Temperature	The kelvin (K), is the SI unit of thermodynamic temperature. It is defined by taking the fixed numerical value of the Boltzmann constant k to be $1{,}380\ 649 \times 10^{-23}$ when expressed in the unit temperature $J \cdot K^{-1}$, equal to $kg \cdot m^2 \cdot s^{-2} \cdot K^{-1}$ where the kilogram, metre and second are defined in terms of h, c and $\Delta\nu_{Cs}$.
Amount of Substance	The mole, (mol), is the SI unit of amount of substance. One mole contains exactly amount of substance $6{,}022\ 140\ 76 \times 10^{23}$ elementary entities. This number is the fixed numerical value of the Avogadro constant, (N_A), when expressed in the unit mol^{-1} and is called the Avogadro number. The amount of substance, symbol n, of a system is a measure of the number of specified elementary entities. An elementary entity may be an atom, a molecule, an ion, an electron, any other particle or specified group of particles.
Luminous Intensity	The candela, (cd), is the SI unit of luminous intensity in a given direction. It is defined by luminous intensity taking the fixed numerical value of the luminous efficacy of monochromatic radiation of frequency $5{,}40 \times 10^{14}$ Hz, (K_{cd}), to be 683 when expressed in the unit $lm \cdot W^{-1}$, which is equal to $cd \cdot sr \cdot W^{-1}$, or $cd \cdot sr \cdot kg^{-1} \cdot m^{-2} \cdot s^3$, where the kilogram, metre and second are defined in terms of h, c and $\Delta\nu_{Cs}$.

The concept of base units and derived units was used to define the SI until 2018. These categories, although no longer essential in the SI, are maintained in view of their convenience and widespread use.

The definitions of the base units, which follow from the definition of the SI in terms of the seven defining constants, are given in Table A.1.

Derived Units

All other quantities may be called *derived quantities* and are measured using derived units, which can be written as products of powers of base units. Twenty-two derived units are given a special name, as listed in Table A.2.

Table A.2
A list of derived SI units.

Definition	Derived Unit	Derived Symbol	SI Unit
plane plane	radian	rad	$m \cdot m^{-1}$
solid angle	stradian[30]	sr	$m^2 \cdot m^{-2}$
frequency	hertz	Hz	s^{-1}
force	newton	N	$kg \cdot m \cdot s^{-2}$
pressure, stress	pascal	Pa	$N \cdot m^{-2}$ or $kg \cdot m^{-1} \cdot s^{-2}$
energy, work, amount of heat	joule	J	$N \cdot m$ or $kg \cdot m^2 \cdot s^{-2}$
power, radiant of flux	watt	W	$J \cdot s^{-1}$ or $kg \cdot m^2 \cdot s^{-3}$
electric charge	coulomb	C	$A \cdot s$
electric potential difference	volt	V	$W \cdot A^{-1}$ or $kg \cdot m^2 \cdot s^{-3} \cdot A^{-1}$
capacitance	farad	F	$C \cdot V^{-1}$ or $kg^{-1} \cdot m^{-2} \cdot s^4 \cdot A^2$
electric resistance	ohm	Ω	$V \cdot A^{-1}$ or $kg \cdot m^2 \cdot s^{-3} \cdot A^{-2}$
electric conductance	siemens	S	$A \cdot V^{-1}$ or $kg^{-1} \cdot m^{-2} \cdot s^3 \cdot A^2$
magnetic flux	weber	Wb	$V \cdot s$ or $kg \cdot m^2 \cdot s^{-2} \cdot A^{-1}$
magnetic flux density	tesla	T	$Wb \cdot m^{-2}$ or $kg \cdot s^{-2} \cdot A^{-1}$
inductance	henry	H	$Wb \cdot A^{-1}$ or $kg \cdot m^2 \cdot s^{-2} \cdot A^{-2}$
Celsius temperature	degree Celcius	°C	K
luminous flux	lumen[32]	lm or $cd \cdot sr$	$cd \cdot sr$
illuminance	lux[34]	lm or $cd \cdot sr \cdot m^{-2}$	$lm \cdot m^{-2}$
activity referred to a radionuclide	becquerel[36]	Bq	s^{-1}
absorbed dose, kerma	gray[38]	Gy	$J \cdot kg^{-1}$
dose equivalent	sievert[40]	Sv	$J \cdot kg^{-1}$ or $m^2 \cdot s^{-2}$
catalytic activity	katal[42]	kat	$mol \cdot s^{-1}$

[30] **stradian:** the unit of solid angle, used in 3D geometry, and is analogous to the radian.

[34] **lux:** the unit used in the measure of the irradiance, as perceived by the spectrally unequally responding human eye, of light that hits or passes through a surface.

[36] **becquerel:** the unit which defines an activity of one per second, on average, for aperiodic activity events referred to a radionuclide. For applications relating to human health this is a small quantity.

[38] **gray:** the unit of radiation quantity absorbed dose which measures the energy deposited by ionizing radiation in a unit mass of absorbing material, and is used for measuring the delivered dose in radiotherapy, food irradiation and radiation sterilization.

[32] **lumen:** the unit which quantifies the perceived power of visible light emitted by a source.

[40] **sievert:** the unit representing the stochastic health risk of ionizing radiation, which is defined as the probability of causing radiation-induced cancer and genetic damage.

[42] **katal:** the unit used for quantifying the catalytic activity of enzymes, that is, measuring the enzymatic activity level in enzyme catalysis, and other catalysts.

Some useful practices regarding unit usage should be mentioned:

- While hertz and becquerel are both equal to the reciprocal second, hertz is used only for periodic phenomena, and becquerel is used only for stochastic processes in radioactive decay.

- The unit of Celsius temperature is the degree Celsius (°C), which is equal in magnitude to the kelvin (K), the unit of thermodynamic temperature. The quantity Celsius temperature t is related to thermodynamic temperature T by the relation

$$\frac{t}{^\circ\mathrm{C}} = \frac{T}{K} - 273.15$$

- The sievert is also used for the quantities "*directional dose equivalent*" and "*personal dose equivalent*".

Ratio Units

There are many more quantities other than units. For each quantity, there is only one SI unit,[43] while the same SI unit may be used to express the values of **several different quantities**.

[43] although this may often be expressed in different ways by using the special names.

> for example, the SI unit $\mathrm{J \cdot K^{-1}}$ may be used to express the value of both **heat capacity** and **entropy**.

It is therefore important **NOT** to use the unit alone to specify the quantity. This applies both to scientific texts and also to measuring instruments.[44]

[44] i.e., an instrument read-out should indicate both the quantity concerned and the unit.

There are quantities with the unit one, 1, such as the ratios of two (**2**) quantities of the same kind.

> For example, refractive index is the ratio of two (**2**) speeds, and relative permittivity is the ratio of the permittivity of a dielectric medium to that of free space.

There are also quantities with the character of a count, for example, the number of cellular or biomolecular entities. These quantities also have the unit one. The unit one is by nature an element of any system of units. Quantities with unit one can therefore be considered as traceable to the SI.

> However, when expressing the values of dimensionless quantities, the unit 1 is not written.

A.1.3 Decimal Multiples and Sub-multiples of SI Units

A set of prefixes have been adopted for use with the SI units to express the values of quantities which are either much larger than, or much smaller than, the SI unit when used without any prefix and can be used with any SI unit.

The SI prefixes are listed in Table A.3.

When the prefixes are used, the prefix name and the unit name are combined to form a single word. Similarly, the prefix symbol and the unit symbol are written without any space to form a single symbol, which may itself be raised to any power.

> For example, we write: kilometer, km; microvolt, (μV); or femtosecond, (fs).

Table A.3
Standard SI Prefixes.

Factor	Name	Symbol	Factor	Name	Symbol
10^{1}	deca	da	10^{-1}	deci	d
10^{2}	hecto	h	10^{-2}	centi	c
10^{3}	kilo	k	10^{-3}	milli	m
10^{6}	mega	M	10^{-6}	pico	μ
10^{9}	giga	G	10^{-9}	nano	n
10^{12}	tera	T	10^{-12}	pico	p
10^{15}	peta	P	10^{-15}	femto	f
10^{18}	exa	E	10^{-18}	atto	a
10^{21}	zeta	Z	10^{-21}	zepto	z
10^{24}	yotta	Y	10^{-24}	yocto	y

When the SI units are used without any prefixes, the resulting set of units is described as being **coherent** in the following sense: when only coherent units are used, equations between the numerical values of quantities take exactly the same form as the equations between the quantities themselves. The use of a coherent set of units has technical advantages, for example in algebraic calculus.

The kilogram (kg), is problematic because the name already includes a prefix, for historical reasons. Multiples and sub-multiples of the kilogram are written by combining prefixes with the gram: thus we write milligram, (mg), not microkilogram, (μkg).

Units Outside SI

The SI is the only system of units that is universally recognized, so it has a distinct advantage for establishing international dialogue. The use of the SI, as a standard system of units, simplifies the teaching of science. For these reasons, the use of SI units is recommended in all fields of science and technology.

> Other units, i.e. non-SI units, are generally defined in terms of SI units using conversion factors. such as feet, inches, yard, etc.

Nonetheless, some non-SI units are still widely used. A few, such as the minute, hour and day as units of time, will always be used because they are part of our culture. Others are used for historical reasons, to meet the needs of special interest groups, or because there is no convenient SI alternative. It will always remain the prerogative of a scientist to use the units that are considered to be best suited to the purpose. However, when non-SI units are used, the correspondence to the SI should always be quoted. A selection of non-SI units is listed in Table A.4 with their conversion factors to

the SI. For a more comprehensive list, see the SI Brochure.

Table A.4
A selection of non-SI units.

Time	Unit	Symbol	Relation to SI
time	minute	min	1 min = 60 s.
time	hour	h	1 h = 3,600 × 10^3 s.
time	day	d	1 d = 8,6400 × 10^4 s.
volume	litre	L or l	1 L = 1 dm^3.
mass	tonne	t	1 t = 1,000 × 10^3 kg.
energy	electronvolt	eV	1 eV = 1,602.176.634 × 10^{-19} J.

An Important Exception When units are named after an individual their symbol should begin with a **capital** letter:[45] In all other cases, with the sole exception being the litre, they begin with a lower case letter:

[45] for example: ampere, A; kelvin, K; hertz, Hz; or coulomb, C

for example: metre, m; second, s; or mole, mol.

A.1 Information

An Exception to the Rule

The symbol for the litre is an exception; either a lower case letter 'l' or a capital 'L' may be used, the capital is allowed in this case to avoid confusion between the lower case letter l and the number one, 1.

A.1.4 Expressing the Values of Quantities

The value of a quantity is written as the **product of a number and a unit**. The number multiplying the unit is the numerical value of the quantity in that unit. A single space is always left between the number and the unit. The numerical value depends on the choice of unit, so that the same value of a quantity can have **different numerical values** when expressed in different units, similar to the examples shown below:

- Speed of a bicycle is approximately $v = 5\,\mathrm{m}\cdot\mathrm{s}^{-1} = 18\,\mathrm{km}\cdot\mathrm{h}^{-1}$,
- Wavelength of one a yellow line of the sodium doublet is $\lambda = 5{,}896 \times 10^{-7}\,\mathrm{m} = 589{,}6\,\mathrm{nm}$.

Quantity symbols are written using *italic (slanting)* type, and they are generally single letters of the Latin or Greek alphabet. Either capital or lower case letters may be used, and additional information on the quantity may be added as a subscript or as information in brackets.

Authorities such as the International Organization for Standardization (ISO) and international scientific unions such as the International Union of Pure and Applied Physics (IUPAP) and the

International Union of Pure and Applied Chemistry (IUPAC) have specified recommended symbols for many quantities. Examples include:

Table A.5 Some examples of recommended symbols.

Symbol	Description
T	for thermodynamic temperature,
C_p	for heat capacity at constant pressure,
x_i	for mole fraction (amount fraction) of species i,
μ_r	for relative permeability.

Unit symbols are written using **roman (upright)** type, regardless of the type used in the surrounding text. They are mathematical entities and **NOT** abbreviations; unit symbols are never followed by a full stop[46] nor by an "s" for the plural. The use of the correct form for unit symbols is mandatory, and is illustrated by examples in the SI Brochure.

[46] except at the end of a sentence.

> Unit symbols may be more than a single letter. They are written in lower case letters, the exception being that the **first letter is a capital when the unit is named after a person**.

However, when the name of a unit is spelled in full, it should begin with a lower case letter ,[47] to distinguish the unit from the person.

[47] except at the beginning of a sentence.

For example a temperature of 293 kelvin.

When writing the value of a quantity as the product of a numerical value and a unit, both the number and the unit may be treated by the ordinary rules of algebra. For example, the equation

$$T = 293\,\mathrm{K} \quad \text{can be written as} \quad \frac{T}{\mathrm{K}} = 293.$$

This procedure is described as the use of *quantity calculus*, or the *algebra of quantities*. It is often useful to use the ratio of a quantity to its unit as the heading for columns in a table, or for labelling the axes of a graph, so the entries in the table or the labels for the tick marks on the axes are all simply denoted by numbers. The example (Table 5) shows a table of the velocity squared versus pressure, with the columns labelled in this way.

Table A.6 Example of column heading in a table of the velocity squared versus pressure.

p/kPa	$v^2/\mathrm{m}^2 \cdot \mathrm{s}^{-2}$
48,73	$9{,}4766 \times 10^4$
72,87	$9{,}4771 \times 10^4$
135,42	$9{,}4784 \times 10^4$

In forming products or quotients of unit symbols the normal rules of algebra apply. In forming products of unit symbols, a space should be left between units.[48] The importance of the space should be noted:

[48] or alternatively a half-high centred dot can be used as a multiplication symbol.

The product of a metre and a second is denoted by $\mathrm{m} \cdot \mathrm{s}$ (with a central dot), but ms, without a space, is used to denote a millisecond.

In addition, when forming complicated products of units, brackets or negative exponents should be used to avoid ambiguities. For example, the molar gas constant R is given by:

$$\begin{aligned} pV_m/T = R &= 8{,}314\,\mathrm{Pa} \cdot \mathrm{m}^3 \cdot \mathrm{mol}^{-1} \cdot \mathrm{K}^{-1} \\ &= 8{,}314\,\mathrm{Pa} \cdot \mathrm{m}^3/(\mathrm{mol} \cdot \mathrm{K}) \end{aligned}$$

When formatting numbers, the decimal marker may be either a point[49] or a comma, depending on the circumstances. For documents in the English language a point is usual, but for many languages and in many countries a comma is usual.

[49] i.e. a full stop.

When a number has many digits, it is customary to group the digits into groups of three (**3**)about the decimal point to aid readability. This is not essential, but it is often done and is generally helpful. When this format is used, the groups of three (**3**) digits should be separated only by a space; neither a point nor a comma should be used. The uncertainty in the numerical value of a quantity may often be conveniently shown by giving the uncertainty in the least significant digits in brackets after the number.

For example: The value of the electron mass is given in the 2014 CODATA listing of fundamental constants as

$$m_e = 9.109\,383\,56\,(11) \times 10^{-31}\,\mathrm{kg}$$

where 11 is the **standard uncertainty** in the final digits quoted for the numerical value.

Appendix B

Dimensional Analysis

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Dimensional analysis offers a method for reducing *complex physical problems* to the simplest[1] form prior to obtaining a quantitative answer. *Bridgman* [1] explains it as follows:

> The principal use of **dimensional analysis** is to deduce from a study of the dimensions of the variables in any physical system certain limitations on the form of any possible relationship between those variables. The method is of great generality and mathematical simplicity.

[1] in this case this means in its most economical form.

At the heart of dimensional analysis is the **concept of similarity**. In physical terms, similarity refers to some equivalence between two things or phenomena which are actually different.[2] The question is;

> What are those conditions, and what is the relationship between the forces?

[2] For example, under some very particular conditions there is a direct relationship between the forces acting on a full-size aircraft and those on a small-scale model of it.

Mathematically, similarity refers to a transformation of variables leading to a reduction in the number of independent variables specifying the problem. Here the question is;

> what kind of transformation actually works?

Dimensional analysis addresses both these questions. Its primary utility derives from its ability to extract the functional form of physical relationships. A problem which at first looks formidable may sometimes be solved with little effort after dimensional analysis.

In problems so well understood that one can write down in mathematical form all the governing laws and boundary conditions, and only the solution is lacking, similarity can also be inferred by

normalizing all the equations and boundary conditions in terms of quantities that specify the problem and identifying the dimensionless groups that appear in the resulting dimensionless equations.

This is an inspectional form of **similarity analysis** as inspectional analysis can take advantage of the problem's full mathematical specification, it may reveal a higher degree of similarity than a "blind" dimensional analysis and in that sense prove more powerful.

Use Case for Dimensional Analysis B.1 Information

Dimensional analysis is, however, the only option in problems where the equations and boundary conditions are not completely articulated, and always useful because it is simple to apply and quick to give insight.

Some of the basic ideas of similarity and dimensional analysis had already surfaced in *Fourier*'s work in the 19th century's first quarter [2], but the subject received more methodical attention only toward the close of that century;

> notably in the works of *Lord Rayleigh*, *Reynolds*, *Maxwell*, and *Froude* in England, and *Carvallo*, *Vaschy* and a number of other scientists and engineers in France [3].



Figure B.1 The study of aerodynamics can be complex and daunting and finding relations between relevant properties can be achieved by the use of dimensional analysis. Picture: A NASA wake turbulence study at Wallops Island is portrayed, 1990. A vortex is created by passage of an aircraft wing, revealed by smoke. Vortices are one of the many phenomena associated with the study of aerodynamics.

By the 1920's the principles were essentially in place: *Buckingham*'s now ubiquitous Π-theorem had appeared [4], and *Bridgman* had published the monograph which still remains the classic in the field. Since then, the literature has grown prodigiously.

Applications now include aerodynamics, hydraulics, ship design, propulsion, heat and mass transfer, combustion, mechanics of elastic and plastic structures, fluid-structure interactions, electromagnetic theory, radiation, astrophysics, underwater and underground explosions, nuclear blasts, impact dynamics, and chemical reactions and processing [5–7], and also biology [8] and even economics [9].

Validity of Dimensional Analysis B.2 Information

Most applications of dimensional analysis are not in question, no doubt because they are well supported by experimental facts. The debate over the method's theoretical-philosophical underpinnings, on the other hand, has never quite stopped festering.

Mathematicians tend to find in the basic arguments a lack of rigor and are tempted to redefine the subject in their own terms, while physicists and engineers often find themselves uncertain about the physical meanings of the words in terms of which the analysis cast.

The problem is that dimensional analysis is based on ideas that originate at such a substratal[3] point in science which most scientists and engineers have lost touch with them. To understand its principles, we will return to some of the very fundamental concepts in science.

[3] being of or relating to a substrate, meaning something basic, underlying, or forming a foundation.

Dimensional analysis is rooted in the nature of the artifices we construct to describe the physical world and explain its functioning in quantitative terms. Einstein [10] has said,

> Pure logical thinking cannot yield us any knowledge of the empirical world; all knowledge starts from experience and ends in it. Propositions arrived at by purely logical means are completely empty as regards reality.[4]

[4] For example, the very first page of the first volume of James Clerk Maxwell's *A treatise on Electricity and Magnetism* [11].

This chapter aims explain dimensional analysis by tracing it back to its physical foundations. Here, we will clarify the terms used in dimensional analysis, explain why and how it works, remark on its utility, and discuss some of the difficulties and questions that typically arise in its application.

B.1 Physical Quantities and Equations

Science begins with the observation and **precise description of things and events**. It is at this very first step that we face the fact upon which dimensional analysis rests:

> Description in absolute terms is **impossible**.

We can do no more than compare one thing with another, to *catch the resemblance of things*.

> Fore example, when we say that something **is** a tree,[5] we mean simply that it has a set of attributes that are in some way shared by certain familiar objects we have agreed to call trees.



[5] From this image it is obvious to call it a tree and not a cat. But on thinking it further when we say it is a tree we mean it contains elements which in total make up an object in which we call tree.

With this, let us begin with understanding how we describe physical properties.

B.1.1 Physical properties

Continuing from our previous example regarding trees, it is fair to say that our brains have evolved to the point where we can recognize trees almost instantly, but describing something like a tree is actually a very complex process.

> Physics starts by breaking the descriptive process down into simpler terms such as **length**, **mass**, **color**, **shape**, **speed**, and **time**.

None of these properties can be defined in **absolute terms**, but only by **reference** to something else:

> For example we would say an object has the length of a meter stick, we say, the color of an orange, the weight of a certain familiar lump of material, or the shape of a sphere.

The references may look to be more precise in nature, but in essence "description" is simply a noting of the similarities between one thing and a set of others that are known to us.

> We can do no more than **compare one thing with another**.

A physical property first arises as a concept based on experience, and is formalized by defining a comparison operation for determining whether two (2) samples of it are equal ($\mathbf{A} = \mathbf{B}$) or unequal ($\mathbf{A} \neq \mathbf{B}$).[6]

This operation, which is an entirely physical procedure, defines the **property**. Properties of the same kind[7] are compared by means of the same comparison operation. Properties of different kinds cannot be compared because there exists no operation that defines equality.

> Asking whether a particular mass is physically equal to a particular length is meaningless.[8]

If a property is defined only in terms of a comparison operation, we have a procedure for establishing whether two samples of it are **equal** or **unequal**, but no concept of what it means for one to be larger or smaller than the other. Shape and color are examples.

> We have procedures for determining if two objects have the same shape, or the same color. But asking whether a square shape is smaller or larger than a circular shape, or whether green is smaller or larger than white, makes no sense.

Properties like shape and color are useful for describing things, but cannot play a role in any quantitative analysis, which deals with relative magnitudes.

[6] We will use bold symbols when we are referring not to numerical values, but to actual physical attributes. This does not mean they are vectors or matrices, however.

[7] or simply, the same properties.

[8] no procedure exists for making the comparison.

B.1.2 Physical Quantities and Base Quantities

Science begins with observation and description, but its ultimate goal is to infer from those observations laws which express the phenomena of the physical world in the simplest and most general terms. That the language of mathematics is ideally suited for expressing those laws is not accidental, but follows from the constraints we put on the types of physical properties that are allowed to appear in quantitative analysis.

> The allowed types of properties are called **physical quantities**.

Physical quantities are of two (2) types:

> base quantities and derived quantities.

The base quantities, which are defined in entirely physical terms, form a complete set of basic

building blocks for an open-ended system of **derived** quantities which can be introduced as necessary. The base and derived quantities together provide a rational basis for describing and analyzing the physical world in quantitative terms.

A base quantity is defined by specifying **two physical operations**:

- A comparison operation for determining whether two samples **A** and **B** of the property are equal ($\mathbf{A} = \mathbf{B}$) or unequal ($\mathbf{A} \neq \mathbf{B}$), and
- An addition operation which defines what is meant by the sum $\mathbf{C} = \mathbf{A} + \mathbf{B}$ of two samples of the property.

Base quantities with the same comparison and addition operations are of the same kind.[9] The addition operation $\mathbf{A} + \mathbf{B}$ defines a physical quantity **C** of the same kind as the quantities being added.

[9] That is, different examples of the same quantity.

Quantities with different comparison and addition operations cannot be compared or added as no procedures exist for executing such operations.

All physical quantities are properties of physical things or events. They are not themselves physical things or events. The comparison and addition operations involve physical manipulations of objects or events that possess the property under consideration.

The comparison and addition operations are **physical**, but are required to have certain properties which mimic those of the corresponding mathematical operations for pure numbers:

i. The comparison operation must obey the **identity law**, and

ii. the addition operation must be commutative ($\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$), associative [$\mathbf{A} + (\mathbf{B} + \mathbf{C}) = (\mathbf{A} + \mathbf{B}) + \mathbf{C}$], and unique.[10]

[10] if $A + B = C$, there exists no finite D such that $A + B + D = C$.

Identity Law B.1 Definition

An **identity** is an equality relating one mathematical expression **A** to another mathematical expression **B**, such that **A** and **B**[11] produce the same value for all values of the variables within a certain domain of discourse.

[11] which might contain some variables

In other words, $\mathbf{A} = \mathbf{B}$ is an identity if **A** and **B** define the same functions, and an identity is an equality between functions that are differently defined.

The two operations together define, in entirely physical terms,

i. the concept of larger and smaller for like quantities. If there exists a finite **B** such that $\mathbf{A} + \mathbf{B} = \mathbf{C}$, then $\mathbf{C} > \mathbf{A}$,

ii. subtraction of like quantities. If $\mathbf{A} + \mathbf{B} = \mathbf{C}$, then $\mathbf{A} = \mathbf{C} - \mathbf{B}$,

iii. multiplication of a physical quantity by a pure number. If $\mathbf{B} = \mathbf{A} + \mathbf{A} + \mathbf{A}$, then $\mathbf{B} \equiv 3\mathbf{A}$), and

iv. division of a physical quantity by a pure number. If $\mathbf{A} = \mathbf{B} + \mathbf{B} + \mathbf{B}$, then $\mathbf{B} \equiv \mathbf{A}/3$).

A base quantity is therefore a property for which the following mathematical operations are defined in physical terms:

> comparison, addition, subtraction, multiplication by a pure number, and division by a pure number.

Each of these operations is performed on physical properties of the same kind and gives a physical property of that kind, and each physical operation obeys the same rules as the corresponding mathematical operation for pure numbers.

This sets the stage for not only **catching the resemblance of things**, but also expressing that resemblance in the language of mathematics. It is important to note that mathematical operations other than the ones listed above are **NOT** defined in physical terms.

> No defining operation exists for forming a tangible entity that represents the product of a mass and a time, for example, or, for that matter, the product of one length and another. Nor can we point to some tangible thing that "is" the cube root of a length, say, or the natural logarithm of a time.

Mathematical Reasoning B.3 Information

Products, ratios, powers, and exponential and other functions such as trigonometric functions and logarithms are defined for numbers, but have no physical correspondence in operations involving actual physical quantities.

Figure 2.1 illustrates the comparison and addition operations of some well-known physical quantities that can be chosen as base quantities.

Figure 2.1a: The comparison and addition operations of length. Figure 2.1b: The comparison and addition operations of mass. Figure 2.1c: The comparison and addition operations of area. Figure 2.1d: The comparison and addition operations of velocity. Figure 2.1e: The comparison and addition operations of force.

The comparison and addition operations for length and mass are familiar. Only one comment is necessary: the mass referred to in the figure is the **gravitational mass**, for which equality is defined by the statement that two masses are equal if a third mass exerts the same gravitational force on each of them separately at the same relative position.

Time is omitted from the figure, largely because its defining operations defy illustration in such simplistic terms. The concept of time is deeply rooted in our biological beings. It is, after all, the warp of our existence, the stuff that keeps everything from happening at once. Time characterizes an event, not a thing. Aristotle referred to time as a "dimension of motion", which pleases the poet in us, but leaves the scientist unmoved. We do best if we adopt a pragmatic notion of time and

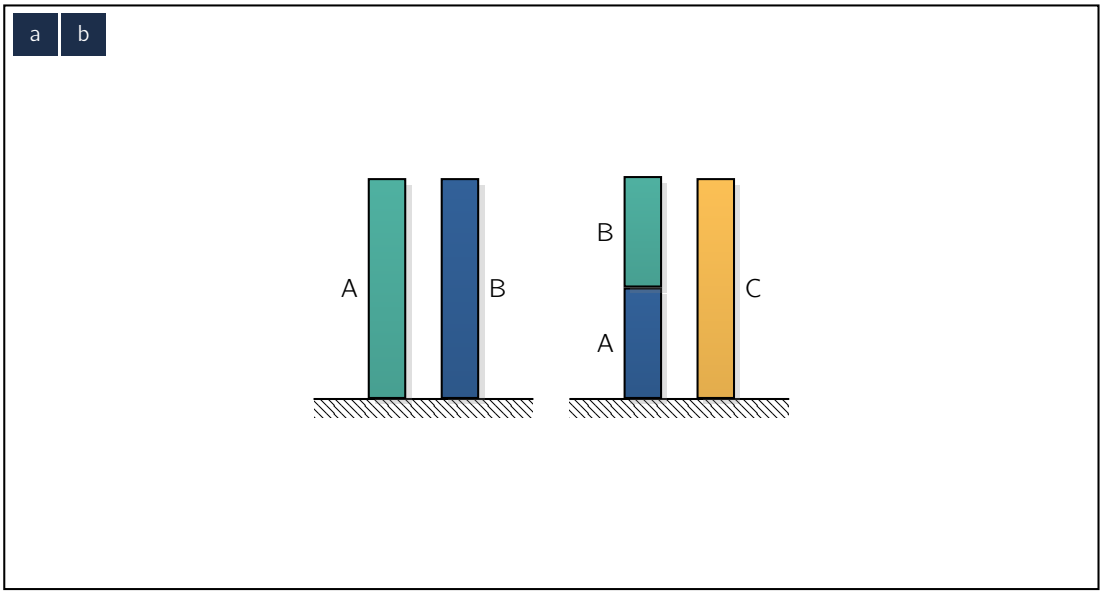


Figure B.2

think of it as being defined in terms of comparison and addition operations involving idealized clocks or stopwatches. Whether those clocks are hourglasses or atomic clocks will affect the precision of the operations, but not their intrinsic character. What time "is" has no relevance in the present context, only the defining operations matter.[12]

The concept of force arises in primitive terms from muscular effort, and is formalized based on the observation that a net force on an object[13] causes a rate of change in its velocity.

> Directionality is important in the definition as force is a **vector quantity**.

For us whom we are accustomed to considering speed as *distance divided by time* and area as *length squared* may be surprised to see them included in figure 2.1 as possible base quantities with their own comparison and addition operations. We include them to show that the set of base quantities is very much a matter of choice. Velocity[14] is a certain property of motion. A self-propelled toy car running across a tabletop has a speed, and we can define acceptable procedures for establishing whether two speeds are equal or unequal and for adding two speeds, as in figure 2.1d. Speed can therefore be taken as a base quantity, should we choose to do so. The same goes for area. Note that two areas may be equal without being congruent, provided one of them can be *cut up* and reassembled into a form which is congruent with the other:

> the addition operation is invoked in making a comparison, and the comparison operation in addition.q

We have said that shape and color, though acceptable physical properties, cannot be base quantities because they lack acceptable addition operations. That shape is disqualified is obvious: what is the sum of a square and a circle? But why is color disqualified? We know that color in the form of light

[12] Sir Arthur Eddinton (1939): "It has come to be accepted practice in introducing new physical quantities that they shall be regarded as defined by the series of measuring operations and calculations of which they are the result. Those who associate with the result a mental picture of some entity disporting itself in a metaphysical realm of existence do so at their own risk; physics can accept no responsibility for this embellishment".

[13] The vector sum of all the forces acting on the object.

[14] or speed, if direction is presumed.

can be added according to well-defined rules, as when red light added to green produces yellow. Is this not an acceptable definition of physical addition? The answer is no. Blue equals blue. But according to this addition operation, blue plus blue also equals blue. The addition operation is not unique: $n\mathbf{A} = \mathbf{A}$ for any color $\mathbf{A}$, where n is any number. This disqualifies color from the ranks of physical quantities.[15] But, we may argue, the color of an object can be identified by the wavelength of light reflected from it, and wavelength can be added.

[15] This makes measurement of color in terms of a unit impossible.

Is this not acceptable for an addition rule?

The answer is again no. This is a rule for adding lengths, not for adding the property we perceive as color.

B.1.3 Unit and numerical value

The two operations which define a base quantity make it possible to express any such quantity as a multiple of a standard sample of its own kind, that is, to "measure it in terms of a unit". The standard sample[16] may be chosen arbitrarily. The comparison operation allows the replication of the unit, and the addition operation allows the identification and replication of fractions of the unit. The measuring process consists of physically adding replicas of the unit and fractions thereof until the sum equals the quantity being measured (figure 2.2). A count of the number of whole and fractional units required yields the numerical value of the quantity being measured. If $\mathbf{a}$ is the unit chosen for quantities of type $\mathbf{A}$, the process of measurement gives a numerical value A, a number, such that

[16] Also called the unit.

$$\mathbf{A} = A\mathbf{a}$$

The measurement process is entirely **physical**. The only mathematics involved is the counting of the number of whole and fractional units once physical equality has been established between the quantity being measured and a sum of replicas of the unit and fractions thereof.

It should be emphasized that numbers can be ascribed to properties in many arbitrary ways, but such numbers will not represent numerical values of physical quantities unless they are assigned by a procedure consistent with the one defined above.

> The numerical value of a base quantity depends on the choice of unit.

A physical quantity exists, independent of the choice of unit. A forefinger has the same length, regardless of whether we measure it in centimeters or inches. A quantity $\mathbf{A}$ can be measured in terms of a unit a or in terms of another unit $\mathbf{a}'$, but the quantity itself remains physically the same, that is,

$$\mathbf{A} = A\mathbf{a} = A'\mathbf{a}' \tag{B.1}$$

If the unit $\mathbf{a}'$ is n times larger than $\mathbf{a}$,

$$\mathbf{a}' = n\mathbf{a},$$

it follows from Eq. (**B.1**) that

$$A' = n^{-1}A. \tag{B.2}$$

If the size of a base quantity's unit is changed by a factor n, the quantity's numerical value changes by a factor n^{-1}.

By convention, all base quantities of the same kind are always measured in terms of the same unit. All base quantities of the same kind thus change by the same factor when the size of that quantity's unit is changed.

The ratio of the numerical values of any two quantities of the same kind is **independent of base unit size**.

Note also that when base quantities of the same kind are added physically ($\mathbf{A}+\mathrm{B}=\mathbf{C}$), the numerical values satisfy an equation of the same form as the physical quantity equation ($A+B=C$), regardless of the size of the chosen unit.

The numerical value equation **mimics the physical equation**, and its form is independent of the unit's size.

B.1.4 Derived quantities, dimensions, and dimensionless quantities

In describing physical things and events quantitatively, we refer to **numerical values** of base quantities, and also introduce numbers **derived** by inserting these values into certain mathematical formulas.

> For example, we determine the distance L that an object moves in time t, and calculate its speed $V = Lt^{-1}$. We can also measure a body's mass m and speed V and calculate its kinetic energy $K = mV^2/2$.

These numbers are **derived quantities of the first kind**.

Not all numbers obtained by inserting base quantities into formulas can be considered physical quantities.[17] Base quantities have a transparently physical origin, which gives rise to the fact that the ratio of any two samples of a base quantity remains constant when the base unit size is changed; an arbitrary choice cannot affect a relative physical magnitude.

P. W. Bridgman[18] postulated that this is in fact a defining attribute of *all* physical quantities. This is Bridgman's principle of absolute significance of relative magnitude:

B.1 Theorem

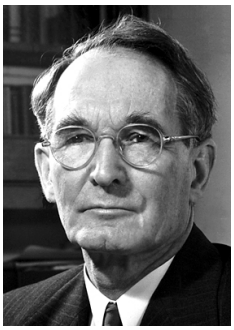
Principle of Absolute Significance of Relative Magnitude

A number Q obtained by inserting the numerical values of base quantities into a formula is a physical quantity if the ratio of any two samples of it remains constant when base unit sizes are changed.

Bridgman went on to show that a monomial formula satisfies the principle absolute significance of relative magnitude only if it has the power-law form

$$Q = \alpha A^a B^b C^c \dots$$

[17] From this point on, the term quantity will be used for the *numerical value* of a physical quantity, unless otherwise noted.



[18] **Percy Williams Bridgman** *(1882 - 1961)* was an American physicist who received the Nobel Prize in Physics in 1946 for his work on the physics of high pressures. He also wrote extensively on the scientific method and on other aspects of the philosophy of science. The **Bridgman effect**, the **BridgmanStockbarger technique**, and the high-pressure mineral

where A, B, C, etc. are **numerical values** of base quantities and the coefficient α and exponents a, b, c, etc. are real numbers whose values distinguish one type of derived quantity from another.

All monomial derived quantities have this power-law form and no other form represents a physical quantity.

A derived quantity of the first kind is defined in terms of a numerical value, which depends on the choice of base units. A derived quantity does not necessarily represent something tangible in the same sense as a base quantity, although it may.

> The square root of time, for example, is a derived quantity because it has the required power-law form, but we cannot point to any physical thing that **is** the square root of time.

To avoid talking of *units* for quantities that may have no physical representation, but whose numerical values nevertheless depend on the choice of base units, we introduce the concept of **dimension**. Each type of base quantity has by definition its own dimension. If A is the numerical value of a length, we say it **has the dimension of length**, and write this as $[A] = L$ where the square brackets imply *the dimension of* and L symbolizes the concept of length. By this we mean simply that if the length unit size is increased by a factor n, the numerical value A will increase by a factor n^{-1}.

The dimension of a derived quantity conveys the same information in generalized form, for derived as well as base quantities. Consider a quantity defined by the formula

$$Q = \alpha L_1^{l_1} L_2^{l_2} \ldots M_1^{m_1} M_2^{m_2} \ldots t_1^{\tau_1} t_2^{\tau_2} \ldots \tag{B.3}$$

where the L_i's are numerical values of certain lengths, M_i's of certain masses, and t_i's of certain times, and α and all exponents are real numbers. If the length unit is changed by a factor n_L, the mass unit by a factor n_m, and the time unit by a factor n_t, it follows from Eq. (**B.2**) and brefeq:dimen-2-6 that the value of Q changes to

$$Q' = n^{-1} Q \qquad \text{where} \qquad n = (n_L)^{\sum l_i} (n_m)^{\sum m_i} (n_t)^{\sum t_i}.$$

This implies that Q transforms like the numerical value of a base quantity with a unit whose size is proportional to $L^{\sum l_i} M^{\sum m_i} t^{\sum \tau_i}$, where L, M, and t represent the sizes of the length, mass, and time units, respectively. By analogy with the meaning of dimension for base quantities, we therefore say that the derived quantity Q has the dimension

$$[Q] = (L)^{\sum l_i} (M)^{\sum m_i} (t)^{\sum t_i}. \tag{B.4}$$

Whether applied to a base or derived quantity, the dimension is simply a formulaic indication of how the quantity's numerical value transforms when the sizes of the base units are changed. A derived quantity's dimension follows from its defining equation. To obtain the result in Eq. (**B.4**) for the quantity defined in Eq. (**B.3**) we simply substitute for each base quantity in Eq. (**B.3**) the symbol for its dimension, omit the numerical coefficient α, and obtain Eq. (**B.4**) by algebra. The dimension of a kinetic energy $mV^2/2$, for example, is $M(L/t)^2 = ML^2t^{-2}$.

Statements to the effect that a quantity's dimension is an *expression of its essential physical nature* are either a meaningless tautology[19] or nonsense[20]. We shall see that a quantity's dimension depends on the choice of the system of units, and is therefore under the control of the observer rather than an inherent attribute of that quantity.

[19] in the case of basic quantities.

By convention, a particular derived quantity is specified by its numerical value followed by the base units upon which that value is based, the latter arranged in a form which reflects the quantity's dimension. The statement $Q = 0{,}021\,\mathrm{kg}\sqrt{\mathrm{s}}$ implies that Q is a quantity with dimension $Mt^{\sqrt{2}}$ and has a magnitude 0,021 if mass is measured in kilograms and time in seconds. Speaking loosely, it may be said that the quantity Q is *measured in units of* $\mathrm{kg}\sqrt{\mathrm{s}}$, which implies both its dimension and the base units on which the indicated value is based. In common parlance the terms unit and dimension are often used synonymously, but such usage is undesirable in a treatise where fundamental understanding is paramount.

Some important points about derived quantities:

i. The dimension of any derived physical quantity is a product of powers of the base quantity dimensions.

ii. Sums of derived quantities with the same dimension are derived quantities of the same dimension. Products and ratios of derived quantities are also derived quantities, with dimensions which are usually different from the original quantities.

iii. All derived quantities with the same dimension change their values by the same factor when the sizes of the base units are changed.

iv. A derived quantity is *dimensionless* if its numerical value remains invariant when the base units are changed. An example is Vt/L, where $V = dx/dt$ is a velocity, t is a time and L is a length. The dimension of a dimensionless quantity is unity, the factor by which the quantity's numerical value changes when base units sizes are changed.

v. Special functions[21] of dimensional derived quantities are in general not derived quantities because their values do not in general transform like derived quantities when base unit size changes. Only when the arguments of these functions are dimensionless will the arguments and therefore the values of the functions remain invariant when units are changed. Special functions with dimensionless arguments are therefore derived quantities with dimension unity.

[21] logarithmic, exponential, trigonometric, etc.

B.1.5 Physical Equations and Constants, Dimensional homogeneity

In **quantitative analysis** of physical events we seek mathematical relationships between the *numerical values* of the physical quantities that describe the **event**.

We are **NOT** interested in just any relationships which can apply between the values of **physical quantities**.[22]

[22] For example, it is possible that some mass in kg is exactly equal to someone's height in cm. We dismiss this kind of *relationship* as an accidental result of the choice of units.

Science is concerned only with expressing a **physical** relationship between one quantity and a set of others, that is, with **physical equations**.

Nature is **indifferent** to the arbitrary choices we make when we pick base units.

We are interested, therefore, only in numerical relationships that remain true independent of base unit size.

This puts certain constraints on the allowable form of physical equations. Let's suppose that, in a specified physical event, the numerical value Q_o of a physical quantity is determined by the numerical values of a set $Q_1, \ldots, Q_n$ other physical quantities, that is,

$$Q_o = f(Q_1, Q_2, \ldots, Q_n), \tag{B.5}$$

The principle of absolute significance of relative values tells us that the relationship implied by Eq. (**B.5**) can be physically relevant only if Q_o and f change by the **same factor** when the magnitudes of any base units are changed.

To put it simply, a physical equation must be **dimensionally homogeneous**.

This dimensional homogeneity imposes the following constraints on any mathematical representation of a relationship given in Eq. (**B.5**):

i. both sides of the equation must have the **same dimension**;

ii. wherever a sum of quantities appears in f, all the terms in the sum must have the same dimension;

iii. all arguments of any exponential, logarithmic, trigonometric or other special functions that appear in f must be dimensionless.

For example, if a physical equation is represented by

$$A = B \exp(-C) - \frac{(D_1 + D_2)}{E} + F,$$

C must be dimensionless, D_1 and D_2 must have the same dimension, and A, B, D/E and F must have the same dimension.

An important consequence of dimensional homogeneity is that the form of a physical equation is **independent of the size of the base units**.

An Example of Acceleration

The following example may help to illustrate the reason for dimensional homogeneity in physical equations and show how conceptual errors that may arise if homogeneity isn't recognized.

Suppose we release an object from rest in a *uniform gravitational field*, in vacuum, and ask what distance x it will fall in a time t. We know of course that elementary *Newtonian mechanics* gives the answer as:

$$x = \frac{1}{2}gt^2, \tag{B.6}$$

where g is the local acceleration of gravity and has the dimension Lt^{-2}. This equation expresses the result of a general physical law, and is clearly **dimensionally homogeneous**.

The physical basis of dimensional homogeneity becomes apparent when we consider the same phenomenon from a different perspective.

Suppose we are ignorant of mechanics and conduct a large variety of experiments,[23] on the time t it takes a body with mass m to fall a distance x from rest in an evacuated chamber. After performing experiments with numerous masses and distances, we find that if t is measured in seconds (s) and x in meters (m), all our data can be accurately represented, regardless of mass, with the single equation

$$x = 4{,}91t^2 \tag{B.7}$$

This is a perfectly correct equation as it describes and predicts all experiments to a very good accuracy.



[23] For example, these examples could be conducted by tossing balls from the top of *Innsbruck* Cathedral.

However, it appears at first glance to be dimensionally non-homogeneous, the two sides seemingly having different dimensions, and therefore appears not to be a true physical equation.

This impression is, however, based on the false presumption that the coefficient 4,91 remains **invariant** when units are changed. In fact, the coefficient 4,91 represents not a dimensionless number, but a particular numerical value of a dimensional physical quantity which characterizes the relationship between x and t. That this must be so becomes clear when we consider how Eq. (**B.7**) must transform when units are changed.

We know that when units are changed, the actual physical distance x remains invariant, and we therefore argue that to obtain the falling distance in feet, for example, the Right Hand Side (RHS) of Eq. (**B.7**), which gives it in meters, must be multiplied by 3,28, the number of feet in one meter.

Thus, if x is measured in feet and t in seconds, the correct version of Eq. (**B.7**) is:

$$x = 16{,}1t^2 \tag{B.8}$$

This same transformation could also have been obtained by arguing that Eq. (**B.7**), being an expression of a general physical law, must, according to Bridgman's principle of absolute significance of relative magnitude, be dimensionally homogeneous, and therefore should properly have been written

$$x = ct^2 \qquad \text{where} \qquad (c = 4{,}91\,\mathrm{m}\cdot\mathrm{s}^{-2}) \tag{B.9}$$

This form makes clear that the coefficient c is a **physical quantity** rather than a numerical coefficient. The units of c indicate its dimension and show that a change of the length unit from meters to feet,

with the time unit remaining **invariant**, changes c by the factor 3,28, the inverse of the factor by which the length unit is changed.

This gives $c = 16{,}1\,\text{ft} \cdot \text{s}^{-2}$, as in Eq. (**B.8**).

Eq. (**B.9**) is the correct way of representing the data of equation Eq. (**B.6**). It is dimensionally homogeneous, and makes the transformation to different base units straightforward.

Every correct physical equation[24] must be dimensionally homogeneous. A fitting formula derived from correct empirical data may at first sight appear dimensionally non-homogeneous because it is intended for particular base units.

[24] that is, every equation that expresses a physically significant relationship between numerical values of physical quantities.

Such formulas can always be rewritten in general, homogeneous form by the following procedure [1]:

i. Replace all the numerical coefficients in the equation with unknown dimensional constants.

ii. Determine the dimensions of these constants by requiring that the new equation be dimensionally homogeneous.

iii. Determine the numerical values of the constants by matching them with those in the original equation when the units are the same.[25]

[25] This is of course how Eq. (**B.9**) was derived from Eq. (**B.7**).

Another example serves to reinforce this point. Suppose it is found that the pressure distribution in the earth's atmosphere can be *approximately* represented by the formula:

$$p = 1{,}01 \times 10^5 \exp\left(-0.00012z\right)$$

where p is the pressure in $\text{N} \cdot \text{m}^{-2}$ and z is the altitude in meters (m). This expression applies only with the described units. The correct, dimensionally homogeneous form of this equation is

$$p = a \exp\left(-bz\right) \qquad \text{where} \qquad a = 1{,}01 \times 10^5\,\text{N} \cdot \text{m}^{-2} \quad \text{and} \quad b = 1{,}2 \times 10^{-4}\,\text{m}^{-1} \qquad (\textbf{B.10})$$

where a and b are physical quantities. In this form the equation is valid independent of the chosen base units.

> The dimensions of a and b indicate **how these quantities change when units are changed**.

The two quantities a and b in Eq. (**B.10**) are physical constants in the sense that their values are **fixed** once the system of units is chosen. In this case the constants characterize a particular environment.[26] Similarly, the acceleration of gravity g in Eq. (**B.6**) is a physical constant that characterizes the local gravitational force field at the earth's surface.

[26] In this case, it is the pressure distribution in the earth's atmosphere.

The basic laws of physics also contain a number of **universal physical constants** whose magnitudes are the same in all problems once the system of units is chosen:

> the speed of light in vacuum c, the universal gravitational constant G, Planck's constant h, Boltzmann's constant k_B,

and many others.

Derived Quantities of the Second Kind

The classification of quantities as base or derived is not unique. There exist general laws which bind different kinds of quantities together in certain relationships, and these laws can be used to transform base quantities into derived ones. Such transformations are useful as they reduce the number of units which must be chosen arbitrarily, and simplify the forms of physical laws.

Area, for example, may be taken as a base quantity with its own comparison and addition operations, and measured in terms of an arbitrarily chosen (base) unit:

> We can use a certain postage stamp, as an absurd example.[27] The floor area of a room may be measured by covering the floor with copies of this postage stamp and parts thereof, and counting the number of whole stamps required.



[27] One could come with a better example but where would then there would not be an opportunity to showcase the **Penny Black**, which is the first postage stamp. This would be an excellent choice to measure the size of a room.

If we adopt this practice we will eventually find that, **regardless** of the unit we have chosen for measuring an area A, its numerical value A will depend on its linear dimensions according to

$$A = c \int \mathrm{d}x\, \mathrm{d}y \tag{B.11}$$

where the integral is taken over the area and c is a dimensional constant the magnitude of which depends on the choice of base units of area and length. Dimensional homogeneity requires that c be a derived quantity,

$$c = \frac{A}{\int \mathrm{d}x\, \mathrm{d}y}$$

with dimension AL^{-2}. Eq. (**B.11**) can be thought of as a **physical law**. A little reflection reveals, however, that it holds simply because the quantity we have symbolized as $\int dxdy$ is defined by comparison and addition rules that happen to be similar to those we have chosen for area. Eq. (**B.11**) is therefore not really a law of nature, but one have crafted ourselves. Dimensional constants like c in Eq. (**B.11**) whose values depend on the choice of units but are entirely independent of the problem being considered are called **universal constant**.

The law given in Eq. (**B.11**) suggests a simplification. If we choose to measure area in terms of a unit that is defined by the area of a square with sides equal to the length unit, the physical coefficient c becomes unity, and Eq. (**B.11**) becomes:

$$A = \int \mathrm{d}x\, \mathrm{d}y \tag{B.12}$$

Area, in effect, has become a **derived quantity** defined in terms of operations involving length. Eq. (**B.12**) does not imply that area now "is", in any physical sense, length squared. The physical property that is area remains what it was before, quite distinct from length, and for that matter quite distinct from "length squared", of which we have no physical concept whatsoever.

We have simply noted that, because our concepts of "integral" and "area" are in fact similar, we may choose to measure area via operations involving length, and have made a decision to do so. The fact that area now has the dimension L^2 simply indicates that the numerical value of any area so defined will change by a factor n^2 when the length unit is changed by a factor n.

By transforming area in this way from a base to a derived quantity, we have accomplished two simplifications:

1. the length unit automatically determines the area unit, and

2. the dimensional physical constant c in Eq. (**B.11**) is replaced with unity.

There is nothing sacred about choosing $c = 1$ in this kind of transformation. We could have adopted as the unit for area circle with diameter equal to one length unit, in which case area would have been defined by $A = (4/\pi)\int \mathrm{d}x\,\mathrm{d}y$ instead of by Eq. (**B.12**). This would have been perfectly acceptable, but less convenient since it would force us to waste a lot of paper writing unnecessary $4/\pi$'s.

In other instances, dimensionless coefficients other than unity are introduced deliberately to make things more convenient. For example, by defining the kinetic energy as $K = mV^2/2$ instead of $K = mV^2$ we make the energy equation for a point mass in a force field read $K + U =$ constant instead of $K/2 + U =$ constant, where U is the force potential.

Similarly, the 4π terms that appear in the **SI system** of units in the integral forms of Gauss's and Ampere's laws are placed there in order to eliminate numerical coefficients in the differential forms such as in **Maxwell's field equations** [12].

Speed offers an example similar to area. We may define speed as a base quantity with its own comparison and addition operations, and choose for it a base unit. If we take this route we would eventually discover that the quantity we have identified as speed is in fact proportional to the distance covered in unit time. We can then choose to **define** speed as the derived quantity $V = \mathrm{d}y/\mathrm{d}x\,xt$, which is equivalent to choosing a speed unit such that unit distance is covered in unit time.

A more interesting example arises with force. Force may be taken as a base quantity with an arbitrarily specified unit. Newton's original law of motion states that if a body of mass m is subjected to a net force F (in the x-direction, say) under non-relativistic conditions, it accelerates in the direction of the force and the acceleration $a = \mathrm{d}^2y/\mathrm{d}x^2\,xt$ is related to the force F and mass m by

$$F = cma \tag{B.13}$$

where the coefficient c is a universal constant with dimension $Ft^2M^{-1}L^{-1}$ if force, length, mass and time are all selected as base quantities. Eq. (**B.13**) is a general physical law which expresses a relationship between the numerical values of three different physical quantities that are involved in the dynamics of a point mass-force, gravitational mass, and acceleration. We are at liberty, however, to choose a force unit such that one unit of force will give unit mass unit acceleration under non-relativistic conditions.

If we do this, we make $c = 1$ in Eq. (**B.13**) and throw *Newton's law* into the coefficient-less form

$$F = ma \tag{B.14}$$

where force has a dimension MLt^2. Eq. (**B.14**) does not imply that force *is* in any physical sense a mass times an acceleration. The product of a mass times an acceleration is in fact not defined

in physical terms, and in any case, a force can exist without there being any mass or acceleration involved, as for example when we push on an immobile brick wall.[28] In contrast with the examples of area and speed, Eq. (**B.14**) does not in all cases provide a recipe for directly evaluating a force by making in situ measurements of base quantities. Instead, we have imparted to force the character of a derived quantity by making the force unit depend in a particular way on the units of mass and length.

[28] and the wall exerts an equal and opposite counter-force on us.

Quantities that are transferred into the derived category by choosing a unit motivated by a general physical law are called derived quantities of the second kind. Force is one example; heat and electric charge are also treated in this way in the SI system.

B.1.6 Systems of units

A system of units is defined by

i. a complete set of base quantities with their defining comparison and addition operations,

ii. the base units, and

iii. all relevant derived quantities, expressed in terms of their defining equations (e.g $V = \mathrm{d}y/\mathrm{d}x\,x\,t$), if they are of the first kind, or the forms of the physical laws that define their units,[29] if they are of the second kind.

[29] e.g. $F = ma$ defines the force unit in the SI system.

The set of derived quantities is open-ended; new ones may be introduced at will in any analysis.

> Systems of units are said to be of the same type if they differ only in the magnitudes of the base units.

In the SI system (Système International) there are six base quantities **Tbl.** B.3:

> length, time, mass, temperature, current, number of elementary particles, and luminous intensity. The units of length, time and mass are the meter (m), the second (s) and the kilogram (kg), respectively.

Force is made a derived quantity by writing *Newton's law* as $F = ma$.

The temperature in any system of units must be a **thermodynamic temperature**. Numbers read from an arbitrarily constructed thermometer scale can be used for establishing whether two temperatures are equal, but they are not numerical values of a physical quantity.[30] The thermodynamic (or absolute) temperature is, however, defined in terms of physical comparison and addition operations appropriate to a base quantity.[31] Measurements via a *Carnot engine* or equivalent device gives the ratio of two absolute temperatures; the temperature unit (i.e. the size of the "degree") must be chosen arbitrarily. The SI temperature unit is the kelvin (K), which is defined as the fraction 1/273,16 of the thermodynamic temperature of the triple point of water.

[30] Is 2 C twice as hot as 1 C? Is 0 C zero temperature in the sense of there being an absence of temperature? For that matter, if temperature is defined in terms of a thermometer with an arbitrarily marked temperature scale, is there any reason why heat should flow from a higher to a lower temperature?

[31] The ratio of the numerical values of two physical thermodynamic temperatures T_1 and T_2 can for example be defined in terms of the (physically measurable) efficiency η_{12} of a Carnot engine which operates between heat reservoirs at the two temperatures. If T_l is the temperature of the reservoir into which heat flows, $T_1/T_2 = 1 - \eta_{12}$, where η_{12} is a pure number. This provides a physical operation for determining the ratio of the numerical

Table B.1
The SI system of Units

Quantity	Symbol	SI Name	SI Symbol
Length	L	meter	m
Time	t	second	s
Mass	M	kilogram	kg
Temperature	T	kelvin	K
Current	I	ampere	A
Number of elementary particles	-	mole	mol
Luminous Intensity	-	candela	cd

Table B.2
An Incomplete set of derived quantities

Quantity	Defining Law	Dimension	Symbol	Name
Area	$A = \int \mathrm{d}x\,\mathrm{d}y$	L^2	m^2	–
Volume	$V = \int \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$	L^3	m^3	–
Frequency	$f = 1/T$	t^{-1}	s^{-1}	Hz
Acceleration	$a = \mathrm{d}^2y/\mathrm{d}x^2xt$	Lt^{-2}	s^{-2}	–
Density	$\rho = M/V$	ML^{-3}	$\mathrm{kg}\cdot\mathrm{m}^{-3}$	–
Force	$F = Ma$	MLt^{-2}	$\mathrm{kg}\cdot\mathrm{m}\cdot\mathrm{s}^2$	N
Stress/Pressure	$p = F/A$	$ML^{-1}t^{-2}$	$\mathrm{N}\cdot\mathrm{m}^{-2} = \mathrm{kg}\cdot\mathrm{m}^{-1}\cdot\mathrm{s}^{-2}$	Pa
Work/Energy	$W = \int F\,\mathrm{d}x$	ML^2t^{-2}	$\mathrm{N}\cdot\mathrm{m} = \mathrm{kg}\cdot\mathrm{m}^2\cdot\mathrm{s}^{-2}$	J
Torque	$T = Fl$	ML^2t^{-2}	$\mathrm{N}\cdot\mathrm{m} = \mathrm{kg}\cdot\mathrm{m}^2\cdot\mathrm{s}^{-2}$	–
Power	$\mathrm{d}y/\mathrm{d}xWt$	ML^2t^{-3}	$\mathrm{J}\cdot\mathrm{s}^{-1} = \mathrm{kg}\cdot\mathrm{m}^2\cdot\mathrm{s}^{-3}$	W
Charge	$Q = \int I\,\mathrm{d}t$	It	$\mathrm{A}\cdot\mathrm{s}$	C

Current In SI the electric current is adopted as a **base quantity** and *charge* is thrown into the derived category using $Q = \int I\,\mathrm{d}t$, as are all other electric and magnetic quantities. The unit for current is the ampere (A), which is defined as the current which, when passed through each of two **infinite**, parallel conductors in vacuum one meter apart, will produce a force of 2×10^{-7} N per unit length on each conductor.[32]

Mole The mole unit is retained in SI as an alternative way of specifying number of things: instead of counting things one by one, one counts them in lots of $6{,}02 \times 10^{23}$ which is also known as Avagadro's number. One must, however, specify what things one is counting.[33]

[32] The conventional factor 2×10^7 is introduced for convenience and historical reasons, and appears in the SI version of *Ampere's law*.

[33] e.g. "one mole of atoms, of molecules, of tennis balls", or whatever.

The SI unit of mole is in fact the old **gram mole**.

Luminous Intensity It does not refer to radiant energy flow per unit solid angle as such, which could be measured in watts per steradian ($W \cdot sr^{-1}$), but only to that portion of it to which "the human eye"[34] is sensitive. At a wavelength of 555 nm, where the human eye is most sensitive to light, one candela is equivalent to $1{,}46 \times 10^{-3}\, W \cdot sr^{-1}$.

[34] as defined by a standard response curve.

B.4 Information

Additional Base (or Derived) Units

Also included sometimes among the base quantities are two dimensionless quantities, plane angle and solid angle, which are measured in radians and steradians, respectively. We consider them derived quantities because, though dimensionless, they are defined in terms of operations involving length, much like area is defined in terms of length operations.

We note again that the dimension of a derived quantity depends on the choice of system of units, which is under the control of the observer and has nothing to do with the quantity's intrinsic nature.

Quantities with quite different physical meaning, like **work** and **torque**, can have the same dimension.

The effect of the system of units on the dimensions of quantities is illustrated in table 2.2, which shows the dimensions of some mechanical quantities in three different types of systems of units. Systems of units are of the same type if they have base units of the same kind but different magnitude, and the derived quantities and the basic physical laws have the same mathematical forms, differing only in the values of the physical constants that appear in them.

In the first type of system illustrated in the table, length, time and mass are base quantities and the force unit is measured in terms of mass and acceleration via Newton's law in the form $F = ma$. The SI and the cgs (centimeter, gram, second) systems are in this category. In the second type, length, time and force are base, and mass is measured via $F = ma$. An example is the British Gravitational System, in which the base units are the foot, the second and the pound-force (lbf). The pound-force is the force exerted by standard gravity (32.2 ft/sš) on a standard mass sample, the pound-mass (hence the term "gravitational system"). Mass in this Table 2.2: Dimensions of some mechanical quantities in different types of systems of units

Table B.3 **Dimensions of some mechanical quantities in different types of systems of units**

Base Quantities and Dimensions	Type 1	Type 2	Type 3
-	L, M, t	L, F, t	L, M, F, t
Velocity $= dy/dx xt$	Lt^{-1}	Lt^{-1}	Lt^{-1}
Acceleration $= d^2y/dx^2 xt$	Lt^{-2}	Lt^{-2}	Lt^{-2}
Mass	M	Ft^2L^{-1}	M
Area $= \int dx\, dy$	L^2	L^2	L^2

Dimensions of some derived quantities Lt^{-1} Lt⁻¹ Lt^{-1} velocity = dx/dt $acceleration = d^2x/dt^2$ Lt^{-2} Lt^{-2} Lt-2 Ft^2L^{-1} mass M M area=fdxdy L^2 L^2 L^2 MLt^{-2} F F force $FM^{-1}L^{-1}t^2$ c=F/ma in Newton's law $c = 1$ $c = 1$

$$ML^2t^{-2}$$

$work = \int Fdx$ FL FL $ML^{-1}t^{-2}$ FL⁻² FL^{-2} $stress = F/A$

$$ML^{-1}t^{-1}$$

$FL^{-2}t$ $FL^{-2}t$ viscosity=$\tau/(\partial u/\partial y)$ system is a derived quantity with dimension Ft²L⁻¹ and a unit (the "slug") which turns out to be $g = 32.2$ times as large as the pound-mass. In the third type, length, time, mass, and force are taken as base quantities, and Newton's law reads $F = cma$, where c is a physical constant with dimension Ft²M⁻¹L⁻¹. The British Engineering System is an example. In this system 32 the base units taken are the foot, the second, the pound-mass (lbm), and the pound-force (lbf), and the constant in Newton's law has the value $c = 1/32.2$ lbf s^2 lbm^{-1} ft^{-1}.

Table 2.2 illustrates the fact that, while an actual physical quantity like force is the same regardless of the (arbitrary) choice of the system of units, its dimension depends on that choice. What is more, depending on how derived quantities are defined, a given physical law may contain a dimensional physical constant the value of which must be specified (as in F=cma), or it may contain no physical constant (as in $F = ma$).

An interesting point to note is that only a few of the available universal laws are usually "used up" to make base quantities into derived ones of the second kind. There are many laws left with universal dimensional physical constants that could in principle be set equal to unity: the gravitational constant G, Planck's constant h, Boltzmann's constant k_B, the speed of light in vacuum c, etc.

Baseless Unit System B.5 Information

This leaves us with some interesting possibilities. For example, it is possible to define systems of units that have no base quantities at all (see Bridgman, 1931). In such systems all units of measurement are related to some of the universal constants that describe our universe. In effect, there exist in the universe **natural** units for all base quantities, based on universal constants such as the speed of light, the quantum of energy, etc. Unfortunately, the choice of such **natural** systems of units turns out to be far from unique, which renders futile any attempt to endow any one of them with unique significance.

B.1.7 Recapitulation

1. A base quantity is a property that is defined in physical terms by two operations: a comparison operation, and an addition operation. The comparison operation is a physical procedure for establishing whether two samples of the quantity are equal or unequal; the addition operation defines what is meant by the sum of two samples of that property. 2. Base quantities are properties for which the following concepts are defined in terms of physical operations: equality, addition,

subtraction, multiplication by a pure number, and division by a pure number. Not defined in terms of physical operations are: product, ratio, power, and logarithmic, exponential, trigonometric and other special functions of physical quantities. 33 3. A base quantity can be measured in terms of an arbitrarily chosen unit of its own kind and a numerical value. 4. A derived quantity of the first kind is a product of various powers of numerical values of base quantities. A derived quantity is defined in terms of numerical value (which depends on base unit size) and does not necessarily have a tangible physical representation. 5. The dimension of any physical quantity, whether base or derived, is a formula that defines how the numerical value of the quantity changes when the base unit sizes are changed. The dimension of a quantity does not by itself provide any information on the quantity's intrinsic nature. The same quantity (e.g. force) may have different dimensions in different systems of units, and quantities that are clearly physically different (e.g. work and torque) may have the same dimension. 6. Relationships between physical quantities may be represented by mathematical relationships between their numerical values. A mathematical equation that correctly describes a physical relationship between quantities is dimensionally homogeneous (see section 2.5). Such equations remain valid when base unit sizes are changed arbitrarily. 7. The categorization of physical quantities as either base or derived is to some extent arbitrary. If a particular base quantity turns out to be uniquely related to some other base quantities via some universal law, then we can, if we so desire, use the law to redefine that quantity as a derived quantity of the second kind whose magnitude depends on the units chosen for the others. All base quantities that are transformed into derived quantities in this way retain their original physical identities (i.e. their comparison and addition operations), but their numerical values are measured in terms of the remaining base quantities, either directly via a defining equation or indirectly by using a unit that is derivable from the remaining base units. 8. A system of units is defined by (a) the base quantities, (b) their units, and (c) the derived quantities, each with either its defining equation or the form of the physical law that has been used to cast the quantity into the derived category. Both the type and the number of base quantities are open to choice. 34

B.2 The Principle of Dimensional Analysis

This chapter introduces the procedure of dimensional analysis and describes Buckingham's[35] Π-theorem, which follows from it. We will start with laying down the procedure in general terms and defines the vocabulary followed by an example. Finally we will discuss the utility of dimensional analysis and some of the pitfalls and questions that arise in its application to real problems.



[35] **Edgar Buckingham** *(1867 - 1940)*

was an American physicist. He graduated from Harvard University with a B.Sc in physics in 1887. He did graduate work at Strasbourg and then studied under the chemist *Wilhelm Ostwald* at Leipzig, from which he was granted a Ph.D in 1893. He worked as a soil physicist at the USDA Bureau of Soils from 1902 to 1906. He worked at the (US) National Bureau of Standards (NIST) 19061937. His fields of expertise included soil physics, gas properties, acoustics, fluid mechanics, and blackbody radiation. He is also the originator of the Buckingham theorem in the field of dimensional analysis.

B.2.1 The steps of dimensional analysis and Buckingham's Π-theorem

The premise of dimensional analysis is that the form of any physically significant equation must be such that the relationship between the actual physical quantities remains valid **independent the magnitudes of the base units**.

> Dimensional analysis derives the logical consequences of this premise.

Suppose we are interested in some particular physical quantity Q_0 that is a **dependent variable** in a well defined physical process or event. By this we mean that, once all the quantities that define the particular process or event are specified, the value of Q_0 follows uniquely.

Step 1: The independent variables

The first and most important step in dimensional analysis is to identify a complete set of independent quantities $Q_1, \ldots, Q_n$ that determine the value of Q_0,

$$Q_0 = f(Q_1, Q_2, \ldots, Q_n). \tag{B.15}$$

A set $Q_1, \ldots, Q_n$ is **complete** if, once the values of the members are specified, no other quantity can affect the value of Q_0, and **independent** if the value of each member can be adjusted arbitrarily without affecting the value of any other member.

Starting with a correct set $Q_1, \ldots, Q_n$ is as important in dimensional analysis as it is in mathematical physics to start with the correct fundamental equations and boundary conditions.

> If the starting point is wrong, so is the answer.

The relationship expressed symbolically in Eq. (**B.15**) is the result of the physical laws which govern the phenomenon of interest. It is our premise that its form must be such that, once the values $Q_1, \ldots, Q_n$ are specified, the equality holds regardless of the sizes of the base units in terms of which the quantities are measured. The steps that follow derive the consequences of this premise.

Step 2: Dimensional considerations

Next we list the dimensions of the dependent variable Q_0 and the independent variables $Q_1, \ldots, Q_n$. As we have discussed previously, the dimension of a quantity depends on the type of system of units (see table 2.2), and we must specify at least the type the system of units before we do this.

For example, if we use a system of **Type 1** in table 2.2 and are dealing with a purely mechanical problem, all quantities have dimensions of the form

$$[Q_i] = L^{l_i} M^{m_i} t^{\tau_i} \tag{B.16}$$

where the exponents l_i, m_i and τ_i are *dimensionless numbers* following from each quantity's definition.

We now pick from the complete set of physically independent variables $Q_1, \ldots, Q_n$ a **complete**, **dimensionally independent subset** $Q_1, \ldots, Q_k$ $(k \leqslant n)$, and express the dimension of each of the remaining independent variables $Q_{k+1}, \ldots, Q_n$ and the dependent variable Q_0 as a product of powers of $Q_1, \ldots, Q_k$.

All physical quantities have dimensions which can be expressed as products of powers of the set of base dimensions. Alternatively, it is possible to express the dimension of one quantity as a product of powers of the dimensions of other quantities which are not necessarily base quantities. A subset $Q_1, \ldots, Q_k$ of the set $Q_1, \ldots, Q_n$ is **dimensionally independent** if none of its members has a dimension that can be expressed in terms of the dimensions of the remaining members, and **complete** if the dimensions of all the remaining quantities $Q_{k+1}, \ldots, Q_n$ of the full set **can** be expressed in terms of the dimensions of the subset $Q_1, \ldots, Q_k$.

Since Eq. (**B.15**) is dimensionally homogeneous, the dimension of the dependent variable Q_0 is also expressible in terms of the dimensions of $Q_1, \ldots, Q_k$

The dimensionally independent subset $Q_1, \ldots, Q_k$ is picked by trial and error. Its members may be picked in different ways, but the number k of dimensionally independent quantities in the full set $Q_1, \ldots, Q_n$ is unique to the set, and cannot exceed the number of base dimensions which appear in the dimensions the quantities in that set.

> For example, if the dimensions of $Q_1, \ldots, Q_n$ involve only length, mass, and time, then $k \leqslant 3$.

Having chosen a complete, dimensionally independent subset $Q_1, \ldots, Q_k$, we express the dimensions of Q_0 and the remaining quantities $Q_{k+1}, \ldots, Q_n$ in terms of the dimensions of $Q_1, \ldots, Q_k$. These will have the form

$$\left[Q_i\right] = \left[Q_1^{N_{i1}} Q_2^{N_{i2}}, \ldots, Q_k^{N_{ik}}\right] \tag{B.17}$$

if $i > k$ or $i = 0$. The exponents N_{ij} are dimensionless real numbers and can in most cases be found quickly by inspection, although a formal algebraic method can be used.

The formal procedure can be illustrated with an example where length, mass and time are the only base quantities, in which case all dimensions have the form of Eq. (**B.16**). Let us take Q_1, Q_2, and

Q_3 as the complete dimensionally independent subset. Equating the dimension given by Eq. (**B.16**) with that of Eq. (**B.17**), we obtain three (**3**) equations:

$$l_i = \sum_{i=1}^{3} N_{ij} l_j, \qquad m_i = \sum_{j=1}^{3} N_{ij} m_j, \qquad \text{and} \qquad t_i = \sum_{i=1}^{3} N_{ij} t_j$$

which can be solved for the three (**3**) unknowns N_{il}, N_{i2}, and N_{i3}.

Step 3: Dimensionless variables

We now define dimensionless forms of the $n-k$ remaining independent variables by dividing each one with the product of powers of $Q_1, \ldots, Q_k$ which has the same dimension,

$$\Pi_i = \frac{Q_{k+i}}{Q_i^{N_{ki1}} Q_2^{N_{ki2}}, \ldots, Q_i^{N_{kik}}},$$

where $i = 1, 2, \ldots, n-k$, and a dimensionless form of the dependent variable Q_0,

$$\Pi_0 = \frac{Q_0}{Q_1^{N_{01}} Q_2^{N_{02}} \ldots Q_t^{N_{0k}}}.$$

Step 4: The End Game and Buckingham's Π-theorem

An alternative form of Eq. (**B.15**) is

$$\Pi_0 = f(Q_1, Q_2, \ldots, Q_n; \Pi_1, \Pi_2, \ldots, \Pi_{n-k}) \tag{B.18}$$

in which all quantities are dimensionless except $Q_1, \ldots, Q_k$. The values of the dimensionless quantities are independent of the sizes of the base units. The values of $Q_1, \ldots, Q_k$, on the other hand, do depend on base unit size. They cannot be put into dimensionless form since they are ,by definition, dimensionally independent of each other.

From the principle that any physically meaningful equation must be dimensionally homogeneous, that is, valid independent of the sizes of the base units, it follows that $Q_1, \ldots, Q_k$ must in fact be absent from Eq. (**B.18**), that is,

$$\Pi_0 = f(\Pi_1, \Pi_2, ..., \Pi_{n-k})$$

This equation is the final result of the dimensional analysis, and contains Buckingham's Π-Theorem:

Buckingham's Π-Theorem-I B.2 Theorem

When a complete relationship between dimensional physical quantities is expressed in dimensionless form, the number of independent quantities that appear in it is reduced from the original n to $n-k$, where k is the maximum number of the original n that are dimensionally independent.

The theorem derives its name from Buckingham's use of the symbol Π for the dimensionless variables in his original 1914 paper. The Π-theorem tells us that, because all complete physical equations must be dimensionally homogeneous, a restatement of any such equation in an appropriate dimensionless form will reduce the number of independent quantities in the problem by k.

This can simplify the problem enormously, as will be evident from the example that follows.

The Π-theorem itself merely tells us the **number** of dimensionless quantities that affect the value of a particular dimensionless dependent variable. It does not tell us the forms of the dimensionless variables. That has to be discovered in the third and fourth steps described above. Nor does the Π-theorem, or for that matter dimensional analysis as such, say anything about the form of the functional relationship expressed by Eq. (**B.18**).

That form has to be discovered by **experimentation** or by **solving the problem theoretically**.

B.2.2 An example: Deformation of an elastic ball striking a wall

Suppose we wish to investigate the *deformation* which occurs in elastic balls when they impact on a wall. We might be interested, for example, in finding out what determines the diameter d of the circular imprint left on the wall after a freshly dyed ball has rebounded from it (figure 3.1).

Step 1: The independent variables

The first step is to identify a complete set of **independent quantities** which determine the imprint radius d. We begin by specifying the problem more clearly.

We agree to restrict our attention to initially spherical, homogeneous balls made of perfectly elastic material, to impacts at perpendicular to the wall, and to walls that are perfectly smooth and flat and so stiff and heavy that they do not deform or move during the impact process.

We also agree to adopt a system of units of **Type 1** in table 2.2, like the SI system.

The numerical value of a dependent variable like d will be depend on the values of all quantities which distinguish one impact event from another. Experience suggests that these should include at least the following:

> the ball's diameter D and velocity V just prior to contact (the initial conditions) and its mass m.

The ball's intrinsic material properties will also play a role. Our theoretical understanding of solid mechanics tells us that the quasi-static response of a perfectly elastic material is characterized by two material properties,

- the modulus of elasticity E and Poisson's ratio γ, and that the inertial effects which inevitably come into play during collision and rebound will also depend on the material's density ρ.

The properties of the wall are irrelevant if it is indeed **perfectly rigid**, as we will assume so. We know, however, by thinking of how the problem would have to be set up as a theoretical one, that

the answer for the numerical value d will also depend on the values of all universal constants that appear in the physical laws that control the ball's impact dynamics.

In this case the process is governed by *Newton*'s law of motion and the law of mass conservation. Having chosen a system of units of **Type 1** in table 2.2, we know that *Newton*'s law has the form $F = ma$ and contains no universal constants[36]. Nor are there any physical constants in the law of mass conservation.

[36] If, on the other hand, we had decided to use a system of **Type 3** in Table 2.1, the value of the dimensional constant $c = F/ma$ for the chosen system of units would have to be specified, and would affect the value of d.

We seem to arrive at the conclusion that d depends on six (**6**) quantities:

D, V, m, E, γ, and ρ.

This is a complete set, as required, but not an **independent** set: once the ball's mass and diameter are specified, its density follows. We must therefore exclude either the density or the mass.[37] We conclude that the following relationship expresses the impact diameter in terms of a complete set of independent variables:

$$d = d(V, \rho, D, E, \gamma). \tag{B.19}$$

[37] Other quantities like V^2, DE^{12}, etcetera, all involving quantities that affect the value of d, are excluded for the same reason. They are not independent of the quantities already included.

Note that the choice of a **complete**, **independent** set for a specified problem is **not unique** except for the number n of its members.[38] One could just as well have chosen V^2, ρ, D, E, and γ, say, or V, m, D, E, γ.

[38] $n = 5$ in this case.

It should also be noted that further assumptions have been taken for granted in Eq. (**B.19**). We have presumed, for example, that the ball's motion is unaffected by the properties of the fluid through which it approaches the wall,[39] and that gravitational effects play a negligible role.

[39] Which is certainly OK if the ball moves through vacuum and a good approximation in air, but may not apply to small balls in viscous liquids.

Step 2: Dimensional considerations

In the type of system of units we have adopted in *Step 1*, the dimensions of the quantities in Eq. (**B.19**) are:

$$\text{Independent} \quad \left[V\right] = Lt^{-1} \qquad \left[\rho\right] = ML^{-3} \qquad \left[D\right] = L$$
$$\left[E\right] = ML^{-1}t^{-2} \qquad \left[\gamma\right] = 1$$

$$\text{Dependent} \quad \left[d\right] = L$$

Inspection of the above shows that the three (**3**) quantities V, ρ, and D, for example, comprise a complete, dimensionally independent subset of the five independent variables.

The dimension of any one of these three cannot be made up of the dimensions of the other two. The dimensions of the remaining independent variables E and γ and the dependent variable d can, however, be made up of those of V, ρ and D as follows:

$[E] = ML^{-1}t^{-2} = (ML^{-3})(Lt)^2 = [\rho V^2]$ independent:

$$[\gamma] = 1 \tag{B.20}$$

$[d] = L = [D]$ dependent:

We have written down these results very simply by inspection. Note again that the dimension of a dimensionless quantity like γ is unity, the factor by which dimensionless numbers change when the sizes of the base units are changed.

Step 3: Dimensionless Similarity Parameters

We non-dimensionalize the remaining independent variables E and γ and the dependent variable d by dividing them by ρV^2, D, and unity, respectively, as suggested by Eq. (**B.20**):

$$\begin{aligned} \text{Independent} \quad \Pi_1 &= \frac{ED^3}{mV^2} \quad \Pi_2 = \gamma \\ \text{Dependent} \quad \Pi_0 &= \frac{d}{D} \end{aligned}$$

Step 4: The End Game

Using the logic that led to Buckingham's Π-theorem, we now conclude that

$$\Pi_o = f(\Pi_1, \Pi_2), \qquad \text{or} \qquad \frac{d}{D} = f\left(\frac{E}{\rho V^2}, \gamma\right). \tag{B.21}$$

The number of independent variables has been reduced from the original $n = 5$ that define the problem to $n - k = 2$.

B.2.3 On the Utility and Difficulties of Dimensional Analysis

Similarity

Dimensional analysis provides a **similarity law** for the phenomenon under consideration. Similarity in this context implies a certain equivalence between two physical phenomena that are actually different. The collisions of two (**2**) different elastic spheres 1 and 2 with a rigid wall, each with its own values of V, ρ, D, E, and γ, may appear to be quite different.

However, under particular conditions where the parameters of the two events are such that Π_1 and Π_2 have the same values, that is, where

$$\frac{E_2}{\rho_2 V_2^2} = \frac{E_1}{\rho_1 V_1^2} \qquad \text{and} \qquad \gamma_2 = \gamma_1, \tag{B.22}$$

where Eq. (**B.21**) informs us that Π_0 has the same value in both cases, that is,

$$\frac{d_2}{D_2} = \frac{d_1}{D_1}. \tag{B.23}$$

When the relationships in Eq. (**B.22**) apply, the two dynamic events are **similar** in the sense of Eq. (**B.23**).

Out-of-scale modeling

Scale modeling deals with the following question:

> If we want to learn something about the performance of a full-scale system 1 by testing a geometrically similar small-scale system model 2,[40] at what conditions should we test the model, and how should we obtain the full-scale performance from measurements at the small scale?

[40] or vice-versa, if the system of interest inaccessibly small.

Dimensional analysis provides the answer.

Suppose we need to know the deformation diameter of a huge, soft rubber ball with a diameter D_1 of 5 m and properties E_1, ρ_1, and γ_1, as it hits the pavement with a speed V_2, of $10\,\mathrm{m}\cdot\mathrm{s}^{-1}$, but are unable to compute it from basic principles. In that case we need only perform one small-scale test with a model 2 of diameter D_2, selecting its properties and test conditions such that Eq. (**B.22**) are satisfied, and measure its imprint diameter d_2. The full-scale value d_1 of the big ball's imprint diameter at its **design conditions** can then be obtained from Eq. (**B.23**).

Dimensional Analysis Reduces the Number of Variables and Minimizes Work

Dimensional analysis reduces the number of variables which must be specified to describe an event, which often leads to an enormous simplification.

In our example of the impacting ball the answer depends on five (**5**) independent variables shown in Eq. (**B.19**), that is, a particular event may be represented as a distribution of d defined in a five-dimensional space of **independent variables**.

Suppose we set out to obtain the answer in a certain region, a certain volume, of this variable-space, by either computation or experimentation, and decide that 10 data points will be required in each variable, with the other four (**4**) being held **constant**. This would require obtaining 10^5 data points. Dimensional analysis, however, shows us that in dimensionless form the answer depends only on two similarity parameters.

This two-dimensional space can be explored with similar resolution with only 100×10^0 data points, that is, with 0.1% of the effort.

Table 3.1 shows some **experimental data** for impacts with balls of three materials and various values of impact velocity.

Table 3.1: Computed "Experimental data" d Material Symbol (m s⁻¹) ρV^2 D (MPa) (kgm^{-3}) (Fig. 3.2) ≙ 3.66E+05 3960 0.22 50133 Alumina 43 0.150 ◅ 3.66E+05 0.170 3960 59 0.22 26937 ▻ 3.66E+05 3960 77 0.22 15511 0.190 6.90E+04 0.250 Aluminum 2705 80 0.33 3973 0 6.90E+04 0.300 2705 126 0.33 1608 ◊ 6.90E+04 2705 345 0.33 215 0.450

Rubber 3.93E+00 1060 5 0.47 127 0.500 3.93E+00 1060 7 79 0.550 * 0.47 * 3.93E+00 12 0.700 1060 0.47 24 1 Computation
+ = 0.5
Œ = 0.30 <u>d</u> 0.5 Œ Œ Œ Œ Œ Œ Œ Œ

Œ Œ Œ Œ Œ Œ Œ Œ Œ Œ Œ 0 10⁵ 10² 10³ 104 10 E
V² Figure 3.2: Plot of "experimental data" in dimensionless form. 45

Fig. 3.2 shows these data as a plot of d/D vs. $E/\rho V^2$ with γ as a parameter. The simulated "experimental scatter" in Fig. 3.2 actually results from the coarseness of the computational grid. Disregarding the scatter, all points fall essentially on a common curve, as predicted by dimensional analysis. The influence of the Poisson's ratio turns out to be virtually negligible, given the scatter, and all the data may be curve-fitted with an equation of the form $d/D = f(E/\rho V^2)$.

An Incomplete Set of Independent Quantities Can Destroy the Analysis

Assuming competence on the part of the analyst, the correctness of the dimensional analysis will depend entirely on whether a complete set of independent quantities $Q_1, \ldots, Q_n$ is in fact properly identified in *Step 1*. Any complete set will provide correct results. If, however, the analysis is based on a set which omits even one independent quantity that affects the value of Q_o, dimensional analysis will give **erroneous results**.

Suppose in our example we had omitted the sphere's modulus of elasticity E in Eq. (**B.19**). Instead of Eq. (**B.21**), we would then have obtained the absurd result

$$\frac{d}{D} = f(\gamma),$$

which implies that the maximum deformation depends on the ball's Poisson ratio, but is independent of its elasticity, mass and approach velocity.

> This single error of omission is clearly fatal to the analysis.

Superfluous Independent Quantities Complicate the Result Unnecessarily

Errors on the side of excess have a less traumatic effect. Over-specification of independent variables does not destroy the analysis, but robs it of its power.

> For every superfluous independent quantity included in the set, there will be in the final dimensionless relationship a superfluous dimensionless similarity parameter.

Suppose we argue quite reasonably that the ball's deformation upon impact will in general also depend on the local gravitational acceleration g.[41] This would change Eq. (**B.21**) to

$$\frac{d}{D} = f\left(\frac{E}{\rho V^2}, \gamma, \frac{gD}{V^2}\right) \qquad \textbf{(B.24)}$$

where gD/V^2 is a dimensionless gravity. Under conditions where the deformation is in fact insensitive to gravity, as we implicitly assumed earlier, Eq. (**B.24**) is "wrong" only in the sense that it suggests a dependence on g that is not noticeably there, and thus unnecessarily complicates our thinking. If by experimentation or computation we eventually discover that there exists a broad range of conditions

[41] Which we assume to be in the direction into the wall on which the impact occurs.

where the similarity parameter involving g has in fact no measurable effect on d/D, and that the conditions of interest fall into this range,[42] we omit the parameter involving g and arrive at the same simpler conclusion as before, but only after due payment in effort for our lack of insight.

[42] To prove this with earthbound experimentation we would have to demonstrate that d/D remains constant when m, D and/or V are changed so as to vary gD/V^2 while keeping ED^3/mV_2 constant.

On the importance of simplifying assumptions

The previous example illustrates an important point about most problems in dimensional analysis:

> Completeness in the set of independent variables is not an **absolute matter**, but depends on how we choose to circumscribe the problem.

It is quite conceivable, for example, that there are *balls* whose deformation upon impact with a wall is gravity-dependent. This would be the case for balls with such low coefficient of elasticity, or large diameter, or low impact velocity, that their deformation at rest on a wall in the gravitational field would be significant compared with their deformation upon impact. When we say **the deformation does not depend on gravity**, we imply that we know with some confidence that there exist conditions where this is true, and choose to confine our attention to those conditions.

Balance within Chaos — B.6 Information

If dimensional analysis depended on a truly complete identification of the independent variables that specify a given physical event, we would in most cases be reduced to impotence. Those familiar with chaos theory and the many-body systems can sympathize with the view that the wind at a certain street corner in Innsbruck, say, may in some measure have been affected by the wing-beat of a butterfly in Bratislava a month earlier. This kind of fastidiousness elicits a shrug from the pragmatist, who proceeds with a problem by saying:

"Based on my experience as a scientist and engineer, I argue that this event should be controlled by the following complete set of independent variables. Assuming tentatively that I am correct, what does dimensional analysis tell me?"

On choosing a complete set of independent variables

Given what has been said above, how does one go about choosing a complete set of independent variables that define a particular problem?

If we know the mathematical forms of all the equations and boundary (and initial) conditions that completely specify a particular type of process or event, one can deduce from them a complete set of independent parameters that define the event. This we do by simply examining all the equations and listing all the quantities whose values would have to be specified to define a particular event through all time. The set may include position and time, if the variable of interest depends on them, universal physical constants (e.g. gravitational constant, universal gas constant R, etc.), material properties (e.g. density, E, etc.) and all other quantities that appear not only in the equations but

also in the boundary and initial conditions that determine the answer to the particular problem at hand. The last point is not as straightforward as it sounds: the question of what constitutes a well-posed mathematical problem is often difficult to answer in non-linear systems.

If the equations and boundary conditions are not well known, as for example when one is trying to use dimensional analysis to help correlate experimental data for a complex phenomenon that is not well understood, one has to proceed by trial and error based on an (educated) guess about the physics of the problem at hand. Support for the analysis (and thus for the guess about the physics) may be obtained à posteriori by showing that all experimental data can indeed be accurately correlated in the dimensionless form suggested by the analysis. There should be sufficient data to be able to show that one has neither missed an important quantity nor included one that is irrelevant.

Result Independence from Chosen Dimensionally Independent Subset

Suppose we had chosen the dimensionally independent subset V, E, and ρ instead of V, D and ρ. Non-dimensionalizing d and D with combinations of V, E and ρ, we might have obtained the result

$$\frac{d}{(\rho V^2/E)^{1/3}} = F\left(\frac{D}{(mV^2/E)^{1/3}}, \gamma\right)$$

This can, however, be rewritten as

$$\frac{d}{D} = \left(\frac{mV^2}{ED^3}\right)^{1/3} F\left(\left(\frac{ED^3}{mV^2}\right)^{1/3}, \gamma\right) = f\left(\frac{ED^3}{mV^2}, \gamma\right) \qquad \textbf{(B.25)}$$

where F and f are different functions of their arguments. Eq. (**B.25**) is of course identical to Eq. (**B.21**).

The Result is Independent of the Type of System of Units

The choice of system of units may affect the dimensions of physical quantities as well as the values of the physical constants that appear in the underlying physical laws.

What effect, if any, does this have on dimensional analysis?

Reason dictates there should be no effect on the **bottom line**, since the observer is free to choose or make up whatever system of units he wants, and his arbitrary choice should not affect the laws of physics.

Consider our example of the dyed ball, but viewed in terms of a system of units like the British Engineering System (type3 in table 2.2), where mass, length, time and force are taken as base units. In such a system Newton's law reads $F = cma$, where c is a physical constant with dimension $Ft^2m^{-1}L^{-1}$. This affects the very first step of the analysis.

Since the impact process is controlled by *Newton*'s law, which now contains the constant c, the value of which must be specified, we now have

$$d = d(V, D, E, m, \gamma, c).$$

The number of independent variable ($n = 6$) has increased by one. The dimensions of the variable are now:

$$\text{Independent} \quad \left[V\right] = Lt^{-1} \quad \left[D\right] = L \quad \left[E\right] = FL^{-2}$$
$$\left[m\right] = M \quad \left[\gamma\right] = 0 \quad \left[c\right] = Ft^2M^{-1}L^{-1}$$

$$\text{Dependent} \quad \left[d\right] = L$$

The quantities V, D, m and c comprise a convenient dimensionally independent subset. The number of this subset has also increased by one ($k = 4$). The dimensions of the remaining quantities can be expressed in terms of these four as

$$\left[d\right] = \left[D\right] \qquad \left[E\right] = \left[cV^2D^{-3}\right] \qquad \left[\gamma\right] = 0$$

and the final result of the analysis is

$$\frac{d}{D} = f\left(\frac{ED^3}{cmV^2}, \gamma\right) \tag{B.26}$$

This differs from our previous result, Eq. (**B.21**), only in that the physical constant c appears in the non-dimensionalization of E. Equations Eq. (**B.21**) and Eq. (**B.26**) are, however, functionally identical. Since $n - k = 2$ in both cases, both analyses imply that d/D depends on two dimensionless parameters, γ and a dimensionless E. It is just that in the second system of units, E must be non-dimensionalized with cmV^2/D^3 instead of mV^2/D^3, since the latter no longer has the same dimension as E. In short, the forms of some of the dimensional parameters may change with the system of units, but the physical content of analysis remains invariant.

This is, of course, as we expected; the choice of system of units is arbitrary, and should not affect the physical **bottom line**.

B.2.4 When Independent Quantities Have Fixed Values

Engineering practice often involves problems where some of the quantities which define the problem have the same fixed values in all the applications being considered.

As a simple example, suppose we are interested in determining the hydrodynamic drag force[43] D on a fully submerged, very long, neutrally buoyant cable being dragged behind a ship, and propose to do this by making small scale experiments in a water tunnel.

[43] A force acting opposite to the direction of motion of any object moving with respect to a surrounding fluid.

Basic fluid mechanics tells us that, barring surface roughness effects, the **drag force** should be completely determined by the cable's length L and diameter d, the ship's, or water's, velocity V, along with the water's density ρ and viscosity μ. Three of these five quantities are **dimensionally independent**, and dimensional analysis (or Buckingham's pi-theorem) tells us that an appropriately defined dimensionless drag is a function of $n - k = 2$ dimensionless similarity parameters.

One way of writing this relationship is

$$\frac{D}{\rho V^2 L^2} = f\left(\text{Re}, \frac{d}{L}\right) \qquad \text{where} \qquad Re = \frac{\rho V L}{\mu} \tag{B.27}$$

is a **Reynolds number** based on cable length and d/L is the cable's **fineness coefficient**, which defines its geometry. Eq. (**B.27**) is a general relationship for the cable-towing problem as stated.



[44] The plume from this candle flame transitions from laminar (low Reynolds number) to turbulent (high Reynolds number).

Reynolds Number — B.2 Definition

In **fluid dynamics**, the Reynolds number (Re) is a **dimensionless quantity** which helps to predict the fluid flow patterns in different situations by measuring the ratio between **inertial** and **viscous** forces.[44] [13] At low Reynolds numbers, flows tend to be dominated by **laminar** (sheet-like) flow, while at high Reynolds numbers, flows tend to be turbulent.

The turbulence results from differences in the fluid's speed and direction, which may sometimes intersect or even move counter to the overall direction of the flow, such as eddy currents. These eddy currents begin to churn the flow, using up energy in the process, which for liquids increases the chances of cavitation.[45]

[45] the phenomenon in which the static pressure of a liquid reduces to below the liquid's vapor pressure, leading to the formation of small vapor-filled cavities in the liquid.

We observe, however, that the density and viscosity have essentially the same values[46] in all the applications that are of interest. The number of quantities that actually vary from case to case are thus actually three, not five.

[46] those of sea water at its typical temperature.

The question arises, does this lead to a simplification of the similarity law expressed by Eq. (**B.27**), that is, to a reduction in the number of independent similarity parameters? Simply omitting the quantities which have fixed values and performing dimensional analysis on the rest cannot answer this question.[47] Dimensional analysis must be based on a **complete** set of independent quantities which determine the quantity of interest. All the quantities whose values determine the quantity of interest must be included, regardless of whether some of them happen to be the same values in the problems that are of interest.

[47] Were we to simply omit the density and viscosity in the present problem, for example, we would imply a relationship $D = f(L, h, V)$ which is not dimensionally homogeneous and therefore unacceptable. There is no way of writing the dimension of force in terms of just length and velocity.

As we have seen previously, omitting even one independent variable can **fatally damage the analysis**.

The general question is the following: What reduction, if any, can be obtained in the number of similarity parameters if a certain number of the independent quantities that define the problem always have the same fixed values? This can be answered by the following analysis.

Suppose we are interested in a quantity Q that is completely determined by the values of n independent quantities Q_i, of which n_F are held at fixed values in all the cases that concern us. Let the quantities that may vary be the first $(n - n_F)$ of Q_i, and designate the n_F quantities with fixed values by F_i:

$$Q = f(Q_1, Q_2, \ldots, Q_{n-n_r}; F_1, F_2, \ldots, F_{k_r}, F_{k_{r1}}, F_{k_{r2}}, \ldots, F_{n_r}) \tag{B.28}$$

Choose a complete, **dimensionally independent** subset of the set F_i. Let these be the first k_F of the fixed set, as indicated in Eq. (**B.28**). Using this subset, non-dimensionalize the remaining

$(n_F - k_F)$ fixed quantities and write the relationship Eq. (**B.28**) in the alternative form

$$Q = f(Q_1, Q_2, \ldots, Q_{n-n_n}; F_1, F_2, \ldots, F_{k_n}, F^*_{k_n}, F^*_{k_n}, \ldots, F^*_{n_n}) \qquad \textbf{(B.29)}$$

where the asterisks $(*)$ indicate dimensionless quantities involving only quantities with fixed values. These dimensionless quantities thus have the same fixed values in the cases that concern us.

For these cases, therefore, we can write Eq. (**B.29**) as:

$$Q = f(Q_1, Q_2, \ldots, Q_{n-n_c}; F_1, F_2, \ldots, F_{k_c}) \qquad \textbf{(B.30)}$$

The value of Q is therefore completely determined by a set of $n - n_F + k_F$ independent quantities consisting of those dependent quantities that are not fixed plus the dimensionally independent subset of the fixed quantities.

Now perform dimensional analysis on the relationship in Eq. (**B.30**). First, select from the set of $n - n_F + k_F$ independent quantities in Eq. (**B.30**) a complete, dimensionally independent subset of k quantities. Let this subset be the first k of the quantities. Since Eq. (**B.30**) contains all the variable independent quantities plus the dimensionally independent subset of the fixed, independent, quantities, the subset we thus obtain is also complete, dimensionally independent subset for the whole original set of n quantities.

According to Eq. (**B.30**), Q depends on $n-n_F+k_F$ independent variables, of which k are independent. Dimensional analysis therefore gives the result:

$$Q^* = f(\Pi_1, \Pi_2, ..., \Pi_N) \qquad \text{where} \qquad N = (n-k) - (n_E - k_E).$$

We have arrived at the following theorem:

Buckingham's Π-Theorem-II — B.3 Theorem

If a quantity Q is **completely determined** by a set of n independent quantities, of which k are dimensionally independent, and if n_F of these quantities have fixed values in all the cases being considered, a number k_F of these being dimensionally independent, then a suitable dimensionless Q will be completely determined by $(n-k) - (n_F - k_F)$ dimensionless similarity parameters.

In other words, the fact that a number n_F of the quantities always have fixed values reduces the number of independent similarity parameters by $(n_F - k_F)$. This theorem is a generalization of Buckingham's Π-theorem, and reduces to it when $n_F = 0$.

Returning to the cable-towing example, in which $n_F = 2$ and $k_F = 2$, we now see immediately that no gain in similarity[48] accrues from the fact that the viscosity and density are essentially the same in all the applications that interest us. The similarity law Eq. (**B.27**) cannot be simplified.

[48] no reduction in the number of similarity parameters.

Simplification occurs only when some of the fixed quantities are dimensionally dependent on the rest.

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Acronyms

A

AC Alternating Current

An electric current that periodically reverses direction and changes its magnitude continuously with time, in contrast to direct current (DC), which flows only in one direction.

B

BJT Bipolar Junction Transistor

Bipolar Junction Transistor

BJT Very Large System Integration

Very Large System Integration

C

CMOS Complimentary MOS

Complimentary MOS

D

DC Direct Current

One-directional flow of electric charge. Direct current may flow through a conductor such as a wire, but can also flow through semiconductors, insulators, or even through a vacuum as in electron or ion beams. The electric current flows in a constant direction, distinguishing it from alternating current (AC). A term formerly used for this type of current was galvanic current.

F

FET Field-Effect Transistor

Field-Effect Transistor

I

IC Integrated Circuit

A small electronic device combining numerous interconnected components, like transistors, resistors, and capacitors, onto a single semiconductor chip, usually silicon

L

LED Light-Emitting Diode

Light-Emitting Diode

M

MOS Metal-Oxide Semiconductor

Metal-Oxide Semiconductor

MOS-FET Metal-Oxide Semiconductor Field-Effect Transistor

In electronics, the metaloxidesemiconductor field-effect transistor (MOSFET, MOS-FET, MOS FET, or MOS transistor) is a type of field-effect transistor (FET), most commonly fabricated by the controlled oxidation of silicon. It has an insulated gate, the voltage of which determines the conductivity of the device. This ability to change conductivity with the amount of applied voltage can be used for amplifying or switching electronic signals. The term metalinsulatorsemiconductor field-effect transistor (MISFET) is almost synonymous with MOSFET. Another near-synonym is insulated-gate field-effect transistor (IGFET).

P

PCB Printed-Circuit Board

Printed-Circuit Board

R

RHS Right Hand Side

Right Hand Side



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